

16 □ VECTOR CALCULUS

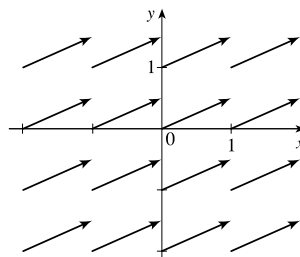
16.1 Vector Fields

1. $\mathbf{F}(x, y) = \mathbf{i} + \frac{1}{2}\mathbf{j}$

All vectors in this field are identical with length

$$\sqrt{1^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2} \text{ and parallel to } \langle 1, \frac{1}{2} \rangle, \text{ or,}$$

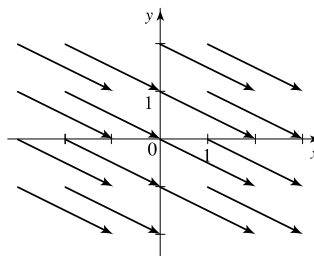
equivalently, $\langle 2, 1 \rangle$.



2. $\mathbf{F}(x, y) = 2\mathbf{i} - \mathbf{j}$

All vectors in this field are identical with length

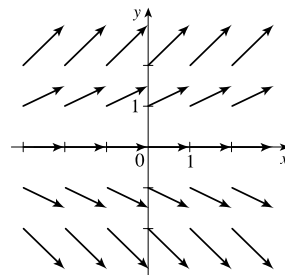
$$\sqrt{2^2 + (-1)^2} = \sqrt{5} \text{ and parallel to } \langle 2, -1 \rangle.$$



3. $\mathbf{F}(x, y) = \mathbf{i} + \frac{1}{2}y\mathbf{j}$

The length of the vector $\mathbf{i} + \frac{1}{2}y\mathbf{j}$ is $\sqrt{1 + \frac{1}{4}y^2}$. Vectors

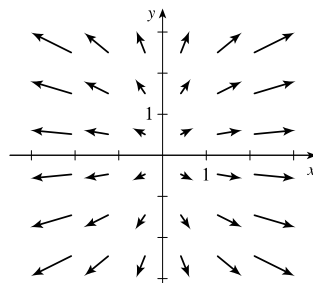
along the line $y = 0$ are horizontal with length 1.



4. $\mathbf{F}(x, y) = x\mathbf{i} + \frac{1}{2}y\mathbf{j}$

The length of the vector $x\mathbf{i} + \frac{1}{2}y\mathbf{j}$ is $\sqrt{x^2 + \frac{1}{4}y^2}$.

Vectors point roughly away from the origin and vectors farther from the origin are longer.

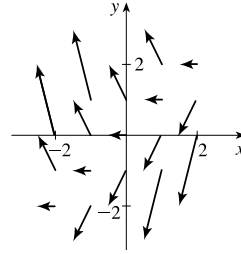


5. $\mathbf{F}(x, y) = -\frac{1}{2}\mathbf{i} + (y - x)\mathbf{j}$

The length of the vector $-\frac{1}{2}\mathbf{i} + (y - x)\mathbf{j}$ is

$$\sqrt{\frac{1}{4} + (y - x)^2}.$$
 Vectors along the line $y = x$ are

horizontal with length $\frac{1}{2}$.



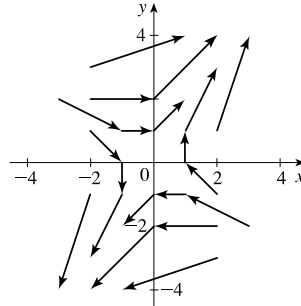
6. $\mathbf{F}(x, y) = y\mathbf{i} + (x + y)\mathbf{j}$

The length of the vector $y\mathbf{i} + (x + y)\mathbf{j}$ is

$$\sqrt{y^2 + (x + y)^2}.$$
 Vectors along the x -axis are vertical,

and vectors along the line $y = -x$ are horizontal with

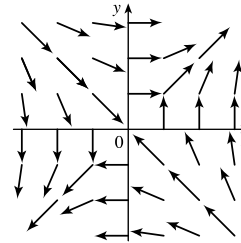
length $|y|$.



7. $\mathbf{F}(x, y) = \frac{y\mathbf{i} + x\mathbf{j}}{\sqrt{x^2 + y^2}}$

The length of the vector $\frac{y\mathbf{i} + x\mathbf{j}}{\sqrt{x^2 + y^2}}$ is

$$\sqrt{\frac{y^2}{x^2 + y^2} + \frac{x^2}{x^2 + y^2}} = 1.$$

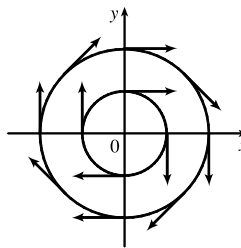


Vectors along the x -axis are vertical, and vectors along the y -axis are horizontal. In general, vectors in Q1 and QIII point away from the origin, whereas vectors in QII and QIV point toward the origin.

8. $\mathbf{F}(x, y) = \frac{y\mathbf{i} - x\mathbf{j}}{\sqrt{x^2 + y^2}}$

All the vectors $\mathbf{F}(x, y)$ are unit vectors tangent to circles

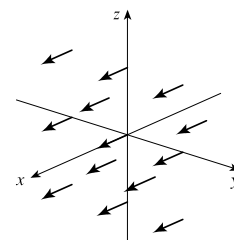
centered at the origin with radius $\sqrt{x^2 + y^2}$.



9. $\mathbf{F}(x, y, z) = \mathbf{i}$

All vectors in this field are identical, with length 1 and

pointing in the direction of the positive x -axis.

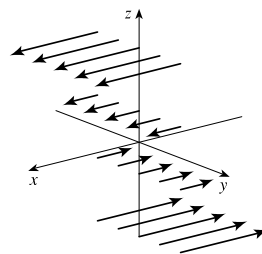


10. $\mathbf{F}(x, y, z) = z \mathbf{i}$

At each point (x, y, z) , $\mathbf{F}(x, y, z)$ is a vector of length $|z|$.

For $z > 0$, all point in the direction of the positive x -axis,

while for $z < 0$, all are in the direction of the negative x -axis. In each plane $z = k$, all the vectors are identical.

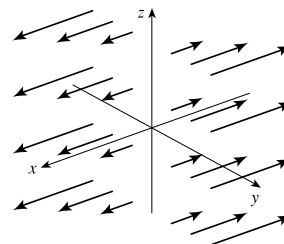


11. $\mathbf{F}(x, y, z) = -y \mathbf{i}$

At each point (x, y, z) , $\mathbf{F}(x, y, z)$ is a vector of length $|y|$.

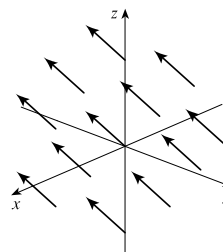
For $y > 0$, all point in the direction of the negative x -axis,

while for $y < 0$, all are in the direction of the positive x -axis. In each plane $y = k$, all the vectors are identical.



12. $\mathbf{F}(x, y, z) = \mathbf{i} + \mathbf{k}$

All vectors in this field have length $\sqrt{2}$ and point in the same direction, parallel to the xz -plane.



13. $\mathbf{F}(x, y) = \langle x, -y \rangle$ corresponds to graph IV. In the first quadrant all the vectors have positive x -components and negative y -components, in the second quadrant all vectors have negative x - and y -components, in the third quadrant all vectors have negative x -components and positive y -components, and in the fourth quadrant all vectors have positive x - and y -components. In addition, the vectors get shorter as we approach the origin.

14. $\mathbf{F}(x, y) = \langle y, x - y \rangle$ corresponds to graph V. All vectors in quadrants I and II have positive x -components while all vectors in quadrants III and IV have negative x -components. In addition, vectors along the line $y = x$ are horizontal, and vectors get shorter as we approach the origin.

15. $\mathbf{F}(x, y) = \langle y, y + 2 \rangle$ corresponds to graph I. As in Exercise 14, all vectors in quadrants I and II have positive x -components while all vectors in quadrants III and IV have negative x -components. Vectors along the line $y = -2$ are horizontal, and the vectors are independent of x (vectors along horizontal lines are identical).

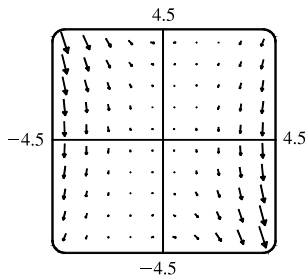
16. $\mathbf{F}(x, y) = \langle y, 2x \rangle$ corresponds to graph VI. In the first quadrant all the vectors have positive x - and y -components. In the second quadrant all vectors have positive x -components and negative y -components. In the third quadrant all vectors have negative x - and y -components. In the fourth quadrant all vectors have negative x -components and positive y -components.

17. $\mathbf{F}(x, y) = \langle \sin y, \cos x \rangle$ corresponds to graph III. Both the x - and y -components oscillate in all four quadrants.

18. $\mathbf{F}(x, y) = \langle \cos(x + y), x \rangle$ corresponds to graph II. All vectors in quadrants I and IV have positive y -components while all vectors in quadrants II and III have negative y -components. Also, the y -components of vectors along any vertical line remain constant while the x -component oscillates.

19. $\mathbf{F}(x, y, z) = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ corresponds to graph IV, since all vectors have identical length and direction.
20. $\mathbf{F}(x, y, z) = \mathbf{i} + 2\mathbf{j} + z\mathbf{k}$ corresponds to graph I, since the horizontal vector components remain constant, but the vectors above the xy -plane point generally upward while the vectors below the xy -plane point generally downward.
21. $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + 3\mathbf{k}$ corresponds to graph III; the projection of each vector onto the xy -plane is $x\mathbf{i} + y\mathbf{j}$, which points away from the origin, and the vectors point generally upward because their z -components are all 3.
22. $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ corresponds to graph II; each vector $\mathbf{F}(x, y, z)$ has the same length and direction as the position vector of the point (x, y, z) , and therefore the vectors all point directly away from the origin.

23.



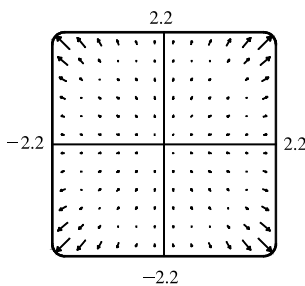
$$\mathbf{F}(x, y) = (y^2 - 2xy)\mathbf{i} + (3xy - 6x^2)\mathbf{j}.$$

The vector field seems to have very short vectors near the line $y = 2x$.

For $\mathbf{F}(x, y) = \langle 0, 0 \rangle$, we must have $y^2 - 2xy = 0$ and $3xy - 6x^2 = 0$.

The first equation holds if $y = 0$ or $y = 2x$, and the second holds if $x = 0$ or $y = 2x$. So both equations hold [and thus $\mathbf{F}(x, y) = \mathbf{0}$] along the line $y = 2x$.

24.



$$\mathbf{F}(\mathbf{x}) = (r^2 - 2r)\mathbf{x}, \text{ where } \mathbf{x} = \langle x, y \rangle \text{ and } r = |\mathbf{x}|.$$

From the graph, it appears that all of the vectors in the field lie on lines through the origin, and that the vectors have very small magnitudes near the circle $|\mathbf{x}| = 2$ and near the origin. Note that $\mathbf{F}(\mathbf{x}) = \mathbf{0} \Leftrightarrow r(r - 2) = 0 \Leftrightarrow r = 0 \text{ or } 2$, so as we suspected, $\mathbf{F}(\mathbf{x}) = \mathbf{0}$ for $|\mathbf{x}| = 2$ and for $|\mathbf{x}| = 0$. Note that where $r^2 - r < 0$, the vectors point towards the origin, and where $r^2 - r > 0$, they point away from the origin.

25. $f(x, y) = y \sin(xy) \Rightarrow$

$$\begin{aligned} \nabla f(x, y) &= f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j} = (y \cos(xy) \cdot y)\mathbf{i} + [y \cdot x \cos(xy) + \sin(xy) \cdot 1]\mathbf{j} \\ &= y^2 \cos(xy)\mathbf{i} + [xy \cos(xy) + \sin(xy)]\mathbf{j} \end{aligned}$$

26. $f(s, t) = \sqrt{2s + 3t} \Rightarrow$

$$\nabla f(s, t) = f_s(s, t)\mathbf{i} + f_t(s, t)\mathbf{j} = \left[\frac{1}{2}(2s + 3t)^{-1/2} \cdot 2 \right]\mathbf{i} + \left[\frac{1}{2}(2s + 3t)^{-1/2} \cdot 3 \right]\mathbf{j} = \frac{1}{\sqrt{2s + 3t}}\mathbf{i} + \frac{3}{2\sqrt{2s + 3t}}\mathbf{j}$$

27. $f(x, y, z) = \sqrt{x^2 + y^2 + z^2} \Rightarrow$

$$\begin{aligned} \nabla f(x, y, z) &= f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k} \\ &= \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2}(2x)\mathbf{i} + \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2}(2y)\mathbf{j} + \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2}(2z)\mathbf{k} \\ &= \frac{x}{\sqrt{x^2 + y^2 + z^2}}\mathbf{i} + \frac{y}{\sqrt{x^2 + y^2 + z^2}}\mathbf{j} + \frac{z}{\sqrt{x^2 + y^2 + z^2}}\mathbf{k} \end{aligned}$$

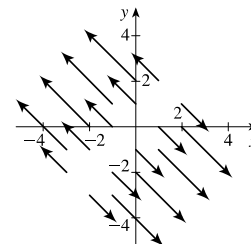
28. $f(x, y, z) = x^2 y e^{y/z} \Rightarrow$

$$\begin{aligned}\nabla f(x, y, z) &= f_x(x, y, z) \mathbf{i} + f_y(x, y, z) \mathbf{j} + f_z(x, y, z) \mathbf{k} \\ &= 2xy e^{y/z} \mathbf{i} + x^2 \left[y \cdot e^{y/z} (1/z) + e^{y/z} \cdot 1 \right] \mathbf{j} + \left[x^2 y e^{y/z} (-y/z^2) \right] \mathbf{k} \\ &= 2xy e^{y/z} \mathbf{i} + x^2 e^{y/z} \left(\frac{y}{z} + 1 \right) \mathbf{j} - \frac{x^2 y^2}{z^2} e^{y/z} \mathbf{k}\end{aligned}$$

29. $f(x, y) = \frac{1}{2}(x - y)^2 \Rightarrow$

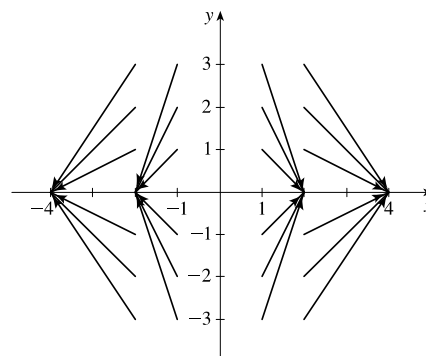
$$\nabla f(x, y) = (x - y)(1) \mathbf{i} + (x - y)(-1) \mathbf{j} = (x - y) \mathbf{i} + (y - x) \mathbf{j}.$$

The length of $\nabla f(x, y)$ is $\sqrt{(x - y)^2 + (y - x)^2} = \sqrt{2} |x - y|$. The vectors are $\mathbf{0}$ along the line $y = x$. Elsewhere the vectors point away from the line $y = x$ with length that increases as the distance from the line increases.



30. $f(x, y) = \frac{1}{2}(x^2 - y^2) \Rightarrow \nabla f(x, y) = x \mathbf{i} - y \mathbf{j}.$

The length of $\nabla f(x, y)$ is $\sqrt{x^2 + y^2}$. The lengths of the vectors increase as the distance from the origin increases, and the terminal point of each vector lies on the x -axis.



31. $f(x, y) = x^2 + y^2 \Rightarrow \nabla f(x, y) = 2x \mathbf{i} + 2y \mathbf{j}$. Thus, each vector $\nabla f(x, y)$ has the same direction and twice the length of the position vector of the point (x, y) , so the vectors all point directly away from the origin and their lengths increase as we move away from the origin. Hence, ∇f is graph III.

32. $f(x, y) = x(x + y) = x^2 + xy \Rightarrow \nabla f(x, y) = (2x + y) \mathbf{i} + x \mathbf{j}$. The y -component of each vector is x , so the vectors point upward in quadrants I and IV and downward in quadrants II and III. Also, the x -component of each vector is 0 along the line $y = -2x$ so the vectors are vertical there. Thus, ∇f is graph IV.

33. $f(x, y) = (x + y)^2 \Rightarrow \nabla f(x, y) = 2(x + y) \mathbf{i} + 2(x + y) \mathbf{j}$. The x - and y -components of each vector are equal, so all vectors are parallel to the line $y = x$. The vectors are $\mathbf{0}$ along the line $y = -x$ and their length increases as the distance from this line increases. Thus, ∇f is graph II.

34. $f(x, y) = \sin \sqrt{x^2 + y^2} \Rightarrow$

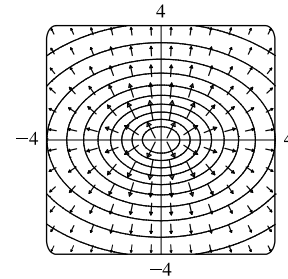
$$\begin{aligned}\nabla f(x, y) &= \left[\cos \sqrt{x^2 + y^2} \cdot \frac{1}{2}(x^2 + y^2)^{-1/2}(2x) \right] \mathbf{i} + \left[\cos \sqrt{x^2 + y^2} \cdot \frac{1}{2}(x^2 + y^2)^{-1/2}(2y) \right] \mathbf{j} \\ &= \frac{\cos \sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2}} x \mathbf{i} + \frac{\cos \sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2}} y \mathbf{j} \text{ or } \frac{\cos \sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2}} (x \mathbf{i} + y \mathbf{j})\end{aligned}$$

Thus each vector is a scalar multiple of its position vector, so the vectors point toward or away from the origin with length that changes in a periodic fashion as we move away from the origin. ∇f is graph I.

35. $f(x, y) = \ln(1 + x^2 + 2y^2)$. We graph

$$\nabla f(x, y) = \frac{2x}{1 + x^2 + 2y^2} \mathbf{i} + \frac{4y}{1 + x^2 + 2y^2} \mathbf{j} \text{ along with a contour map}$$

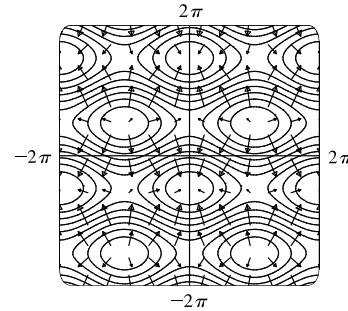
of f . The graph shows that the gradient vectors are perpendicular to the level curves. Also, the gradient vectors point in the direction in which f is increasing and are longer where the level curves are closer together.



36. $f(x, y) = \cos x - 2 \sin y$.

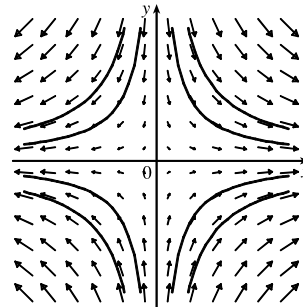
We graph $\nabla f(x, y) = -\sin x \mathbf{i} - 2 \cos y \mathbf{j}$ along with a contour map of f .

The graph shows that the gradient vectors are perpendicular to the level curves. Also, the gradient vectors point in the direction in which f is increasing and are longer where the level curves are closer together.



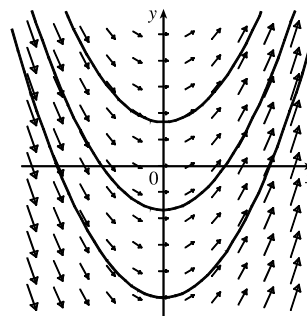
37. $\mathbf{V}(x, y) = \langle x^2, x + y^2 \rangle$. At $t = 3$ the particle is at $(2, 1)$ so its velocity is $\mathbf{V}(2, 1) = \langle 4, 3 \rangle$. After 0.01 units of time, the particle's change in location should be approximately $0.01 \mathbf{V}(2, 1) = 0.01 \langle 4, 3 \rangle = \langle 0.04, 0.03 \rangle$, so the particle should be approximately at the point $(2.04, 1.03)$.
38. $\mathbf{F}(x, y) = \langle xy - 2, y^2 - 10 \rangle$. At $t = 1$ the particle is at $(1, 3)$ so its velocity is $\mathbf{F}(1, 3) = \langle 1, -1 \rangle$. After 0.05 units of time, the particle's change in location should be approximately $0.05 \mathbf{F}(1, 3) = 0.05 \langle 1, -1 \rangle = \langle 0.05, -0.05 \rangle$, so the particle should be approximately at the point $(1.05, 2.95)$.

39. (a) We sketch the vector field $\mathbf{F}(x, y) = x \mathbf{i} - y \mathbf{j}$ along with several approximate flow lines. The flow lines appear to be hyperbolas with shape similar to the graph of $y = \pm 1/x$, so we might guess that the flow lines have equations $y = C/x$.



- (b) If $x = x(t)$ and $y = y(t)$ are parametric equations of a flow line, then the velocity vector of the flow line at the point (x, y) is $x'(t) \mathbf{i} + y'(t) \mathbf{j}$. Since the velocity vectors coincide with the vectors in the vector field, we have $x'(t) \mathbf{i} + y'(t) \mathbf{j} = x \mathbf{i} - y \mathbf{j} \Rightarrow dx/dt = x, dy/dt = -y$. To solve these differential equations, we know $dx/dt = x \Rightarrow dx/x = dt \Rightarrow \ln|x| = t + C \Rightarrow x = \pm e^{t+C} = Ae^t$ for some constant A , and $dy/dt = -y \Rightarrow dy/y = -dt \Rightarrow \ln|y| = -t + K \Rightarrow y = \pm e^{-t+K} = Be^{-t}$ for some constant B . Therefore $xy = Ae^t Be^{-t} = AB = \text{constant}$. If the flow line passes through $(1, 1)$ then $(1)(1) = \text{constant} = 1 \Rightarrow xy = 1 \Rightarrow y = 1/x, x > 0$.

40. (a) We sketch the vector field $\mathbf{F}(x, y) = \mathbf{i} + x\mathbf{j}$ along with several approximate flow lines. The flow lines appear to be parabolas.
- (b) If $x = x(t)$ and $y = y(t)$ are parametric equations of a flow line, then the velocity vector of the flow line at the point (x, y) is $x'(t)\mathbf{i} + y'(t)\mathbf{j}$. Since the velocity vectors coincide with the vectors in the vector field, we have $x'(t)\mathbf{i} + y'(t)\mathbf{j} = \mathbf{i} + x\mathbf{j} \Rightarrow \frac{dx}{dt} = 1, \frac{dy}{dt} = x$. Thus,
- $$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{x}{1} = x.$$



- (c) From part (b), $dy/dx = x$. Integrating, we have $y = \frac{1}{2}x^2 + c$. Since the particle starts at the origin, we know $(0, 0)$ is on the curve, so $0 = 0 + c \Rightarrow c = 0$ and the path the particle follows is $y = \frac{1}{2}x^2$.

16.2 Line Integrals

1. $x = t^2$ and $y = 2t, 0 \leq t \leq 3$, so by Formula 3

$$\begin{aligned} \int_C y \, ds &= \int_0^3 2t \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \, dt = \int_0^3 2t \sqrt{(2t)^2 + (2)^2} \, dt = \int_0^3 2t \sqrt{4t^2 + 4} \, dt \\ &= \int_0^3 4t \sqrt{t^2 + 1} \, dt = \left[2 \cdot \frac{2}{3}(t^2 + 1)^{3/2}\right]_0^3 = \frac{4}{3}(10^{3/2} - 1) \end{aligned}$$

2. $x = t^3$ and $y = t^4, 1 \leq t \leq 2$, so by Formula 3

$$\begin{aligned} \int_C (x/y) \, ds &= \int_1^2 (t^3/t^4) \sqrt{(3t^2)^2 + (4t^3)^2} \, dt = \int_1^2 (1/t) \cdot t^2 \sqrt{9 + 16t^2} \, dt = \int_1^2 t \sqrt{9 + 16t^2} \, dt \\ &= \left[\frac{1}{32} \cdot \frac{2}{3}(9 + 16t^2)^{3/2}\right]_1^2 = \frac{1}{48}(73^{3/2} - 25^{3/2}) \text{ or } \frac{1}{48}(73\sqrt{73} - 125) \end{aligned}$$

3. Parametric equations for C are $x = 4 \cos t, y = 4 \sin t, -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$. Then

$$\begin{aligned} \int_C xy^4 \, ds &= \int_{-\pi/2}^{\pi/2} (4 \cos t)(4 \sin t)^4 \sqrt{(-4 \sin t)^2 + (4 \cos t)^2} \, dt = \int_{-\pi/2}^{\pi/2} 4^5 \cos t \sin^4 t \sqrt{16(\sin^2 t + \cos^2 t)} \, dt \\ &= 4^5 \int_{-\pi/2}^{\pi/2} (\sin^4 t \cos t)(4) \, dt = (4)^6 \left[\frac{1}{5} \sin^5 t\right]_{-\pi/2}^{\pi/2} = 4^6 \cdot \frac{2}{5} = 1638.4 \end{aligned}$$

4. Parametric equations for C are $x = 2 + 3t, y = 4t, 0 \leq t \leq 1$. Then

$$\int_C xe^y \, ds = \int_0^1 (2 + 3t) e^{4t} \sqrt{3^2 + 4^2} \, dt = 5 \int_0^1 (2 + 3t) e^{4t} \, dt$$

Integrating by parts with $u = 2 + 3t \Rightarrow du = 3 \, dt, dv = e^{4t} \, dt \Rightarrow v = \frac{1}{4}e^{4t}$ gives

$$\int_C xe^y \, ds = 5 \left[\frac{1}{4}(2 + 3t)e^{4t} - \frac{3}{16}e^{4t} \right]_0^1 = 5 \left[\frac{5}{4}e^4 - \frac{3}{16}e^4 - \frac{1}{2} + \frac{3}{16} \right] = \frac{85}{16}e^4 - \frac{25}{16}$$

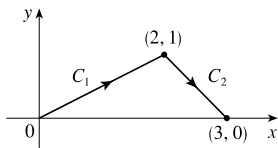
5. If we choose x as the parameter, parametric equations for C are $x = x, y = x^2$ for $0 \leq x \leq \pi$ and by Equations 7

$$\begin{aligned} \int_C (x^2 y + \sin x) \, dy &= \int_0^\pi [x^2(x^2) + \sin x] \cdot 2x \, dx = 2 \int_0^\pi (x^5 + x \sin x) \, dx \\ &= 2 \left[\frac{1}{6}x^6 - x \cos x + \sin x \right]_0^\pi \quad \left[\text{where we integrated by parts} \right. \\ &\quad \left. \text{in the second term} \right] \\ &= 2 \left[\frac{1}{6}\pi^6 + \pi + 0 - 0 \right] = \frac{1}{3}\pi^6 + 2\pi \end{aligned}$$

6. Choosing y as the parameter, we have $x = y^3$, $y = y$, $-1 \leq y \leq 1$. Then

$$\int_C e^x dx = \int_{-1}^1 e^{y^3} \cdot 3y^2 dy = e^{y^3} \Big|_{-1}^1 = e^1 - e^{-1} = e - \frac{1}{e}.$$

7.



$$C = C_1 + C_2$$

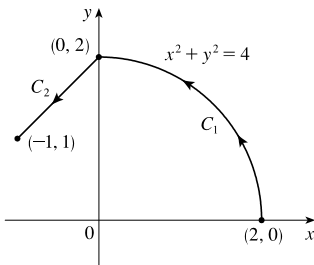
$$\text{On } C_1: x = x, y = \frac{1}{2}x \Rightarrow dy = \frac{1}{2}dx, \quad 0 \leq x \leq 2.$$

$$\text{On } C_2: x = x, y = 3 - x \Rightarrow dy = -dx, \quad 2 \leq x \leq 3.$$

Then

$$\begin{aligned} \int_C (x + 2y) dx + x^2 dy &= \int_{C_1} (x + 2y) dx + x^2 dy + \int_{C_2} (x + 2y) dx + x^2 dy \\ &= \int_0^2 \left[x + 2\left(\frac{1}{2}x\right) + x^2\left(\frac{1}{2}\right) \right] dx + \int_2^3 \left[x + 2(3 - x) + x^2(-1) \right] dx \\ &= \int_0^2 \left(2x + \frac{1}{2}x^2 \right) dx + \int_2^3 (6 - x - x^2) dx \\ &= \left[x^2 + \frac{1}{6}x^3 \right]_0^2 + \left[6x - \frac{1}{2}x^2 - \frac{1}{3}x^3 \right]_2^3 = \frac{16}{3} - 0 + \frac{9}{2} - \frac{22}{3} = \frac{5}{2} \end{aligned}$$

8.



$$C = C_1 + C_2$$

$$\text{On } C_1: x = 2 \cos t \Rightarrow dx = -2 \sin t dt,$$

$$y = 2 \sin t \Rightarrow dy = 2 \cos t dt, \quad 0 \leq t \leq \frac{\pi}{2}.$$

$$\text{On } C_2: x = -t \Rightarrow dx = -dt,$$

$$y = 2 - t \Rightarrow dy = -dt, \quad 0 \leq t \leq 1.$$

Then

$$\begin{aligned} \int_C x^2 dx + y^2 dy &= \int_{C_1} x^2 dx + y^2 dy + \int_{C_2} x^2 dx + y^2 dy \\ &= \int_0^{\pi/2} (2 \cos t)^2 (-2 \sin t dt) + (2 \sin t)^2 (2 \cos t dt) + \int_0^1 (-t)^2 (-dt) + (2 - t)^2 (-dt) \\ &= \int_0^{\pi/2} (-8 \cos^2 t \sin t + 8 \sin^2 t \cos t) dt - 2 \int_0^1 (t^2 - 2t + 2) dt \\ &= 8 \left[\frac{1}{3} \cos^3 t + \frac{1}{3} \sin^3 t \right]_0^{\pi/2} - 2 \left[\frac{1}{3} t^3 - t^2 + 2t \right]_0^1 = 8 \left(\frac{1}{3} - \frac{1}{3} \right) - 2 \left(\frac{1}{3} - 1 + 2 \right) = -\frac{8}{3} \end{aligned}$$

9. $x = \cos t$, $y = \sin t$, $z = t$, $0 \leq t \leq \pi/2$. Then by Formula 9,

$$\begin{aligned} \int_C x^2 y ds &= \int_0^{\pi/2} (\cos t)^2 (\sin t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \\ &= \int_0^{\pi/2} \cos^2 t \sin t \sqrt{(-\sin t)^2 + (\cos t)^2 + (1)^2} dt = \int_0^{\pi/2} \cos^2 t \sin t \sqrt{\sin^2 t + \cos^2 t + 1} dt \\ &= \sqrt{2} \int_0^{\pi/2} \cos^2 t \sin t dt = \sqrt{2} \left[-\frac{1}{3} \cos^3 t \right]_0^{\pi/2} = \sqrt{2} \left(0 + \frac{1}{3} \right) = \frac{\sqrt{2}}{3} \end{aligned}$$

10. Parametric equations for the line segment C from $(3, 1, 2)$ to $(1, 2, 5)$ are $x = 3 - 2t$, $y = 1 + t$, $z = 2 + 3t$, $0 \leq t \leq 1$.

Then by Formula 9,

$$\begin{aligned} \int_C y^2 z ds &= \int_0^1 (1 + t)^2 (2 + 3t) \sqrt{(-2)^2 + 1^2 + 3^2} dt = \sqrt{14} \int_0^1 (3t^3 + 8t^2 + 7t + 2) dt \\ &= \sqrt{14} \left[\frac{3}{4} t^4 + \frac{8}{3} t^3 + \frac{7}{2} t^2 + 2t \right]_0^1 = \sqrt{14} \left(\frac{3}{4} + \frac{8}{3} + \frac{7}{2} + 2 \right) = \frac{107}{12} \sqrt{14} \end{aligned}$$

11. Parametric equations for the line segment C from $(0, 0, 0)$ to $(1, 2, 3)$ are $x = t$, $y = 2t$, $z = 3t$, $0 \leq t \leq 1$. Then by

$$\text{Formula 9, } \int_C x e^{yz} ds = \int_0^1 t e^{(2t)(3t)} \sqrt{1^2 + 2^2 + 3^2} dt = \sqrt{14} \int_0^1 t e^{6t^2} dt = \sqrt{14} \left[\frac{1}{12} e^{6t^2} \right]_0^1 = \frac{\sqrt{14}}{12} (e^6 - 1).$$

12. $C: x = t, y = \cos 2t, z = \sin 2t, 0 \leq t \leq 2\pi$.

$$\sqrt{(dx/dt)^2 + (dy/dt)^2 + (dz/dt)^2} = \sqrt{1^2 + (-2 \sin 2t)^2 + (2 \cos 2t)^2} = \sqrt{1 + 4(\sin^2 2t + \cos^2 2t)} = \sqrt{5}.$$

Then by Formula 9,

$$\begin{aligned} \int_C (x^2 + y^2 + z^2) ds &= \int_0^{2\pi} (t^2 + \cos^2 2t + \sin^2 2t) \sqrt{5} dt = \sqrt{5} \int_0^{2\pi} (t^2 + 1) dt \\ &= \sqrt{5} \left[\frac{1}{3} t^3 + t \right]_0^{2\pi} = \sqrt{5} \left[\frac{1}{3} (8\pi^3) + 2\pi \right] = \sqrt{5} \left(\frac{8}{3} \pi^3 + 2\pi \right) \end{aligned}$$

13. $C: x = t, y = t^2, z = t^3, 0 \leq t \leq 1$.

$$\int_C x y e^{yz} dy = \int_0^1 (t)(t^2) e^{(t^2)(t^3)} \cdot 2t dt = \int_0^1 2t^4 e^{t^5} dt = \left[\frac{2}{5} e^{t^5} \right]_0^1 = \frac{2}{5} (e^1 - e^0) = \frac{2}{5} (e - 1)$$

14. $C: x = e^t, y = 2t, z = \ln t, 1 \leq t \leq 2$.

$$\begin{aligned} \int_C y e^z dz + x \ln x dy - y dx &= \int_1^2 2t e^{\ln t} \frac{1}{t} dt + e^t \ln e^t \cdot 2 dt - 2t e^t dt = \int_1^2 (2t + 2t e^t - 2t e^t) dt \\ &= \int_1^2 2t dt = [t^2]_1^2 = 4 - 1 = 3 \end{aligned}$$

15. $C: x = \sin t, y = \cos t, z = \tan t, -\pi/4 \leq t \leq \pi/4$.

$$\begin{aligned} \int_C z dx + xy dy + y^2 dz &= \int_{-\pi/4}^{\pi/4} (\tan t)(\cos t dt) + (\sin t)(\cos t)(-\sin t dt) + (\cos^2 t)(\sec^2 t dt) \\ &= \int_{-\pi/4}^{\pi/4} (\sin t - \sin^2 t \cos t + 1) dt = \left[-\cos t - \frac{1}{3} \sin^3 t + t \right]_{-\pi/4}^{\pi/4} \\ &= \left[-\frac{\sqrt{2}}{2} - \frac{1}{3} \left(\frac{\sqrt{2}}{2} \right)^3 + \frac{\pi}{4} \right] - \left[-\frac{\sqrt{2}}{2} - \frac{1}{3} \left(-\frac{\sqrt{2}}{2} \right)^3 - \frac{\pi}{4} \right] = \frac{\pi}{2} - \frac{\sqrt{2}}{6} \end{aligned}$$

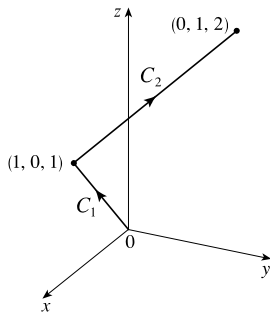
16. $C: x = \sqrt{t}, y = t, z = t^2, 1 \leq t \leq 4$.

$$\begin{aligned} \int_C y dx + z dy + x dz &= \int_1^4 t \cdot \frac{1}{2} t^{-1/2} dt + t^2 \cdot dt + \sqrt{t} \cdot 2t dt = \int_1^4 \left(\frac{1}{2} t^{1/2} + t^2 + 2t^{3/2} \right) dt \\ &= \left[\frac{1}{3} t^{3/2} + \frac{1}{3} t^3 + \frac{4}{5} t^{5/2} \right]_1^4 = \frac{8}{3} + \frac{64}{3} + \frac{128}{5} - \frac{1}{3} - \frac{1}{3} - \frac{4}{5} = \frac{722}{15} \end{aligned}$$

17. Parametric equations for the line segment C from $(1, 0, 0)$ to $(4, 1, 2)$ are $x = 1 + 3t$, $y = t$, $z = 2t$, $0 \leq t \leq 1$. Then

$$\begin{aligned} \int_C z^2 dx + x^2 dy + y^2 dz &= \int_0^1 (2t)^2 \cdot 3 dt + (1 + 3t)^2 dt + t^2 \cdot 2 dt = \int_0^1 (23t^2 + 6t + 1) dt \\ &= \left[\frac{23}{3} t^3 + 3t^2 + t \right]_0^1 = \frac{23}{3} + 3 + 1 = \frac{35}{3} \end{aligned}$$

- 18.



$$C = C_1 + C_2$$

$$\text{On } C_1 \text{ from } (0, 0, 0) \text{ to } (1, 0, 1): x = t \Rightarrow dx = dt,$$

$$y = 0 \Rightarrow dy = 0 dt,$$

$$z = t \Rightarrow dz = dt, 0 \leq t \leq 1.$$

$$\text{On } C_2 \text{ from } (1, 0, 1) \text{ to } (0, 1, 2): x = 1 - t \Rightarrow dx = -dt,$$

$$y = t \Rightarrow dy = dt,$$

$$z = 1 + t \Rightarrow dz = dt, 0 \leq t \leq 1. \quad [\text{continued}]$$

Then

$$\begin{aligned}
 & \int_C (y+z) dx + (x+z) dy + (x+y) dz \\
 &= \int_{C_1} (y+z) dx + (x+z) dy + (x+y) dz + \int_{C_2} (y+z) dx + (x+z) dy + (x+y) dz \\
 &= \int_0^1 (0+t) dt + (t+t) \cdot 0 dt + (t+0) dt + \int_0^1 (t+1+t)(-dt) + (1-t+1+t) dt + (1-t+t) dt \\
 &= \int_0^1 2t dt + \int_0^1 (-2t+2) dt = [t^2]_0^1 + [-t^2+2t]_0^1 = 1+1=2
 \end{aligned}$$

19. (a) Along the line $x = -3$, the vectors of $\mathbf{F}$ have positive y -components, so since the path goes upward, the integrand $\mathbf{F} \cdot \mathbf{T}$ is always positive. Therefore $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot \mathbf{T} ds$ is positive.

(b) All of the (nonzero) field vectors along the circle with radius 3 are pointed in the clockwise direction, that is, opposite the direction to the path. So $\mathbf{F} \cdot \mathbf{T}$ is negative, and therefore $\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot \mathbf{T} ds$ is negative.

20. Vectors starting on C_1 point in roughly the same direction as C_1 , so the tangential component $\mathbf{F} \cdot \mathbf{T}$ is positive. Then

$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot \mathbf{T} ds$ is positive. On the other hand, no vectors starting on C_2 point in the same direction as C_2 , while some vectors point in roughly the opposite direction, so we would expect $\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot \mathbf{T} ds$ to be negative.

21. $\mathbf{F}(x, y) = xy^2 \mathbf{i} - x^2 \mathbf{j}$ and $\mathbf{r}(t) = t^3 \mathbf{i} + t^2 \mathbf{j}$, $0 \leq t \leq 1 \Rightarrow$

$\mathbf{F}(\mathbf{r}(t)) = (t^3)(t^2)^2 \mathbf{i} - (t^3)^2 \mathbf{j} = t^7 \mathbf{i} - t^6 \mathbf{j}$ and $\mathbf{r}'(t) = 3t^2 \mathbf{i} + 2t \mathbf{j}$. Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^1 (t^7 \cdot 3t^2 - t^6 \cdot 2t) dt = \int_0^1 (3t^9 - 2t^7) dt = \left[\frac{3}{10} t^{10} - \frac{1}{4} t^8 \right]_0^1 = \frac{3}{10} - \frac{1}{4} = \frac{1}{20}.$$

22. $\mathbf{F}(x, y, z) = (x+y^2) \mathbf{i} + xz \mathbf{j} + (y+z) \mathbf{k}$ and $\mathbf{r}(t) = t^2 \mathbf{i} + t^3 \mathbf{j} - 2t \mathbf{k}$, $0 \leq t \leq 2 \Rightarrow$

$\mathbf{F}(\mathbf{r}(t)) = (t^2 + (t^3)^2) \mathbf{i} + (t^2)(-2t) \mathbf{j} + (t^3 - 2t) \mathbf{k} = (t^2 + t^6) \mathbf{i} - 2t^3 \mathbf{j} + (t^3 - 2t) \mathbf{k}$ and $\mathbf{r}'(t) = 2t \mathbf{i} + 3t^2 \mathbf{j} - 2 \mathbf{k}$.

Then

$$\begin{aligned}
 \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^2 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^2 (2t^3 + 2t^7 - 6t^5 - 2t^3 + 4t) dt = \int_0^2 (2t^7 - 6t^5 + 4t) dt \\
 &= \left[\frac{1}{4} t^8 - t^6 + 2t^2 \right]_0^2 = 64 - 64 + 8 = 8
 \end{aligned}$$

23. $\mathbf{F}(x, y, z) = \sin x \mathbf{i} + \cos y \mathbf{j} + xz \mathbf{k}$ and $\mathbf{r}(t) = t^3 \mathbf{i} - t^2 \mathbf{j} + t \mathbf{k}$, $0 \leq t \leq 1 \Rightarrow$

$\mathbf{F}(\mathbf{r}(t)) = \sin t^3 \mathbf{i} + \cos(-t^2) \mathbf{j} + t^3 \cdot t \mathbf{k}$ and $\mathbf{r}'(t) = 3t^2 \mathbf{i} - 2t \mathbf{j} + \mathbf{k}$. Then

$$\begin{aligned}
 \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^1 (3t^2 \sin t^3 - 2t \cos t^2 + t^4) dt \\
 &= \left[-\cos t^3 - \sin t^2 + \frac{1}{5} t^5 \right]_0^1 = \frac{6}{5} - \cos 1 - \sin 1
 \end{aligned}$$

24. $\mathbf{F}(x, y, z) = xz \mathbf{i} + z^3 \mathbf{j} + y \mathbf{k}$ and $\mathbf{r}(t) = e^t \mathbf{i} + e^{2t} \mathbf{j} + e^{-t} \mathbf{k}$, $-1 \leq t \leq 1 \Rightarrow$

$\mathbf{F}(\mathbf{r}(t)) = e^t e^{-t} \mathbf{i} + (e^{-t})^3 \mathbf{j} + e^{2t} \mathbf{k} = \mathbf{i} + e^{-3t} \mathbf{j} + e^{2t} \mathbf{k}$ and $\mathbf{r}'(t) = e^t \mathbf{i} + 2e^{2t} \mathbf{j} - e^{-t} \mathbf{k}$. Then

$$\begin{aligned}
 \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_{-1}^1 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_{-1}^1 (1 \cdot e^t + e^{-3t} \cdot 2e^{2t} - e^{2t} e^{-t}) dt \\
 &= \int_{-1}^1 2e^{-t} dt = -2[e^{-t}]_{-1}^1 = -2(e^{-1} - e)
 \end{aligned}$$

$$25. \mathbf{F}(\mathbf{r}(t)) = \sqrt{\sin^2 t + \sin t \cos t} \mathbf{i} + [(\sin t \cos t)/\sin^2 t] \mathbf{j} = \sqrt{\sin^2 t + \sin t \cos t} \mathbf{i} + \cot t \mathbf{j},$$

$$\mathbf{r}'(t) = 2 \sin t \cos t \mathbf{i} + (\cos^2 t - \sin^2 t) \mathbf{j}. \text{ Then}$$

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_{\pi/6}^{\pi/3} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \\ &= \int_{\pi/6}^{\pi/3} [2 \sin t \cos t \sqrt{\sin^2 t + \sin t \cos t} + (\cot t)(\cos^2 t - \sin^2 t)] dt \approx 0.5424 \end{aligned}$$

$$26. \mathbf{F}(\mathbf{r}(t)) = (\cos t \tan t) e^{\sin t} \mathbf{i} + (\tan t \sin t) e^{\cos t} \mathbf{j} + (\sin t \cos t) e^{\tan t} \mathbf{k}$$

$$= (\sin t) e^{\sin t} \mathbf{i} + (\tan t \sin t) e^{\cos t} \mathbf{j} + (\sin t \cos t) e^{\tan t} \mathbf{k},$$

$$\mathbf{r}'(t) = \cos t \mathbf{i} - \sin t \mathbf{j} + \sec^2 t \mathbf{k}. \text{ Then}$$

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{\pi/4} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \\ &= \int_0^{\pi/4} [(\sin t \cos t) e^{\sin t} - (\tan t \sin^2 t) e^{\cos t} + (\tan t) e^{\tan t}] dt \approx 0.8527 \end{aligned}$$

$$27. x = t^2, \quad y = t^3, \quad z = \sqrt{t} \text{ so by Formula 9,}$$

$$\begin{aligned} \int_C xy \arctan z \, ds &= \int_1^2 (t^2)(t^3) \arctan \sqrt{t} \cdot \sqrt{(2t)^2 + (3t^2)^2 + [1/(2\sqrt{t})]^2} dt \\ &= \int_1^2 t^5 \sqrt{4t^2 + 9t^4 + 1/(4t)} \arctan \sqrt{t} \, dt \approx 94.8231 \end{aligned}$$

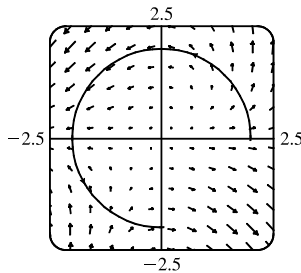
$$28. x = 1 + 3t, \quad y = 2 + t^2, \quad z = t^4 \text{ so by Formula 9,}$$

$$\begin{aligned} \int_C z \ln(x + y) \, ds &= \int_{-1}^1 t^4 \ln(1 + 3t + 2 + t^2) \cdot \sqrt{(3)^2 + (2t)^2 + (4t^3)^2} dt \\ &= \int_{-1}^1 t^4 \sqrt{9 + 4t^2 + 16t^6} \ln(3 + 3t + t^2) \, dt \approx 1.7260 \end{aligned}$$

29. We graph $\mathbf{F}(x, y) = (x - y) \mathbf{i} + xy \mathbf{j}$ and the curve C . We see that most of the vectors starting on C point in roughly the same direction as C , so for these portions of C the tangential component $\mathbf{F} \cdot \mathbf{T}$ is positive. Although some vectors in the third quadrant which start on C point in roughly the opposite direction, and hence give negative tangential components, it seems reasonable that the effect of these portions of C is outweighed by the positive tangential components. Thus, we would expect $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T} \, ds$ to be positive.

To verify, we evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$. The curve C can be represented by $\mathbf{r}(t) = 2 \cos t \mathbf{i} + 2 \sin t \mathbf{j}$, $0 \leq t \leq \frac{3\pi}{2}$,

so $\mathbf{F}(\mathbf{r}(t)) = (2 \cos t - 2 \sin t) \mathbf{i} + 4 \cos t \sin t \mathbf{j}$ and $\mathbf{r}'(t) = -2 \sin t \mathbf{i} + 2 \cos t \mathbf{j}$. Then



$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{3\pi/2} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \\ &= \int_0^{3\pi/2} [-2 \sin t (2 \cos t - 2 \sin t) + 2 \cos t (4 \cos t \sin t)] dt \\ &= 4 \int_0^{3\pi/2} (\sin^2 t - \sin t \cos t + 2 \sin t \cos^2 t) dt \\ &= 3\pi + \frac{2}{3} \quad [\text{using a CAS}] \end{aligned}$$

30. We graph $\mathbf{F}(x, y) = \frac{x}{\sqrt{x^2 + y^2}} \mathbf{i} + \frac{y}{\sqrt{x^2 + y^2}} \mathbf{j}$ and the curve C . In the

first quadrant, each vector starting on C points in roughly the same direction as C , so the tangential component $\mathbf{F} \cdot \mathbf{T}$ is positive. In the second quadrant, each vector starting on C points in roughly the direction opposite to C , so $\mathbf{F} \cdot \mathbf{T}$ is negative. Here, it appears that the tangential components in the first and second quadrants counteract each other, so it seems reasonable to guess

that $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T} ds$ is zero. To verify, we evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$. The curve C can be represented by

$$\mathbf{r}(t) = t \mathbf{i} + (1 + t^2) \mathbf{j}, \quad -1 \leq t \leq 1, \text{ so } \mathbf{F}(\mathbf{r}(t)) = \frac{t}{\sqrt{t^2 + (1 + t^2)^2}} \mathbf{i} + \frac{1 + t^2}{\sqrt{t^2 + (1 + t^2)^2}} \mathbf{j} \text{ and } \mathbf{r}'(t) = \mathbf{i} + 2t \mathbf{j}. \text{ Then}$$

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_{-1}^1 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_{-1}^1 \left(\frac{t}{\sqrt{t^2 + (1 + t^2)^2}} + \frac{2t(1 + t^2)}{\sqrt{t^2 + (1 + t^2)^2}} \right) dt \\ &= \int_{-1}^1 \frac{t(3 + 2t^2)}{\sqrt{t^4 + 3t^2 + 1}} dt = 0 \quad [\text{since the integrand is an odd function}] \end{aligned}$$

31. (a) $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \langle e^{t^2-1}, (t^2)(t^3) \rangle \cdot \langle 2t, 3t^2 \rangle dt = \int_0^1 (2te^{t^2-1} + 3t^7) dt = [e^{t^2-1} + \frac{3}{8}t^8]_0^1 = \frac{11}{8} - 1/e$

(b) $\mathbf{r}(0) = \mathbf{0}, \quad \mathbf{F}(\mathbf{r}(0)) = \langle e^{-1}, 0 \rangle;$

$$\mathbf{r}\left(\frac{1}{\sqrt{2}}\right) = \left\langle \frac{1}{2}, \frac{1}{2\sqrt{2}} \right\rangle, \quad \mathbf{F}\left(\mathbf{r}\left(\frac{1}{\sqrt{2}}\right)\right) = \left\langle e^{-1/2}, \frac{1}{4\sqrt{2}} \right\rangle;$$

$$\mathbf{r}(1) = \langle 1, 1 \rangle, \quad \mathbf{F}(\mathbf{r}(1)) = \langle 1, 1 \rangle.$$

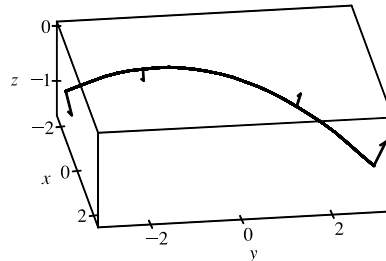
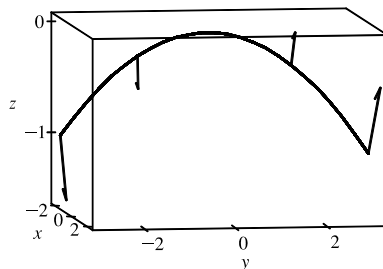
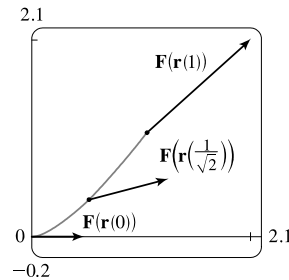
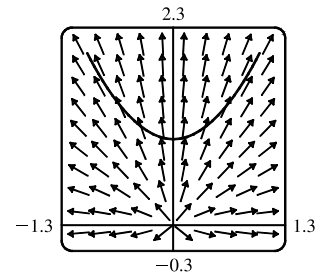
In order to generate the graph with Maple, we use the `line` command in the `plottools` package to define each of the vectors. For example,

`v1:=line([0,0],[exp(-1),0]):`

generates the vector from the vector field at the point $(0, 0)$ (but without an arrowhead) and gives it the name `v1`. To show everything on the same screen, we use the `display` command. In Mathematica, we use `ListPlot` (with the `PlotJoined -> True` option) to generate the vectors, and then `Show` to show everything on the same screen.

32. (a) $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{-1}^1 \langle 2t, t^2, 3t \rangle \cdot \langle 2, 3, -2t \rangle dt = \int_{-1}^1 (4t + 3t^2 - 6t^2) dt = [2t^2 - t^3]_{-1}^1 = -2$

(b) Now $\mathbf{F}(\mathbf{r}(t)) = \langle 2t, t^2, 3t \rangle$, so $\mathbf{F}(\mathbf{r}(-1)) = \langle -2, 1, -3 \rangle$, $\mathbf{F}(\mathbf{r}(-\frac{1}{2})) = \langle -1, \frac{1}{4}, -\frac{3}{2} \rangle$, $\mathbf{F}(\mathbf{r}(\frac{1}{2})) = \langle 1, \frac{1}{4}, \frac{3}{2} \rangle$, and $\mathbf{F}(\mathbf{r}(1)) = \langle 2, 1, 3 \rangle$.



33. $x = e^{-t} \cos 4t$, $y = e^{-t} \sin 4t$, $z = e^{-t}$, $0 \leq t \leq 2\pi$.

Then $\frac{dx}{dt} = e^{-t}(-\sin 4t)(4) - e^{-t} \cos 4t = -e^{-t}(4 \sin 4t + \cos 4t)$,

$\frac{dy}{dt} = e^{-t}(\cos 4t)(4) - e^{-t} \sin 4t = -e^{-t}(-4 \cos 4t + \sin 4t)$, and $\frac{dz}{dt} = -e^{-t}$, so

$$\begin{aligned} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} &= \sqrt{(-e^{-t})^2[(4 \sin 4t + \cos 4t)^2 + (-4 \cos 4t + \sin 4t)^2 + 1]} \\ &= e^{-t} \sqrt{16(\sin^2 4t + \cos^2 4t) + \sin^2 4t + \cos^2 4t + 1} = 3\sqrt{2}e^{-t} \end{aligned}$$

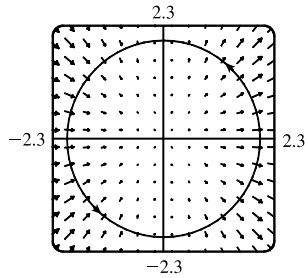
So by Formula 9,

$$\begin{aligned} \int_C x^3 y^2 z \, ds &= \int_0^{2\pi} (e^{-t} \cos 4t)^3 (e^{-t} \sin 4t)^2 (e^{-t}) (3\sqrt{2}e^{-t}) \, dt \\ &= \int_0^{2\pi} 3\sqrt{2}e^{-7t} \cos^3 4t \sin^2 4t \, dt = \frac{172,704}{5,632,705} \sqrt{2} (1 - e^{-14\pi}) \end{aligned}$$

34. (a) We parametrize the circle C as $\mathbf{r}(t) = 2 \cos t \mathbf{i} + 2 \sin t \mathbf{j}$, $0 \leq t \leq 2\pi$. So $\mathbf{F}(\mathbf{r}(t)) = \langle 4 \cos^2 t, 4 \cos t \sin t \rangle$,

$\mathbf{r}'(t) = \langle -2 \sin t, 2 \cos t \rangle$, and $W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} (-8 \cos^2 t \sin t + 8 \cos^2 t \sin t) \, dt = 0$.

(b)



From the graph, we see that all of the vectors in the field are perpendicular to the path. This indicates that the field does no work on the particle, since the field never pulls the particle in the direction in which it is going. In other words, at any point along C , $\mathbf{F} \cdot \mathbf{T} = 0$, and so certainly $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$.

35. We use the parametrization $x = 2 \cos t$, $y = 2 \sin t$, $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$. Then

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \sqrt{(-2 \sin t)^2 + (2 \cos t)^2} dt = 2 dt, \text{ so } m = \int_C k \, ds = 2k \int_{-\pi/2}^{\pi/2} dt = 2k(\pi),$$

$$\bar{x} = \frac{1}{2\pi k} \int_C xk \, ds = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} (2 \cos t) 2 \, dt = \frac{1}{2\pi} [4 \sin t]_{-\pi/2}^{\pi/2} = \frac{4}{\pi}, \bar{y} = \frac{1}{2\pi k} \int_C yk \, ds = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} (2 \sin t) 2 \, dt = 0.$$

Hence $(\bar{x}, \bar{y}) = (\frac{4}{\pi}, 0)$.

36. We use the parametrization $x = a \cos t$, $y = a \sin t$, $0 \leq t \leq \frac{\pi}{2}$. Then

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \sqrt{(-a \sin t)^2 + (a \cos t)^2} dt = a \, dt, \text{ so}$$

$$m = \int_C \rho(x, y) \, ds = \int_C kxy \, ds = \int_0^{\pi/2} k(a \cos t)(a \sin t) a \, dt = ka^3 \int_0^{\pi/2} \cos t \sin t \, dt = ka^3 \left[\frac{1}{2} \sin^2 t\right]_0^{\pi/2} = \frac{1}{2} ka^3,$$

$$\begin{aligned} \bar{x} &= \frac{1}{ka^3/2} \int_C x(kxy) \, ds = \frac{2}{ka^3} \int_0^{\pi/2} k(a \cos t)^2 (a \sin t) a \, dt = \frac{2}{ka^3} \cdot ka^4 \int_0^{\pi/2} \cos^2 t \sin t \, dt \\ &= 2a \left[-\frac{1}{3} \cos^3 t\right]_0^{\pi/2} = 2a \left(0 + \frac{1}{3}\right) = \frac{2}{3}a, \text{ and} \end{aligned}$$

$$\begin{aligned} \bar{y} &= \frac{1}{ka^3/2} \int_C y(kxy) \, ds = \frac{2}{ka^3} \int_0^{\pi/2} k(a \cos t)(a \sin t)^2 a \, dt = \frac{2}{ka^3} \cdot ka^4 \int_0^{\pi/2} \sin^2 t \cos t \, dt \\ &= 2a \left[\frac{1}{3} \sin^3 t\right]_0^{\pi/2} = 2a \left(\frac{1}{3} - 0\right) = \frac{2}{3}a. \end{aligned}$$

Therefore the mass is $\frac{1}{2} ka^3$ and the center of mass is $(\bar{x}, \bar{y}) = (\frac{2}{3}a, \frac{2}{3}a)$.

37. (a) $\bar{x} = \frac{1}{m} \int_C x \rho(x, y, z) ds$, $\bar{y} = \frac{1}{m} \int_C y \rho(x, y, z) ds$, $\bar{z} = \frac{1}{m} \int_C z \rho(x, y, z) ds$, where $m = \int_C \rho(x, y, z) ds$.

(b) $ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt = \sqrt{(2 \cos t)^2 + (-2 \sin t)^2 + 3^2} dt = \sqrt{4(\cos^2 t + \sin^2 t) + 9} dt = \sqrt{13} dt$.

$$m = \int_C k ds = k \int_0^{2\pi} \sqrt{13} dt = k \sqrt{13} \int_0^{2\pi} dt = 2\pi k \sqrt{13},$$

$$\bar{x} = \frac{1}{m} \int_C x \rho(x, y, z) ds = \frac{1}{2\pi k \sqrt{13}} \int_0^{2\pi} 2k \sqrt{13} \sin t dt = 0,$$

$$\bar{y} = \frac{1}{m} \int_C y \rho(x, y, z) ds = \frac{1}{2\pi k \sqrt{13}} \int_0^{2\pi} 2k \sqrt{13} \cos t dt = 0,$$

$$\bar{z} = \frac{1}{m} \int_C z \rho(x, y, z) ds = \frac{1}{2\pi k \sqrt{13}} \int_0^{2\pi} (k \sqrt{13})(3t) dt = \frac{3}{2\pi} (2\pi^2) = 3\pi. \text{ Hence, } (\bar{x}, \bar{y}, \bar{z}) = (0, 0, 3\pi).$$

38. $m = \int_C (x^2 + y^2 + z^2) ds = \int_0^{2\pi} (t^2 + \cos^2 t + \sin^2 t) \sqrt{(1)^2 + (-\sin t)^2 + (\cos t)^2} dt$
 $= \int_0^{2\pi} (t^2 + 1) \sqrt{2} dt = \sqrt{2} \left[\frac{1}{3} t^3 + t \right]_0^{2\pi} = \sqrt{2} \left(\frac{8}{3} \pi^3 + 2\pi \right),$

$$\bar{x} = \frac{1}{\sqrt{2} \left(\frac{8}{3} \pi^3 + 2\pi \right)} \int_0^{2\pi} \sqrt{2} (t^3 + t) dt = \frac{1}{\frac{8}{3} \pi^3 + 2\pi} \left[\frac{1}{4} t^4 + \frac{1}{2} t^2 \right]_0^{2\pi} = \frac{4\pi^4 + 2\pi^2}{\frac{8}{3} \pi^3 + 2\pi} \cdot \frac{3/(2\pi)}{3/(2\pi)}$$

$$= \frac{3\pi(2\pi^2 + 1)}{4\pi^2 + 3},$$

$$\bar{y} = \frac{1}{\sqrt{2} \left(\frac{8}{3} \pi^3 + 2\pi \right)} \int_0^{2\pi} (\sqrt{2} \cos t)(t^2 + 1) dt = 0, \text{ and}$$

$$\bar{z} = \frac{1}{\sqrt{2} \left(\frac{8}{3} \pi^3 + 2\pi \right)} \int_0^{2\pi} (\sqrt{2} \sin t)(t^2 + 1) dt = 0. \text{ Hence, } (\bar{x}, \bar{y}, \bar{z}) = \left(\frac{3\pi(2\pi^2 + 1)}{4\pi^2 + 3}, 0, 0 \right).$$

39. From Example 3, $\rho(x, y) = k(1 - y)$, $x = \cos t$, $y = \sin t$, and $ds = dt$, $0 \leq t \leq \pi \Rightarrow$

$$I_x = \int_C y^2 \rho(x, y) ds = \int_0^\pi \sin^2 t [k(1 - \sin t)] dt = k \int_0^\pi (\sin^2 t - \sin^3 t) dt$$

$$= \frac{1}{2} k \int_0^\pi (1 - \cos 2t) dt - k \int_0^\pi (1 - \cos^2 t) \sin t dt \quad \left[\begin{array}{l} \text{Let } u = \cos t, du = -\sin t dt \\ \text{in the second integral} \end{array} \right]$$

$$= k \left[\frac{t}{2} + \int_1^{-1} (1 - u^2) du \right] = k \left(\frac{\pi}{2} - \frac{4}{3} \right)$$

$$I_y = \int_C x^2 \rho(x, y) ds = k \int_0^\pi \cos^2 t (1 - \sin t) dt = \frac{k}{2} \int_0^\pi (1 + \cos 2t) dt - k \int_0^\pi \cos^2 t \sin t dt$$

$$= k \left(\frac{\pi}{2} - \frac{2}{3} \right), \text{ using the same substitution as above.}$$

40. The wire is given as $x = 2 \sin t$, $y = 2 \cos t$, $z = 3t$, $0 \leq t \leq 2\pi$ with $\rho(x, y, z) = k$. Then

$$ds = \sqrt{(2 \cos t)^2 + (-2 \sin t)^2 + 3^2} dt = \sqrt{4(\cos^2 t + \sin^2 t) + 9} dt = \sqrt{13} dt \text{ and}$$

$$I_x = \int_C (y^2 + z^2) \rho(x, y, z) ds = \int_0^{2\pi} (4 \cos^2 t + 9t^2)(k) \sqrt{13} dt = \sqrt{13} k \left[4 \left(\frac{1}{2} t + \frac{1}{4} \sin 2t \right) + 3t^3 \right]_0^{2\pi}$$

$$= \sqrt{13} k (4\pi + 24\pi^3) = 4\sqrt{13} \pi k (1 + 6\pi^2)$$

[continued]

$$\begin{aligned}
 I_y &= \int_C (x^2 + z^2) \rho(x, y, z) \, ds = \int_0^{2\pi} (4 \sin^2 t + 9t^2) (k) \sqrt{13} \, dt = \sqrt{13} k \left[4 \left(\frac{1}{2} t - \frac{1}{4} \sin 2t \right) + 3t^3 \right]_0^{2\pi} \\
 &= \sqrt{13} k (4\pi + 24\pi^3) = 4\sqrt{13} \pi k (1 + 6\pi^2)
 \end{aligned}$$

$$I_z = \int_C (x^2 + y^2) \rho(x, y, z) \, ds = \int_0^{2\pi} (4 \sin^2 t + 4 \cos^2 t) (k) \sqrt{13} \, dt = 4\sqrt{13} k \int_0^{2\pi} dt = 8\pi \sqrt{13} k$$

$$\begin{aligned}
 41. \quad W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \langle t - \sin t, (1 - \cos t) + 2 \rangle \cdot \langle 1 - \cos t, \sin t \rangle \, dt \\
 &= \int_0^{2\pi} (t - t \cos t - \sin t + \sin t \cos t + 3 \sin t - \sin t \cos t) \, dt \\
 &= \int_0^{2\pi} (t - t \cos t + 2 \sin t) \, dt = \left[\frac{1}{2} t^2 - (t \sin t + \cos t) - 2 \cos t \right]_0^{2\pi} \quad \left[\begin{array}{l} \text{integrate by parts} \\ \text{in the second term} \end{array} \right] \\
 &= 2\pi^2
 \end{aligned}$$

42. Choosing y as the parameter, the curve C is parametrized by $x = y^2 + 1$, $y = y$, $0 \leq y \leq 1$. Then

$$\begin{aligned}
 W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \langle (y^2 + 1)^2, ye^{y^2+1} \rangle \cdot \langle 2y, 1 \rangle \, dy = \int_0^1 [2y(y^2 + 1)^2 + ye^{y^2+1}] \, dy \\
 &= \left[\frac{1}{3} (y^2 + 1)^3 + \frac{1}{2} e^{y^2+1} \right]_0^1 = \frac{8}{3} + \frac{1}{2} e^2 - \frac{1}{3} - \frac{1}{2} e = \frac{1}{2} e^2 - \frac{1}{2} e + \frac{7}{3}
 \end{aligned}$$

43. $\mathbf{r}(t) = \langle 2t, t, 1 - t \rangle$, $0 \leq t \leq 1$.

$$\begin{aligned}
 W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \langle 2t - t^2, t - (1 - t)^2, 1 - t - (2t)^2 \rangle \cdot \langle 2, 1, -1 \rangle \, dt \\
 &= \int_0^1 (4t - 2t^2 + t - 1 + 2t - t^2 - 1 + t + 4t^2) \, dt = \int_0^1 (t^2 + 8t - 2) \, dt = \left[\frac{1}{3} t^3 + 4t^2 - 2t \right]_0^1 = \frac{7}{3}
 \end{aligned}$$

44. $\mathbf{r}(t) = 2\mathbf{i} + t\mathbf{j} + 5t\mathbf{k}$, $0 \leq t \leq 1$. Therefore

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \frac{K \langle 2, t, 5t \rangle}{(4 + 26t^2)^{3/2}} \cdot \langle 0, 1, 5 \rangle \, dt = K \int_0^1 \frac{26t}{(4 + 26t^2)^{3/2}} \, dt = K \left[-(4 + 26t^2)^{-1/2} \right]_0^1 = K \left(\frac{1}{2} - \frac{1}{\sqrt{30}} \right).$$

45. (a) $\mathbf{r}(t) = at^2 \mathbf{i} + bt^3 \mathbf{j} \Rightarrow \mathbf{v}(t) = \mathbf{r}'(t) = 2at \mathbf{i} + 3bt^2 \mathbf{j} \Rightarrow \mathbf{a}(t) = \mathbf{v}'(t) = 2a \mathbf{i} + 6bt \mathbf{j}$, and force is mass times acceleration: $\mathbf{F}(t) = m \mathbf{a}(t) = 2ma \mathbf{i} + 6mbt \mathbf{j}$, $0 \leq t \leq 1$.

$$\begin{aligned}
 (b) \quad W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 (2ma \mathbf{i} + 6mbt \mathbf{j}) \cdot (2at \mathbf{i} + 3bt^2 \mathbf{j}) \, dt = \int_0^1 (4ma^2 t + 18mb^2 t^3) \, dt \\
 &= \left[2ma^2 t^2 + \frac{9}{2} mb^2 t^4 \right]_0^1 = 2ma^2 + \frac{9}{2} mb^2
 \end{aligned}$$

46. $\mathbf{r}(t) = a \sin t \mathbf{i} + b \cos t \mathbf{j} + ct \mathbf{k} \Rightarrow \mathbf{v}(t) = \mathbf{r}'(t) = a \cos t \mathbf{i} - b \sin t \mathbf{j} + c \mathbf{k} \Rightarrow \mathbf{a}(t) = \mathbf{v}'(t) = -a \sin t \mathbf{i} - b \cos t \mathbf{j}$ and $\mathbf{F}(t) = m \mathbf{a}(t) = -ma \sin t \mathbf{i} - mb \cos t \mathbf{j}$. Thus

$$\begin{aligned}
 W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{\pi/2} (-ma \sin t \mathbf{i} - mb \cos t \mathbf{j}) \cdot (a \cos t \mathbf{i} - b \sin t \mathbf{j} + c \mathbf{k}) \, dt \\
 &= \int_0^{\pi/2} (-ma^2 \sin t \cos t + mb^2 \sin t \cos t) \, dt = m(b^2 - a^2) \left[\frac{1}{2} \sin^2 t \right]_0^{\pi/2} = \frac{1}{2} m(b^2 - a^2)
 \end{aligned}$$

47. The combined weight of the man and the paint is 185 lb, so the force exerted (equal and opposite to that exerted by gravity) is

$\mathbf{F} = 185 \mathbf{k}$. To parametrize the staircase, let $x = 20 \cos t$, $y = 20 \sin t$, $z = \frac{90}{6\pi} t = \frac{15}{\pi} t$, $0 \leq t \leq 6\pi$. Then the work done is

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{6\pi} \langle 0, 0, 185 \rangle \cdot \langle -20 \sin t, 20 \cos t, \frac{15}{\pi} \rangle \, dt = (185) \frac{15}{\pi} \int_0^{6\pi} dt = (185) \left(\frac{15}{\pi} \right) (6\pi) \approx 1.67 \times 10^4 \text{ ft}\cdot\text{lb}$$

48. This time m is a function of t : $m = 185 - \frac{9}{6\pi}t = 185 - \frac{3}{2\pi}t$. So let $\mathbf{F} = (185 - \frac{3}{2\pi}t)\mathbf{k}$. To parametrize the staircase, let $x = 20 \cos t$, $y = 20 \sin t$, $z = \frac{90}{6\pi}t = \frac{15}{\pi}t$, $0 \leq t \leq 6\pi$. Therefore

$$\begin{aligned} W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{6\pi} \langle 0, 0, 185 - \frac{3}{2\pi}t \rangle \cdot \langle -20 \sin t, 20 \cos t, \frac{15}{\pi} \rangle dt = \frac{15}{\pi} \int_0^{6\pi} (185 - \frac{3}{2\pi}t) dt \\ &= \frac{15}{\pi} \left[185t - \frac{3}{4\pi}t^2 \right]_0^{6\pi} = 90 \left(185 - \frac{9}{2} \right) \approx 1.62 \times 10^4 \text{ ft}\cdot\text{lb} \end{aligned}$$

49. (a) $\mathbf{r}(t) = \langle \cos t, \sin t \rangle$, $0 \leq t \leq 2\pi$, and let $\mathbf{F} = \langle a, b \rangle$. Then

$$\begin{aligned} W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \langle a, b \rangle \cdot \langle -\sin t, \cos t \rangle dt = \int_0^{2\pi} (-a \sin t + b \cos t) dt = [a \cos t + b \sin t]_0^{2\pi} \\ &= a + 0 - a + 0 = 0 \end{aligned}$$

- (b) Yes. $\mathbf{F}(x, y) = k\mathbf{x} = \langle kx, ky \rangle$ and

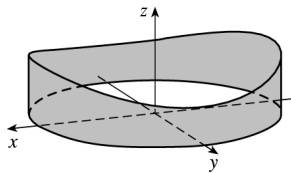
$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \langle k \cos t, k \sin t \rangle \cdot \langle -\sin t, \cos t \rangle dt = \int_0^{2\pi} (-k \sin t \cos t + k \sin t \cos t) dt = \int_0^{2\pi} 0 dt = 0.$$

50. Consider the base of the fence in the xy -plane, centered at the origin, with the height given by

$z = h(x, y) = 4 + 0.01(x^2 - y^2)$. To graph the fence, observe that the fence is highest when $y = 0$ (where the height is 5 m) and lowest when $x = 0$ (a height of 3 m). When $y = \pm x$, the height is 4 m.

Also, the fence can be graphed using parametric equations (see Section 16.6): $x = 10 \cos u$, $y = 10 \sin u$,

$$\begin{aligned} z &= v[4 + 0.01((10 \cos u)^2 - (10 \sin u)^2)] = v(4 + \cos^2 u - \sin^2 u) \\ &= v(4 + \cos 2u), \quad 0 \leq u \leq 2\pi, \quad 0 \leq v \leq 1. \end{aligned}$$



The surface area of one side of the fence is $\int_C h(x, y) ds$, where the base C of the fence is given by

$x = 10 \cos t$, $y = 10 \sin t$, $0 \leq t \leq 2\pi$. Then

$$\begin{aligned} \int_C h(x, y) ds &= \int_0^{2\pi} [4 + 0.01((10 \cos t)^2 - (10 \sin t)^2)] \sqrt{(-10 \sin t)^2 + (10 \cos t)^2} dt \\ &= \int_0^{2\pi} (4 + \cos 2t) \sqrt{100} dt = 10 \left[4t + \frac{1}{2} \sin 2t \right]_0^{2\pi} = 10(8\pi) = 80\pi \text{ m}^2 \end{aligned}$$

If we paint both sides of the fence, the total surface area to cover is $160\pi \text{ m}^2$, and since 1 L of paint covers 100 m^2 , we require

$$\frac{160\pi}{100} = 1.6\pi \approx 5.03 \text{ L of paint.}$$

51. Let $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$. Then

$$\begin{aligned} \int_C \mathbf{v} \cdot d\mathbf{r} &= \int_a^b \langle v_1, v_2, v_3 \rangle \cdot \langle x'(t), y'(t), z'(t) \rangle dt = \int_a^b [v_1 x'(t) + v_2 y'(t) + v_3 z'(t)] dt \\ &= [v_1 x(t) + v_2 y(t) + v_3 z(t)]_a^b = [v_1 x(b) + v_2 y(b) + v_3 z(b)] - [v_1 x(a) + v_2 y(a) + v_3 z(a)] \\ &= v_1 [x(b) - x(a)] + v_2 [y(b) - y(a)] + v_3 [z(b) - z(a)] \\ &= \langle v_1, v_2, v_3 \rangle \cdot \langle x(b) - x(a), y(b) - y(a), z(b) - z(a) \rangle \\ &= \langle v_1, v_2, v_3 \rangle \cdot [\langle x(b), y(b), z(b) \rangle - \langle x(a), y(a), z(a) \rangle] = \mathbf{v} \cdot [\mathbf{r}(b) - \mathbf{r}(a)] \end{aligned}$$

52. If $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ then

$$\begin{aligned}\int_C \mathbf{r} \cdot d\mathbf{r} &= \int_a^b \langle x(t), y(t), z(t) \rangle \cdot \langle x'(t), y'(t), z'(t) \rangle dt = \int_a^b [x(t)x'(t) + y(t)y'(t) + z(t)z'(t)] dt \\ &= \left[\frac{1}{2}[x(t)]^2 + \frac{1}{2}[y(t)]^2 + \frac{1}{2}[z(t)]^2 \right]_a^b \\ &= \frac{1}{2} \{ [x(b)]^2 + [y(b)]^2 + [z(b)]^2 - ([x(a)]^2 + [y(a)]^2 + [z(a)]^2) \} \\ &= \frac{1}{2} [|\mathbf{r}(b)|^2 - |\mathbf{r}(a)|^2]\end{aligned}$$

53. The work done in moving the object is $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T} ds$. We can approximate this integral by dividing C into 7 segments of equal length $\Delta s = 2$ and approximating $\mathbf{F} \cdot \mathbf{T}$, that is, the tangential component of force, at a point (x_i^*, y_i^*) on each segment. Since C is composed of straight line segments, $\mathbf{F} \cdot \mathbf{T}$ is the scalar projection of each force vector onto C . If we choose (x_i^*, y_i^*) to be the point on the segment closest to the origin, then the work done is

$$\int_C \mathbf{F} \cdot \mathbf{T} ds \approx \sum_{i=1}^7 [\mathbf{F}(x_i^*, y_i^*) \cdot \mathbf{T}(x_i^*, y_i^*)] \Delta s = [2 + 2 + 2 + 2 + 1 + 1 + 1](2) = 22$$

Thus, we estimate the work done to be approximately 22 J.

54. Use the orientation pictured in the figure. Then since $\mathbf{B}$ is tangent to any circle that lies in the plane perpendicular to the wire, $\mathbf{B} = |\mathbf{B}| \mathbf{T}$ where $\mathbf{T}$ is the unit tangent to the circle $C: x = r \cos \theta, y = r \sin \theta$. Thus $\mathbf{B} = |\mathbf{B}| \langle -\sin \theta, \cos \theta \rangle$. Then $\int_C \mathbf{B} \cdot d\mathbf{r} = \int_0^{2\pi} |\mathbf{B}| \langle -\sin \theta, \cos \theta \rangle \cdot \langle -r \sin \theta, r \cos \theta \rangle d\theta = \int_0^{2\pi} |\mathbf{B}| r d\theta = 2\pi r |\mathbf{B}|$. (Note that $|\mathbf{B}|$ here is the magnitude of the field at a distance r from the wire's center.) But by Ampere's Law $\int_C \mathbf{B} \cdot d\mathbf{r} = \mu_0 I$. Hence $|\mathbf{B}| = \mu_0 I / (2\pi r)$.

16.3 The Fundamental Theorem for Line Integrals

- C appears to be a smooth curve, and since ∇f is continuous, we know f is differentiable. Then Theorem 2 says that the value of $\int_C \nabla f \cdot d\mathbf{r}$ is simply the difference of the values of f at the terminal and initial points of C . From the graph, this is $50 - 10 = 40$.
- C is represented by the vector function $\mathbf{r}(t) = (t^2 + 1)\mathbf{i} + (t^3 + t)\mathbf{j}$, $0 \leq t \leq 1$, so $\mathbf{r}'(t) = 2t\mathbf{i} + (3t^2 + 1)\mathbf{j}$. Since $3t^2 + 1 \neq 0$, we have $\mathbf{r}'(t) \neq \mathbf{0}$, thus C is a smooth curve. ∇f is continuous, and hence f is differentiable, so by Theorem 2 we have $\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(1)) - f(\mathbf{r}(0)) = f(2, 2) - f(1, 0) = 9 - 3 = 6$.
- Let $P(x, y) = xy + y^2$ and $Q(x, y) = x^2 + 2xy$. Then $\partial P / \partial y = x + 2y$ and $\partial Q / \partial x = 2x + 2y$. Since $\partial P / \partial y \neq \partial Q / \partial x$, $\mathbf{F}(x, y) = P\mathbf{i} + Q\mathbf{j}$ is not conservative by Theorem 5.
- $\partial(y^2 - 2x) / \partial y = 2y = \partial(2xy) / \partial x$ and the domain of $\mathbf{F}$ is $\mathbb{R}^2$ which is open and simply-connected, so $\mathbf{F}$ is conservative by Theorem 6. Thus, there exists a function f such that $\nabla f = \mathbf{F}$, that is, $f_x(x, y) = y^2 - 2x$ and $f_y(x, y) = 2xy$. But $f_x(x, y) = y^2 - 2x$ implies $f(x, y) = xy^2 - x^2 + g(y)$ and differentiating both sides of this equation with respect to y gives $f_y(x, y) = 2xy + g'(y)$. Thus $2xy = 2xy + g'(y)$ so $g'(y) = 0$ and $g(y) = K$ where K is a constant. Hence $f(x, y) = xy^2 - x^2 + K$ is a potential function for $\mathbf{F}$.

$$5. \frac{\partial}{\partial y} (y^2 e^{xy}) = y^2 \cdot x e^{xy} + 2y e^{xy} = (xy^2 + 2y) e^{xy},$$

$$\frac{\partial}{\partial x} [(1 + xy) e^{xy}] = (1 + xy) \cdot y e^{xy} + y e^{xy} = y e^{xy} + xy^2 e^{xy} + y e^{xy} = (xy^2 + 2y) e^{xy}.$$

Since these partial derivatives are equal and the domain of $\mathbf{F}$ is $\mathbb{R}^2$ which is open and simply-connected, $\mathbf{F}$ is conservative by Theorem 6. Thus, there exists a function f such that $\nabla f = \mathbf{F}$, that is, $f_x(x, y) = y^2 e^{xy}$ and $f_y(x, y) = (1 + xy) e^{xy}$. But $f_x(x, y) = y^2 e^{xy}$ implies $f(x, y) = y e^{xy} + g(y)$ and differentiating both sides of this equation with respect to y gives $f_y(x, y) = (1 + xy) e^{xy} + g'(y)$. Thus $(1 + xy) e^{xy} = (1 + xy) e^{xy} + g'(y)$ so $g'(y) = 0$ and $g(y) = K$ where K is a constant. Hence $f(x, y) = y e^{xy} + K$ is a potential function for $\mathbf{F}$.

$$6. \partial(y e^x)/\partial y = e^x = \partial(e^x + e^y)/\partial x \text{ and the domain of } \mathbf{F} \text{ is } \mathbb{R}^2 \text{ which is open and simply-connected, so } \mathbf{F} \text{ is conservative.}$$

Hence there exists a function f such that $\nabla f = \mathbf{F}$. Here $f_x(x, y) = y e^x$ implies $f(x, y) = y e^x + g(y)$ and then $f_y(x, y) = e^x + g'(y)$. But $f_y(x, y) = e^x + e^y$ so $g'(y) = e^y \Rightarrow g(y) = e^y + K$ and $f(x, y) = y e^x + e^y + K$ is a potential function for $\mathbf{F}$.

$$7. \partial(y e^x + \sin y)/\partial y = e^x + \cos y = \partial(e^x + x \cos y)/\partial x \text{ and the domain of } \mathbf{F} \text{ is } \mathbb{R}^2. \text{ Hence } \mathbf{F} \text{ is conservative so there exists a function } f \text{ such that } \nabla f = \mathbf{F}. \text{ Then } f_x(x, y) = y e^x + \sin y \text{ implies } f(x, y) = y e^x + x \sin y + g(y) \text{ and } f_y(x, y) = e^x + x \cos y + g'(y). \text{ But } f_y(x, y) = e^x + x \cos y \text{ so } g(y) = K \text{ and } f(x, y) = y e^x + x \sin y + K \text{ is a potential function for } \mathbf{F}.$$

$$8. \partial(2xy + y^{-2})/\partial y = 2x - 2y^{-3} = \partial(x^2 - 2xy^{-3})/\partial x \text{ and the domain of } \mathbf{F} \text{ is } \{(x, y) \mid y > 0\} \text{ which is open and simply-connected. Hence } \mathbf{F} \text{ is conservative, so there exists a function } f \text{ such that } \nabla f = \mathbf{F}. \text{ Then } f_x(x, y) = 2xy + y^{-2} \text{ implies } f(x, y) = x^2 y + xy^{-2} + g(y) \text{ and } f_y(x, y) = x^2 - 2xy^{-3} + g'(y). \text{ But } f_y(x, y) = x^2 - 2xy^{-3} \text{ so } g'(y) = 0 \Rightarrow g(y) = K. \text{ Then } f(x, y) = x^2 y + xy^{-2} + K \text{ is a potential function for } \mathbf{F}.$$

$$9. \partial(y^2 \cos x + \cos y)/\partial y = 2y \cos x - \sin y = \partial(2y \sin x - x \sin y)/\partial x \text{ and the domain of } \mathbf{F} \text{ is } \mathbb{R}^2 \text{ which is open and simply-connected. Hence } \mathbf{F} \text{ is conservative so there exists a function } f \text{ such that } \nabla f = \mathbf{F}. \text{ Then } f_x(x, y) = y^2 \cos x + \cos y \text{ implies } f(x, y) = y^2 \sin x + x \cos y + g(y) \text{ and } f_y(x, y) = 2y \sin x - x \sin y + g'(y). \text{ But } f_y(x, y) = 2y \sin x - x \sin y \text{ so } g'(y) = 0 \Rightarrow g(y) = K \text{ and } f(x, y) = y^2 \sin x + x \cos y + K \text{ is a potential function for } \mathbf{F}.$$

$$10. \partial(\ln y + y/x)/\partial y = 1/y + 1/x = \partial(\ln x + x/y)/\partial x \text{ and the domain of } \mathbf{F} \text{ is } \{(x, y) \mid x > 0, y > 0\} \text{ which is open and simply-connected. Hence } \mathbf{F} \text{ is conservative so there exists a function } f \text{ such that } \nabla f = \mathbf{F}. \text{ Then } f_x(x, y) = \ln y + y/x \text{ implies } f(x, y) = x \ln y + y \ln x + g(y) \text{ and } f_y(x, y) = x/y + \ln x + g'(y). \text{ But } f_y(x, y) = \ln x + x/y \text{ so } g'(y) = 0 \Rightarrow g(y) = K \text{ and } f(x, y) = x \ln y + y \ln x + K \text{ is a potential function for } \mathbf{F}.$$

$$11. (a) \mathbf{F} \text{ has continuous first-order partial derivatives and } \frac{\partial}{\partial y} (2xy) = 2x = \frac{\partial}{\partial x} (x^2) \text{ on } \mathbb{R}^2, \text{ which is open and}$$

simply-connected. Thus, $\mathbf{F}$ is conservative by Theorem 6. Then we know that the line integral of $\mathbf{F}$ is independent of path;

in particular, the value of $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends only on the endpoints of C . Since all three curves have the same initial and terminal points, $\int_C \mathbf{F} \cdot d\mathbf{r}$ will have the same value for each curve.

- (b) We first find a potential function f , so that $\nabla f = \mathbf{F}$. We know $f_x(x, y) = 2xy$ and $f_y(x, y) = x^2$. Integrating

$f_x(x, y)$ with respect to x , we have $f(x, y) = x^2y + g(y)$. Differentiating both sides with respect to y gives

$f_y(x, y) = x^2 + g'(y)$, so we must have $x^2 + g'(y) = x^2 \Rightarrow g'(y) = 0 \Rightarrow g(y) = K$, a constant.

Thus $f(x, y) = x^2y + K$, and we can take $K = 0$. All three curves start at $(1, 2)$ and end at $(3, 2)$, so by Theorem 2,

$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(3, 2) - f(1, 2) = 3^2(2) - 1^2(2) = 16$ for each curve.

12. (a) $\mathbf{F}(x, y) = 2xy \mathbf{i} + (x^2 + \sin y) \mathbf{j}$.

Solution 1: $\mathbf{F}$ has continuous first-order partial derivatives and $\frac{\partial(2xy)}{\partial y} = 2x = \frac{\partial(x^2 + \sin y)}{\partial x}$ on $\mathbb{R}^2$, which is open and

simply-connected. Therefore, the vector field is conservative and there exists a function $f(x, y)$ such that $\nabla f = \mathbf{F}$. Here,

$f_x(x, y) = 2xy$ implies $f(x, y) = x^2y + g(y)$ and $f_y(x, y) = x^2 + g'(y)$, but $f_y(x, y) = x^2 + \sin y$, which implies

$g'(y) = \sin y \Rightarrow g(y) = -\cos y + K$ and $f(x, y) = x^2y - \cos y + K$ is a potential function for $\mathbf{F}$. Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(2, \frac{\pi}{2}) - f(0, 0) = 2^2(\frac{\pi}{2}) - \cos(\frac{\pi}{2}) + K - 0 + \cos 0 - K = 2\pi + 1$$

Solution 2: As in Example 4, since $\mathbf{F}$ is conservative, we know that $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path, so we replace the

curve C by the simpler curve C_1 consisting of the line segment connecting the two endpoints of C . Thus, C_1 can be

represented by $\mathbf{r}(t) = 2t \mathbf{i} + \frac{\pi}{2}t \mathbf{j}$, $0 \leq t \leq 1$. Then

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \\ &= \int_0^1 \{2(2t)(\frac{\pi}{2}t)(2) + [(2t)^2 + \sin(\frac{\pi}{2}t)](\frac{\pi}{2})\} dt = \left[\int_0^1 6\pi t^2 + \frac{\pi}{2} \sin(\frac{\pi}{2}t) \right] dt \\ &= [2\pi t^3 - \cos(\frac{\pi}{2}t)]_0^1 = (2\pi - 0) - (0 - 1) = 2\pi + 1 \end{aligned}$$

- (b) From part (a), we know that $\mathbf{F}$ is conservative and therefore independent of path. Thus, since C is a closed path, by

Theorem 3, $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$.

13. (a) $\mathbf{F}(x, y) = (3x^2 + y^2) \mathbf{i} + 2xy \mathbf{j}$. $\mathbf{r}(t) = 2 \cos t \mathbf{i} + 2 \sin t \mathbf{j}$, $\pi \leq t \leq 2\pi$. Then

$$\mathbf{F}(\mathbf{r}(t)) = [3(2 \cos t)^2 + (2 \sin t)^2] \mathbf{i} + 2(2 \cos t)(2 \sin t) \mathbf{j} = (12 \cos^2 t + 4 \sin^2 t) \mathbf{i} + 8 \cos t \sin t \mathbf{j}$$

and $\mathbf{r}'(t) = -2 \sin t \mathbf{i} + 2 \cos t \mathbf{j}$, so

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_{\pi}^{2\pi} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_{\pi}^{2\pi} [-2 \sin t(12 \cos^2 t + 4 \sin^2 t) + 2 \cos t(8 \cos t \sin t)] dt \\ &= \int_{\pi}^{2\pi} (-8 \sin t \cos^2 t - 8 \sin^3 t) dt = -8 \int_{\pi}^{2\pi} \sin t dt = 8 [\cos t]_{\pi}^{2\pi} = 8[1 - (-1)] = 16 \end{aligned}$$

- (b) $\frac{\partial(3x^2 + y^2)}{\partial y} = 2y = \frac{\partial(2xy)}{\partial x}$ and the domain of $\mathbf{F}$ is $\mathbb{R}^2$, which is open and simply-connected. Thus, $\mathbf{F}$ is conservative

and there exists a function $f(x, y)$ such that $\nabla f = \mathbf{F}$. Then $f_x(x, y) = 3x^2 + y^2$ implies $f(x, y) = x^3 + xy^2 + g(y)$.

[continued]

Differentiating both sides with respect to y gives $f_y(x, y) = 2xy + g'(y)$, so we must have $2xy + g'(y) = 2xy \Rightarrow g'(y) = 0 \Rightarrow g(y) = K$, a constant. Thus, $f(x, y) = x^3 + xy^2 + K$ is a potential function for $\mathbf{F}$.

(c) From part (b), $f(x, y) = x^3 + xy^2 + K$. Thus,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(2, 0) - f(-2, 0) = 2^3 + 0 + K - (-2)^3 - 0 - K = 16$$

(d) We replace C with the line segment from $(-2, 0)$ to $(2, 0)$ so $\mathbf{r}(t) = \langle t, 0 \rangle$, $-2 \leq t \leq 2$, which implies $\mathbf{r}'(t) = \langle 1, 0 \rangle$ and

$$\mathbf{F}(\mathbf{r}(t)) = \langle 3t^2, 0 \rangle. \text{ Thus, } \int_C \mathbf{F} \cdot d\mathbf{r} = \int_{-2}^2 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_{-2}^2 3t^2 dt = [t^3]_{-2}^2 = 2^3 - (-2)^3 = 16.$$

14. (a) $\mathbf{F}(x, y) = \langle \sin y + e^x, x \cos y \rangle$ and $C: x = t, y = t(3 - t), 0 \leq t \leq 3$.

$\frac{\partial(\sin y + e^x)}{\partial y} = \cos y = \frac{\partial(x \cos y)}{\partial x}$ and the domain of $\mathbf{F}$ is $\mathbb{R}^2$, which is open and simply-connected. Thus, $\mathbf{F}$ is

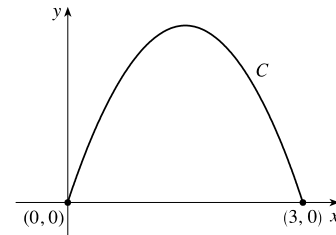
conservative and there exists a function $f(x, y)$ such that $\nabla f = \mathbf{F}$. Then $f_x(x, y) = \sin y + e^x$ implies

$f(x, y) = x \sin y + e^x + g(y)$, so $f_y(x, y) = x \cos y + g'(y)$, which implies $g'(y) = 0 \Rightarrow g(y) = K$. Therefore,

$f(x, y) = x \sin y + e^x + K$ is a potential function for $\mathbf{F}$.

(b) The endpoints of C are $(0, 0)$ and $(3, 0)$. Thus, by Theorem 2,

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C \nabla f \cdot d\mathbf{r} = f(3, 0) - f(0, 0) \\ &= 3 \cdot \sin 0 + e^3 + K - (0 \cdot \sin 0 + e^0 + K) \\ &= e^3 - 1 \end{aligned}$$



(c) We replace C with the line segment that connects the endpoints of C

along the x -axis. So $x = t, y = 0, 0 \leq t \leq 3$. Then $\mathbf{r}(t) = \langle t, 0 \rangle$, $\mathbf{r}'(t) = \langle 1, 0 \rangle$, and

$$\mathbf{F}(\mathbf{r}(t)) = \langle \sin 0 + e^t, t \cos 0 \rangle = \langle e^t, t \rangle. \text{ Therefore, } \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^3 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^3 e^t dt = [e^t]_0^3 = e^3 - 1.$$

15. (a) $\mathbf{F}(x, y) = \langle ye^{xy}, xe^{xy} \rangle$ and $C: x = \sin \frac{\pi}{2}t, y = e^{t-1}(1 - \cos \pi t), 0 \leq t \leq 1$.

$\frac{\partial(ye^{xy})}{\partial y} = e^{xy} + xye^{xy} = \frac{\partial(xe^{xy})}{\partial x}$ and the domain of $\mathbf{F}$ is $\mathbb{R}^2$, which is open and simply connected. Thus, $\mathbf{F}$ is

conservative and there exists a function $f(x, y)$ such that $\nabla f = \mathbf{F}$. Then $f_x(x, y) = ye^{xy}$ implies $f(x, y) = e^{xy} + g(y)$,

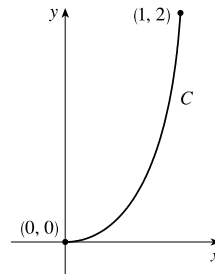
so $f_y(x, y) = xe^{xy} + g'(y)$, which implies $g'(y) = 0 \Rightarrow g(y) = K$. Therefore, $f(x, y) = e^{xy} + K$ is a potential function for $\mathbf{F}$.

(b) The endpoints of C are $(0, 0)$ and $(1, 2)$. Thus, by Theorem 2,

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C \nabla f \cdot d\mathbf{r} = f(1, 2) - f(0, 0) \\ &= e^2 + K - e^0 - K = e^2 - 1 \end{aligned}$$

(c) We replace C with the line segment that connects the endpoints of C .

So $x = t$ and $y = 2t, 0 \leq t \leq 1$. Then $\mathbf{r}(t) = \langle t, 2t \rangle$, $\mathbf{r}'(t) = \langle 1, 2 \rangle$, and



$\mathbf{F}(\mathbf{r}(t)) = \langle 2te^{2t^2}, te^{2t^2} \rangle$. Therefore,

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^1 4te^{2t^2} dt \\ &= \int_0^2 e^u du \quad [u = 2t^2, du = 4t dt] \\ &= [e^u]_0^2 = e^2 - 1\end{aligned}$$

16. $f(x, y, z) = xy^2z + x^2$ and $C: x = t^2, y = e^{t^2-1}, z = t^2 + t, -1 \leq t \leq 1$. ∇f is the gradient vector field of f and therefore conservative. Thus, by Theorem 2, $\int_C \nabla f \cdot d\mathbf{r} = f(1, 1, 2) - f(1, 1, 0) = [(1)(1^2)(2) + 1^2] - (0 + 1^2) = 2$.

17. (a) $\mathbf{F}(x, y) = \langle 2x, 4y \rangle$. If $\mathbf{F} = \nabla f$, then $f_x(x, y) = 2x$ and $f_y(x, y) = 4y$. $f_x(x, y) = 2x$ implies that $f(x, y) = x^2 + g(y)$ and $f_y(x, y) = g'(y) = 4y$, so $g(y) = 2y^2 + K$. We can take $K = 0$, so $f(x, y) = x^2 + 2y^2$.

- (b) C is a smooth curve with initial point $(4, -2)$ and terminal point $(1, 1)$, so by Theorem 2,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(1, 1) - f(4, -2) = (1 + 2) - (16 + 8) = -21.$$

18. (a) $\mathbf{F}(x, y) = (3 + 2xy^2)\mathbf{i} + 2x^2y\mathbf{j}$. If $\mathbf{F} = \nabla f$, then $f_x(x, y) = 3 + 2xy^2$ and $f_y(x, y) = 2x^2y$.

$f_x(x, y) = 3 + 2xy^2$ implies $f(x, y) = 3x + x^2y^2 + g(y)$ and $f_y(x, y) = 2x^2y + g'(y)$. But $f_y(x, y) = 2x^2y$ so $g'(y) = 0 \Rightarrow g(y) = K$. We can take $K = 0$, so $f(x, y) = 3x + x^2y^2$.

- (b) C is a smooth curve with initial point $(1, 1)$ and terminal point $(4, \frac{1}{4})$, so by Theorem 2,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(4, \frac{1}{4}) - f(1, 1) = (12 + 1) - (3 + 1) = 9.$$

19. (a) $\mathbf{F}(x, y) = x^2y^3\mathbf{i} + x^3y^2\mathbf{j}$. If $\mathbf{F} = \nabla f$, then $f_x(x, y) = x^2y^3$ and $f_y(x, y) = x^3y^2$.

$f_x(x, y) = x^2y^3$ implies $f(x, y) = \frac{1}{3}x^3y^3 + g(y)$ and $f_y(x, y) = x^3y^2 + g'(y)$. But $f_y(x, y) = x^3y^2$, so $g'(y) = 0 \Rightarrow g(y) = K$, a constant. We can take $K = 0$, so $f(x, y) = \frac{1}{3}x^3y^3$.

- (b) C is a smooth curve with initial point $\mathbf{r}(0) = (0, 0)$ and terminal point $\mathbf{r}(1) = (-1, 3)$, so by Theorem 2,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(-1, 3) - f(0, 0) = -9 - 0 = -9.$$

20. (a) $\mathbf{F}(x, y) = (1 + xy)e^{xy}\mathbf{i} + x^2e^{xy}\mathbf{j}$. $f_y(x, y) = x^2e^{xy}$ implies $f(x, y) = xe^{xy} + g(x) \Rightarrow$

$f_x(x, y) = xye^{xy} + e^{xy} + g'(x) = (1 + xy)e^{xy} + g'(x)$. But $f_x(x, y) = (1 + xy)e^{xy}$ so $g'(x) = 0 \Rightarrow g(x) = K$. We can take $K = 0$, so $f(x, y) = xe^{xy}$.

- (b) The initial point of C is $\mathbf{r}(0) = (1, 0)$ and the terminal point is $\mathbf{r}(\pi/2) = (0, 2)$, so by Theorem 2,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(0, 2) - f(1, 0) = 0 - e^0 = -1.$$

21. (a) $\mathbf{F}(x, y, z) = 2xy\mathbf{i} + (x^2 + 2yz)\mathbf{j} + y^2\mathbf{k}$. $f_x(x, y, z) = 2xy$ implies that $f(x, y, z) = x^2y + g(y, z)$ and so

$f_y(x, y, z) = x^2 + g_y(y, z)$. But $f_y(x, y, z) = x^2 + 2yz$, which implies $g_y(y, z) = 2yz + h_y(z) \Rightarrow$

$g(y, z) = y^2z + h(z)$. So $f(x, y, z) = x^2y + y^2z + h(z)$ and $f_z(x, y, z) = y^2 + h'(z)$. But $f_z(x, y, z) = y^2 \Rightarrow$

$h'(z) = 0 \Rightarrow h(z) = K$. We can take $K = 0$, so $f(x, y, z) = x^2y + y^2z$.

(b) C is a smooth curve with initial point $(2, -3, 1)$ and terminal point $(-5, 1, 2)$, so by Theorem 2,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(-5, 1, 2) - f(2, -3, 1) = (25 + 2) - (-12 + 9) = 30.$$

22. (a) $\mathbf{F}(x, y, z) = (y^2z + 2xz^2)\mathbf{i} + 2xyz\mathbf{j} + (xy^2 + 2x^2z)\mathbf{k}$. $f_x(x, y, z) = y^2z + 2xz^2$ implies

$$f(x, y, z) = xy^2z + x^2z^2 + g(y, z) \text{ and so } f_y(x, y, z) = 2xyz + g_y(y, z). \text{ But } f_y(x, y, z) = 2xyz \text{ so}$$

$$g_y(y, z) = 0 \Rightarrow g(y, z) = h(z). \text{ Thus, } f(x, y, z) = xy^2z + x^2z^2 + h(z) \text{ and } f_z(x, y, z) = xy^2 + 2x^2z + h'(z).$$

$$\text{But } f_z(x, y, z) = xy^2 + 2x^2z, \text{ so } h'(z) = 0 \Rightarrow h(z) = K. \text{ Hence, } f(x, y, z) = xy^2z + x^2z^2 \text{ (taking } K = 0\text{).}$$

(b) $t = 0$ corresponds to the point $(0, 1, 0)$ and $t = 1$ corresponds to $(1, 2, 1)$, so by Theorem 2,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(1, 2, 1) - f(0, 1, 0) = 5 - 0 = 5.$$

23. (a) $\mathbf{F}(x, y, z) = yze^{xz}\mathbf{i} + e^{xz}\mathbf{j} + xye^{xz}\mathbf{k}$. $f_x(x, y, z) = yze^{xz}$ implies $f(x, y, z) = ye^{xz} + g(y, z)$ and so

$$f_y(x, y, z) = e^{xz} + g_y(y, z). \text{ But } f_y(x, y, z) = e^{xz} \text{ so } g_y(y, z) = 0 \Rightarrow g(y, z) = h(z). \text{ Thus,}$$

$$f(x, y, z) = ye^{xz} + h(z) \text{ and } f_z(x, y, z) = xye^{xz} + h'(z). \text{ But } f_z(x, y, z) = xye^{xz}, \text{ so } h'(z) = 0 \Rightarrow h(z) = K.$$

$$\text{Hence } f(x, y, z) = ye^{xz} \text{ (taking } K = 0\text{).}$$

(b) $\mathbf{r}(0) = \langle 1, -1, 0 \rangle$, $\mathbf{r}(2) = \langle 5, 3, 0 \rangle$ so $\int_C \mathbf{F} \cdot d\mathbf{r} = f(5, 3, 0) - f(1, -1, 0) = 3e^0 + e^0 = 4$.

24. (a) $\mathbf{F}(x, y, z) = \sin y\mathbf{i} + (x \cos y + \cos z)\mathbf{j} - y \sin z\mathbf{k}$. $f_x(x, y, z) = \sin y$ implies $f(x, y, z) = x \sin y + g(y, z)$ and so

$$f_y(x, y, z) = x \cos y + g_y(y, z). \text{ But } f_y(x, y, z) = x \cos y + \cos z \text{ so } g_y(y, z) = \cos z \Rightarrow g(y, z) = y \cos z + h(z).$$

$$\text{Thus, } f(x, y, z) = x \sin y + y \cos z + h(z) \text{ and } f_z(x, y, z) = -y \sin z + h'(z). \text{ But } f_z(x, y, z) = -y \sin z, \text{ so}$$

$$h'(z) = 0 \Rightarrow h(z) = K. \text{ Hence, } f(x, y, z) = x \sin y + y \cos z \text{ (taking } K = 0\text{).}$$

(b) $\mathbf{r}(0) = \langle 0, 0, 0 \rangle$, $\mathbf{r}(\pi/2) = \langle 1, \pi/2, \pi \rangle$ so $\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, \pi/2, \pi) - f(0, 0, 0) = 1 - \frac{\pi}{2} - 0 = 1 - \frac{\pi}{2}$.

25. The functions $2xe^{-y}$ and $2y - x^2e^{-y}$ have continuous first-order derivatives on $\mathbb{R}^2$ and

$$\frac{\partial}{\partial y}(2xe^{-y}) = -2xe^{-y} = \frac{\partial}{\partial x}(2y - x^2e^{-y}), \text{ so } \mathbf{F}(x, y) = 2xe^{-y}\mathbf{i} + (2y - x^2e^{-y})\mathbf{j} \text{ is a conservative vector field by}$$

Theorem 6 and hence the line integral is independent of path. Thus a potential function f exists, and $f_x(x, y) = 2xe^{-y}$

implies $f(x, y) = x^2e^{-y} + g(y)$ and $f_y(x, y) = -x^2e^{-y} + g'(y)$. But $f_y(x, y) = 2y - x^2e^{-y}$ so

$$g'(y) = 2y \Rightarrow g(y) = y^2 + K. \text{ We can take } K = 0, \text{ so } f(x, y) = x^2e^{-y} + y^2. \text{ Then}$$

$$\int_C 2xe^{-y} dx + (2y - x^2e^{-y}) dy = f(2, 1) - f(1, 0) = 4e^{-1} + 1 - 1 = 4/e.$$

26. The functions $\sin y$ and $x \cos y - \sin y$ have continuous first-order derivatives on $\mathbb{R}^2$ and

$$\frac{\partial}{\partial y}(\sin y) = \cos y = \frac{\partial}{\partial x}(x \cos y - \sin y), \text{ so } \mathbf{F}(x, y) = \sin y\mathbf{i} + (x \cos y - \sin y)\mathbf{j} \text{ is a conservative vector field by}$$

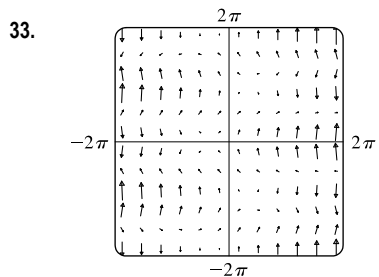
Theorem 6 and hence the line integral is independent of path. Thus a potential function f exists, and $f_x(x, y) = \sin y$ implies

$$f(x, y) = x \sin y + g(y) \text{ and } f_y(x, y) = x \cos y + g'(y). \text{ But } f_y(x, y) = x \cos y - \sin y, \text{ so } g'(y) = -\sin y \Rightarrow$$

$$g(y) = \cos y + K. \text{ We can take } K = 0, \text{ so } f(x, y) = x \sin y + \cos y. \text{ Then}$$

$$\int_C \sin y dx + (x \cos y - \sin y) dy = f(1, \pi) - f(2, 0) = -1 - 1 = -2.$$

27. If $\mathbf{F}$ is conservative, then $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path. This means that the work done along all piecewise-smooth curves that have the described initial and terminal points is the same. Your reply: It doesn't matter which curve is chosen.
28. The curves C_1 and C_2 connect the same two points but $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} \neq \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$. Thus $\mathbf{F}$ is not independent of path, and therefore is not conservative.
29. $\mathbf{F}(x, y) = x^3 \mathbf{i} + y^3 \mathbf{j}$, $W = \int_C \mathbf{F} \cdot d\mathbf{r}$. Since $\partial(x^3)/\partial y = 0 = \partial(y^3)/\partial x$, there exists a function f such that $\nabla f = \mathbf{F}$. In fact, $f_x(x, y) = x^3 \Rightarrow f(x, y) = \frac{1}{4}x^4 + g(y) \Rightarrow f_y(x, y) = 0 + g'(y)$. But $f_y(x, y) = y^3$ so $g'(y) = y^3 \Rightarrow g(y) = \frac{1}{4}y^4 + K$. We can take $K = 0$, so $f(x, y) = \frac{1}{4}x^4 + \frac{1}{4}y^4$. Thus $W = \int_C \mathbf{F} \cdot d\mathbf{r} = f(2, 2) - f(1, 0) = (4 + 4) - (\frac{1}{4} + 0) = \frac{31}{4}$.
30. $\mathbf{F}(x, y) = (2x + y) \mathbf{i} + x \mathbf{j}$, $W = \int_C \mathbf{F} \cdot d\mathbf{r}$. Since $\partial(2x + y)/\partial y = 1 = \partial(x)/\partial x$, there exists a function f such that $\nabla f = \mathbf{F}$. In fact, $f_x(x, y) = 2x + y \Rightarrow f(x, y) = x^2 + xy + g(y) \Rightarrow f_y(x, y) = x + g'(y)$. But $f_y(x, y) = x$ so $g'(y) = 0 \Rightarrow g(y) = K$. We can take $K = 0$, so $f(x, y) = x^2 + xy$. Thus $W = \int_C \mathbf{F} \cdot d\mathbf{r} = f(4, 3) - f(1, 1) = (16 + 12) - (1 + 1) = 26$.
31. We know that if the vector field (call it $\mathbf{F}$) is conservative, then around any closed path C , $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$. But take C to be a circle centered at the origin, oriented counterclockwise. All of the field vectors that start on C are roughly in the direction of motion along C , so the integral around C will be positive. Therefore the field is not conservative.
32. We know that if the vector field (call it $\mathbf{F}$) is conservative, then around any closed path C , $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$. For any closed path we draw in the field, it appears that some vectors on the curve point in approximately the same direction as the curve and a similar number point in roughly the opposite direction. (Some appear perpendicular to the curve as well.) Therefore it is plausible that $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve C which means $\mathbf{F}$ is conservative.



From the graph, it appears that $\mathbf{F}$ is conservative, since around all closed paths, the number and size of the field vectors pointing in directions similar to that of the path seem to be roughly the same as the number and size of the vectors pointing in the opposite direction. To check, we calculate

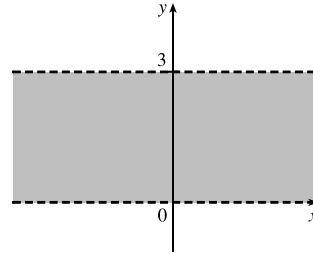
$$\frac{\partial}{\partial y} (\sin y) = \cos y = \frac{\partial}{\partial x} (1 + x \cos y). \text{ Thus, } \mathbf{F} \text{ is conservative, by Theorem 6.}$$

34. $f(x, y) = \sin(x - 2y) \Rightarrow \mathbf{F} = \nabla f(x, y) = \cos(x - 2y) \mathbf{i} - 2 \cos(x - 2y) \mathbf{j}$
- (a) We use Theorem 2: $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$ where C_1 starts at $t = a$ and ends at $t = b$. So because $f(0, 0) = \sin 0 = 0$ and $f(\pi, \pi) = \sin(\pi - 2\pi) = 0$, one possible curve C_1 is the straight line from $(0, 0)$ to (π, π) ; that is, $\mathbf{r}(t) = \pi t \mathbf{i} + \pi t \mathbf{j}$, $0 \leq t \leq 1$.
- (b) From (a), $\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$. So because $f(0, 0) = \sin 0 = 0$ and $f(\frac{\pi}{2}, 0) = 1$, one possible curve C_2 is $\mathbf{r}(t) = \frac{\pi}{2} t \mathbf{i}$, $0 \leq t \leq 1$, the straight line from $(0, 0)$ to $(\frac{\pi}{2}, 0)$.

35. Since $\mathbf{F}$ is conservative, there exists a function f such that $\mathbf{F} = \nabla f$, that is, $P = f_x$, $Q = f_y$, and $R = f_z$. Since P , Q , and R have continuous first-order partial derivatives, Clairaut's Theorem says that $\partial P/\partial y = f_{xy} = f_{yx} = \partial Q/\partial x$, $\partial P/\partial z = f_{xz} = f_{zx} = \partial R/\partial x$, and $\partial Q/\partial z = f_{yz} = f_{zy} = \partial R/\partial y$.

36. Here $\mathbf{F}(x, y, z) = y\mathbf{i} + x\mathbf{j} + xyz\mathbf{k}$. Then using the notation of Exercise 35, $\partial P/\partial z = 0$ while $\partial R/\partial x = yz$. Since these aren't equal, $\mathbf{F}$ is not conservative. Thus by Theorem 4, the line integral of $\mathbf{F}$ is not independent of path.

37. $D = \{(x, y) \mid 0 < y < 3\}$ consists of those points between, but not on, the horizontal lines $y = 0$ and $y = 3$.

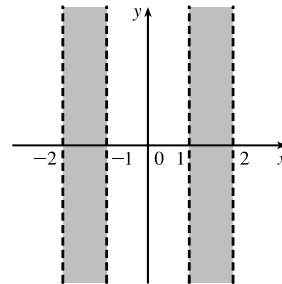


(a) Since D does not include any of its boundary points, it is open. More formally, at any point in D there is a disk centered at that point that lies entirely in D .

(b) Any two points chosen in D can always be joined by a path that lies entirely in D , so D is connected. (D consists of just one “piece.”)

(c) D is connected and it has no holes, so it's simply-connected. (Every simple closed curve in D encloses only points that are in D .)

38. $D = \{(x, y) \mid 1 < |x| < 2\}$ consists of those points between, but not on, the vertical lines $x = 1$ and $x = 2$, together with the points between the vertical lines $x = -1$ and $x = -2$.

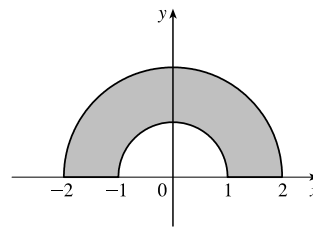


(a) The region does not include any of its boundary points, so it is open.

(b) D consists of two separate pieces, so it is not connected. [For instance, both the points $(-1.5, 0)$ and $(1.5, 0)$ lie in D but they cannot be joined by a path that lies entirely in D .]

(c) Because D is not connected, it's not simply-connected.

39. $D = \{(x, y) \mid 1 \leq x^2 + y^2 \leq 4, y \geq 0\}$ is the semiannular region in the upper half-plane between circles centered at the origin of radii 1 and 2 (including all boundary points).



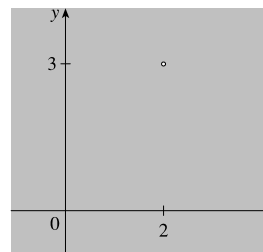
(a) D includes boundary points, so it is not open. [Note that at any boundary point, $(1, 0)$ for instance, any disk centered there cannot lie entirely in D .]

(b) The region consists of one piece, so it's connected.

(c) D is connected and has no holes, so it's simply-connected.

40. $D = \{(x, y) \mid (x, y) \neq (2, 3)\}$ consists of all points in the xy -plane except for $(2, 3)$.

- (a) D has only one boundary point, namely $(2, 3)$, which is not included, so the region is open.
- (b) D is connected, as it consists of only one piece.
- (c) D is not simply-connected, as it has a hole at $(2, 3)$. Thus any simple closed curve that encloses $(2, 3)$ lies in D but includes a point that is not in D .



41. (a) $P = -\frac{y}{x^2 + y^2}$, $\frac{\partial P}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$ and $Q = \frac{x}{x^2 + y^2}$, $\frac{\partial Q}{\partial x} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$. Thus $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

- (b) $C_1: x = \cos t, y = \sin t, 0 \leq t \leq \pi$, $C_2: x = \cos t, y = \sin t, t = 2\pi$ to $t = \pi$. Then

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_0^\pi \frac{(-\sin t)(-\sin t) + (\cos t)(\cos t)}{\cos^2 t + \sin^2 t} dt = \int_0^\pi dt = \pi \text{ and } \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{2\pi}^\pi dt = -\pi$$

Since these aren't equal, the line integral of $\mathbf{F}$ isn't independent of path. (Or notice that $\int_{C_3} \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} dt = 2\pi$ where C_3 is the circle $x^2 + y^2 = 1$, and apply the contrapositive of Theorem 3.) This doesn't contradict Theorem 6, since the domain of $\mathbf{F}$, which is $\mathbb{R}^2$ except the origin, isn't simply-connected.

42. (a) Here $\mathbf{F}(\mathbf{r}) = c\mathbf{r}/|\mathbf{r}|^3$ and $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Then $f(\mathbf{r}) = -c/|\mathbf{r}|$ is a potential function for $\mathbf{F}$, that is, $\nabla f = \mathbf{F}$.

(See the discussion of gradient fields in Section 16.1.) Hence $\mathbf{F}$ is conservative and its line integral is independent of path.

Let $P_1 = (x_1, y_1, z_1)$ and $P_2 = (x_2, y_2, z_2)$.

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = f(P_2) - f(P_1) = -\frac{c}{(x_2^2 + y_2^2 + z_2^2)^{1/2}} + \frac{c}{(x_1^2 + y_1^2 + z_1^2)^{1/2}} = c\left(\frac{1}{d_1} - \frac{1}{d_2}\right).$$

- (b) In this case, $c = -(mMG) \Rightarrow$

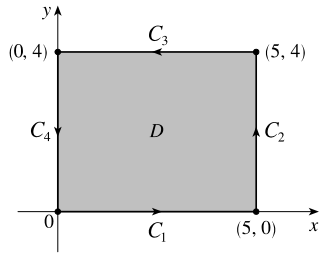
$$\begin{aligned} W &= -mMG \left(\frac{1}{1.52 \times 10^{11}} - \frac{1}{1.47 \times 10^{11}} \right) \\ &= -(5.97 \times 10^{24})(1.99 \times 10^{30})(6.67 \times 10^{-11})(-2.2377 \times 10^{-13}) \\ &\approx 1.77 \times 10^{32} \text{ J} \end{aligned}$$

- (c) In this case, $c = \epsilon qQ \Rightarrow$

$$W = \epsilon qQ \left(\frac{1}{10^{-12}} - \frac{1}{5 \times 10^{-13}} \right) = (8.985 \times 10^9)(1)(-1.6 \times 10^{-19})(-10^{12}) \approx 1400 \text{ J}.$$

16.4 Green's Theorem

1. (a)



$$C_1: x = t \Rightarrow dx = dt, y = 0 \Rightarrow dy = 0 dt, 0 \leq t \leq 5.$$

$$C_2: x = 5 \Rightarrow dx = 0 dt, y = t \Rightarrow dy = dt, 0 \leq t \leq 4.$$

$$C_3: x = 5 - t \Rightarrow dx = -dt, y = 4 \Rightarrow dy = 0 dt, 0 \leq t \leq 5.$$

$$C_4: x = 0 \Rightarrow dx = 0 dt, y = 4 - t \Rightarrow dy = -dt, 0 \leq t \leq 4$$

$$\begin{aligned} \text{Thus } \oint_C y^2 dx + x^2 y dy &= \oint_{C_1 + C_2 + C_3 + C_4} y^2 dx + x^2 y dy = \int_0^5 0 dt + \int_0^4 25t dt + \int_0^5 (-16 + 0) dt + \int_0^4 0 dt \\ &= 0 + \left[\frac{25}{2} t^2 \right]_0^4 + [-16t]_0^5 + 0 = 200 + (-80) = 120 \end{aligned}$$

(b) Note that C as given in part (a) is a positively oriented, piecewise-smooth, simple closed curve. Then by Green's Theorem,

$$\begin{aligned} \oint_C y^2 dx + x^2 y dy &= \iint_D \left[\frac{\partial}{\partial x} (x^2 y) - \frac{\partial}{\partial y} (y^2) \right] dA = \int_0^5 \int_0^4 (2xy - 2y) dy dx = \int_0^5 [xy^2 - y^2]_{y=0}^{y=4} dx \\ &= \int_0^5 (16x - 16) dx = [8x^2 - 16x]_0^5 = 200 - 80 = 120 \end{aligned}$$

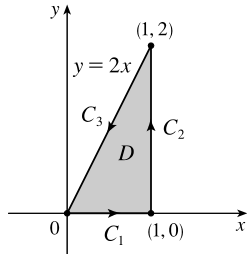
2. (a) Parametric equations for C are $x = 4 \cos t$, $y = 4 \sin t$, $0 \leq t \leq 2\pi$. Then $dx = -4 \sin t dt$, $dy = 4 \cos t dt$ and

$$\begin{aligned} \oint_C y dx - x dy &= \int_0^{2\pi} [(4 \sin t)(-4 \sin t) - (4 \cos t)(4 \cos t)] dt \\ &= -16 \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt = -16 \int_0^{2\pi} 1 dt = -16(2\pi) = -32\pi \end{aligned}$$

(b) Note that C as given in part (a) is a positively oriented, smooth, simple closed curve. Then by Green's Theorem,

$$\begin{aligned} \oint_C y dx - x dy &= \iint_D \left[\frac{\partial}{\partial x} (-x) - \frac{\partial}{\partial y} (y) \right] dA = \iint_D (-1 - 1) dA = -2 \iint_D dA \\ &= -2(\text{area of } D) = -2 \cdot \pi(4)^2 = -32\pi \end{aligned}$$

3. (a)



$$C_1: x = t \Rightarrow dx = dt, y = 0 \Rightarrow dy = 0 dt, 0 \leq t \leq 1.$$

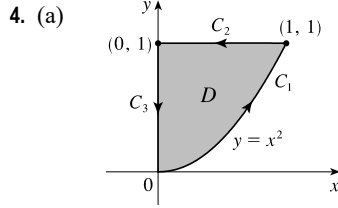
$$C_2: x = 1 \Rightarrow dx = 0 dt, y = t \Rightarrow dy = dt, 0 \leq t \leq 2.$$

$$C_3: x = 1 - t \Rightarrow dx = -dt, y = 2 - 2t \Rightarrow dy = -2 dt, 0 \leq t \leq 1.$$

Thus

$$\begin{aligned} \oint_C xy dx + x^2 y^3 dy &= \oint_{C_1 + C_2 + C_3} xy dx + x^2 y^3 dy \\ &= \int_0^1 0 dt + \int_0^2 t^3 dt + \int_0^1 [-(1-t)(2-2t) - 2(1-t)^2(2-2t)^3] dt \\ &= 0 + \left[\frac{1}{4} t^4 \right]_0^2 + \int_0^1 [-2(1-t)^2 - 16(1-t)^5] dt \\ &= 4 + \left[\frac{2}{3} (1-t)^3 + \frac{8}{3} (1-t)^6 \right]_0^1 = 4 + 0 - \frac{10}{3} = \frac{2}{3} \end{aligned}$$

$$\begin{aligned} \text{(b) } \oint_C xy dx + x^2 y^3 dy &= \iint_D \left[\frac{\partial}{\partial x} (x^2 y^3) - \frac{\partial}{\partial y} (xy) \right] dA = \int_0^1 \int_0^{2x} (2xy^3 - x) dy dx \\ &= \int_0^1 \left[\frac{1}{2} xy^4 - xy \right]_{y=0}^{y=2x} dx = \int_0^1 (8x^5 - 2x^2) dx = \frac{4}{3} - \frac{2}{3} = \frac{2}{3} \end{aligned}$$



$$C_1: x = t \Rightarrow dx = dt, y = t^2 \Rightarrow dy = 2t dt, 0 \leq t \leq 1$$

$$C_2: x = 1 - t \Rightarrow dx = -dt, y = 1 \Rightarrow dy = 0 dt, 0 \leq t \leq 1$$

$$C_3: x = 0 \Rightarrow dx = 0 dt, y = 1 - t \Rightarrow dy = -dt, 0 \leq t \leq 1$$

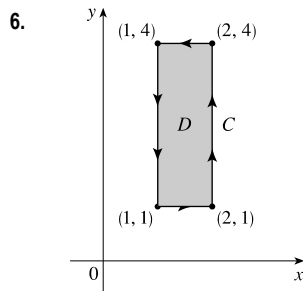
Thus

$$\begin{aligned} \oint_C x^2 y^2 dx + xy dy &= \oint_{C_1+C_2+C_3} x^2 y^2 dx + xy dy \\ &= \int_0^1 [t^2(t^2)^2 dt + t(t^2)(2t dt)] + \int_0^1 [(1-t)^2(1)^2(-dt) + (1-t)(1)(0 dt)] \\ &\quad + \int_0^1 [(0)^2(1-t)^2(0 dt) + (0)(1-t)(-dt)] \\ &= \int_0^1 (t^6 + 2t^4) dt + \int_0^1 (-1 + 2t - t^2) dt + \int_0^1 0 dt \\ &= \left[\frac{1}{7}t^7 + \frac{2}{5}t^5\right]_0^1 + \left[-t + t^2 - \frac{1}{3}t^3\right]_0^1 + 0 = \left(\frac{1}{7} + \frac{2}{5}\right) + (-1 + 1 - \frac{1}{3}) = \frac{22}{105} \end{aligned}$$

$$\begin{aligned} \text{(b)} \oint_C x^2 y^2 dx + xy dy &= \iint_D \left[\frac{\partial}{\partial x}(xy) - \frac{\partial}{\partial y}(x^2 y^2) \right] dA = \int_0^1 \int_{x^2}^1 (y - 2x^2 y) dy dx \\ &= \int_0^1 \left[\frac{1}{2}y^2 - x^2 y^2 \right]_{y=x^2}^{y=1} dx = \int_0^1 \left(\frac{1}{2} - x^2 - \frac{1}{2}x^4 + x^6 \right) dx \\ &= \left[\frac{1}{2}x - \frac{1}{3}x^3 - \frac{1}{10}x^5 + \frac{1}{7}x^7 \right]_0^1 = \frac{1}{2} - \frac{1}{3} - \frac{1}{10} + \frac{1}{7} = \frac{22}{105} \end{aligned}$$

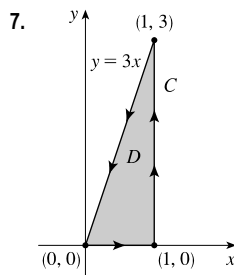
5. The region D enclosed by C is $[0, 3] \times [0, 4]$, so

$$\begin{aligned} \int_C ye^x dx + 2e^x dy &= \iint_D \left[\frac{\partial}{\partial x}(2e^x) - \frac{\partial}{\partial y}(ye^x) \right] dA = \int_0^3 \int_0^4 (2e^x - e^x) dy dx \\ &= \int_0^3 e^x dx \int_0^4 dy = [e^x]_0^3 [y]_0^4 = (e^3 - e^0)(4 - 0) = 4(e^3 - 1) \end{aligned}$$



The region D enclosed by C can be given by $\{(x, y) \mid 1 \leq x \leq 2, 1 \leq y \leq 4\}$, so

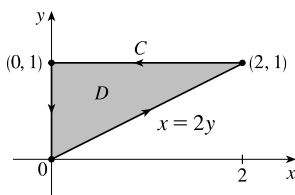
$$\begin{aligned} \int_C \ln(xy) dx + \frac{y}{x} dy &= \iint_D \left[\frac{\partial}{\partial x} \left(\frac{y}{x} \right) - \frac{\partial}{\partial y} \ln(xy) \right] dA \\ &= \int_1^4 \int_1^2 \left(-\frac{y}{x^2} - \frac{1}{y} \right) dx dy = \int_1^4 \left[\frac{y}{x} - \frac{x}{y} \right]_{x=1}^{x=2} dy \\ &= \int_1^4 \left(\frac{y}{2} - \frac{2}{y} - y + \frac{1}{y} \right) dy = \int_1^4 \left(-\frac{y}{2} - \frac{1}{y} \right) dy \\ &= \left[-\frac{y^2}{4} - \ln y \right]_1^4 = -\frac{16}{4} - \ln 4 + \frac{1}{4} + 0 = -\frac{15}{4} - \ln 4 \end{aligned}$$



The region D enclosed by C can be given by $\{(x, y) \mid 0 \leq y \leq 3x, 0 \leq x \leq 1\}$, so

$$\begin{aligned} \int_C x^2 y^2 dx + y \tan^{-1} y dy &= \iint_D \left[\frac{\partial}{\partial x}(y \tan^{-1} y) - \frac{\partial}{\partial y}(x^2 y^2) \right] dA \\ &= \int_0^1 \int_0^{3x} (-2x^2 y) dy dx = -\int_0^1 [x^2 y^2]_{y=0}^{y=3x} dx \\ &= -9 \int_0^1 x^4 dx = -\frac{9}{5} [x^5]_0^1 = -\frac{9}{5} \end{aligned}$$

8.



The region D enclosed by C is given by $\{(x, y) \mid 0 \leq y \leq 1, 0 \leq x \leq 2y\}$, so

$$\begin{aligned} \int_C (x^2 + y^2) dx + (x^2 - y^2) dy &= \iint_D \left[\frac{\partial}{\partial x} (x^2 - y^2) - \frac{\partial}{\partial y} (x^2 + y^2) \right] dA \\ &= \int_0^1 \int_0^{2y} (2x - 2y) dx dy \\ &= \int_0^1 [x^2 - 2xy]_{x=0}^{x=2y} dy \\ &= \int_0^1 (4y^2 - 4y^2) dy = \int_0^1 0 dy = 0 \end{aligned}$$

$$\begin{aligned} 9. \int_C (y + e^{\sqrt{x}}) dx + (2x + \cos y^2) dy &= \iint_D \left[\frac{\partial}{\partial x} (2x + \cos y^2) - \frac{\partial}{\partial y} (y + e^{\sqrt{x}}) \right] dA \\ &= \int_0^1 \int_{x^2}^{\sqrt{x}} (2 - 1) dy dx = \int_0^1 (\sqrt{x} - x^2) dx = \left[\frac{2}{3} x^{3/2} - \frac{1}{3} x^3 \right]_0^1 = \frac{1}{3} \end{aligned}$$

$$\begin{aligned} 10. \int_C y^4 dx + 2xy^3 dy &= \iint_D \left[\frac{\partial}{\partial x} (2xy^3) - \frac{\partial}{\partial y} (y^4) \right] dA = \iint_D (2y^3 - 4y^3) dA \\ &= -2 \iint_D y^3 dA = 0 \end{aligned}$$

because $f(x, y) = y^3$ is an odd function with respect to y and D is symmetric about the x -axis.

$$\begin{aligned} 11. \int_C y^3 dx - x^3 dy &= \iint_D \left[\frac{\partial}{\partial x} (-x^3) - \frac{\partial}{\partial y} (y^3) \right] dA = \iint_D (-3x^2 - 3y^2) dA = \int_0^{2\pi} \int_0^2 (-3r^2) r dr d\theta \\ &= -3 \int_0^{2\pi} d\theta \int_0^2 r^3 dr = -3 [\theta]_0^{2\pi} \left[\frac{1}{4} r^4 \right]_0^2 = -3(2\pi)(4) = -24\pi \end{aligned}$$

$$\begin{aligned} 12. \int_C (1 - y^3) dx + (x^3 + e^{y^2}) dy &= \iint_D \left[\frac{\partial}{\partial x} (x^3 + e^{y^2}) - \frac{\partial}{\partial y} (1 - y^3) \right] dA = \iint_D (3x^2 + 3y^2) dA \\ &= \int_0^{2\pi} \int_2^3 (3r^2) r dr d\theta = 3 \int_0^{2\pi} d\theta \int_2^3 r^3 dr \\ &= 3 [\theta]_0^{2\pi} \left[\frac{1}{4} r^4 \right]_2^3 = 3(2\pi) \cdot \frac{1}{4} (81 - 16) = \frac{195}{2} \pi \end{aligned}$$

13. The region D enclosed by C is given by $\{(r, \theta) \mid 1 \leq r \leq 2, 0 \leq \theta \leq \pi/2\}$ (in polar coordinates), which is traversed counterclockwise, so C has positive orientation. Thus,

$$\begin{aligned} \int_C (3 + e^{x^2}) dx + (\tan^{-1} y + 3x^2) dy &= \iint_D \left[\frac{\partial}{\partial x} (\tan^{-1} y + 3x^2) - \frac{\partial}{\partial y} (3 + e^{x^2}) \right] dA \\ &= \iint_D 6x dA = 6 \int_0^{\pi/2} \int_1^2 r \cos \theta r dr d\theta \quad [\text{Switching to polar coordinates}] \\ &= 6 \int_0^{\pi/2} \cos \theta d\theta \int_1^2 r^2 dr = 6 [\sin \theta]_{\theta=0}^{\theta=\pi/2} \left[\frac{r^3}{3} \right]_{r=1}^{r=2} \\ &= 6(1 - 0) \left(\frac{8}{3} - \frac{1}{3} \right) = 14 \end{aligned}$$

14. The region D enclosed by C is given by $\{(x, y) \mid y^2 \leq x \leq 4, 0 \leq y \leq 2\}$. C is traversed clockwise, so $-C$ gives the positive orientation. Then

$$\begin{aligned} - \int_{-C} (x^{2/3} + y^2) dx + (y^{4/3} - x^2) dy &= - \iint_D \left[\frac{\partial}{\partial x} (y^{4/3} - x^2) - \frac{\partial}{\partial y} (x^{2/3} + y^2) \right] dA \\ &= - \int_0^2 \int_{y^2}^4 (-2x - 2y) dx dy = \int_0^2 \int_{y^2}^4 (2x + 2y) dx dy \\ &= \int_0^2 [x^2 + 2xy]_{x=y^2}^{x=4} dy = \int_0^2 (16 + 8y - y^4 - 2y^3) dy \\ &= \left[16y + 4y^2 - \frac{y^5}{5} - \frac{y^4}{2} \right]_0^2 = 16(2) + 4(2^2) - \frac{2^5}{5} - \frac{2^4}{2} - 0 = \frac{168}{5} \end{aligned}$$

15. $\mathbf{F}(x, y) = \langle y \cos x - xy \sin x, xy + x \cos x \rangle$ and the region D enclosed by C is given by

$\{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq 4 - 2x\}$. C is traversed clockwise, so $-C$ gives the positive orientation.

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= - \int_{-C} (y \cos x - xy \sin x) dx + (xy + x \cos x) dy \\ &= - \iint_D \left[\frac{\partial}{\partial x} (xy + x \cos x) - \frac{\partial}{\partial y} (y \cos x - xy \sin x) \right] dA \\ &= - \iint_D (y - x \sin x + \cos x - \cos x + x \sin x) dA = - \int_0^2 \int_0^{4-2x} y dy dx \\ &= - \int_0^2 \left[\frac{1}{2} y^2 \right]_{y=0}^{y=4-2x} dx = - \int_0^2 \frac{1}{2} (4 - 2x)^2 dx = - \int_0^2 (8 - 8x + 2x^2) dx \\ &= - \left[8x - 4x^2 + \frac{2}{3} x^3 \right]_0^2 = - \left(16 - 16 + \frac{16}{3} - 0 \right) = -\frac{16}{3} \end{aligned}$$

16. $\mathbf{F}(x, y) = \langle e^{-x} + y^2, e^{-y} + x^2 \rangle$ and the region D enclosed by C is given by $\{(x, y) \mid -\pi/2 \leq x \leq \pi/2, 0 \leq y \leq \cos x\}$.

C is traversed clockwise, so $-C$ gives the positive orientation.

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= - \int_{-C} (e^{-x} + y^2) dx + (e^{-y} + x^2) dy = - \iint_D \left[\frac{\partial}{\partial x} (e^{-y} + x^2) - \frac{\partial}{\partial y} (e^{-x} + y^2) \right] dA \\ &= - \int_{-\pi/2}^{\pi/2} \int_0^{\cos x} (2x - 2y) dy dx = - \int_{-\pi/2}^{\pi/2} [2xy - y^2]_{y=0}^{y=\cos x} dx \\ &= - \int_{-\pi/2}^{\pi/2} (2x \cos x - \cos^2 x) dx = - \int_{-\pi/2}^{\pi/2} \left[2x \cos x - \frac{1}{2}(1 + \cos 2x) \right] dx \\ &= - \left[2x \sin x + 2 \cos x - \frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right) \right]_{-\pi/2}^{\pi/2} \quad [\text{integrate by parts in the first term}] \\ &= - \left(\pi - \frac{1}{4}\pi - \pi - \frac{1}{4}\pi \right) = \frac{1}{2}\pi \end{aligned}$$

17. $\mathbf{F}(x, y) = \langle y - \cos y, x \sin y \rangle$ and the region D enclosed by C is the disk with radius 2 centered at $(3, -4)$.

C is traversed clockwise, so $-C$ gives the positive orientation.

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= - \int_{-C} (y - \cos y) dx + (x \sin y) dy = - \iint_D \left[\frac{\partial}{\partial x} (x \sin y) - \frac{\partial}{\partial y} (y - \cos y) \right] dA \\ &= - \iint_D (\sin y - 1 - \sin y) dA = \iint_D dA = \text{area of } D = \pi(2)^2 = 4\pi \end{aligned}$$

18. $\mathbf{F}(x, y) = \langle \sqrt{x^2 + 1}, \tan^{-1} x \rangle$ and the region D enclosed by C is given by $\{(x, y) \mid 0 \leq x \leq 1, x \leq y \leq 1\}$.

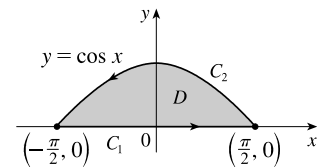
C is oriented positively, so

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C \sqrt{x^2 + 1} dx + \tan^{-1} x dy = \iint_D \left[\frac{\partial}{\partial x} (\tan^{-1} x) - \frac{\partial}{\partial y} (\sqrt{x^2 + 1}) \right] dA \\ &= \int_0^1 \int_x^1 \left(\frac{1}{1+x^2} - 0 \right) dy dx = \int_0^1 \frac{1}{1+x^2} [y]_{y=x}^{y=1} dx = \int_0^1 \frac{1}{1+x^2} (1-x) dx \\ &= \int_0^1 \left(\frac{1}{1+x^2} - \frac{x}{1+x^2} \right) dx = \left[\tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]_0^1 = \frac{\pi}{4} - \frac{1}{2} \ln 2 \end{aligned}$$

19. Here $C = C_1 + C_2$ where

C_1 can be parametrized as $x = t, y = 0, -\pi/2 \leq t \leq \pi/2$, and

C_2 is given by $x = -t, y = \cos t, -\pi/2 \leq t \leq \pi/2$.



[continued]

Then the line integral is

$$\begin{aligned}\oint_{C_1+C_2} x^3 y^4 dx + x^5 y^4 dy &= \int_{-\pi/2}^{\pi/2} (0+0) dt + \int_{-\pi/2}^{\pi/2} [(-t)^3 (\cos t)^4 (-1) + (-t)^5 (\cos t)^4 (-\sin t)] dt \\ &= 0 + \int_{-\pi/2}^{\pi/2} (t^3 \cos^4 t + t^5 \cos^4 t \sin t) dt = \frac{1}{15} \pi^4 - \frac{4144}{1125} \pi^2 + \frac{7,578,368}{253,125} \approx 0.0779\end{aligned}$$

according to a CAS. The double integral is

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_{-\pi/2}^{\pi/2} \int_0^{\cos x} (5x^4 y^4 - 4x^3 y^3) dy dx = \frac{1}{15} \pi^4 - \frac{4144}{1125} \pi^2 + \frac{7,578,368}{253,125} \approx 0.0779,$$

verifying Green's Theorem in this case.

20. We can parametrize C as $x = \cos \theta$, $y = 2 \sin \theta$, $0 \leq \theta \leq 2\pi$. Then the line integral is

$$\begin{aligned}\oint_C P dx + Q dy &= \int_0^{2\pi} [2 \cos \theta - (\cos \theta)^3 (2 \sin \theta)^5] (-\sin \theta) d\theta + \int_0^{2\pi} (\cos \theta)^3 (2 \sin \theta)^8 \cdot 2 \cos \theta d\theta \\ &= \int_0^{2\pi} (-2 \cos \theta \sin \theta + 32 \cos^3 \theta \sin^6 \theta + 512 \cos^4 \theta \sin^8 \theta) d\theta = 7\pi,\end{aligned}$$

according to a CAS. The double integral is $\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_{-1}^1 \int_{-\sqrt{4-4x^2}}^{\sqrt{4-4x^2}} (3x^2 y^8 + 5x^3 y^4) dy dx = 7\pi$.

21. By Green's Theorem, $W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C x(x+y) dx + xy^2 dy = \iint_D (y^2 - x) dA$ where C is the path described in the question and D is the triangle bounded by C . So

$$\begin{aligned}W &= \int_0^1 \int_0^{1-x} (y^2 - x) dy dx = \int_0^1 \left[\frac{1}{3} y^3 - xy \right]_{y=0}^{y=1-x} dx = \int_0^1 \left(\frac{1}{3} (1-x)^3 - x(1-x) \right) dx \\ &= \left[-\frac{1}{12} (1-x)^4 - \frac{1}{2} x^2 + \frac{1}{3} x^3 \right]_0^1 = \left(-\frac{1}{2} + \frac{1}{3} \right) - \left(-\frac{1}{12} \right) = -\frac{1}{12}\end{aligned}$$

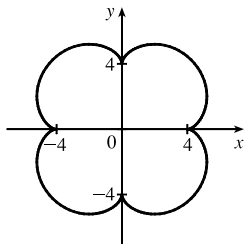
22. By Green's Theorem, $W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \sin x dx + (\sin y + xy^2 + \frac{1}{3} x^3) dy = \iint_D (y^2 + x^2 - 0) dA$, where D is the region (a quarter-disk) bounded by C . Converting to polar coordinates, we have

$$W = \int_0^{\pi/2} \int_0^5 r^2 \cdot r dr d\theta = \left[\theta \right]_0^{\pi/2} \left[\frac{1}{4} r^4 \right]_0^5 = \frac{1}{2} \pi \left(\frac{625}{4} \right) = \frac{625}{8} \pi.$$

23. Let C_1 be the arch of the cycloid from $(0, 0)$ to $(2\pi, 0)$, which corresponds to $0 \leq t \leq 2\pi$, and let C_2 be the segment from $(2\pi, 0)$ to $(0, 0)$, so C_2 is given by $x = 2\pi - t$, $y = 0$, $0 \leq t \leq 2\pi$. Then $C = C_1 \cup C_2$ is traversed clockwise, so $-C$ is oriented positively. Thus $-C$ encloses the area under one arch of the cycloid and from (5) we have

$$\begin{aligned}A &= -\oint_{-C} y dx = \int_{C_1} y dx + \int_{C_2} y dx = \int_0^{2\pi} (1 - \cos t)(1 - \cos t) dt + \int_0^{2\pi} 0 (-dt) \\ &= \int_0^{2\pi} (1 - 2 \cos t + \cos^2 t) dt + 0 = \left[t - 2 \sin t + \frac{1}{2} t + \frac{1}{4} \sin 2t \right]_0^{2\pi} = 3\pi\end{aligned}$$

- 24.



$$\begin{aligned}A &= \oint_C x dy = \int_0^{2\pi} (5 \cos t - \cos 5t)(5 \cos t - 5 \cos 5t) dt \\ &= \int_0^{2\pi} (25 \cos^2 t - 30 \cos t \cos 5t + 5 \cos^2 5t) dt \\ &= \left[25 \left(\frac{1}{2} t + \frac{1}{4} \sin 2t \right) - 30 \left(\frac{1}{8} \sin 4t + \frac{1}{12} \sin 6t \right) + 5 \left(\frac{1}{2} t + \frac{1}{20} \sin 10t \right) \right]_0^{2\pi} \\ &= 30\pi\end{aligned}$$

[Use Formula 80 in the Table of Integrals]

25. (a) Using Equation 16.2.8, we write parametric equations of the line segment as $x = (1-t)x_1 + tx_2$, $y = (1-t)y_1 + ty_2$, $0 \leq t \leq 1$. Then $dx = (x_2 - x_1) dt$ and $dy = (y_2 - y_1) dt$, so

$$\begin{aligned}\int_C x dy - y dx &= \int_0^1 [(1-t)x_1 + tx_2](y_2 - y_1) dt + [(1-t)y_1 + ty_2](x_2 - x_1) dt \\ &= \int_0^1 (x_1(y_2 - y_1) - y_1(x_2 - x_1) + t[(y_2 - y_1)(x_2 - x_1) - (x_2 - x_1)(y_2 - y_1)]) dt \\ &= \int_0^1 (x_1 y_2 - x_2 y_1) dt = x_1 y_2 - x_2 y_1\end{aligned}$$

- (b) We apply Green's Theorem to the path $C = C_1 \cup C_2 \cup \cdots \cup C_n$, where C_i is the line segment that joins (x_i, y_i) to (x_{i+1}, y_{i+1}) for $i = 1, 2, \dots, n-1$, and C_n is the line segment that joins (x_n, y_n) to (x_1, y_1) . From (5),

$$\frac{1}{2} \int_C x dy - y dx = \iint_D dA, \text{ where } D \text{ is the polygon bounded by } C. \text{ Therefore}$$

$$\begin{aligned}\text{area of polygon} = A(D) &= \iint_D dA = \frac{1}{2} \int_C x dy - y dx \\ &= \frac{1}{2} \left(\int_{C_1} x dy - y dx + \int_{C_2} x dy - y dx + \cdots + \int_{C_{n-1}} x dy - y dx + \int_{C_n} x dy - y dx \right)\end{aligned}$$

To evaluate these integrals we use the formula from (a) to get

$$A(D) = \frac{1}{2} [(x_1 y_2 - x_2 y_1) + (x_2 y_3 - x_3 y_2) + \cdots + (x_{n-1} y_n - x_n y_{n-1}) + (x_n y_1 - x_1 y_n)].$$

- (c) $A = \frac{1}{2} [(0 \cdot 1 - 2 \cdot 0) + (2 \cdot 3 - 1 \cdot 1) + (1 \cdot 2 - 0 \cdot 3) + (0 \cdot 1 - (-1) \cdot 2) + (-1 \cdot 0 - 0 \cdot 1)]$
 $= \frac{1}{2} (0 + 5 + 2 + 2) = \frac{9}{2}$

26. By Green's Theorem, $\frac{1}{2A} \oint_C x^2 dy = \frac{1}{2A} \iint_D 2x dA = \frac{1}{A} \iint_D x dA = \bar{x}$ and
 $-\frac{1}{2A} \oint_C y^2 dx = -\frac{1}{2A} \iint_D (-2y) dA = \frac{1}{A} \iint_D y dA = \bar{y}.$

27. We orient the quarter-circular region as shown in the figure.

$$A = \frac{1}{4} \pi a^2 \text{ so } \bar{x} = \frac{1}{\pi a^2/2} \oint_C x^2 dy \text{ and } \bar{y} = -\frac{1}{\pi a^2/2} \oint_C y^2 dx.$$

Here $C = C_1 + C_2 + C_3$ where

$$C_1: x = t, y = 0, 0 \leq t \leq a;$$

$$C_2: x = a \cos t, y = a \sin t, 0 \leq t \leq \frac{\pi}{2}; \text{ and}$$

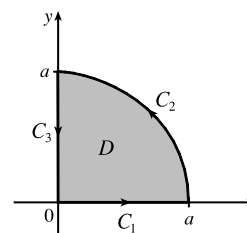
$$C_3: x = 0, y = a - t, 0 \leq t \leq a. \text{ Then}$$

$$\begin{aligned}\oint_C x^2 dy &= \int_{C_1} x^2 dy + \int_{C_2} x^2 dy + \int_{C_3} x^2 dy = \int_0^a 0 dt + \int_0^{\pi/2} (a \cos t)^2 (a \cos t) dt + \int_0^a 0 dt \\ &= \int_0^{\pi/2} a^3 \cos^3 t dt = a^3 \int_0^{\pi/2} (1 - \sin^2 t) \cos t dt = a^3 \left[\sin t - \frac{1}{3} \sin^3 t \right]_0^{\pi/2} = \frac{2}{3} a^3\end{aligned}$$

$$\text{so } \bar{x} = \frac{1}{\pi a^2/2} \oint_C x^2 dy = \frac{4a}{3\pi}.$$

$$\begin{aligned}\oint_C y^2 dx &= \int_{C_1} y^2 dx + \int_{C_2} y^2 dx + \int_{C_3} y^2 dx = \int_0^a 0 dt + \int_0^{\pi/2} (a \sin t)^2 (-a \sin t) dt + \int_0^a 0 dt \\ &= \int_0^{\pi/2} (-a^3 \sin^3 t) dt = -a^3 \int_0^{\pi/2} (1 - \cos^2 t) \sin t dt = -a^3 \left[\frac{1}{3} \cos^3 t - \cos t \right]_0^{\pi/2} = -\frac{2}{3} a^3,\end{aligned}$$

$$\text{so } \bar{y} = -\frac{1}{\pi a^2/2} \oint_C y^2 dx = \frac{4a}{3\pi}. \text{ Thus } (\bar{x}, \bar{y}) = \left(\frac{4a}{3\pi}, \frac{4a}{3\pi} \right).$$



28. Here $A = \frac{1}{2}ab$ and $C = C_1 + C_2 + C_3$, where $C_1: x = x, y = 0, 0 \leq x \leq a$;

$C_2: x = a, y = y, 0 \leq y \leq b$; and $C_3: x = x, y = \frac{b}{a}x, x = a$ to $x = 0$. Then

$$\begin{aligned}\oint_C x^2 dy &= \int_{C_1} x^2 dy + \int_{C_2} x^2 dy + \int_{C_3} x^2 dy = 0 + \int_0^b a^2 dy + \int_a^0 (x^2) \left(\frac{b}{a} dx\right) \\ &= a^2 b + \frac{b}{a} \left[\frac{1}{3} x^3\right]_a^0 = a^2 b - \frac{1}{3} a^2 b = \frac{2}{3} a^2 b.\end{aligned}$$

Similarly, $\oint_C y^2 dx = \int_{C_1} y^2 dx + \int_{C_2} y^2 dx + \int_{C_3} y^2 dx = 0 + 0 + \int_a^0 \left(\frac{b}{a}x\right)^2 dx = \frac{b^2}{a^2} \cdot \frac{1}{3} x^3 \Big|_a^0 = -\frac{1}{3} ab^2$. Thus

$$\bar{x} = \frac{1}{2A} \oint_C x^2 dy = \frac{1}{ab} \cdot \frac{2}{3} a^2 b = \frac{2}{3} a \text{ and } \bar{y} = -\frac{1}{2A} \oint_C y^2 dx = -\frac{1}{ab} \left(-\frac{1}{3} ab^2\right) = \frac{1}{3} b, \text{ so } (\bar{x}, \bar{y}) = \left(\frac{2}{3} a, \frac{1}{3} b\right).$$

29. By Green's Theorem, $-\frac{1}{3} \rho \oint_C y^3 dx = -\frac{1}{3} \rho \iint_D (-3y^2) dA = \iint_D y^2 \rho dA = I_x$ and

$$\frac{1}{3} \rho \oint_C x^3 dy = \frac{1}{3} \rho \iint_D (3x^2) dA = \iint_D x^2 \rho dA = I_y.$$

30. By symmetry the moments of inertia about any two diameters are equal. Centering the disk at the origin, the moment of inertia about a diameter equals

$$\begin{aligned}I_y &= \frac{1}{3} \rho \oint_C x^3 dy = \frac{1}{3} \rho \int_0^{2\pi} (a \cos t)^3 (a \cos t dt) = \frac{1}{3} \rho \int_0^{2\pi} (a^4 \cos^4 t) dt \\ &= \frac{1}{3} a^4 \rho \int_0^{2\pi} \left[\frac{1}{2}(1 + \cos 2t)\right]^2 dt = \frac{1}{3} a^4 \rho \int_0^{2\pi} \left(\frac{3}{8} + \frac{1}{2} \cos 2t + \frac{1}{8} \cos 4t\right) dt \\ &= \frac{1}{3} a^4 \rho \left[\frac{3}{8} t + \frac{1}{4} \sin 2t + \frac{1}{32} \sin 4t\right]_0^{2\pi} = \frac{1}{3} a^4 \rho \cdot \frac{3(2\pi)}{8} = \frac{1}{4} \pi a^4 \rho\end{aligned}$$

31. As in Example 5, let C' be a counterclockwise-oriented circle with center the origin and radius a , where a is chosen to be small enough so that C' lies inside C , and D the region bounded by C and C' . Here

$$P = \frac{2xy}{(x^2 + y^2)^2} \Rightarrow \frac{\partial P}{\partial y} = \frac{2x(x^2 + y^2)^2 - 2xy \cdot 2(x^2 + y^2) \cdot 2y}{(x^2 + y^2)^4} = \frac{2x^3 - 6xy^2}{(x^2 + y^2)^3} \text{ and}$$

$$Q = \frac{y^2 - x^2}{(x^2 + y^2)^2} \Rightarrow \frac{\partial Q}{\partial x} = \frac{-2x(x^2 + y^2)^2 - (y^2 - x^2) \cdot 2(x^2 + y^2) \cdot 2x}{(x^2 + y^2)^4} = \frac{2x^3 - 6xy^2}{(x^2 + y^2)^3}. \text{ Thus, as in the example,}$$

$$\int_C P dx + Q dy + \int_{-C'} P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA = \iint_D 0 dA = 0$$

and $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C'} \mathbf{F} \cdot d\mathbf{r}$. We parametrize C' as $\mathbf{r}(t) = a \cos t \mathbf{i} + a \sin t \mathbf{j}, 0 \leq t \leq 2\pi$. Then

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_{C'} \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \frac{2(a \cos t)(a \sin t) \mathbf{i} + (a^2 \sin^2 t - a^2 \cos^2 t) \mathbf{j}}{(a^2 \cos^2 t + a^2 \sin^2 t)^2} \cdot (-a \sin t \mathbf{i} + a \cos t \mathbf{j}) dt \\ &= \frac{1}{a} \int_0^{2\pi} (-\cos t \sin^2 t - \cos^3 t) dt = \frac{1}{a} \int_0^{2\pi} (-\cos t \sin^2 t - \cos t (1 - \sin^2 t)) dt \\ &= -\frac{1}{a} \int_0^{2\pi} \cos t dt = -\frac{1}{a} \sin t \Big|_0^{2\pi} = 0\end{aligned}$$

32. P and Q have continuous partial derivatives on $\mathbb{R}^2$, so by Green's Theorem we have

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA = \iint_D \left[\frac{\partial}{\partial x}(3x - y^2) - \frac{\partial}{\partial y}(x^2 + y)\right] dA \\ &= \iint_D (3 - 1) dA = 2 \iint_D dA = 2 \cdot A(D) = 2 \cdot 6 = 12\end{aligned}$$

33. Since C is a simple closed path which doesn't pass through or enclose the origin, there exists an open region that doesn't contain the origin but does contain D . Thus $P = -y/(x^2 + y^2)$ and $Q = x/(x^2 + y^2)$ have continuous partial derivatives on this open region containing D and we can apply Green's Theorem. But by Exercise 16.3.41(a), $\partial P/\partial y = \partial Q/\partial x$, so

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D 0 \, dA = 0.$$

34. We express D as a type II region: $D = \{(x, y) \mid f_1(y) \leq x \leq f_2(y), c \leq y \leq d\}$ where f_1 and f_2 are continuous functions.

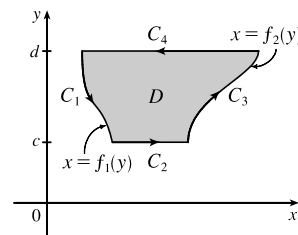
Then $\iint_D \frac{\partial Q}{\partial x} \, dA = \int_c^d \int_{f_1(y)}^{f_2(y)} \frac{\partial Q}{\partial x} \, dx \, dy = \int_c^d [Q(f_2(y), y) - Q(f_1(y), y)] \, dy$ by the Fundamental Theorem of

Calculus. But referring to the figure, $\oint_C Q \, dy = \int_{C_1+C_2+C_3+C_4} Q \, dy$.

Then $\int_{C_1} Q \, dy = \int_d^c Q(f_1(y), y) \, dy$, $\int_{C_2} Q \, dy = \int_c^d Q \, dy = 0$,

and $\int_{C_3} Q \, dy = \int_c^d Q(f_2(y), y) \, dy$. Hence

$\oint_C Q \, dy = \int_c^d [Q(f_2(y), y) - Q(f_1(y), y)] \, dy = \iint_D (\partial Q/\partial x) \, dA$.



35. Using the first part of Equation 5, we have that $\iint_R dx \, dy = A(R) = \int_{\partial R} x \, dy$. But $x = g(u, v)$, and $dy = \frac{\partial h}{\partial u} du + \frac{\partial h}{\partial v} dv$, and we orient ∂S by taking the positive direction to be that which corresponds, under the mapping, to the positive direction along ∂R , so

$$\begin{aligned} \int_{\partial R} x \, dy &= \int_{\partial S} g(u, v) \left(\frac{\partial h}{\partial u} du + \frac{\partial h}{\partial v} dv \right) = \int_{\partial S} g(u, v) \frac{\partial h}{\partial u} du + g(u, v) \frac{\partial h}{\partial v} dv \\ &= \pm \iint_S \left[\frac{\partial}{\partial u} \left(g(u, v) \frac{\partial h}{\partial v} \right) - \frac{\partial}{\partial v} \left(g(u, v) \frac{\partial h}{\partial u} \right) \right] dA \quad [\text{using Green's Theorem in the } uv\text{-plane}] \\ &= \pm \iint_S \left(\frac{\partial g}{\partial u} \frac{\partial h}{\partial v} + g(u, v) \frac{\partial^2 h}{\partial u \partial v} - \frac{\partial g}{\partial v} \frac{\partial h}{\partial u} - g(u, v) \frac{\partial^2 h}{\partial v \partial u} \right) dA \quad [\text{using the Chain Rule}] \\ &= \pm \iint_S \left(\frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \right) dA \quad [\text{by the equality of mixed partials}] = \pm \iint_S \frac{\partial(x, y)}{\partial(u, v)} du \, dv \end{aligned}$$

The sign is chosen to be positive if the orientation that we gave to ∂S corresponds to the usual positive orientation, and it is negative otherwise. In either case, since $A(R)$ is positive, the sign chosen must be the same as the sign of $\frac{\partial(x, y)}{\partial(u, v)}$.

Therefore, $A(R) = \iint_R dx \, dy = \iint_S \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv$.

16.5 Curl and Divergence

$$\begin{aligned} 1. \text{ (a) } \operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy^2z^2 & x^2yz^2 & x^2y^2z \end{vmatrix} \\ &= \left[\frac{\partial}{\partial y}(x^2y^2z) - \frac{\partial}{\partial z}(x^2yz^2) \right] \mathbf{i} - \left[\frac{\partial}{\partial x}(x^2y^2z) - \frac{\partial}{\partial z}(xy^2z^2) \right] \mathbf{j} + \left[\frac{\partial}{\partial x}(x^2yz^2) - \frac{\partial}{\partial y}(xy^2z^2) \right] \mathbf{k} \\ &= (2x^2yz - 2x^2yz) \mathbf{i} - (2xy^2z - 2xy^2z) \mathbf{j} + (2xyz^2 - 2xyz^2) \mathbf{k} = \mathbf{0} \end{aligned}$$

$$(b) \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial}{\partial x} (xy^2z^2) + \frac{\partial}{\partial y} (x^2yz^2) + \frac{\partial}{\partial z} (x^2y^2z) = y^2z^2 + x^2z^2 + x^2y^2$$

$$\begin{aligned} 2. (a) \operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0 & x^3yz^2 & y^4z^3 \end{vmatrix} \\ &= \left[\frac{\partial}{\partial y} (y^4z^3) - \frac{\partial}{\partial z} (x^3yz^2) \right] \mathbf{i} - \left[\frac{\partial}{\partial x} (y^4z^3) - \frac{\partial}{\partial z} (0) \right] \mathbf{j} + \left[\frac{\partial}{\partial x} (x^3yz^2) - \frac{\partial}{\partial y} (0) \right] \mathbf{k} \\ &= (4y^3z^3 - 2x^3yz) \mathbf{i} - (0 - 0) \mathbf{j} + (3x^2yz^2 - 0) \mathbf{k} \\ &= (4y^3z^3 - 2x^3yz) \mathbf{i} + 3x^2yz^2 \mathbf{k} \end{aligned}$$

$$(b) \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial}{\partial x} (0) + \frac{\partial}{\partial y} (x^3yz^2) + \frac{\partial}{\partial z} (y^4z^3) = 0 + x^3z^2 + 3y^4z^2 = x^3z^2 + 3y^4z^2$$

$$\begin{aligned} 3. (a) \operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xye^z & 0 & yze^x \end{vmatrix} = (ze^x - 0) \mathbf{i} - (yze^x - xye^z) \mathbf{j} + (0 - xe^z) \mathbf{k} \\ &= ze^x \mathbf{i} + (xye^z - yze^x) \mathbf{j} - xe^z \mathbf{k} \end{aligned}$$

$$(b) \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial}{\partial x} (xye^z) + \frac{\partial}{\partial y} (0) + \frac{\partial}{\partial z} (yze^x) = ye^z + 0 + ye^x = y(e^z + e^x)$$

$$\begin{aligned} 4. (a) \operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \sin yz & \sin zx & \sin xy \end{vmatrix} \\ &= (x \cos xy - x \cos zx) \mathbf{i} - (y \cos xy - y \cos yz) \mathbf{j} + (z \cos zx - z \cos yz) \mathbf{k} \\ &= x(\cos xy - \cos zx) \mathbf{i} + y(\cos yz - \cos xy) \mathbf{j} + z(\cos zx - \cos yz) \mathbf{k} \end{aligned}$$

$$(b) \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial}{\partial x} (\sin yz) + \frac{\partial}{\partial y} (\sin zx) + \frac{\partial}{\partial z} (\sin xy) = 0 + 0 + 0 = 0$$

$$\begin{aligned} 5. (a) \operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \frac{\sqrt{x}}{1+z} & \frac{\sqrt{y}}{1+x} & \frac{\sqrt{z}}{1+y} \end{vmatrix} \\ &= [\sqrt{z}(-1)(1+y)^{-2} - 0] \mathbf{i} - [0 - \sqrt{x}(-1)(1+z)^{-2}] \mathbf{j} + [\sqrt{y}(-1)(1+x)^{-2} - 0] \mathbf{k} \\ &= -\frac{\sqrt{z}}{(1+y)^2} \mathbf{i} - \frac{\sqrt{x}}{(1+z)^2} \mathbf{j} - \frac{\sqrt{y}}{(1+x)^2} \mathbf{k} \end{aligned}$$

$$\begin{aligned} (b) \operatorname{div} \mathbf{F} &= \nabla \cdot \mathbf{F} = \frac{\partial}{\partial x} \left(\frac{\sqrt{x}}{1+z} \right) + \frac{\partial}{\partial y} \left(\frac{\sqrt{y}}{1+x} \right) + \frac{\partial}{\partial z} \left(\frac{\sqrt{z}}{1+y} \right) \\ &= \frac{1}{2\sqrt{x}(1+z)} + \frac{1}{2\sqrt{y}(1+x)} + \frac{1}{2\sqrt{z}(1+y)} \end{aligned}$$

$$\begin{aligned}
 6. \text{ (a) } \operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \ln(2y+3z) & \ln(x+3z) & \ln(x+2y) \end{vmatrix} \\
 &= \left(\frac{2}{x+2y} - \frac{3}{x+3z} \right) \mathbf{i} - \left(\frac{1}{x+2y} - \frac{3}{2y+3z} \right) \mathbf{j} + \left(\frac{1}{x+3z} - \frac{2}{2y+3z} \right) \mathbf{k} \\
 &= \left(\frac{2}{x+2y} - \frac{3}{x+3z} \right) \mathbf{i} + \left(\frac{3}{2y+3z} - \frac{1}{x+2y} \right) \mathbf{j} + \left(\frac{1}{x+3z} - \frac{2}{2y+3z} \right) \mathbf{k}
 \end{aligned}$$

$$\text{(b) } \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial}{\partial x} [\ln(2y+3z)] + \frac{\partial}{\partial y} [\ln(x+3z)] + \frac{\partial}{\partial z} [\ln(x+2y)] = 0 + 0 + 0 = 0$$

$$\begin{aligned}
 7. \text{ (a) } \operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ e^x \sin y & e^y \sin z & e^z \sin x \end{vmatrix} = (0 - e^y \cos z) \mathbf{i} - (e^z \cos x - 0) \mathbf{j} + (0 - e^x \cos y) \mathbf{k} \\
 &= \langle -e^y \cos z, -e^z \cos x, -e^x \cos y \rangle
 \end{aligned}$$

$$\text{(b) } \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial}{\partial x} (e^x \sin y) + \frac{\partial}{\partial y} (e^y \sin z) + \frac{\partial}{\partial z} (e^z \sin x) = e^x \sin y + e^y \sin z + e^z \sin x$$

$$\begin{aligned}
 8. \text{ (a) } \operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \arctan(xy) & \arctan(yz) & \arctan(zx) \end{vmatrix} \\
 &= \left(0 - \frac{y}{1+(yz)^2} \right) \mathbf{i} - \left(\frac{z}{1+(zx)^2} - 0 \right) \mathbf{j} + \left(0 - \frac{x}{1+(xy)^2} \right) \mathbf{k} \\
 &= \left\langle -\frac{y}{1+y^2z^2}, -\frac{z}{1+x^2z^2}, -\frac{x}{1+x^2y^2} \right\rangle
 \end{aligned}$$

$$\text{(b) } \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial}{\partial x} [\arctan(xy)] + \frac{\partial}{\partial y} [\arctan(yz)] + \frac{\partial}{\partial z} [\arctan(zx)] = \frac{y}{1+x^2y^2} + \frac{z}{1+y^2z^2} + \frac{x}{1+x^2z^2}$$

9. (a) $\operatorname{div} \mathbf{F}$ is negative because the vectors that start near P are shorter than those that end near P . Intuitively, if $\mathbf{F}$ represents a velocity field of fluid flow, then the net flow at P is inward.

(b) $\operatorname{curl} \mathbf{F}$ is zero because we can see that if $\mathbf{F}$ represents a velocity field of fluid flow, then a paddle wheel placed at P moves with the fluid, but does not rotate.

10. (a) $\operatorname{div} \mathbf{F}$ is positive because the vectors that start near P are longer than those that end near P . Intuitively, if $\mathbf{F}$ represents a velocity field of fluid flow, then the net flow at P is outward.

(b) $\operatorname{curl} \mathbf{F}$ is zero because we can see that if $\mathbf{F}$ represents a velocity field of fluid flow, then a paddle wheel placed at P moves with the fluid, but does not rotate.

11. (a) $\operatorname{div} \mathbf{F}$ is zero because the vectors that start near P are the same length as those that end near P . Intuitively, if $\mathbf{F}$ represents a velocity field of fluid flow, then the net flow at P is zero.

(b) $\operatorname{curl} \mathbf{F} \neq 0$ because we can see that if $\mathbf{F}$ represents a velocity field of fluid flow, then a paddle wheel placed at P would rotate clockwise about its axis, and hence the curl vector there points in the direction of $-\mathbf{k}$.

12. (a) $\operatorname{div} \mathbf{F}$ is zero because the vectors that start near P are the same length as those that end near P . Intuitively, if $\mathbf{F}$ represents a velocity field of fluid flow, then the net flow at P is zero.
- (b) $\operatorname{curl} \mathbf{F} \neq 0$ because we can see that if $\mathbf{F}$ represents a velocity field of fluid flow, then a paddle wheel placed at P would rotate counterclockwise about its axis, and hence the curl vector there points in the direction of $\mathbf{k}$.
13. (a) We need to verify $\operatorname{curl}(\nabla f) = \mathbf{0}$ for $f(x, y, z) = \sin xyz$. First, $\nabla f = yz \cos xyz \mathbf{i} + xz \cos xyz \mathbf{j} + xy \cos xyz \mathbf{k}$. Then

$$\begin{aligned} \operatorname{curl}(\nabla f) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz \cos xyz & xz \cos xyz & xy \cos xyz \end{vmatrix} \\ &= [-x^2 yz \sin xyz + x \cos xyz - (-x^2 yz \sin xyz + x \cos xyz)] \mathbf{i} \\ &\quad - [-xy^2 z \sin xyz + y \cos xyz - (-xy^2 z \sin xyz + y \cos xyz)] \mathbf{j} \\ &\quad + [-xyz^2 \sin xyz + z \cos xyz - (-xyz^2 \sin xyz + z \cos xyz)] \mathbf{k} = \mathbf{0} \end{aligned}$$

- (b) We need to verify that $\operatorname{div} \operatorname{curl} \mathbf{F} = 0$ for $\mathbf{F}(x, y, z) = xyz^2 \mathbf{i} + x^2 yz^3 \mathbf{j} + y^2 \mathbf{k}$. First,

$$\begin{aligned} \operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xyz^2 & x^2 yz^3 & y^2 \end{vmatrix} \\ &= (2y - 3x^2 yz^2) \mathbf{i} - (0 - 2xyz) \mathbf{j} + (2xyz^3 - xz^2) \mathbf{k} \\ &= (2y - 3x^2 yz^2) \mathbf{i} + 2xyz \mathbf{j} + (2xyz^3 - xz^2) \mathbf{k} \end{aligned}$$

Then

$$\begin{aligned} \operatorname{div} \operatorname{curl} \mathbf{F} &= \nabla \cdot (\nabla \times \mathbf{F}) = \frac{\partial}{\partial x}(2y - 3x^2 yz^2) + \frac{\partial}{\partial y}(2xyz) + \frac{\partial}{\partial z}(2xyz^3 - xz^2) \\ &= -6xyz^2 + 2xz + 6xyz^2 - 2xz = 0 \end{aligned}$$

14. (a) $\operatorname{curl} f = \nabla \times f$ is meaningless because f is a scalar field.
- (b) $\operatorname{grad} f$ is a vector field.
- (c) $\operatorname{div} \mathbf{F}$ is a scalar field.
- (d) $\operatorname{curl}(\operatorname{grad} f)$ is a vector field.
- (e) $\operatorname{grad} \mathbf{F}$ is meaningless because $\mathbf{F}$ is not a scalar field.
- (f) $\operatorname{grad}(\operatorname{div} \mathbf{F})$ is a vector field.
- (g) $\operatorname{div}(\operatorname{grad} f)$ is a scalar field.
- (h) $\operatorname{grad}(\operatorname{div} f)$ is meaningless because f is a scalar field.
- (i) $\operatorname{curl}(\operatorname{curl} \mathbf{F})$ is a vector field.
- (j) $\operatorname{div}(\operatorname{div} \mathbf{F})$ is meaningless because $\operatorname{div} \mathbf{F}$ is a scalar field.
- (k) $(\operatorname{grad} f) \times (\operatorname{div} \mathbf{F})$ is meaningless because $\operatorname{div} \mathbf{F}$ is a scalar field.
- (l) $\operatorname{div}(\operatorname{curl}(\operatorname{grad} f))$ is a scalar field.

$$\begin{aligned}
 15. \operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2xy^3z^2 & 3x^2y^2z^2 & 2x^2y^3z \end{vmatrix} \\
 &= (6x^2y^2z - 6x^2y^2z)\mathbf{i} - (4xy^3z - 4xy^3z)\mathbf{j} + (6xy^2z^2 - 6xy^2z^2)\mathbf{k} = \mathbf{0}
 \end{aligned}$$

and $\mathbf{F}$ is defined on all of $\mathbb{R}^3$ whose component functions have continuous partial derivatives, so, by Theorem 4, $\mathbf{F}$ is

conservative. Thus, there exists a function f such that $\nabla f = \mathbf{F}$. Now $f_x(x, y, z) = 2xy^3z^2$ implies that

$f(x, y, z) = x^2y^3z^2 + g(y, z)$ and then $f_y(x, y, z) = 3x^2y^2z^2 + g_y(y, z)$. But $f_y(x, y, z) = 3x^2y^2z^2$, so $g(y, z) = h(z)$

and $f(x, y, z) = x^2y^3z^2 + h(z)$. Thus, $f_z(x, y, z) = 2x^2y^3z + h'(z)$, but $f_z(x, y, z) = 2x^2y^3z$, so $h(z) = K$, a constant.

Hence, a potential function for $\mathbf{F}$ is $f(x, y, z) = x^2y^3z^2 + K$.

$$16. \operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz & xz + y & xy - x \end{vmatrix} = (x - x)\mathbf{i} - (y - 1 - y)\mathbf{j} + (z - z)\mathbf{k} = \mathbf{j} \neq \mathbf{0},$$

so $\mathbf{F}$ is not conservative.

$$17. \operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \ln y & (x/y) + \ln z & y/z \end{vmatrix} = (1/z - 1/z)\mathbf{i} - (0 - 0)\mathbf{j} + (1/y - 1/y)\mathbf{k} = \mathbf{0}$$

and the partial derivatives of the component functions are defined and continuous on the open set $\{(x, y, z) \mid y, z > 0\}$, so, by

Theorem 4, $\mathbf{F}$ is conservative. Thus, there exists a function f such that $\nabla f = \mathbf{F}$. Now $f_x(x, y, z) = \ln y$ implies that

$f(x, y, z) = x \ln y + g(y, z)$ and then $f_y(x, y, z) = (x/y) + g_y(y, z)$. But $f_y(x, y, z) = (x/y) + \ln z$, so

$g(y, z) = y \ln z + h(z)$ and $f(x, y, z) = x \ln y + y \ln z + h(z)$. Thus, $f_z(x, y, z) = (y/z) + h'(z)$, but $f_z(x, y, z) = y/z$,

so $h(z) = K$, a constant. Hence, a potential function for $\mathbf{F}$ is $f(x, y, z) = x \ln y + y \ln z + K$.

$$\begin{aligned}
 18. \operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz \sin(xy) & xz \sin(xy) & -\cos(xy) \end{vmatrix} \\
 &= [x \sin(xy) - x \sin(xy)]\mathbf{i} - [y \sin(xy) - y \sin(xy)]\mathbf{j} \\
 &\quad + [z \sin(xy) + xyz \cos(xy) - z \sin(xy) - xyz \cos(xy)]\mathbf{k} \\
 &= \mathbf{0}
 \end{aligned}$$

and $\mathbf{F}$ is defined on all of $\mathbb{R}^3$ whose component functions have continuous partial derivatives, so, by Theorem 4, $\mathbf{F}$ is

conservative. Thus, there exists a function f such that $\nabla f = \mathbf{F}$. Now $f_x(x, y, z) = yz \sin(xy)$ implies that

$f(x, y, z) = -z \cos(xy) + g(y, z)$ and then $f_y(x, y, z) = xz \sin(xy) + g_y(y, z)$. But $f_y(x, y, z) = xz \sin(xy)$, so

$g(y, z) = h(z)$ and $f(x, y, z) = -z \cos(xy) + h(z)$. Thus, $f_z(x, y, z) = -\cos(xy) + h'(z)$, but $f_z(x, y, z) = -\cos(xy)$,

so $h(z) = K$, a constant. Hence, a potential function for $\mathbf{F}$ is $f(x, y, z) = -z \cos(xy) + K$.

$$\begin{aligned}
 19. \operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz^2e^{xz} & ze^{xz} & xyz e^{xz} \end{vmatrix} \\
 &= (xz e^{xz} - e^{xz} - xze^{xz})\mathbf{i} - (yze^{xz} + xyz^2e^{xz} - 2yze^{xz} - xyz^2e^{xz})\mathbf{j} + (z^2e^{xz} - z^2e^{xz})\mathbf{k} \\
 &= -e^{xz}\mathbf{i} + yze^{xz}\mathbf{j} \neq \mathbf{0},
 \end{aligned}$$

so $\mathbf{F}$ is not conservative.

$$\begin{aligned}
 20. \operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ e^z \cos x & e^y \cos z & e^z \sin x - e^y \sin z \end{vmatrix} \\
 &= [-e^y \sin z - (-e^y \sin z)]\mathbf{i} - (e^z \cos x - e^z \cos x)\mathbf{j} + (0 - 0)\mathbf{k} = \mathbf{0}
 \end{aligned}$$

and $\mathbf{F}$ is defined on all of $\mathbb{R}^3$ whose component functions have continuous partial derivatives, so, by Theorem 4, $\mathbf{F}$ is

conservative. Thus, there exists a function f such that $\nabla f = \mathbf{F}$. Now $f_x(x, y, z) = e^z \cos x$ implies that

$f(x, y, z) = e^z \sin x + g(y, z)$ and then $f_y(x, y, z) = g_y(y, z)$. But $f_y(x, y, z) = e^y \cos z$, so $g(y, z) = e^y \cos z + h(z)$ and

$f(x, y, z) = e^z \sin x + e^y \cos z + h(z)$. Thus, $f_z(x, y, z) = e^z \sin x - e^y \sin z + h'(z)$, but

$f_z(x, y, z) = e^z \sin x - e^y \sin z$, so $h(z) = K$, a constant. Hence, a potential function for $\mathbf{F}$ is

$$f(x, y, z) = e^z \sin x + e^y \cos z + K.$$

$$21. \text{ No. Assume there is such a } \mathbf{G}. \text{ Then } \operatorname{div}(\operatorname{curl} \mathbf{G}) = \frac{\partial}{\partial x}(x \sin y) + \frac{\partial}{\partial y}(\cos y) + \frac{\partial}{\partial z}(z - xy) = \sin y - \sin y + 1 \neq 0,$$

which contradicts Theorem 11.

$$22. \text{ No. Assume there is such a } \mathbf{G}. \text{ Then } \operatorname{div}(\operatorname{curl} \mathbf{G}) = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 \neq 0 \text{ which contradicts}$$

Theorem 11.

$$23. \operatorname{curl} \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f(x) & g(y) & h(z) \end{vmatrix} = (0 - 0)\mathbf{i} + (0 - 0)\mathbf{j} + (0 - 0)\mathbf{k} = \mathbf{0}. \text{ Hence } \mathbf{F} = f(x)\mathbf{i} + g(y)\mathbf{j} + h(z)\mathbf{k}$$

is irrotational.

$$24. \operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(f(y, z)) + \frac{\partial}{\partial y}(g(x, z)) + \frac{\partial}{\partial z}(h(x, y)) = 0 \text{ so } \mathbf{F} \text{ is incompressible.}$$

For Exercises 25–31, let $\mathbf{F}(x, y, z) = P_1\mathbf{i} + Q_1\mathbf{j} + R_1\mathbf{k}$ and $\mathbf{G}(x, y, z) = P_2\mathbf{i} + Q_2\mathbf{j} + R_2\mathbf{k}$.

$$\begin{aligned}
 25. \operatorname{div}(\mathbf{F} + \mathbf{G}) &= \operatorname{div}\langle P_1 + P_2, Q_1 + Q_2, R_1 + R_2 \rangle = \frac{\partial(P_1 + P_2)}{\partial x} + \frac{\partial(Q_1 + Q_2)}{\partial y} + \frac{\partial(R_1 + R_2)}{\partial z} \\
 &= \frac{\partial P_1}{\partial x} + \frac{\partial P_2}{\partial x} + \frac{\partial Q_1}{\partial y} + \frac{\partial Q_2}{\partial y} + \frac{\partial R_1}{\partial z} + \frac{\partial R_2}{\partial z} = \left(\frac{\partial P_1}{\partial x} + \frac{\partial Q_1}{\partial y} + \frac{\partial R_1}{\partial z} \right) + \left(\frac{\partial P_2}{\partial x} + \frac{\partial Q_2}{\partial y} + \frac{\partial R_2}{\partial z} \right) \\
 &= \operatorname{div}\langle P_1, Q_1, R_1 \rangle + \operatorname{div}\langle P_2, Q_2, R_2 \rangle = \operatorname{div} \mathbf{F} + \operatorname{div} \mathbf{G}
 \end{aligned}$$

$$\begin{aligned}
26. \operatorname{curl} \mathbf{F} + \operatorname{curl} \mathbf{G} &= \left[\left(\frac{\partial R_1}{\partial y} - \frac{\partial Q_1}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P_1}{\partial z} - \frac{\partial R_1}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q_1}{\partial x} - \frac{\partial P_1}{\partial y} \right) \mathbf{k} \right] \\
&\quad + \left[\left(\frac{\partial R_2}{\partial y} - \frac{\partial Q_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P_2}{\partial z} - \frac{\partial R_2}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q_2}{\partial x} - \frac{\partial P_2}{\partial y} \right) \mathbf{k} \right] \\
&= \left[\frac{\partial(R_1 + R_2)}{\partial y} - \frac{\partial(Q_1 + Q_2)}{\partial z} \right] \mathbf{i} + \left[\frac{\partial(P_1 + P_2)}{\partial z} - \frac{\partial(R_1 + R_2)}{\partial x} \right] \mathbf{j} \\
&\quad + \left[\frac{\partial(Q_1 + Q_2)}{\partial x} - \frac{\partial(P_1 + P_2)}{\partial y} \right] \mathbf{k} = \operatorname{curl}(\mathbf{F} + \mathbf{G})
\end{aligned}$$

$$\begin{aligned}
27. \operatorname{div}(f\mathbf{F}) &= \operatorname{div}(f \langle P_1, Q_1, R_1 \rangle) = \operatorname{div}\langle fP_1, fQ_1, fR_1 \rangle = \frac{\partial(fP_1)}{\partial x} + \frac{\partial(fQ_1)}{\partial y} + \frac{\partial(fR_1)}{\partial z} \\
&= \left(f \frac{\partial P_1}{\partial x} + P_1 \frac{\partial f}{\partial x} \right) + \left(f \frac{\partial Q_1}{\partial y} + Q_1 \frac{\partial f}{\partial y} \right) + \left(f \frac{\partial R_1}{\partial z} + R_1 \frac{\partial f}{\partial z} \right) \\
&= f \left(\frac{\partial P_1}{\partial x} + \frac{\partial Q_1}{\partial y} + \frac{\partial R_1}{\partial z} \right) + \langle P_1, Q_1, R_1 \rangle \cdot \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle = f \operatorname{div} \mathbf{F} + \mathbf{F} \cdot \nabla f
\end{aligned}$$

$$\begin{aligned}
28. \operatorname{curl}(f\mathbf{F}) &= \left[\frac{\partial(fR_1)}{\partial y} - \frac{\partial(fQ_1)}{\partial z} \right] \mathbf{i} + \left[\frac{\partial(fP_1)}{\partial z} - \frac{\partial(fR_1)}{\partial x} \right] \mathbf{j} + \left[\frac{\partial(fQ_1)}{\partial x} - \frac{\partial(fP_1)}{\partial y} \right] \mathbf{k} \\
&= \left[f \frac{\partial R_1}{\partial y} + R_1 \frac{\partial f}{\partial y} - f \frac{\partial Q_1}{\partial z} - Q_1 \frac{\partial f}{\partial z} \right] \mathbf{i} + \left[f \frac{\partial P_1}{\partial z} + P_1 \frac{\partial f}{\partial z} - f \frac{\partial R_1}{\partial x} - R_1 \frac{\partial f}{\partial x} \right] \mathbf{j} \\
&\quad + \left[f \frac{\partial Q_1}{\partial x} + Q_1 \frac{\partial f}{\partial x} - f \frac{\partial P_1}{\partial y} - P_1 \frac{\partial f}{\partial y} \right] \mathbf{k} \\
&= f \left[\frac{\partial R_1}{\partial y} - \frac{\partial Q_1}{\partial z} \right] \mathbf{i} + f \left[\frac{\partial P_1}{\partial z} - \frac{\partial R_1}{\partial x} \right] \mathbf{j} + f \left[\frac{\partial Q_1}{\partial x} - \frac{\partial P_1}{\partial y} \right] \mathbf{k} \\
&\quad + \left[R_1 \frac{\partial f}{\partial y} - Q_1 \frac{\partial f}{\partial z} \right] \mathbf{i} + \left[P_1 \frac{\partial f}{\partial z} - R_1 \frac{\partial f}{\partial x} \right] \mathbf{j} + \left[Q_1 \frac{\partial f}{\partial x} - P_1 \frac{\partial f}{\partial y} \right] \mathbf{k} \\
&= f \operatorname{curl} \mathbf{F} + (\nabla f) \times \mathbf{F}
\end{aligned}$$

$$\begin{aligned}
29. \operatorname{div}(\mathbf{F} \times \mathbf{G}) &= \nabla \cdot (\mathbf{F} \times \mathbf{G}) = \begin{vmatrix} \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P_1 & Q_1 & R_1 \\ P_2 & Q_2 & R_2 \end{vmatrix} = \frac{\partial}{\partial x} \begin{vmatrix} Q_1 & R_1 \\ Q_2 & R_2 \end{vmatrix} - \frac{\partial}{\partial y} \begin{vmatrix} P_1 & R_1 \\ P_2 & R_2 \end{vmatrix} + \frac{\partial}{\partial z} \begin{vmatrix} P_1 & Q_1 \\ P_2 & Q_2 \end{vmatrix} \\
&= \left[Q_1 \frac{\partial R_2}{\partial x} + R_2 \frac{\partial Q_1}{\partial x} - Q_2 \frac{\partial R_1}{\partial x} - R_1 \frac{\partial Q_2}{\partial x} \right] - \left[P_1 \frac{\partial R_2}{\partial y} + R_2 \frac{\partial P_1}{\partial y} - P_2 \frac{\partial R_1}{\partial y} - R_1 \frac{\partial P_2}{\partial y} \right] \\
&\quad + \left[P_1 \frac{\partial Q_2}{\partial z} + Q_2 \frac{\partial P_1}{\partial z} - P_2 \frac{\partial Q_1}{\partial z} - Q_1 \frac{\partial P_2}{\partial z} \right] \\
&= \left[P_2 \left(\frac{\partial R_1}{\partial y} - \frac{\partial Q_1}{\partial z} \right) + Q_2 \left(\frac{\partial P_1}{\partial z} - \frac{\partial R_1}{\partial x} \right) + R_2 \left(\frac{\partial Q_1}{\partial x} - \frac{\partial P_1}{\partial y} \right) \right] \\
&\quad - \left[P_1 \left(\frac{\partial R_2}{\partial y} - \frac{\partial Q_2}{\partial z} \right) + Q_1 \left(\frac{\partial P_2}{\partial z} - \frac{\partial R_2}{\partial x} \right) + R_1 \left(\frac{\partial Q_2}{\partial x} - \frac{\partial P_2}{\partial y} \right) \right] \\
&= \mathbf{G} \cdot \operatorname{curl} \mathbf{F} - \mathbf{F} \cdot \operatorname{curl} \mathbf{G}
\end{aligned}$$

$$30. \operatorname{div}(\nabla f \times \nabla g) = \nabla g \cdot \operatorname{curl}(\nabla f) - \nabla f \cdot \operatorname{curl}(\nabla g) \quad [\text{by Exercise 29}] = 0 \quad [\text{by Theorem 3}]$$

$$\begin{aligned}
31. \operatorname{curl}(\operatorname{curl} \mathbf{F}) &= \nabla \times (\nabla \times \mathbf{F}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial R_1/\partial y - \partial Q_1/\partial z & \partial P_1/\partial z - \partial R_1/\partial x & \partial Q_1/\partial x - \partial P_1/\partial y \end{vmatrix} \\
&= \left(\frac{\partial^2 Q_1}{\partial y \partial x} - \frac{\partial^2 P_1}{\partial y^2} - \frac{\partial^2 P_1}{\partial z^2} + \frac{\partial^2 R_1}{\partial z \partial x} \right) \mathbf{i} + \left(\frac{\partial^2 R_1}{\partial z \partial y} - \frac{\partial^2 Q_1}{\partial z^2} - \frac{\partial^2 Q_1}{\partial x^2} + \frac{\partial^2 P_1}{\partial x \partial y} \right) \mathbf{j} \\
&\quad + \left(\frac{\partial^2 P_1}{\partial x \partial z} - \frac{\partial^2 R_1}{\partial x^2} - \frac{\partial^2 R_1}{\partial y^2} + \frac{\partial^2 Q_1}{\partial y \partial z} \right) \mathbf{k}
\end{aligned}$$

Now let's consider $\operatorname{grad}(\operatorname{div} \mathbf{F}) - \nabla^2 \mathbf{F}$ and compare with the above. (Note that $\nabla^2 \mathbf{F}$ is defined in the discussion following Example 16.5.5.)

$$\begin{aligned}
\operatorname{grad}(\operatorname{div} \mathbf{F}) - \nabla^2 \mathbf{F} &= \left[\left(\frac{\partial^2 P_1}{\partial x^2} + \frac{\partial^2 Q_1}{\partial x \partial y} + \frac{\partial^2 R_1}{\partial x \partial z} \right) \mathbf{i} + \left(\frac{\partial^2 P_1}{\partial y \partial x} + \frac{\partial^2 Q_1}{\partial y^2} + \frac{\partial^2 R_1}{\partial y \partial z} \right) \mathbf{j} + \left(\frac{\partial^2 P_1}{\partial z \partial x} + \frac{\partial^2 Q_1}{\partial z \partial y} + \frac{\partial^2 R_1}{\partial z^2} \right) \mathbf{k} \right] \\
&\quad - \left[\left(\frac{\partial^2 P_1}{\partial x^2} + \frac{\partial^2 P_1}{\partial y^2} + \frac{\partial^2 P_1}{\partial z^2} \right) \mathbf{i} + \left(\frac{\partial^2 Q_1}{\partial x^2} + \frac{\partial^2 Q_1}{\partial y^2} + \frac{\partial^2 Q_1}{\partial z^2} \right) \mathbf{j} \right. \\
&\quad \left. + \left(\frac{\partial^2 R_1}{\partial x^2} + \frac{\partial^2 R_1}{\partial y^2} + \frac{\partial^2 R_1}{\partial z^2} \right) \mathbf{k} \right] \\
&= \left(\frac{\partial^2 Q_1}{\partial x \partial y} + \frac{\partial^2 R_1}{\partial x \partial z} - \frac{\partial^2 P_1}{\partial y^2} - \frac{\partial^2 P_1}{\partial z^2} \right) \mathbf{i} + \left(\frac{\partial^2 P_1}{\partial y \partial x} + \frac{\partial^2 R_1}{\partial y \partial z} - \frac{\partial^2 Q_1}{\partial x^2} - \frac{\partial^2 Q_1}{\partial z^2} \right) \mathbf{j} \\
&\quad + \left(\frac{\partial^2 P_1}{\partial z \partial x} + \frac{\partial^2 Q_1}{\partial z \partial y} - \frac{\partial^2 R_1}{\partial x^2} - \frac{\partial^2 R_1}{\partial y^2} \right) \mathbf{k}
\end{aligned}$$

Then applying Clairaut's Theorem to reverse the order of differentiation in the second partial derivatives as needed and comparing, we have $\operatorname{curl} \operatorname{curl} \mathbf{F} = \operatorname{grad} \operatorname{div} \mathbf{F} - \nabla^2 \mathbf{F}$ as desired.

$$32. (a) \nabla \cdot \mathbf{r} = \left(\frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) \cdot (x \mathbf{i} + y \mathbf{j} + z \mathbf{k}) = 1 + 1 + 1 = 3$$

$$\begin{aligned}
(b) \nabla \cdot (r\mathbf{r}) &= \nabla \cdot \sqrt{x^2 + y^2 + z^2} (x \mathbf{i} + y \mathbf{j} + z \mathbf{k}) \\
&= \left(\frac{x^2}{\sqrt{x^2 + y^2 + z^2}} + \sqrt{x^2 + y^2 + z^2} \right) + \left(\frac{y^2}{\sqrt{x^2 + y^2 + z^2}} + \sqrt{x^2 + y^2 + z^2} \right) \\
&\quad + \left(\frac{z^2}{\sqrt{x^2 + y^2 + z^2}} + \sqrt{x^2 + y^2 + z^2} \right) \\
&= \frac{1}{\sqrt{x^2 + y^2 + z^2}} (4x^2 + 4y^2 + 4z^2) = 4 \sqrt{x^2 + y^2 + z^2} = 4r
\end{aligned}$$

Another method:

$$\text{By Exercise 27, } \nabla \cdot (r\mathbf{r}) = \operatorname{div}(r\mathbf{r}) = r \operatorname{div} \mathbf{r} + \mathbf{r} \cdot \nabla r = 3r + \mathbf{r} \cdot \frac{\mathbf{r}}{r} \quad [\text{see Exercise 33(a) below}] = 4r.$$

$$\begin{aligned}
\text{(c) } \nabla^2 r^3 &= \nabla^2 (x^2 + y^2 + z^2)^{3/2} \\
&= \frac{\partial}{\partial x} \left[\frac{3}{2} (x^2 + y^2 + z^2)^{1/2} (2x) \right] + \frac{\partial}{\partial y} \left[\frac{3}{2} (x^2 + y^2 + z^2)^{1/2} (2y) \right] + \frac{\partial}{\partial z} \left[\frac{3}{2} (x^2 + y^2 + z^2)^{1/2} (2z) \right] \\
&= 3 \left[\frac{1}{2} (x^2 + y^2 + z^2)^{-1/2} (2x)(x) + (x^2 + y^2 + z^2)^{1/2} \right] \\
&\quad + 3 \left[\frac{1}{2} (x^2 + y^2 + z^2)^{-1/2} (2y)(y) + (x^2 + y^2 + z^2)^{1/2} \right] \\
&\quad + 3 \left[\frac{1}{2} (x^2 + y^2 + z^2)^{-1/2} (2z)(z) + (x^2 + y^2 + z^2)^{1/2} \right] \\
&= 3(x^2 + y^2 + z^2)^{-1/2} (4x^2 + 4y^2 + 4z^2) = 12(x^2 + y^2 + z^2)^{1/2} = 12r
\end{aligned}$$

Another method: $\frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{3/2} = 3x \sqrt{x^2 + y^2 + z^2} \Rightarrow \nabla r^3 = 3r(x \mathbf{i} + y \mathbf{j} + z \mathbf{k}) = 3r \mathbf{r}$,

so $\nabla^2 r^3 = \nabla \cdot \nabla r^3 = \nabla \cdot (3r \mathbf{r}) = 3(4r) = 12r$ by part (b).

33. (a) $\nabla r = \nabla \sqrt{x^2 + y^2 + z^2} = \frac{x}{\sqrt{x^2 + y^2 + z^2}} \mathbf{i} + \frac{y}{\sqrt{x^2 + y^2 + z^2}} \mathbf{j} + \frac{z}{\sqrt{x^2 + y^2 + z^2}} \mathbf{k} = \frac{x \mathbf{i} + y \mathbf{j} + z \mathbf{k}}{\sqrt{x^2 + y^2 + z^2}} = \frac{\mathbf{r}}{r}$

(b) $\nabla \times \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \left[\frac{\partial}{\partial y} (z) - \frac{\partial}{\partial z} (y) \right] \mathbf{i} + \left[\frac{\partial}{\partial z} (x) - \frac{\partial}{\partial x} (z) \right] \mathbf{j} + \left[\frac{\partial}{\partial x} (y) - \frac{\partial}{\partial y} (x) \right] \mathbf{k} = \mathbf{0}$

(c) $\nabla \left(\frac{1}{r} \right) = \nabla \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right)$

$$\begin{aligned}
&= -\frac{1}{2\sqrt{x^2 + y^2 + z^2}} (2x) \mathbf{i} - \frac{1}{2\sqrt{x^2 + y^2 + z^2}} (2y) \mathbf{j} - \frac{1}{2\sqrt{x^2 + y^2 + z^2}} (2z) \mathbf{k} \\
&= -\frac{x \mathbf{i} + y \mathbf{j} + z \mathbf{k}}{(x^2 + y^2 + z^2)^{3/2}} = -\frac{\mathbf{r}}{r^3}
\end{aligned}$$

(d) $\nabla \ln r = \nabla \ln(x^2 + y^2 + z^2)^{1/2} = \frac{1}{2} \nabla \ln(x^2 + y^2 + z^2)$

$$= \frac{x}{x^2 + y^2 + z^2} \mathbf{i} + \frac{y}{x^2 + y^2 + z^2} \mathbf{j} + \frac{z}{x^2 + y^2 + z^2} \mathbf{k} = \frac{x \mathbf{i} + y \mathbf{j} + z \mathbf{k}}{x^2 + y^2 + z^2} = \frac{\mathbf{r}}{r^2}$$

34. $\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k} \Rightarrow r = |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$, so

$$\mathbf{F} = \frac{\mathbf{r}}{r^p} = \frac{x}{(x^2 + y^2 + z^2)^{p/2}} \mathbf{i} + \frac{y}{(x^2 + y^2 + z^2)^{p/2}} \mathbf{j} + \frac{z}{(x^2 + y^2 + z^2)^{p/2}} \mathbf{k}$$

Then $\frac{\partial}{\partial x} \frac{x}{(x^2 + y^2 + z^2)^{p/2}} = \frac{(x^2 + y^2 + z^2) - px^2}{(x^2 + y^2 + z^2)^{1+p/2}} = \frac{r^2 - px^2}{r^{p+2}}$. Similarly,

$\frac{\partial}{\partial y} \frac{y}{(x^2 + y^2 + z^2)^{p/2}} = \frac{r^2 - py^2}{r^{p+2}}$ and $\frac{\partial}{\partial z} \frac{z}{(x^2 + y^2 + z^2)^{p/2}} = \frac{r^2 - pz^2}{r^{p+2}}$. Thus

$$\begin{aligned}
\operatorname{div} \mathbf{F} &= \nabla \cdot \mathbf{F} = \frac{r^2 - px^2}{r^{p+2}} + \frac{r^2 - py^2}{r^{p+2}} + \frac{r^2 - pz^2}{r^{p+2}} = \frac{3r^2 - px^2 - py^2 - pz^2}{r^{p+2}} \\
&= \frac{3r^2 - p(x^2 + y^2 + z^2)}{r^{p+2}} = \frac{3r^2 - pr^2}{r^{p+2}} = \frac{3-p}{r^p}
\end{aligned}$$

Consequently, if $p = 3$ we have $\operatorname{div} \mathbf{F} = 0$.

35. By (13), $\oint_C f(\nabla g) \cdot \mathbf{n} \, ds = \iint_D \operatorname{div}(f\nabla g) \, dA = \iint_D [f \operatorname{div}(\nabla g) + \nabla g \cdot \nabla f] \, dA$ by Exercise 27. But $\operatorname{div}(\nabla g) = \nabla^2 g$.

$$\text{Hence } \iint_D f \nabla^2 g \, dA = \oint_C f(\nabla g) \cdot \mathbf{n} \, ds - \iint_D \nabla g \cdot \nabla f \, dA.$$

36. By Exercise 35, $\iint_D f \nabla^2 g \, dA = \oint_C f(\nabla g) \cdot \mathbf{n} \, ds - \iint_D \nabla g \cdot \nabla f \, dA$ and

$$\iint_D g \nabla^2 f \, dA = \oint_C g(\nabla f) \cdot \mathbf{n} \, ds - \iint_D \nabla f \cdot \nabla g \, dA. \text{ Hence}$$

$$\iint_D (f \nabla^2 g - g \nabla^2 f) \, dA = \oint_C [f(\nabla g) \cdot \mathbf{n} - g(\nabla f) \cdot \mathbf{n}] \, ds + \iint_D (\nabla f \cdot \nabla g - \nabla g \cdot \nabla f) \, dA = \oint_C [f \nabla g - g \nabla f] \cdot \mathbf{n} \, ds.$$

37. Let $f(x, y) = 1$. Then $\nabla f = \mathbf{0}$ and Green's first identity (see Exercise 35) says

$$\iint_D \nabla^2 g \, dA = \oint_C (\nabla g) \cdot \mathbf{n} \, ds - \iint_D \mathbf{0} \cdot \nabla g \, dA \Rightarrow \iint_D \nabla^2 g \, dA = \oint_C \nabla g \cdot \mathbf{n} \, ds. \text{ But } g \text{ is harmonic on } D, \text{ so}$$

$$\nabla^2 g = 0 \Rightarrow \oint_C \nabla g \cdot \mathbf{n} \, ds = 0 \text{ and } \oint_C D_{\mathbf{n}} g \, ds = \oint_C (\nabla g \cdot \mathbf{n}) \, ds = 0.$$

38. Let $g = f$. Then Green's first identity (see Exercise 35) says $\iint_D f \nabla^2 f \, dA = \oint_C (f)(\nabla f) \cdot \mathbf{n} \, ds - \iint_D \nabla f \cdot \nabla f \, dA$.

$$\text{But } f \text{ is harmonic, so } \nabla^2 f = 0, \text{ and } \nabla f \cdot \nabla f = |\nabla f|^2, \text{ so we have } 0 = \oint_C (f)(\nabla f) \cdot \mathbf{n} \, ds - \iint_D |\nabla f|^2 \, dA \Rightarrow$$

$$\iint_D |\nabla f|^2 \, dA = \oint_C (f)(\nabla f) \cdot \mathbf{n} \, ds = 0 \text{ since } f(x, y) = 0 \text{ on } C.$$

39. (a) We know that $\omega = v/d$ (where v is the tangential speed), and from the diagram $\sin \theta = d/r$ (where $r = |\mathbf{r}|$) $\Rightarrow$

$$v = d\omega = (\sin \theta)r\omega = |\mathbf{w} \times \mathbf{r}| \text{ (by 12.4.9). But } \mathbf{v} \text{ is perpendicular to both } \mathbf{w} \text{ and } \mathbf{r}, \text{ so that } \mathbf{v} = \mathbf{w} \times \mathbf{r}.$$

$$(b) \text{ From part (a), } \mathbf{v} = \mathbf{w} \times \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & \omega \\ x & y & z \end{vmatrix} = (0 \cdot z - \omega y)\mathbf{i} - (0 \cdot z - \omega x)\mathbf{j} + (0 \cdot y - 0 \cdot x)\mathbf{k} = -\omega y\mathbf{i} + \omega x\mathbf{j}.$$

$$(c) \operatorname{curl} \mathbf{v} = \nabla \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -\omega y & \omega x & 0 \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(\omega x) \right] \mathbf{i} + \left[\frac{\partial}{\partial z}(-\omega y) - \frac{\partial}{\partial x}(0) \right] \mathbf{j} + \left[\frac{\partial}{\partial x}(\omega x) - \frac{\partial}{\partial y}(-\omega y) \right] \mathbf{k}$$

$$= [\omega - (-\omega)] \mathbf{k} = 2\omega \mathbf{k} = 2\mathbf{w}$$

40. Let $\mathbf{H} = \langle h_1, h_2, h_3 \rangle$ and $\mathbf{E} = \langle E_1, E_2, E_3 \rangle$.

$$(a) \nabla \times (\nabla \times \mathbf{E}) = \nabla \times (\operatorname{curl} \mathbf{E}) = \nabla \times \left(-\frac{1}{c} \frac{\partial \mathbf{H}}{\partial t} \right) = -\frac{1}{c} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial h_1/\partial t & \partial h_2/\partial t & \partial h_3/\partial t \end{vmatrix}$$

$$= -\frac{1}{c} \left[\left(\frac{\partial^2 h_3}{\partial y \partial t} - \frac{\partial^2 h_2}{\partial z \partial t} \right) \mathbf{i} + \left(\frac{\partial^2 h_1}{\partial z \partial t} - \frac{\partial^2 h_3}{\partial x \partial t} \right) \mathbf{j} + \left(\frac{\partial^2 h_2}{\partial x \partial t} - \frac{\partial^2 h_1}{\partial y \partial t} \right) \mathbf{k} \right]$$

$$= -\frac{1}{c} \frac{\partial}{\partial t} \left[\left(\frac{\partial h_3}{\partial y} - \frac{\partial h_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial h_1}{\partial z} - \frac{\partial h_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial h_2}{\partial x} - \frac{\partial h_1}{\partial y} \right) \mathbf{k} \right] \quad \begin{array}{l} \text{[assuming that the partial derivatives} \\ \text{are continuous so that the order of} \\ \text{differentiation does not matter]} \end{array}$$

$$= -\frac{1}{c} \frac{\partial}{\partial t} \operatorname{curl} \mathbf{H} = -\frac{1}{c} \frac{\partial}{\partial t} \left(\frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} \right) = -\frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\begin{aligned}
\text{(b) } \nabla \times (\nabla \times \mathbf{H}) &= \nabla \times (\text{curl } \mathbf{H}) = \nabla \times \left(\frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} \right) = \frac{1}{c} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial E_1/\partial t & \partial E_2/\partial t & \partial E_3/\partial t \end{vmatrix} \\
&= \frac{1}{c} \left[\left(\frac{\partial^2 E_3}{\partial y \partial t} - \frac{\partial^2 E_2}{\partial z \partial t} \right) \mathbf{i} + \left(\frac{\partial^2 E_1}{\partial z \partial t} - \frac{\partial^2 E_3}{\partial x \partial t} \right) \mathbf{j} + \left(\frac{\partial^2 E_2}{\partial x \partial t} - \frac{\partial^2 E_1}{\partial y \partial t} \right) \mathbf{k} \right] \\
&= \frac{1}{c} \frac{\partial}{\partial t} \left[\left(\frac{\partial E_3}{\partial y} - \frac{\partial E_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial E_1}{\partial z} - \frac{\partial E_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial E_2}{\partial x} - \frac{\partial E_1}{\partial y} \right) \mathbf{k} \right] \quad \text{[assuming that the partial derivatives} \\
&\quad \text{are continuous so that the order of} \\
&\quad \text{differentiation does not matter]} \\
&= \frac{1}{c} \frac{\partial}{\partial t} \text{curl } \mathbf{E} = \frac{1}{c} \frac{\partial}{\partial t} \left(-\frac{1}{c} \frac{\partial \mathbf{H}}{\partial t} \right) = -\frac{1}{c^2} \frac{\partial^2 \mathbf{H}}{\partial t^2}
\end{aligned}$$

(c) Using Exercise 31, we have that $\text{curl curl } \mathbf{E} = \text{grad div } \mathbf{E} - \nabla^2 \mathbf{E} \Rightarrow$

$$\nabla^2 \mathbf{E} = \text{grad div } \mathbf{E} - \text{curl curl } \mathbf{E} = \text{grad } 0 + \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \text{[from part (a)]} = \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

(d) As in part (c), $\nabla^2 \mathbf{H} = \text{grad div } \mathbf{H} - \text{curl curl } \mathbf{H} = \text{grad } 0 + \frac{1}{c^2} \frac{\partial^2 \mathbf{H}}{\partial t^2}$ [using part (b)] $= \frac{1}{c^2} \frac{\partial^2 \mathbf{H}}{\partial t^2}$.

41. For any continuous function f on $\mathbb{R}^3$, define a vector field $\mathbf{G}(x, y, z) = \langle g(x, y, z), 0, 0 \rangle$ where $g(x, y, z) = \int_0^x f(t, y, z) dt$.

Then $\text{div } \mathbf{G} = \frac{\partial}{\partial x} (g(x, y, z)) + \frac{\partial}{\partial y} (0) + \frac{\partial}{\partial z} (0) = \frac{\partial}{\partial x} \int_0^x f(t, y, z) dt = f(x, y, z)$ by the Fundamental Theorem of

Calculus. Thus every continuous function f on $\mathbb{R}^3$ is the divergence of some vector field.

16.6 Parametric Surfaces and Their Areas

1. $P(4, -5, 1)$ lies on the parametric surface $\mathbf{r}(u, v) = \langle u + v, u - 2v, 3 + u - v \rangle$ if and only if there are values for u and v where $u + v = 4$, $u - 2v = -5$, and $3 + u - v = 1$. From the first equation we have $u = 4 - v$ and substituting into the second equation gives $4 - v - 2v = -5 \Leftrightarrow v = 3$. Then $u = 1$, and these values satisfy the third equation, so P does lie on the surface.

$Q(0, 4, 6)$ lies on $\mathbf{r}(u, v)$ if and only if $u + v = 0$, $u - 2v = 4$, and $3 + u - v = 6$, but solving the first two equations simultaneously gives $u = \frac{4}{3}$, $v = -\frac{4}{3}$ and these values do not satisfy the third equation, so Q does not lie on the surface.

2. $P(1, 2, 1)$ lies on the parametric surface $\mathbf{r}(u, v) = \langle 1 + u - v, u + v^2, u^2 - v^2 \rangle$ if and only if there are values for u and v where $1 + u - v = 1$, $u + v^2 = 2$, and $u^2 - v^2 = 1$. From the first equation we have $u = v$ and substituting into the third equation gives $0 = 1$, an impossibility, so P does not lie on the surface.

$Q(2, 3, 3)$ lies on $\mathbf{r}(u, v)$ if and only if $1 + u - v = 2$, $u + v^2 = 3$, and $u^2 - v^2 = 3$. From the first equation we have $u = v + 1$ and substituting into the second equation gives $v + 1 + v^2 = 3 \Leftrightarrow v^2 + v - 2 = 0 \Leftrightarrow (v + 2)(v - 1) = 0$, so $v = -2 \Rightarrow u = -1$ or $v = 1 \Rightarrow u = 2$. The third equation is satisfied by $u = 2$, $v = 1$ so Q does lie on the surface.

3. $\mathbf{r}(u, v) = (u + v)\mathbf{i} + (3 - v)\mathbf{j} + (1 + 4u + 5v)\mathbf{k} = \langle 0, 3, 1 \rangle + u\langle 1, 0, 4 \rangle + v\langle 1, -1, 5 \rangle$. From Example 3, we recognize this as a vector equation of a plane through the point $(0, 3, 1)$ and containing vectors $\mathbf{a} = \langle 1, 0, 4 \rangle$ and $\mathbf{b} = \langle 1, -1, 5 \rangle$. If we

wish to find a more conventional equation for the plane, a normal vector to the plane is $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 4 \\ 1 & -1 & 5 \end{vmatrix} = 4\mathbf{i} - \mathbf{j} - \mathbf{k}$

and an equation of the plane is $4(x - 0) - (y - 3) - (z - 1) = 0$ or $4x - y - z = -4$.

4. $\mathbf{r}(u, v) = u^2\mathbf{i} + u\cos v\mathbf{j} + u\sin v\mathbf{k}$, so the corresponding parametric equations for the surface are $x = u^2$, $y = u\cos v$, $z = u\sin v$. For any point (x, y, z) on the surface, we have $y^2 + z^2 = u^2\cos^2 v + u^2\sin^2 v = u^2 = x$. Since no restrictions are placed on the parameters, the surface is $x = y^2 + z^2$, which we recognize as a circular paraboloid whose axis is the x -axis.
5. $\mathbf{r}(s, t) = \langle s\cos t, s\sin t, s \rangle$, so the corresponding parametric equations for the surface are $x = s\cos t$, $y = s\sin t$, $z = s$. For any point (x, y, z) on the surface, we have $x^2 + y^2 = s^2\cos^2 t + s^2\sin^2 t = s^2 = z^2$. Since no restrictions are placed on the parameters, the surface is $z^2 = x^2 + y^2$, which we recognize as a circular cone with axis the z -axis.
6. $\mathbf{r}(s, t) = \langle 3\cos t, s, \sin t \rangle$, so the corresponding parametric equations for the surface are $x = 3\cos t$, $y = s$, $z = \sin t$. For any point (x, y, z) on the surface, we have $(x/3)^2 + z^2 = \cos^2 t + \sin^2 t = 1$, so vertical cross-sections parallel to the xz -plane are all identical ellipses. Since $y = s$ and $-1 \leq s \leq 1$, the surface is the portion of the elliptic cylinder $\frac{1}{9}x^2 + z^2 = 1$ corresponding to $-1 \leq y \leq 1$.
7. $\mathbf{r}(u, v) = \langle u^2, v^2, u + v \rangle$, $-1 \leq u \leq 1$, $-1 \leq v \leq 1$.

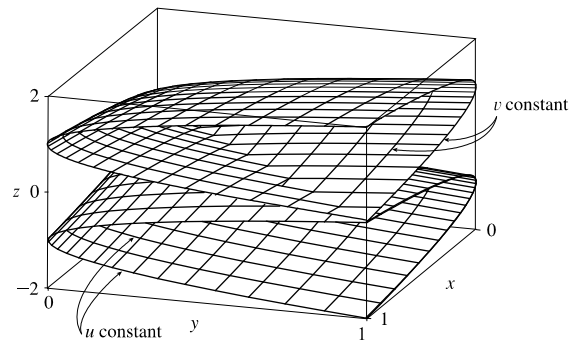
The surface has parametric equations $x = u^2$, $y = v^2$, $z = u + v$, $-1 \leq u \leq 1$, $-1 \leq v \leq 1$.

In Maple, the surface can be graphed by entering

```
plot3d([u^2, v^2, u+v], u=-1..1, v=-1..1);
```

In Mathematica we use the `ParametricPlot3D` command.

If we keep u constant at u_0 , $x = u_0^2$, a constant, so the corresponding grid curves must be the curves parallel to the yz -plane. If v is constant, we have $y = v_0^2$, a constant, so these grid curves are the curves parallel to the xz -plane.



8. $\mathbf{r}(u, v) = \langle u, v^3, -v \rangle$, $-2 \leq u \leq 2$, $-2 \leq v \leq 2$.

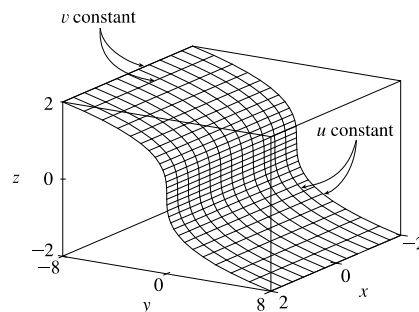
The surface has parametric equations $x = u$, $y = v^3$, $z = -v$,

$-2 \leq u \leq 2$, $-2 \leq v \leq 2$. If $u = u_0$ is constant,

$x = u_0 = \text{constant}$, so the corresponding grid curves are the curves

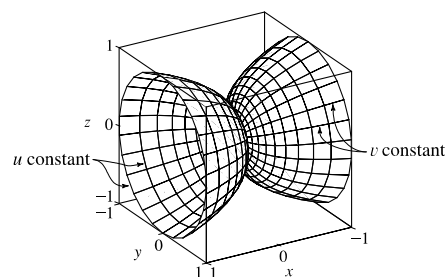
parallel to the yz -plane. If $v = v_0$ is constant, $y = v_0^3 = \text{constant}$, so

the corresponding grid curves are the curves parallel to the xz -plane.



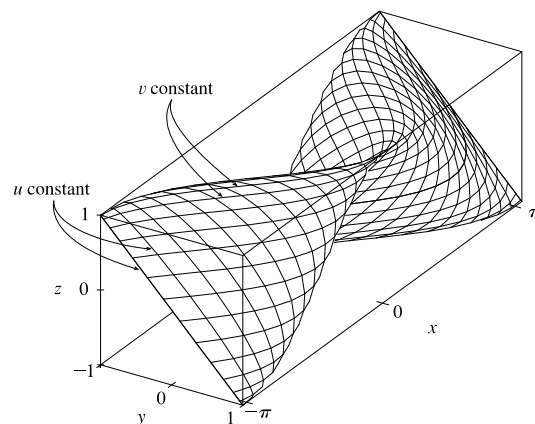
9. $\mathbf{r}(u, v) = \langle u^3, u \sin v, u \cos v \rangle$, $-1 \leq u \leq 1$, $0 \leq v \leq 2\pi$

The surface has parametric equations $x = u^3$, $y = u \sin v$, $z = u \cos v$, $-1 \leq u \leq 1$, $0 \leq v \leq 2\pi$. Note that if $u = u_0$ is constant then $x = u_0^3$ is constant and $y = u_0 \sin v$, $z = u_0 \cos v$ describe a circle in y, z of radius $|u_0|$, so the corresponding grid curves are circles parallel to the yz -plane. If $v = v_0$, a constant, the parametric equations become $x = u^3$, $y = u \sin v_0$, $z = u \cos v_0$. Then $y = (\tan v_0)z$, so these are the grid curves we see that lie in planes $y = kz$ that pass through the x -axis.



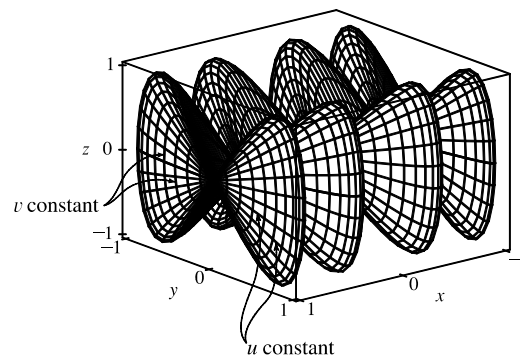
10. $\mathbf{r}(u, v) = \langle u, \sin(u + v), \sin v \rangle$, $-\pi \leq u \leq \pi$, $-\pi \leq v \leq \pi$.

The surface has parametric equations $x = u$, $y = \sin(u + v)$, $z = \sin v$, $-\pi \leq u \leq \pi$, $-\pi \leq v \leq \pi$. If $u = u_0$ is constant, $x = u_0 = \text{constant}$, so the corresponding grid curves are the curves parallel to the yz -plane. If $v = v_0$ is constant, $z = \sin v_0 = \text{constant}$, so the corresponding grid curves are the curves parallel to the xy -plane.



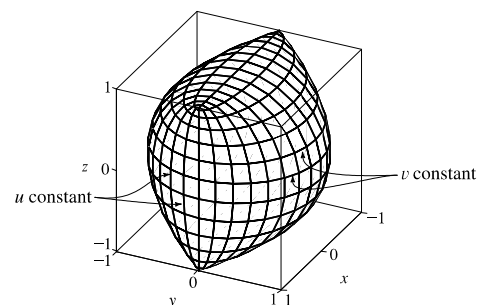
11. $x = \sin v$, $y = \cos u \sin 4v$, $z = \sin 2u \sin 4v$, $0 \leq u \leq 2\pi$, $-\frac{\pi}{2} \leq v \leq \frac{\pi}{2}$.

Note that if $v = v_0$ is constant, then $x = \sin v_0$ is constant, so the corresponding grid curves must be parallel to the yz -plane. These are the vertically oriented grid curves we see, each shaped like a “figure-eight.” When $u = u_0$ is held constant, the parametric equations become $x = \sin v$, $y = \cos u_0 \sin 4v$, $z = \sin 2u_0 \sin 4v$. Since z is a constant multiple of y , the corresponding grid curves are the curves contained in planes $z = ky$ that pass through the x -axis.



12. $x = \cos u$, $y = \sin u \sin v$, $z = \cos v$, $0 \leq u \leq 2\pi$, $0 \leq v \leq 2\pi$.

If $u = u_0$ is constant, then $x = \cos u_0 = \text{constant}$, so the corresponding grid curves are the curves parallel to the yz -plane. If $v = v_0$ is constant, then $z = \cos v_0 = \text{constant}$, so the corresponding grid curves are the curves parallel to the xy -plane.

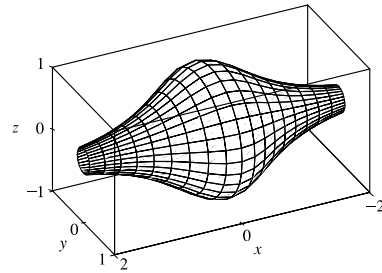


13. $\mathbf{r}(u, v) = u \cos v \mathbf{i} + u \sin v \mathbf{j} + v \mathbf{k}$. The parametric equations for the surface are $x = u \cos v$, $y = u \sin v$, $z = v$. We look at the grid curves first; if we fix v , then x and y parametrize a straight line in the plane $z = v$ which intersects the z -axis. If u is held constant, the projection onto the xy -plane is circular; with $z = v$, each grid curve is a helix. The surface is a spiraling ramp, graph IV.
14. $\mathbf{r}(u, v) = uv^2 \mathbf{i} + u^2v \mathbf{j} + (u^2 - v^2) \mathbf{k}$. The parametric equations for the surface are $x = uv^2$, $y = u^2v$, $z = u^2 - v^2$. If $u = u_0$ is held constant, then $x = u_0v^2$, $y = u_0^2v$ so $x = u_0(y/u_0^2)^2 = (1/u_0^3)y^2$, and $z = u_0^2 - v^2 = u_0^2 - (1/u_0)x$. Thus each grid curve corresponding to $u = u_0$ lies in the plane $z = u_0^2 - (1/u_0)x$ and its projection onto the xy -plane is a parabola $x = ky^2$ with axis the x -axis. Similarly, if $v = v_0$ is held constant, then $x = uv_0^2$, $y = u^2v_0 \Rightarrow y = (x/v_0^2)^2v_0 = (1/v_0^3)x^2$, and $z = u^2 - v_0^2 = (1/v_0)y - v_0^2$. Each grid curve lies in the plane $z = (1/v_0)y - v_0^2$ and its projection onto the xy -plane is a parabola $y = kx^2$ with axis the y -axis. The surface is graph VI.
15. $\mathbf{r}(u, v) = (u^3 - u) \mathbf{i} + v^2 \mathbf{j} + u^2 \mathbf{k}$. The parametric equations for the surface are $x = u^3 - u$, $y = v^2$, $z = u^2$. If we fix u then x and z are constant so each corresponding grid curve is contained in a line parallel to the y -axis. (Since $y = v^2 \geq 0$, the grid curves are half-lines.) If v is held constant, then $y = v^2 = \text{constant}$, so each grid curve is contained in a plane parallel to the xz -plane. Since x and z are functions of u only, the grid curves all have the same shape. The surface is the cylinder shown in graph I.
16. $x = (1 - u)(3 + \cos v) \cos 4\pi u$, $y = (1 - u)(3 + \cos v) \sin 4\pi u$, $z = 3u + (1 - u) \sin v$. These equations correspond to graph V: when $u = 0$, then $x = 3 + \cos v$, $y = 0$, and $z = \sin v$, which are equations of a circle with radius 1 in the xz -plane centered at $(3, 0, 0)$. When $u = \frac{1}{2}$, then $x = \frac{3}{2} + \frac{1}{2} \cos v$, $y = 0$, and $z = \frac{3}{2} + \frac{1}{2} \sin v$, which are equations of a circle with radius $\frac{1}{2}$ in the xz -plane centered at $(\frac{3}{2}, 0, \frac{3}{2})$. When $u = 1$, then $x = y = 0$ and $z = 3$, giving the topmost point shown in the graph. This suggests that the grid curves with u constant are the vertically oriented circles visible on the surface. The spiraling grid curves correspond to keeping v constant.
17. $x = \cos^3 u \cos^3 v$, $y = \sin^3 u \cos^3 v$, $z = \sin^3 v$. If $v = v_0$ is held constant then $z = \sin^3 v_0$ is constant, so the corresponding grid curve lies in a horizontal plane. Several of the graphs exhibit horizontal grid curves, but the curves for this surface are neither ellipses nor straight lines, so graph III is the only possibility. (In fact, the horizontal grid curves here are members of the family $x = a \cos^3 u$, $y = a \sin^3 u$ and are called astroids.) The vertical grid curves we see on the surface correspond to $u = u_0$ held constant, as then we have $x = \cos^3 u_0 \cos^3 v$, $y = \sin^3 u_0 \cos^3 v$ so the corresponding grid curve lies in the vertical plane $y = (\tan^3 u_0)x$ through the z -axis.
18. $x = \sin u$, $y = \cos u \sin v$, $z = \sin v$. If $v = v_0$ is fixed, then $z = \sin v_0$ is constant, and $x = \sin u$, $y = (\sin v_0) \cos u$ describe an ellipse that is contained in the horizontal plane $z = \sin v_0$. If $u = u_0$ is fixed, then $x = \sin u_0$ is constant, and $y = (\cos u_0) \sin v$, $z = \sin v \Rightarrow y = (\cos u_0)z$, so the grid curves are portions of lines through the x -axis contained in the plane $x = \sin u_0$ (parallel to the yz -plane). The surface is graph II.

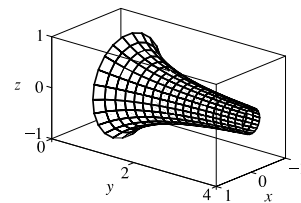
19. From Example 3, parametric equations for the plane through the point $(0, 0, 0)$ that contains the vectors $\mathbf{a} = \langle 1, -1, 0 \rangle$ and $\mathbf{b} = \langle 0, 1, -1 \rangle$ are $x = 0 + u(1) + v(0) = u$, $y = 0 + u(-1) + v(1) = v - u$, $z = 0 + u(0) + v(-1) = -v$.
20. From Example 3, parametric equations for the plane through the point $(0, -1, 5)$ that contains the vectors $\mathbf{a} = \langle 2, 1, 4 \rangle$ and $\mathbf{b} = \langle -3, 2, 5 \rangle$ are $x = 0 + u(2) + v(-3) = 2u - 3v$, $y = -1 + u(1) + v(2) = -1 + u + 2v$, $z = 5 + u(4) + v(5) = 5 + 4u + 5v$.
21. Solving the equation $4x^2 - 4y^2 - z^2 = 4$ for x gives $x^2 = 1 + y^2 + \frac{1}{4}z^2 \Rightarrow x = \sqrt{1 + y^2 + \frac{1}{4}z^2}$. (We choose the positive root since we want the part of the hyperboloid that corresponds to $x \geq 0$.) If we let y and z be the parameters, parametric equations are $y = y$, $z = z$, $x = \sqrt{1 + y^2 + \frac{1}{4}z^2}$.
22. Solving the equation $x^2 + 2y^2 + 3z^2 = 1$ for y gives $y^2 = \frac{1}{2}(1 - x^2 - 3z^2) \Rightarrow y = -\sqrt{\frac{1}{2}(1 - x^2 - 3z^2)}$ (since we want the part of the ellipsoid that corresponds to $y \leq 0$). If we let x and z be the parameters, parametric equations are $x = x$, $z = z$, $y = -\sqrt{\frac{1}{2}(1 - x^2 - 3z^2)}$.
- Alternate solution:* The equation can be rewritten as $x^2 + \frac{y^2}{(1/\sqrt{2})^2} + \frac{z^2}{(1/\sqrt{3})^2} = 1$, and if we let $x = u \cos v$ and $z = \frac{1}{\sqrt{3}} u \sin v$, then $y = -\sqrt{\frac{1}{2}(1 - x^2 - 3z^2)} = -\sqrt{\frac{1}{2}(1 - u^2 \cos^2 v - u^2 \sin^2 v)} = -\sqrt{\frac{1}{2}(1 - u^2)}$, where $0 \leq u \leq 1$ and $0 \leq v \leq 2\pi$.
- Second alternate solution:* We can adapt the formulas for converting from spherical to rectangular coordinates as follows. We let $x = \sin \phi \cos \theta$, $y = \frac{1}{\sqrt{2}} \sin \phi \sin \theta$, $z = \frac{1}{\sqrt{3}} \cos \phi$; the surface is generated for $0 \leq \phi \leq \pi$, $\pi \leq \theta \leq 2\pi$.
23. Since the cone $z = \sqrt{x^2 + y^2}$ intersects the sphere $x^2 + y^2 + z^2 = 4$ in the circle $x^2 + y^2 = 2$, $z = \sqrt{2}$ and we want the portion of the sphere above this, we can parametrize the surface as $x = x$, $y = y$, $z = \sqrt{4 - x^2 - y^2}$ where $x^2 + y^2 \leq 2$.
- Alternate solution:* Using spherical coordinates, $x = 2 \sin \phi \cos \theta$, $y = 2 \sin \phi \sin \theta$, $z = 2 \cos \phi$ where $0 \leq \phi \leq \frac{\pi}{4}$ and $0 \leq \theta \leq 2\pi$.
24. We can parametrize the cylinder as $x = 3 \cos \theta$, $y = y$, $z = 3 \sin \theta$. To restrict the surface to that portion above the xy -plane and between the planes $y = -4$ and $y = 4$ we require $0 \leq \theta \leq \pi$, $-4 \leq y \leq 4$.
25. In spherical coordinates, parametric equations are $x = 6 \sin \phi \cos \theta$, $y = 6 \sin \phi \sin \theta$, $z = 6 \cos \phi$. The intersection of the sphere with the plane $z = 3\sqrt{3}$ corresponds to $z = 6 \cos \phi = 3\sqrt{3} \Rightarrow \cos \phi = \frac{\sqrt{3}}{2} \Rightarrow \phi = \frac{\pi}{6}$, and the plane $z = 0$ (the xy -plane) corresponds to $\phi = \frac{\pi}{2}$. Thus the surface is described by $\frac{\pi}{6} \leq \phi \leq \frac{\pi}{2}$, $0 \leq \theta \leq 2\pi$.

26. Using x and y as the parameters, $x = x$, $y = y$, $z = x + 3$ where $0 \leq x^2 + y^2 \leq 1$. Also, since the plane intersects the cylinder in an ellipse, the surface is a planar ellipse in the plane $z = x + 3$. Thus, parametrizing with respect to s and θ , we have $x = s \cos \theta$, $y = s \sin \theta$, $z = 3 + s \cos \theta$ where $0 \leq s \leq 1$ and $0 \leq \theta \leq 2\pi$.
27. The surface appears to be a portion of a circular cylinder of radius 3 with axis the x -axis. An equation of the cylinder is $y^2 + z^2 = 9$, and we can impose the restrictions $0 \leq x \leq 5$, $y \geq 0$ to obtain the portion shown. To graph the surface on a CAS, we can use parametric equations $x = u$, $y = 3 \cos v$, $z = 3 \sin v$ with the parameter domain $0 \leq u \leq 5$, $\frac{\pi}{2} \leq v \leq \frac{3\pi}{2}$. Alternatively, we can regard x and z as parameters. Then parametric equations are $x = x$, $z = z$, $y = -\sqrt{9 - z^2}$, where $0 \leq x \leq 5$ and $-3 \leq z \leq 3$.
28. The surface appears to be a portion of a sphere of radius 1 centered at the origin. In spherical coordinates, the sphere has equation $\rho = 1$, and imposing the restrictions $\frac{\pi}{2} \leq \theta \leq 2\pi$, $\frac{\pi}{4} \leq \phi \leq \pi$ will give the portion of the sphere shown. Thus, to graph the surface on a CAS we can either use spherical coordinates with the stated restrictions, or we can use parametric equations: $x = \sin \phi \cos \theta$, $y = \sin \phi \sin \theta$, $z = \cos \phi$, $\frac{\pi}{2} \leq \theta \leq 2\pi$, $\frac{\pi}{4} \leq \phi \leq \pi$.
29. Using Equations 3, we have the parametrization $x = x$,

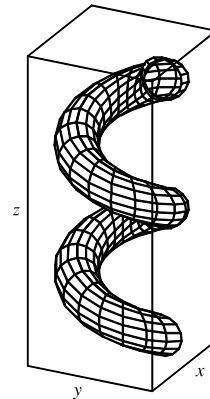
$$y = \frac{1}{1+x^2} \cos \theta, \quad z = \frac{1}{1+x^2} \sin \theta, \quad -2 \leq x \leq 2, \quad 0 \leq \theta \leq 2\pi.$$



30. Letting θ be the angle of rotation about the y -axis (adapting Equations 3), we have the parametrization $x = (1/y) \cos \theta$, $y = y$, $z = (1/y) \sin \theta$, $y \geq 1$, $0 \leq \theta \leq 2\pi$.

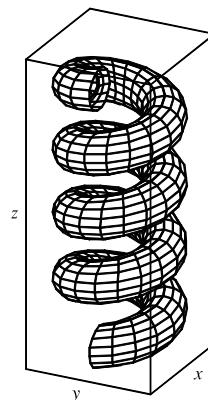


31. (a) Replacing $\cos u$ by $\sin u$ and $\sin u$ by $\cos u$ gives parametric equations $x = (2 + \sin v) \sin u$, $y = (2 + \sin v) \cos u$, $z = u + \cos v$. From the graph, it appears that the direction of the spiral is reversed. We can verify this observation by noting that the projection of the spiral grid curves onto the xy -plane, given by $x = (2 + \sin v) \sin u$, $y = (2 + \sin v) \cos u$, $z = 0$, draws a circle in the clockwise direction for each value of v . The original equations, on the other hand, give circular projections drawn in the counterclockwise direction. The equation for z is identical in both surfaces, so as z increases, these grid curves spiral up in opposite directions for the two surfaces.

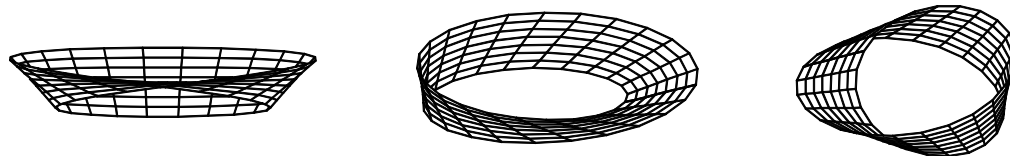


(b) Replacing $\cos u$ by $\cos 2u$ and $\sin u$ by $\sin 2u$ gives parametric equations

$x = (2 + \sin v) \cos 2u$, $y = (2 + \sin v) \sin 2u$, $z = u + \cos v$. From the graph, it appears that the number of coils in the surface doubles within the same parametric domain. We can verify this observation by noting that the projection of the spiral grid curves onto the xy -plane, given by $x = (2 + \sin v) \cos 2u$, $y = (2 + \sin v) \sin 2u$, $z = 0$ (where v is constant), complete circular revolutions for $0 \leq u \leq \pi$ while the original surface requires $0 \leq u \leq 2\pi$ for a complete revolution. Thus, the new surface winds around twice as fast as the original surface, and since the equation for z is identical in both surfaces, we observe twice as many circular coils in the same z -interval.



32. First we graph the surface as viewed from the front, then from two additional viewpoints.



The surface appears as a twisted sheet, and is unusual because it has only one side. (The Möbius strip is discussed in more detail in Section 16.7.)

33. $\mathbf{r}(u, v) = (u + v) \mathbf{i} + 3u^2 \mathbf{j} + (u - v) \mathbf{k}$.

$\mathbf{r}_u = \mathbf{i} + 6u \mathbf{j} + \mathbf{k}$ and $\mathbf{r}_v = \mathbf{i} - \mathbf{k}$, so $\mathbf{r}_u \times \mathbf{r}_v = -6u \mathbf{i} + 2 \mathbf{j} - 6u \mathbf{k}$. Since the point $(2, 3, 0)$ corresponds to $u = 1$, $v = 1$, a normal vector to the surface at $(2, 3, 0)$ is $-6 \mathbf{i} + 2 \mathbf{j} - 6 \mathbf{k}$, and an equation of the tangent plane is $-6x + 2y - 6z = -6$ or $3x - y + 3z = 3$.

34. $\mathbf{r}(u, v) = (u^2 + 1) \mathbf{i} + (v^3 + 1) \mathbf{j} + (u + v) \mathbf{k}$.

$\mathbf{r}_u = 2u \mathbf{i} + \mathbf{k}$ and $\mathbf{r}_v = 3v^2 \mathbf{j} + \mathbf{k}$, so $\mathbf{r}_u \times \mathbf{r}_v = -3v^2 \mathbf{i} - 2u \mathbf{j} + 6uv^2 \mathbf{k}$. Since the point $(5, 2, 3)$ corresponds to $u = 2$, $v = 1$, a normal vector to the surface at $(5, 2, 3)$ is $-3 \mathbf{i} - 4 \mathbf{j} + 12 \mathbf{k}$, and an equation of the tangent plane is $-3(x - 5) - 4(y - 2) + 12(z - 3) = 0$ or $3x + 4y - 12z = -13$.

35. $\mathbf{r}(u, v) = u \cos v \mathbf{i} + u \sin v \mathbf{j} + v \mathbf{k} \Rightarrow \mathbf{r}(1, \frac{\pi}{3}) = (\frac{1}{2}, \frac{\sqrt{3}}{2}, \frac{\pi}{3})$.

$\mathbf{r}_u = \cos v \mathbf{i} + \sin v \mathbf{j}$ and $\mathbf{r}_v = -u \sin v \mathbf{i} + u \cos v \mathbf{j} + \mathbf{k}$, so a normal vector to the surface at the point $(\frac{1}{2}, \frac{\sqrt{3}}{2}, \frac{\pi}{3})$ is

$\mathbf{r}_u(1, \frac{\pi}{3}) \times \mathbf{r}_v(1, \frac{\pi}{3}) = (\frac{1}{2} \mathbf{i} + \frac{\sqrt{3}}{2} \mathbf{j}) \times (-\frac{\sqrt{3}}{2} \mathbf{i} + \frac{1}{2} \mathbf{j} + \mathbf{k}) = \frac{\sqrt{3}}{2} \mathbf{i} - \frac{1}{2} \mathbf{j} + \mathbf{k}$. Thus an equation of the tangent plane at

$(\frac{1}{2}, \frac{\sqrt{3}}{2}, \frac{\pi}{3})$ is $\frac{\sqrt{3}}{2}(x - \frac{1}{2}) - \frac{1}{2}(y - \frac{\sqrt{3}}{2}) + 1(z - \frac{\pi}{3}) = 0$ or $\frac{\sqrt{3}}{2}x - \frac{1}{2}y + z = \frac{\pi}{3}$.

36. $\mathbf{r}(u, v) = \sin u \mathbf{i} + \cos u \sin v \mathbf{j} + \sin v \mathbf{k} \Rightarrow \mathbf{r}(\frac{\pi}{6}, \frac{\pi}{6}) = (\frac{1}{2}, \frac{\sqrt{3}}{4}, \frac{1}{2})$.

$\mathbf{r}_u = \cos u \mathbf{i} - \sin u \sin v \mathbf{j}$ and $\mathbf{r}_v = \cos u \cos v \mathbf{j} + \cos v \mathbf{k}$, so a normal vector to the surface at the point $(\frac{1}{2}, \frac{\sqrt{3}}{4}, \frac{1}{2})$ is

$$\mathbf{r}_u\left(\frac{\pi}{6}, \frac{\pi}{6}\right) \times \mathbf{r}_v\left(\frac{\pi}{6}, \frac{\pi}{6}\right) = \left(\frac{\sqrt{3}}{2}\mathbf{i} - \frac{1}{4}\mathbf{j}\right) \times \left(\frac{3}{4}\mathbf{j} + \frac{\sqrt{3}}{2}\mathbf{k}\right) = -\frac{\sqrt{3}}{8}\mathbf{i} - \frac{3}{4}\mathbf{j} + \frac{3\sqrt{3}}{8}\mathbf{k}.$$

Thus an equation of the tangent plane at $\left(\frac{1}{2}, \frac{\sqrt{3}}{4}, \frac{1}{2}\right)$ is $-\frac{\sqrt{3}}{8}\left(x - \frac{1}{2}\right) - \frac{3}{4}\left(y - \frac{\sqrt{3}}{4}\right) + \frac{3\sqrt{3}}{8}\left(z - \frac{1}{2}\right) = 0$ or

$$\sqrt{3}x + 6y - 3\sqrt{3}z = \frac{\sqrt{3}}{2} \quad \text{or} \quad 2x + 4\sqrt{3}y - 6z = 1.$$

37. $\mathbf{r}(u, v) = u^2\mathbf{i} + 2u\sin v\mathbf{j} + u\cos v\mathbf{k} \Rightarrow \mathbf{r}(1, 0) = (1, 0, 1).$

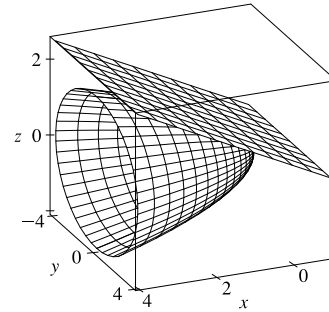
$$\mathbf{r}_u = 2u\mathbf{i} + 2\sin v\mathbf{j} + \cos v\mathbf{k} \quad \text{and} \quad \mathbf{r}_v = 2u\cos v\mathbf{j} - u\sin v\mathbf{k},$$

so a normal vector to the surface at the point $(1, 0, 1)$ is

$$\mathbf{r}_u(1, 0) \times \mathbf{r}_v(1, 0) = (2\mathbf{i} + \mathbf{k}) \times (2\mathbf{j}) = -2\mathbf{i} + 4\mathbf{k}.$$

Thus an equation of the tangent plane at $(1, 0, 1)$ is

$$-2(x - 1) + 0(y - 0) + 4(z - 1) = 0 \quad \text{or} \quad -x + 2z = 1.$$



38. $\mathbf{r}(u, v) = (1 - u^2 - v^2)\mathbf{i} - v\mathbf{j} - u\mathbf{k}.$

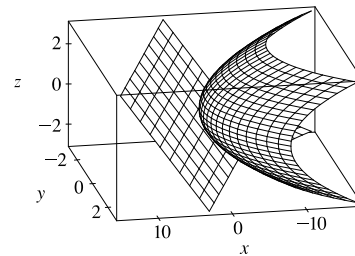
$$\mathbf{r}_u = -2u\mathbf{i} - \mathbf{k} \quad \text{and} \quad \mathbf{r}_v = -2v\mathbf{i} - \mathbf{j}. \quad \text{Since the point } (-1, -1, -1)$$

corresponds to $u = 1, v = 1$, a normal vector to the surface at

$(-1, -1, -1)$ is

$$\mathbf{r}_u(1, 1) \times \mathbf{r}_v(1, 1) = (-2\mathbf{i} - \mathbf{k}) \times (-2\mathbf{i} - \mathbf{j}) = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}.$$

Thus an equation of the tangent plane is $-1(x + 1) + 2(y + 1) + 2(z + 1) = 0$ or $-x + 2y + 2z = -3.$



39. The surface S is given by $z = f(x, y) = 6 - 3x - 2y$ which intersects the xy -plane in the line $3x + 2y = 6$, so D is the triangular region given by $\{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq 3 - \frac{3}{2}x\}$. By Formula 9, the surface area of S is

$$\begin{aligned} A(S) &= \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA \\ &= \iint_D \sqrt{1 + (-3)^2 + (-2)^2} dA = \sqrt{14} \iint_D dA = \sqrt{14} A(D) = \sqrt{14} \left(\frac{1}{2} \cdot 2 \cdot 3\right) = 3\sqrt{14}. \end{aligned}$$

40. $\mathbf{r}(u, v) = \langle u + v, 2 - 3u, 1 + u - v \rangle \Rightarrow \mathbf{r}_u = \langle 1, -3, 1 \rangle, \mathbf{r}_v = \langle 1, 0, -1 \rangle$, and $\mathbf{r}_u \times \mathbf{r}_v = \langle 3, 2, 3 \rangle$. Then by Definition 6,

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| dA = \int_0^2 \int_{-1}^1 |\langle 3, 2, 3 \rangle| dv du = \sqrt{22} \int_0^2 du \int_{-1}^1 dv = \sqrt{22} (2)(2) = 4\sqrt{22}$$

41. Here we can write $z = f(x, y) = \frac{1}{3} - \frac{1}{3}x - \frac{2}{3}y$ and D is the disk $x^2 + y^2 \leq 3$, so by Formula 9 the area of the surface is

$$\begin{aligned} A(S) &= \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA = \iint_D \sqrt{1 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{2}{3}\right)^2} dA = \frac{\sqrt{14}}{3} \iint_D dA \\ &= \frac{\sqrt{14}}{3} A(D) = \frac{\sqrt{14}}{3} \cdot \pi(\sqrt{3})^2 = \sqrt{14}\pi \end{aligned}$$

42. $z = f(x, y) = \sqrt{x^2 + y^2} \Rightarrow \frac{\partial z}{\partial x} = \frac{1}{2}(x^2 + y^2)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2 + y^2}}, \frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}},$ and

$$\sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} = \sqrt{1 + \frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2}} = \sqrt{1 + \frac{x^2 + y^2}{x^2 + y^2}} = \sqrt{2}$$

[continued]

Here D is given by $\{(x, y) \mid 0 \leq x \leq 1, x^2 \leq y \leq x\}$, so by Formula 9, the surface area of S is

$$A(S) = \iint_D \sqrt{2} \, dA = \int_0^1 \int_{x^2}^x \sqrt{2} \, dy \, dx = \sqrt{2} \int_0^1 (x - x^2) \, dx = \sqrt{2} \left[\frac{1}{2}x^2 - \frac{1}{3}x^3 \right]_0^1 = \sqrt{2} \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{\sqrt{2}}{6}$$

43. $z = f(x, y) = \frac{2}{3}(x^{3/2} + y^{3/2})$ and $D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}$. Then $f_x = x^{1/2}$, $f_y = y^{1/2}$ and

$$\begin{aligned} A(S) &= \iint_D \sqrt{1 + (\sqrt{x})^2 + (\sqrt{y})^2} \, dA = \int_0^1 \int_0^1 \sqrt{1 + x + y} \, dy \, dx \\ &= \int_0^1 \left[\frac{2}{3}(x + y + 1)^{3/2} \right]_{y=0}^{y=1} dx = \frac{2}{3} \int_0^1 \left[(x + 2)^{3/2} - (x + 1)^{3/2} \right] dx \\ &= \frac{2}{3} \left[\frac{2}{5}(x + 2)^{5/2} - \frac{2}{5}(x + 1)^{5/2} \right]_0^1 = \frac{4}{15}(3^{5/2} - 2^{5/2} - 2^{5/2} + 1) = \frac{4}{15}(3^{5/2} - 2^{7/2} + 1) \end{aligned}$$

44. $z = f(x, y) = 4 - 2x^2 + y$ and $D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq x\}$. Thus, by Formula 9,

$$\begin{aligned} A(S) &= \iint_D \sqrt{1 + (-4x)^2 + (1)^2} \, dA = \int_0^1 \int_0^x \sqrt{16x^2 + 2} \, dy \, dx = \int_0^1 x \sqrt{16x^2 + 2} \, dx \\ &= \frac{1}{32} \cdot \frac{2}{3} (16x^2 + 2)^{3/2} \Big|_0^1 = \frac{1}{48} (18^{3/2} - 2^{3/2}) = \frac{1}{48} (54\sqrt{2} - 2\sqrt{2}) = \frac{13}{12}\sqrt{2} \end{aligned}$$

45. $z = f(x, y) = xy$ with $x^2 + y^2 \leq 1$, so $f_x = y$, $f_y = x \Rightarrow$

$$\begin{aligned} A(S) &= \iint_D \sqrt{1 + y^2 + x^2} \, dA = \int_0^{2\pi} \int_0^1 \sqrt{r^2 + 1} \, r \, dr \, d\theta = \int_0^{2\pi} \left[\frac{1}{3}(r^2 + 1)^{3/2} \right]_{r=0}^{r=1} d\theta \\ &= \int_0^{2\pi} \frac{1}{3}(2\sqrt{2} - 1) \, d\theta = \frac{2\pi}{3}(2\sqrt{2} - 1) \end{aligned}$$

46. A parametric representation of the surface is $x = z^2 + y$, $y = y$, $z = z$ with $0 \leq y \leq 2$, $0 \leq z \leq 2$.

Hence $\mathbf{r}_y \times \mathbf{r}_z = (\mathbf{i} + \mathbf{j}) \times (2z\mathbf{i} + \mathbf{k}) = \mathbf{i} - \mathbf{j} - 2z\mathbf{k}$. Then

$$\begin{aligned} A(S) &= \iint_D |\mathbf{r}_y \times \mathbf{r}_z| \, dA = \int_0^2 \int_0^2 \sqrt{1 + 1 + 4z^2} \, dy \, dz = \int_0^2 2\sqrt{2 + 4z^2} \, dz \\ &= \left[2 \cdot \frac{1}{2} (z\sqrt{2 + 4z^2} + \ln(2z + \sqrt{2 + 4z^2})) \right]_0^2 \quad \left[\begin{array}{l} \text{Use trigonometric substitution} \\ \text{or Formula 21 in the Table of Integrals} \end{array} \right] \\ &= 6\sqrt{2} + \ln(4 + 3\sqrt{2}) - \ln\sqrt{2} \text{ or } 6\sqrt{2} + \ln \frac{4 + 3\sqrt{2}}{\sqrt{2}} = 6\sqrt{2} + \ln(2\sqrt{2} + 3) \end{aligned}$$

Note: In general, if $x = f(y, z)$ then $\mathbf{r}_y \times \mathbf{r}_z = \mathbf{i} - \frac{\partial f}{\partial y} \mathbf{j} - \frac{\partial f}{\partial z} \mathbf{k}$ and $A(S) = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial y}\right)^2 + \left(\frac{\partial f}{\partial z}\right)^2} \, dA$.

47. A parametric representation of the surface is $x = x$, $y = x^2 + z^2$, $z = z$ with $0 \leq x^2 + z^2 \leq 16$.

Hence $\mathbf{r}_x \times \mathbf{r}_z = (\mathbf{i} + 2x\mathbf{j}) \times (2z\mathbf{j} + \mathbf{k}) = 2x\mathbf{i} - \mathbf{j} + 2z\mathbf{k}$.

Note: In general, if $y = f(x, z)$ then $\mathbf{r}_x \times \mathbf{r}_z = \frac{\partial f}{\partial x} \mathbf{i} - \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$, and $A(S) = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial z}\right)^2} \, dA$.

Then

$$\begin{aligned} A(S) &= \iint_{0 \leq x^2 + z^2 \leq 16} \sqrt{1 + 4x^2 + 4z^2} \, dA = \int_0^{2\pi} \int_0^4 \sqrt{1 + 4r^2} \, r \, dr \, d\theta \\ &= \int_0^{2\pi} d\theta \int_0^4 r \sqrt{1 + 4r^2} \, dr = 2\pi \left[\frac{1}{12}(1 + 4r^2)^{3/2} \right]_0^4 = \frac{\pi}{6} (65^{3/2} - 1) \end{aligned}$$

48. $\mathbf{r}(u, v) = u \cos v \mathbf{i} + u \sin v \mathbf{j} + v \mathbf{k}$, $0 \leq u \leq 1$, $0 \leq v \leq \pi \Rightarrow$

$\mathbf{r}_u = \langle \cos v, \sin v, 0 \rangle$, $\mathbf{r}_v = \langle -u \sin v, u \cos v, 1 \rangle$, and $\mathbf{r}_u \times \mathbf{r}_v = \langle \sin v, -\cos v, u \rangle$. Then

$$\begin{aligned} A(S) &= \int_0^\pi \int_0^1 \sqrt{1+u^2} \, du \, dv = \int_0^\pi dv \int_0^1 \sqrt{1+u^2} \, du \\ &= \pi \left[\frac{u}{2} \sqrt{u^2+1} + \frac{1}{2} \ln |u + \sqrt{u^2+1}| \right]_0^1 = \frac{\pi}{2} [\sqrt{2} + \ln(1 + \sqrt{2})] \end{aligned}$$

49. $x = u^2$, $y = uv$, $z = \frac{1}{2}v^2$, $0 \leq u \leq 1$, $0 \leq v \leq 2 \Rightarrow \mathbf{r}_u = \langle 2u, v, 0 \rangle$, $\mathbf{r}_v = \langle 0, u, v \rangle$, and $\mathbf{r}_u \times \mathbf{r}_v = \langle v^2, -2uv, 2u^2 \rangle$.

Then

$$\begin{aligned} A(S) &= \iint_D |\mathbf{r}_u \times \mathbf{r}_v| \, dA = \int_0^1 \int_0^2 \sqrt{v^4 + 4u^2v^2 + 4u^4} \, dv \, du = \int_0^1 \int_0^2 \sqrt{(v^2 + 2u^2)^2} \, dv \, du \\ &= \int_0^1 \int_0^2 (v^2 + 2u^2) \, dv \, du = \int_0^1 \left[\frac{1}{3}v^3 + 2u^2v \right]_{v=0}^{v=2} du = \int_0^1 \left(\frac{8}{3} + 4u^2 \right) du = \left[\frac{8}{3}u + \frac{4}{3}u^3 \right]_0^1 = 4 \end{aligned}$$

50. The cylinder encloses separate portions of the sphere in the upper and lower halves. The top half of the sphere is

$z = f(x, y) = \sqrt{b^2 - x^2 - y^2}$ and D is given by $\{(x, y) \mid x^2 + y^2 \leq a^2\}$. By Formula 9, the surface area of the upper enclosed portion is

$$\begin{aligned} A &= \iint_D \sqrt{1 + \left(\frac{-x}{\sqrt{b^2 - x^2 - y^2}} \right)^2 + \left(\frac{-y}{\sqrt{b^2 - x^2 - y^2}} \right)^2} \, dA = \iint_D \sqrt{1 + \frac{x^2 + y^2}{b^2 - x^2 - y^2}} \, dA \\ &= \iint_D \sqrt{\frac{b^2}{b^2 - x^2 - y^2}} \, dA = \int_0^{2\pi} \int_0^a \frac{b}{\sqrt{b^2 - r^2}} r \, dr \, d\theta = b \int_0^{2\pi} d\theta \int_0^a \frac{r}{\sqrt{b^2 - r^2}} \, dr \\ &= b [\theta]_0^{2\pi} [-\sqrt{b^2 - r^2}]_0^a = 2\pi b (-\sqrt{b^2 - a^2} + \sqrt{b^2 - 0}) = 2\pi b (b - \sqrt{b^2 - a^2}) \end{aligned}$$

The lower portion of the sphere enclosed by the cylinder has identical shape, so the total area is $2A = 4\pi b(b - \sqrt{b^2 - a^2})$.

51. From Formula 9 with $z = f(x, y)$, we have $A(S) = \iint_D \sqrt{1 + (f_x)^2 + (f_y)^2} \, dA$. Since $|f_x| \leq 1$ and $|f_y| \leq 1$, we know

$0 \leq (f_x)^2 \leq 1$ and $0 \leq (f_y)^2 \leq 1$, so $1 \leq 1 + (f_x)^2 + (f_y)^2 \leq 3 \Rightarrow 1 \leq \sqrt{1 + (f_x)^2 + (f_y)^2} \leq \sqrt{3}$. By

Property 15.2.10, $\iint_D 1 \, dA \leq \iint_D \sqrt{1 + (f_x)^2 + (f_y)^2} \, dA \leq \iint_D \sqrt{3} \, dA \Rightarrow A(D) \leq A(S) \leq \sqrt{3} A(D) \Rightarrow$

$$\pi R^2 \leq A(S) \leq \sqrt{3} \pi R^2.$$

52. $z = f(x, y) = \cos(x^2 + y^2)$ with $x^2 + y^2 \leq 1$.

$$\begin{aligned} A(S) &= \iint_D \sqrt{1 + (-2x \sin(x^2 + y^2))^2 + (-2y \sin(x^2 + y^2))^2} \, dA \\ &= \iint_D \sqrt{1 + 4x^2 \sin^2(x^2 + y^2) + 4y^2 \sin^2(x^2 + y^2)} \, dA = \iint_D \sqrt{1 + 4(x^2 + y^2) \sin^2(x^2 + y^2)} \, dA \\ &= \int_0^{2\pi} \int_0^1 \sqrt{1 + 4r^2 \sin^2(r^2)} \, r \, dr \, d\theta = \int_0^{2\pi} d\theta \int_0^1 r \sqrt{1 + 4r^2 \sin^2(r^2)} \, dr \\ &= 2\pi \int_0^1 r \sqrt{1 + 4r^2 \sin^2(r^2)} \, dr \approx 4.1073 \end{aligned}$$

53. $z = f(x, y) = \ln(x^2 + y^2 + 2)$ with $x^2 + y^2 \leq 1$.

$$\begin{aligned} A(S) &= \iint_D \sqrt{1 + \left(\frac{2x}{x^2 + y^2 + 2} \right)^2 + \left(\frac{2y}{x^2 + y^2 + 2} \right)^2} \, dA = \iint_D \sqrt{1 + \frac{4x^2 + 4y^2}{(x^2 + y^2 + 2)^2}} \, dA \\ &= \int_0^{2\pi} \int_0^1 \sqrt{1 + \frac{4r^2}{(r^2 + 2)^2}} \, r \, dr \, d\theta = \int_0^{2\pi} d\theta \int_0^1 r \sqrt{\frac{(r^2 + 2)^2 + 4r^2}{(r^2 + 2)^2}} \, dr = 2\pi \int_0^1 \frac{r \sqrt{r^4 + 8r^2 + 4}}{r^2 + 2} \, dr \\ &\approx 3.5618 \end{aligned}$$

54. Let $f(x, y) = \frac{1+x^2}{1+y^2}$. Then $f_x = \frac{2x}{1+y^2}$,

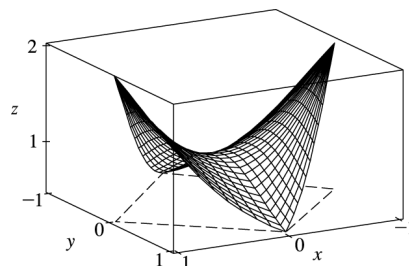
$$f_y = (1+x^2) \left[-\frac{2y}{(1+y^2)^2} \right] = -\frac{2y(1+x^2)}{(1+y^2)^2}.$$

We use a CAS to estimate

$$\int_{-1}^1 \int_{-(1-|x|)}^{1-|x|} \sqrt{1+f_x^2+f_y^2} dy dx \approx 2.6959.$$

In order to graph only the part of the surface above the square, we

use $-(1-|x|) \leq y \leq 1-|x|$ as the y -range in our plot command.



55. (a) $z = \frac{1}{1+x^2+y^2} \Rightarrow \frac{\partial z}{\partial x} = \frac{-2x}{(1+x^2+y^2)^2}$ and $\frac{\partial z}{\partial y} = \frac{-2y}{(1+x^2+y^2)^2}$.

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA = \int_0^6 \int_0^4 \sqrt{1 + \frac{4x^2 + 4y^2}{(1+x^2+y^2)^4}} dy dx.$$

Using the Midpoint Rule with $f(x, y) = \sqrt{1 + \frac{4x^2 + 4y^2}{(1+x^2+y^2)^4}}$, $m = 3$, $n = 2$ we have

$$A(S) \approx \sum_{i=1}^3 \sum_{j=1}^2 f(\bar{x}_i, \bar{y}_j) \Delta A = 4[f(1, 1) + f(1, 3) + f(3, 1) + f(3, 3) + f(5, 1) + f(5, 3)] \approx 24.2055$$

(b) Using a CAS, we have $A(S) = \int_0^6 \int_0^4 \sqrt{1 + \frac{4x^2 + 4y^2}{(1+x^2+y^2)^4}} dy dx \approx 24.2476$. This agrees with the estimate in part (a)

to the first decimal place.

56. $\mathbf{r}(u, v) = \langle \cos^3 u \cos^3 v, \sin^3 u \cos^3 v, \sin^3 v \rangle$, so $\mathbf{r}_u = \langle -3\cos^2 u \sin u \cos^3 v, 3\sin^2 u \cos u \cos^3 v, 0 \rangle$,

$\mathbf{r}_v = \langle -3\cos^3 u \cos^2 v \sin v, -3\sin^3 u \cos^2 v \sin v, 3\sin^2 v \cos v \rangle$, and

$\mathbf{r}_u \times \mathbf{r}_v = \langle 9\cos u \sin^2 u \cos^4 v \sin^2 v, 9\cos^2 u \sin u \cos^4 v \sin^2 v, 9\cos^2 u \sin^2 u \cos^5 v \sin v \rangle$. Then

$$\begin{aligned} |\mathbf{r}_u \times \mathbf{r}_v| &= 9 \sqrt{\cos^2 u \sin^4 u \cos^8 v \sin^4 v + \cos^4 u \sin^2 u \cos^8 v \sin^4 v + \cos^4 u \sin^4 u \cos^{10} v \sin^2 v} \\ &= 9 \sqrt{\cos^2 u \sin^2 u \cos^8 v \sin^2 v (\sin^2 v + \cos^2 u \sin^2 u \cos^2 v)} \\ &= 9 \cos^4 v |\cos u \sin u \sin v| \sqrt{\sin^2 v + \cos^2 u \sin^2 u \cos^2 v} \end{aligned}$$

Using a CAS, we have $A(S) = \int_0^\pi \int_0^{2\pi} 9 \cos^4 v |\cos u \sin u \sin v| \sqrt{\sin^2 v + \cos^2 u \sin^2 u \cos^2 v} dv du \approx 4.4506$.

57. $z = 1 + 2x + 3y + 4y^2$, so

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA = \int_1^4 \int_0^1 \sqrt{1 + 4 + (3 + 8y)^2} dy dx = \int_1^4 \int_0^1 \sqrt{14 + 48y + 64y^2} dy dx.$$

Using a CAS, we have

$$\int_1^4 \int_0^1 \sqrt{14 + 48y + 64y^2} dy dx = \frac{45}{8} \sqrt{14} + \frac{15}{16} \ln(11\sqrt{5} + 3\sqrt{14}\sqrt{5}) - \frac{15}{16} \ln(3\sqrt{5} + \sqrt{14}\sqrt{5})$$

or $\frac{45}{8} \sqrt{14} + \frac{15}{16} \ln \frac{11\sqrt{5} + 3\sqrt{70}}{3\sqrt{5} + \sqrt{70}}$.

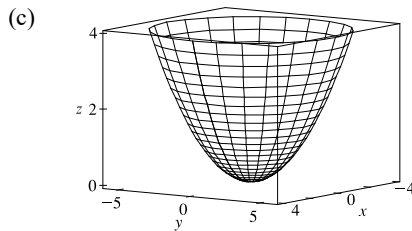
58. (a) $x = au \cos v, y = bu \sin v, z = u^2, 0 \leq u \leq 2, 0 \leq v \leq 2\pi \Rightarrow \mathbf{r}_u = a \cos v \mathbf{i} + b \sin v \mathbf{j} + 2u \mathbf{k},$

$$\mathbf{r}_v = -au \sin v \mathbf{i} + bu \cos v \mathbf{j} + 0 \mathbf{k}, \text{ and } \mathbf{r}_u \times \mathbf{r}_v = -2bu^2 \cos v \mathbf{i} - 2au^2 \sin v \mathbf{j} + abu \mathbf{k}.$$

$$A(S) = \int_0^{2\pi} \int_0^2 |\mathbf{r}_u \times \mathbf{r}_v| du dv = \int_0^{2\pi} \int_0^2 \sqrt{4b^2u^4 \cos^2 v + 4a^2u^4 \sin^2 v + a^2b^2u^2} du dv$$

- (b) $x^2 = a^2u^2 \cos^2 v, y^2 = b^2u^2 \sin^2 v, z = u^2 \Rightarrow x^2/a^2 + y^2/b^2 = u^2 = z$ which is an elliptic paraboloid. To find D , notice that $0 \leq u \leq 2 \Rightarrow 0 \leq z \leq 4 \Rightarrow 0 \leq x^2/a^2 + y^2/b^2 \leq 4$. Therefore, using Formula 9, we have

$$A(S) = \int_{-2a}^{2a} \int_{-b\sqrt{4-(x^2/a^2)}}^{b\sqrt{4-(x^2/a^2)}} \sqrt{1 + (2x/a^2)^2 + (2y/b^2)^2} dy dx.$$



- (d) We substitute $a = 2, b = 3$ in the integral in part (a) to get

$A(S) = \int_0^{2\pi} \int_0^2 2u \sqrt{9u^2 \cos^2 v + 4u^2 \sin^2 v + 9} du dv$. We use a CAS to estimate the integral accurate to four decimal places. To speed up the calculation, we can set `Digits:=7`; (in Maple) or use the approximation command `N` (in Mathematica). We find that $A(S) \approx 115.6596$.

59. (a) $x = a \sin u \cos v, y = b \sin u \sin v, z = c \cos u \Rightarrow$

$$\begin{aligned} \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} &= (\sin u \cos v)^2 + (\sin u \sin v)^2 + (\cos u)^2 \\ &= \sin^2 u + \cos^2 u = 1 \end{aligned}$$

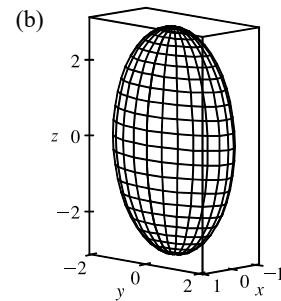
and since the ranges of u and v are sufficient to generate the entire graph, the parametric equations represent an ellipsoid.

- (c) From the parametric equations (with $a = 1, b = 2$, and $c = 3$),

we calculate $\mathbf{r}_u = \cos u \cos v \mathbf{i} + 2 \cos u \sin v \mathbf{j} - 3 \sin u \mathbf{k}$ and

$\mathbf{r}_v = -\sin u \sin v \mathbf{i} + 2 \sin u \cos v \mathbf{j}$. So $\mathbf{r}_u \times \mathbf{r}_v = 6 \sin^2 u \cos v \mathbf{i} + 3 \sin^2 u \sin v \mathbf{j} + 2 \sin u \cos u \mathbf{k}$, and the surface

area is given by $A(S) = \int_0^{2\pi} \int_0^\pi |\mathbf{r}_u \times \mathbf{r}_v| du dv = \int_0^{2\pi} \int_0^\pi \sqrt{36 \sin^4 u \cos^2 v + 9 \sin^4 u \sin^2 v + 4 \cos^2 u \sin^2 u} du dv$



60. (a) $x = a \cosh u \cos v, y = b \cosh u \sin v, z = c \sinh u \Rightarrow$

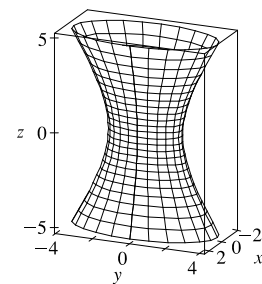
$$\begin{aligned} \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} &= \cosh^2 u \cos^2 v + \cosh^2 u \sin^2 v - \sinh^2 u \\ &= \cosh^2 u - \sinh^2 u = 1 \end{aligned}$$

and the parametric equations represent a hyperboloid of one sheet.

- (c) $\mathbf{r}_u = \sinh u \cos v \mathbf{i} + 2 \sinh u \sin v \mathbf{j} + 3 \cosh u \mathbf{k}$ and

$\mathbf{r}_v = -\cosh u \sin v \mathbf{i} + 2 \cosh u \cos v \mathbf{j}$, so $\mathbf{r}_u \times \mathbf{r}_v = -6 \cosh^2 u \cos v \mathbf{i} - 3 \cosh^2 u \sin v \mathbf{j} + 2 \cosh u \sinh u \mathbf{k}$.

We integrate between $u = \sinh^{-1}(-1) = -\ln(1 + \sqrt{2})$ and $u = \sinh^{-1} 1 = \ln(1 + \sqrt{2})$, since then z varies between



−3 and 3, as desired. So the surface area is

$$\begin{aligned} A(S) &= \int_0^{2\pi} \int_{-\ln(1+\sqrt{2})}^{\ln(1+\sqrt{2})} |\mathbf{r}_u \times \mathbf{r}_v| \, du \, dv \\ &= \int_0^{2\pi} \int_{-\ln(1+\sqrt{2})}^{\ln(1+\sqrt{2})} \sqrt{36 \cosh^4 u \cos^2 v + 9 \cosh^4 u \sin^2 v + 4 \cosh^2 u \sinh^2 u} \, du \, dv \end{aligned}$$

61. To find the region D : $z = x^2 + y^2$ implies $z + z^2 = 4z$ or $z^2 - 3z = 0$. Thus $z = 0$ or $z = 3$ are the planes where the surfaces intersect. But $x^2 + y^2 + z^2 = 4z$ implies $x^2 + y^2 + (z - 2)^2 = 4$, so $z = 3$ intersects the upper hemisphere. Thus $(z - 2)^2 = 4 - x^2 - y^2$ or $z = 2 + \sqrt{4 - x^2 - y^2}$. Therefore D is the region inside the circle $x^2 + y^2 + (3 - 2)^2 = 4$, that is, $D = \{(x, y) \mid x^2 + y^2 \leq 3\}$.

$$\begin{aligned} A(S) &= \iint_D \sqrt{1 + [(-x)(4 - x^2 - y^2)^{-1/2}]^2 + [(-y)(4 - x^2 - y^2)^{-1/2}]^2} \, dA \\ &= \int_0^{2\pi} \int_0^{\sqrt{3}} \sqrt{1 + \frac{r^2}{4 - r^2}} \, r \, dr \, d\theta = \int_0^{2\pi} \int_0^{\sqrt{3}} \frac{2r \, dr}{\sqrt{4 - r^2}} \, d\theta = \int_0^{2\pi} \left[-2(4 - r^2)^{1/2} \right]_{r=0}^{r=\sqrt{3}} d\theta \\ &= \int_0^{2\pi} (-2 + 4) \, d\theta = 2\theta \Big|_0^{2\pi} = 4\pi \end{aligned}$$

62. We first find the area of the face of the surface that intersects the positive y -axis. A parametric representation of the surface is

$$x = x, \quad y = \sqrt{1 - z^2}, \quad z = z \text{ with } x^2 + z^2 \leq 1. \text{ Then } \mathbf{r}(x, z) = \langle x, \sqrt{1 - z^2}, z \rangle \Rightarrow \mathbf{r}_x = \langle 1, 0, 0 \rangle,$$

$$\mathbf{r}_z = \langle 0, -z/\sqrt{1 - z^2}, 1 \rangle \text{ and } \mathbf{r}_x \times \mathbf{r}_z = \langle 0, -1, -z/\sqrt{1 - z^2} \rangle \Rightarrow |\mathbf{r}_x \times \mathbf{r}_z| = \sqrt{1 + \frac{z^2}{1 - z^2}} = \frac{1}{\sqrt{1 - z^2}}.$$

$$A(S) = \iint_{x^2 + z^2 \leq 1} |\mathbf{r}_x \times \mathbf{r}_z| \, dA = \int_{-1}^1 \int_{-\sqrt{1 - z^2}}^{\sqrt{1 - z^2}} \frac{1}{\sqrt{1 - z^2}} \, dx \, dz = 4 \int_0^1 \int_0^{\sqrt{1 - z^2}} \frac{1}{\sqrt{1 - z^2}} \, dx \, dz \quad \left[\begin{array}{l} \text{by the symmetry} \\ \text{of the surface} \end{array} \right]$$

This integral is improper [when $z = 1$], so

$$A(S) = \lim_{t \rightarrow 1^-} 4 \int_0^t \int_0^{\sqrt{1 - z^2}} \frac{1}{\sqrt{1 - z^2}} \, dx \, dz = \lim_{t \rightarrow 1^-} 4 \int_0^t \frac{\sqrt{1 - z^2}}{\sqrt{1 - z^2}} \, dz = \lim_{t \rightarrow 1^-} 4 \int_0^t 1 \, dz = \lim_{t \rightarrow 1^-} 4t = 4$$

Since the complete surface consists of four congruent faces, the total surface area is $4(4) = 16$.

Alternate solution: The face of the surface that intersects the positive y -axis can also be parametrized as

$$\mathbf{r}(x, \theta) = \langle x, \cos \theta, \sin \theta \rangle \text{ for } -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \text{ and } x^2 + z^2 \leq 1 \Leftrightarrow x^2 + \sin^2 \theta \leq 1 \Leftrightarrow$$

$$-\sqrt{1 - \sin^2 \theta} \leq x \leq \sqrt{1 - \sin^2 \theta} \Leftrightarrow -\cos \theta \leq x \leq \cos \theta. \text{ Then } \mathbf{r}_x = \langle 1, 0, 0 \rangle, \mathbf{r}_\theta = \langle 0, -\sin \theta, \cos \theta \rangle \text{ and}$$

$$\mathbf{r}_x \times \mathbf{r}_\theta = \langle 0, -\cos \theta, -\sin \theta \rangle \Rightarrow |\mathbf{r}_x \times \mathbf{r}_\theta| = 1, \text{ so}$$

$$A(S) = \int_{-\pi/2}^{\pi/2} \int_{-\cos \theta}^{\cos \theta} 1 \, dx \, d\theta = \int_{-\pi/2}^{\pi/2} 2 \cos \theta \, d\theta = 2 \sin \theta \Big|_{-\pi/2}^{\pi/2} = 4. \text{ Again, the area of the complete surface}$$

is $4(4) = 16$.

63. Let $A(S_1)$ be the surface area of that portion of the surface which lies above the plane $z = 0$. Then $A(S) = 2A(S_1)$.

Following Example 10, a parametric representation of S_1 is $x = a \sin \phi \cos \theta$, $y = a \sin \phi \sin \theta$,

$z = a \cos \phi$ and $|\mathbf{r}_\phi \times \mathbf{r}_\theta| = a^2 \sin \phi$. For D , $0 \leq \phi \leq \frac{\pi}{2}$ and for each fixed ϕ , $(x - \frac{1}{2}a)^2 + y^2 \leq (\frac{1}{2}a)^2$ or

$[a \sin \phi \cos \theta - \frac{1}{2}a]^2 + a^2 \sin^2 \phi \sin^2 \theta \leq (a/2)^2$ implies $a^2 \sin^2 \phi - a^2 \sin \phi \cos \theta \leq 0$ or

$\sin \phi (\sin \phi - \cos \theta) \leq 0$. But $0 \leq \phi \leq \frac{\pi}{2}$, so $\cos \theta \geq \sin \phi$ or $\sin(\frac{\pi}{2} + \theta) \geq \sin \phi$ or $\phi - \frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} - \phi$.

Hence $D = \{(\phi, \theta) \mid 0 \leq \phi \leq \frac{\pi}{2}, \phi - \frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} - \phi\}$. Then

$$\begin{aligned} A(S_1) &= \int_0^{\pi/2} \int_{\phi - (\pi/2)}^{(\pi/2) - \phi} a^2 \sin \phi \, d\theta \, d\phi = a^2 \int_0^{\pi/2} (\pi - 2\phi) \sin \phi \, d\phi \\ &= a^2 [(-\pi \cos \phi) - 2(-\phi \cos \phi + \sin \phi)]_0^{\pi/2} = a^2(\pi - 2) \end{aligned}$$

Thus $A(S) = 2a^2(\pi - 2)$.

Alternate solution: Working on S_1 we could parametrize the portion of the sphere by $x = x$, $y = y$, $z = \sqrt{a^2 - x^2 - y^2}$.

Then $|\mathbf{r}_x \times \mathbf{r}_y| = \sqrt{1 + \frac{x^2}{a^2 - x^2 - y^2} + \frac{y^2}{a^2 - x^2 - y^2}} = \frac{a}{\sqrt{a^2 - x^2 - y^2}}$ and

$$\begin{aligned} A(S_1) &= \iint_{0 \leq (x - (a/2))^2 + y^2 \leq (a/2)^2} \frac{a}{\sqrt{a^2 - x^2 - y^2}} \, dA = \int_{-\pi/2}^{\pi/2} \int_0^{a \cos \theta} \frac{a}{\sqrt{a^2 - r^2}} r \, dr \, d\theta \\ &= \int_{-\pi/2}^{\pi/2} -a(a^2 - r^2)^{1/2} \Big|_{r=0}^{r=a \cos \theta} d\theta = \int_{-\pi/2}^{\pi/2} a^2 [1 - (1 - \cos^2 \theta)^{1/2}] d\theta \\ &= \int_{-\pi/2}^{\pi/2} a^2 (1 - |\sin \theta|) d\theta = 2a^2 \int_0^{\pi/2} (1 - \sin \theta) d\theta = 2a^2 \left(\frac{\pi}{2} - 1\right) \end{aligned}$$

Thus $A(S) = 4a^2(\frac{\pi}{2} - 1) = 2a^2(\pi - 2)$.

Notes:

- (1) Perhaps working in spherical coordinates is the most obvious approach here. However, you must be careful in setting up D .
- (2) In the alternate solution, you can avoid having to use $|\sin \theta|$ by working in the first octant and then multiplying by 4. However, if you set up S_1 as above and arrived at $A(S_1) = a^2\pi$, you now see your error.

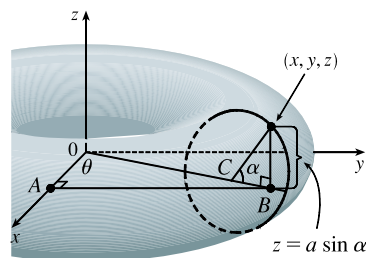
64. (a) Here $z = a \sin \alpha$, $y = |AB|$, and $x = |OA|$. But

$|OB| = |OC| + |CB| = b + a \cos \alpha$ and $\sin \theta = \frac{|AB|}{|OB|}$ so that

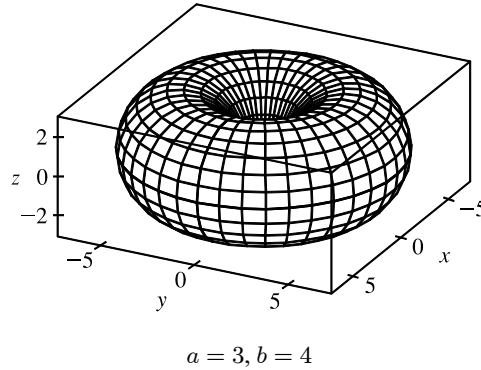
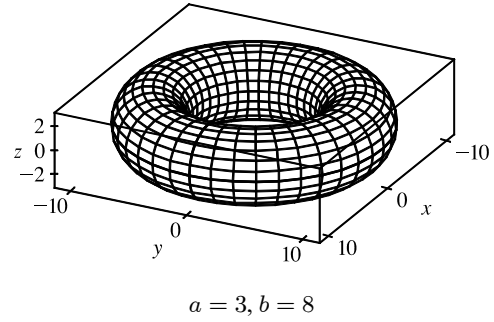
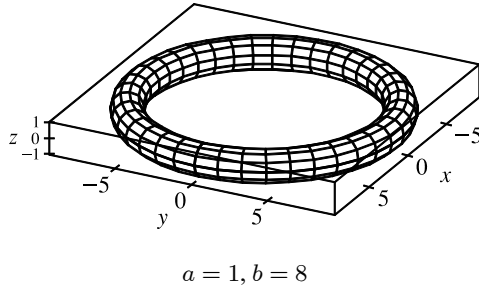
$y = |OB| \sin \theta = (b + a \cos \alpha) \sin \theta$. Similarly $\cos \theta = \frac{|OA|}{|OB|}$ so

$x = (b + a \cos \alpha) \cos \theta$. Hence a parametric representation for the torus is $x = b \cos \theta + a \cos \alpha \cos \theta$, $y = b \sin \theta + a \cos \alpha \sin \theta$,

$z = a \sin \alpha$, where $0 \leq \alpha \leq 2\pi$, $0 \leq \theta \leq 2\pi$.



(b)



(c) $x = b \cos \theta + a \cos \alpha \cos \theta$, $y = b \sin \theta + a \cos \alpha \sin \theta$, $z = a \sin \alpha$, so $\mathbf{r}_\alpha = \langle -a \sin \alpha \cos \theta, -a \sin \alpha \sin \theta, a \cos \alpha \rangle$,

$\mathbf{r}_\theta = \langle -(b + a \cos \alpha) \sin \theta, (b + a \cos \alpha) \cos \theta, 0 \rangle$ and

$$\begin{aligned} \mathbf{r}_\alpha \times \mathbf{r}_\theta &= (-ab \cos \alpha \cos \theta - a^2 \cos \alpha \cos^2 \theta) \mathbf{i} + (-ab \sin \alpha \cos \theta - a^2 \sin \alpha \cos^2 \theta) \mathbf{j} \\ &\quad + (-ab \cos^2 \alpha \sin \theta - a^2 \cos^2 \alpha \sin \theta \cos \theta - ab \sin^2 \alpha \sin \theta - a^2 \sin^2 \alpha \sin \theta \cos \theta) \mathbf{k} \\ &= -a(b + a \cos \alpha) [(\cos \theta \cos \alpha) \mathbf{i} + (\sin \theta \cos \alpha) \mathbf{j} + (\sin \alpha) \mathbf{k}] \end{aligned}$$

Then $|\mathbf{r}_\alpha \times \mathbf{r}_\theta| = a(b + a \cos \alpha) \sqrt{\cos^2 \theta \cos^2 \alpha + \sin^2 \theta \cos^2 \alpha + \sin^2 \alpha} = a(b + a \cos \alpha)$.

Note: $b > a$, $-1 \leq \cos \alpha \leq 1$ so $|b + a \cos \alpha| = b + a \cos \alpha$. Hence

$$A(S) = \int_0^{2\pi} \int_0^{2\pi} a(b + a \cos \alpha) d\alpha d\theta = 2\pi [ab\alpha + a^2 \sin \alpha]_0^{2\pi} = 4\pi^2 ab.$$

16.7 Surface Integrals

1. The box is a cube where each face has surface area 4. The centers of the faces are $(\pm 1, 0, 0)$, $(0, \pm 1, 0)$, $(0, 0, \pm 1)$. For each face we take the point P_{ij}^* to be the center of the face and $f(x, y, z) = \cos(x + 2y + 3z)$, so by Definition 1,

$$\begin{aligned} \iint_S f(x, y, z) dS &\approx [f(1, 0, 0)](4) + [f(-1, 0, 0)](4) + [f(0, 1, 0)](4) \\ &\quad + [f(0, -1, 0)](4) + [f(0, 0, 1)](4) + [f(0, 0, -1)](4) \\ &= 4 [\cos 1 + \cos(-1) + \cos 2 + \cos(-2) + \cos 3 + \cos(-3)] \approx -6.93 \end{aligned}$$

2. Each quarter-cylinder has surface area $\frac{1}{4}[2\pi(1)(2)] = \pi$, and the top and bottom disks have surface area $\pi(1)^2 = \pi$. We can take $(0, 0, 1)$ as a sample point in the top disk, $(0, 0, -1)$ in the bottom disk, and $(\pm 1, 0, 0)$, $(0, \pm 1, 0)$ in the four

quarter-cylinders. Then $\iint_S f(x, y, z) dS$ can be approximated by the Riemann sum

$$\begin{aligned} f(1, 0, 0)(\pi) + f(-1, 0, 0)(\pi) + f(0, 1, 0)(\pi) + f(0, -1, 0)(\pi) + f(0, 0, 1)(\pi) + f(0, 0, -1)(\pi) \\ = (2 + 2 + 3 + 3 + 4 + 4)\pi = 18\pi \approx 56.5. \end{aligned}$$

3. We can use the xz - and yz -planes to divide H into four patches of equal size, each with surface area equal to $\frac{1}{8}$ the surface area of a sphere with radius $\sqrt{50}$, so $\Delta S = \frac{1}{8}(4)\pi(\sqrt{50})^2 = 25\pi$. Then $(\pm 3, \pm 4, 5)$ are sample points in the four patches, and using a Riemann sum as in Definition 1, we have

$$\begin{aligned} \iint_H f(x, y, z) dS &\approx f(3, 4, 5) \Delta S + f(3, -4, 5) \Delta S + f(-3, 4, 5) \Delta S + f(-3, -4, 5) \Delta S \\ &= (7 + 8 + 9 + 12)(25\pi) = 900\pi \approx 2827 \end{aligned}$$

4. On the surface, $f(x, y, z) = g(\sqrt{x^2 + y^2 + z^2}) = g(2) = -5$. So since the area of a sphere is $4\pi r^2$,

$$\iint_S f(x, y, z) dS = \iint_S g(2) dS = -5 \iint_S dS = -5[4\pi(2)^2] = -80\pi.$$

5. $\mathbf{r}(u, v) = (u + v)\mathbf{i} + (u - v)\mathbf{j} + (1 + 2u + v)\mathbf{k}$, $0 \leq u \leq 2$, $0 \leq v \leq 1$ and

$$\mathbf{r}_u \times \mathbf{r}_v = (\mathbf{i} + \mathbf{j} + 2\mathbf{k}) \times (\mathbf{i} - \mathbf{j} + \mathbf{k}) = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k} \Rightarrow |\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{3^2 + 1^2 + (-2)^2} = \sqrt{14}. \text{ Then by Formula 2,}$$

$$\begin{aligned} \iint_S (x + y + z) dS &= \iint_D (u + v + u - v + 1 + 2u + v) |\mathbf{r}_u \times \mathbf{r}_v| dA = \int_0^1 \int_0^2 (4u + v + 1) \cdot \sqrt{14} du dv \\ &= \sqrt{14} \int_0^1 [2u^2 + uv + u]_{u=0}^{u=2} dv = \sqrt{14} \int_0^1 (2v + 10) dv = \sqrt{14} [v^2 + 10v]_0^1 = 11\sqrt{14} \end{aligned}$$

6. $\mathbf{r}(u, v) = u \cos v \mathbf{i} + u \sin v \mathbf{j} + u \mathbf{k}$, $0 \leq u \leq 1$, $0 \leq v \leq \pi/2$ and

$$\mathbf{r}_u \times \mathbf{r}_v = (\cos v \mathbf{i} + \sin v \mathbf{j} + \mathbf{k}) \times (-u \sin v \mathbf{i} + u \cos v \mathbf{j}) = -u \cos v \mathbf{i} - u \sin v \mathbf{j} + u \mathbf{k} \Rightarrow$$

$$|\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{u^2 \cos^2 v + u^2 \sin^2 v + u^2} = \sqrt{2}u \text{ [since } u \geq 0\text{]}. \text{ Then by Formula 2,}$$

$$\begin{aligned} \iint_S xyz dS &= \iint_D (u \cos v)(u \sin v)(u) |\mathbf{r}_u \times \mathbf{r}_v| dA = \int_0^1 \int_0^{\pi/2} (u^3 \sin v \cos v) \cdot \sqrt{2} u dv du \\ &= \sqrt{2} \int_0^1 u^4 du \int_0^{\pi/2} \sin v \cos v dv = \sqrt{2} \left[\frac{1}{5} u^5 \right]_0^1 \left[\frac{1}{2} \sin^2 v \right]_0^{\pi/2} = \sqrt{2} \cdot \frac{1}{5} \cdot \frac{1}{2} = \frac{1}{10} \sqrt{2} \end{aligned}$$

7. $\mathbf{r}(u, v) = \langle u \cos v, u \sin v, v \rangle$, $0 \leq u \leq 1$, $0 \leq v \leq \pi$ and

$$\mathbf{r}_u \times \mathbf{r}_v = \langle \cos v, \sin v, 0 \rangle \times \langle -u \sin v, u \cos v, 1 \rangle = \langle \sin v, -\cos v, u \rangle \Rightarrow$$

$$|\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{\sin^2 v + \cos^2 v + u^2} = \sqrt{u^2 + 1}. \text{ Then}$$

$$\begin{aligned} \iint_S y dS &= \iint_D (u \sin v) |\mathbf{r}_u \times \mathbf{r}_v| dA = \int_0^1 \int_0^\pi (u \sin v) \cdot \sqrt{u^2 + 1} dv du = \int_0^1 u \sqrt{u^2 + 1} du \int_0^\pi \sin v dv \\ &= \left[\frac{1}{3} (u^2 + 1)^{3/2} \right]_0^1 [-\cos v]_0^\pi = \frac{1}{3} (2^{3/2} - 1) \cdot 2 = \frac{2}{3} (2\sqrt{2} - 1) \end{aligned}$$

8. $\mathbf{r}(u, v) = \langle 2uv, u^2 - v^2, u^2 + v^2 \rangle$, $u^2 + v^2 \leq 1$ and

$$\mathbf{r}_u \times \mathbf{r}_v = \langle 2v, 2u, 2u \rangle \times \langle 2u, -2v, 2v \rangle = \langle 8uv, 4u^2 - 4v^2, -4u^2 - 4v^2 \rangle, \text{ so}$$

$$\begin{aligned} |\mathbf{r}_u \times \mathbf{r}_v| &= \sqrt{(8uv)^2 + (4u^2 - 4v^2)^2 + (-4u^2 - 4v^2)^2} = \sqrt{64u^2v^2 + 32u^4 + 32v^4} \\ &= \sqrt{32(u^2 + v^2)^2} = 4\sqrt{2}(u^2 + v^2) \end{aligned}$$

[continued]

Then

$$\begin{aligned}\iint_S (x^2 + y^2) dS &= \iint_D [(2uv)^2 + (u^2 - v^2)^2] |\mathbf{r}_u \times \mathbf{r}_v| dA = \iint_D (4u^2v^2 + u^4 - 2u^2v^2 + v^4) \cdot 4\sqrt{2}(u^2 + v^2) dA \\ &= 4\sqrt{2} \iint_D (u^4 + 2u^2v^2 + v^4)(u^2 + v^2) dA = 4\sqrt{2} \iint_D (u^2 + v^2)^3 dA = 4\sqrt{2} \int_0^{2\pi} \int_0^1 (r^2)^3 r dr d\theta \\ &= 4\sqrt{2} \int_0^{2\pi} d\theta \int_0^1 r^7 dr = 4\sqrt{2} [\theta]_0^{2\pi} \left[\frac{1}{8}r^8\right]_0^1 = 4\sqrt{2} \cdot 2\pi \cdot \frac{1}{8} = \sqrt{2}\pi\end{aligned}$$

9. $z = 1 + 2x + 3y$, so $\frac{\partial z}{\partial x} = 2$ and $\frac{\partial z}{\partial y} = 3$. The surface is the graph of a function, so by Formula 4,

$$\begin{aligned}\iint_S x^2 y z dS &= \iint_D x^2 y z \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA = \int_0^3 \int_0^2 x^2 y (1 + 2x + 3y) \sqrt{4 + 9 + 1} dy dx \\ &= \sqrt{14} \int_0^3 \int_0^2 (x^2 y + 2x^3 y + 3x^2 y^2) dy dx = \sqrt{14} \int_0^3 \left[\frac{1}{2}x^2 y^2 + x^3 y^2 + x^2 y^3\right]_{y=0}^{y=2} dx \\ &= \sqrt{14} \int_0^3 (10x^2 + 4x^3) dx = \sqrt{14} \left[\frac{10}{3}x^3 + x^4\right]_0^3 = 171\sqrt{14}\end{aligned}$$

10. The surface S is given by $z = 4 - 2x - 2y$, which intersects the xy -plane in the line $2x + 2y = 4$, so

$D = \{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq 2 - x\}$. The surface is the graph of a function, so by Formula 4,

$$\begin{aligned}\iint_S xz dS &= \iint_D x(4 - 2x - 2y) \sqrt{(-2)^2 + (-2)^2 + 1} dA = 3 \int_0^2 \int_0^{2-x} (4x - 2x^2 - 2xy) dy dx \\ &= 3 \int_0^2 [4xy - 2x^2 y - xy^2]_{y=0}^{y=2-x} dx = 3 \int_0^2 [4x(2-x) - 2x^2(2-x) - x(2-x)^2] dx \\ &= 3 \int_0^2 (x^3 - 4x^2 + 4x) dx = 3 \left[\frac{1}{4}x^4 - \frac{4}{3}x^3 + 2x^2\right]_0^2 = 3 \left(4 - \frac{32}{3} + 8\right) = 4\end{aligned}$$

11. An equation of the plane through the points $(1, 0, 0)$, $(0, -2, 0)$, and $(0, 0, 4)$ is $4x - 2y + z = 4$ (see Example 12.5.5), so

S is the region in the plane $z = 4 - 4x + 2y$ over $D = \{(x, y) \mid 0 \leq x \leq 1, 2x - 2 \leq y \leq 0\}$. Thus, by Formula 4,

$$\begin{aligned}\iint_S x dS &= \iint_D x \sqrt{(-4)^2 + (2)^2 + 1} dA = \sqrt{21} \int_0^1 \int_{2x-2}^0 x dy dx = \sqrt{21} \int_0^1 [xy]_{y=2x-2}^{y=0} dx \\ &= \sqrt{21} \int_0^1 (-2x^2 + 2x) dx = \sqrt{21} \left[-\frac{2}{3}x^3 + x^2\right]_0^1 = \sqrt{21} \left(-\frac{2}{3} + 1\right) = \frac{\sqrt{21}}{3}\end{aligned}$$

12. $z = \frac{2}{3}(x^{3/2} + y^{3/2})$ and by Formula 4,

$$\begin{aligned}\iint_S y dS &= \iint_D y \sqrt{(\sqrt{x})^2 + (\sqrt{y})^2 + 1} dA = \int_0^1 \int_0^1 y \sqrt{x + y + 1} dx dy \\ &= \int_0^1 y \left[\frac{2}{3}(x + y + 1)^{3/2}\right]_{x=0}^{x=1} dy = \int_0^1 \frac{2}{3} y [(y + 2)^{3/2} - (y + 1)^{3/2}] dy\end{aligned}$$

Substituting $u = y + 2$ in the first term and $t = y + 1$ in the second, we have

$$\begin{aligned}\iint_S y dS &= \frac{2}{3} \int_2^3 (u - 2)u^{3/2} du - \frac{2}{3} \int_1^2 (t - 1)t^{3/2} dt = \frac{2}{3} \left[\frac{2}{7}u^{7/2} - \frac{4}{5}u^{5/2}\right]_2^3 - \frac{2}{3} \left[\frac{2}{7}t^{7/2} - \frac{2}{5}t^{5/2}\right]_1^2 \\ &= \frac{2}{3} \left[\frac{2}{7}(3^{7/2} - 2^{7/2}) - \frac{4}{5}(3^{5/2} - 2^{5/2}) - \frac{2}{7}(2^{7/2} - 1) + \frac{2}{5}(2^{5/2} - 1)\right] \\ &= \frac{2}{3} \left(\frac{18}{35}\sqrt{3} + \frac{8}{35}\sqrt{2} - \frac{4}{35}\right) = \frac{4}{105}(9\sqrt{3} + 4\sqrt{2} - 2)\end{aligned}$$

13. Using y and z as parameters, we have $\mathbf{r}(y, z) = (y^2 + z^2)\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, $y^2 + z^2 \leq 1$. Then

$\mathbf{r}_y \times \mathbf{r}_z = (2y\mathbf{i} + \mathbf{j}) \times (2z\mathbf{i} + \mathbf{k}) = \mathbf{i} - 2y\mathbf{j} - 2z\mathbf{k}$ and $|\mathbf{r}_y \times \mathbf{r}_z| = \sqrt{1 + 4y^2 + 4z^2} = \sqrt{1 + 4(y^2 + z^2)}$. Thus, by

Formula 2,

$$\begin{aligned}
 \iint_S z^2 dS &= \iint_{y^2+z^2 \leq 1} z^2 \sqrt{1+4(y^2+z^2)} dA = \int_0^{2\pi} \int_0^1 (r \sin \theta)^2 \sqrt{1+4r^2} r dr d\theta \\
 &= \int_0^{2\pi} \sin^2 \theta d\theta \int_0^1 r^3 \sqrt{1+4r^2} dr \quad \left[\text{let } u = 1+4r^2 \Rightarrow r^2 = \frac{1}{4}(u-1) \text{ and } r dr = \frac{1}{8} du \right] \\
 &= \left[\frac{1}{2}\theta - \frac{1}{4} \sin 2\theta \right]_0^{2\pi} \int_1^5 \frac{1}{4}(u-1) \sqrt{u} \cdot \frac{1}{8} du = \pi \cdot \frac{1}{32} \int_1^5 (u^{3/2} - u^{1/2}) du = \frac{1}{32} \pi \left[\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right]_1^5 \\
 &= \frac{1}{32} \pi \left[\frac{2}{5} (5)^{5/2} - \frac{2}{3} (5)^{3/2} - \frac{2}{5} + \frac{2}{3} \right] = \frac{1}{32} \pi \left(\frac{20}{3} \sqrt{5} + \frac{4}{15} \right) = \frac{1}{120} \pi (25\sqrt{5} + 1)
 \end{aligned}$$

14. Using x and z as parameters, we have $\mathbf{r}(x, z) = x\mathbf{i} + \sqrt{x^2 + z^2}\mathbf{j} + z\mathbf{k}$, $x^2 + z^2 \leq 25$. Then

$$\mathbf{r}_x \times \mathbf{r}_z = \left(\mathbf{i} + \frac{x}{\sqrt{x^2 + z^2}} \mathbf{j} \right) \times \left(\frac{z}{\sqrt{x^2 + z^2}} \mathbf{j} + \mathbf{k} \right) = \frac{x}{\sqrt{x^2 + z^2}} \mathbf{i} - \mathbf{j} + \frac{z}{\sqrt{x^2 + z^2}} \mathbf{k} \text{ and}$$

$$|\mathbf{r}_x \times \mathbf{r}_z| = \sqrt{\frac{x^2}{x^2 + z^2} + 1 + \frac{z^2}{x^2 + z^2}} = \sqrt{\frac{x^2 + z^2}{x^2 + z^2} + 1} = \sqrt{2}. \text{ Thus, by Formula 2,}$$

$$\begin{aligned}
 \iint_S y^2 z^2 dS &= \iint_{x^2+z^2 \leq 25} (x^2 + z^2) z^2 \sqrt{2} dA = \sqrt{2} \int_0^{2\pi} \int_0^5 r^2 (r \sin \theta)^2 r dr d\theta \\
 &= \sqrt{2} \int_0^{2\pi} \sin^2 \theta d\theta \int_0^5 r^5 dr = \sqrt{2} \left[\frac{1}{2}\theta - \frac{1}{4} \sin 2\theta \right]_0^{2\pi} \left[\frac{1}{6} r^6 \right]_0^5 \\
 &= \sqrt{2} (\pi) \cdot \frac{1}{6} (15,625 - 0) = \frac{15,625\sqrt{2}}{6} \pi
 \end{aligned}$$

15. Using x and z as parameters, we have $\mathbf{r}(x, z) = x\mathbf{i} + (x^2 + 4z)\mathbf{j} + z\mathbf{k}$, $0 \leq x \leq 1$, $0 \leq z \leq 1$. Then

$$\mathbf{r}_x \times \mathbf{r}_z = (\mathbf{i} + 2x\mathbf{j}) \times (4\mathbf{j} + \mathbf{k}) = 2x\mathbf{i} - \mathbf{j} + 4\mathbf{k} \text{ and } |\mathbf{r}_x \times \mathbf{r}_z| = \sqrt{4x^2 + 1 + 16} = \sqrt{4x^2 + 17}. \text{ Thus, by Formula 2,}$$

$$\begin{aligned}
 \iint_S x dS &= \int_0^1 \int_0^1 x \sqrt{4x^2 + 17} dz dx = \int_0^1 x \sqrt{4x^2 + 17} dx = \left[\frac{1}{8} \cdot \frac{2}{3} (4x^2 + 17)^{3/2} \right]_0^1 \\
 &= \frac{1}{12} (21^{3/2} - 17^{3/2}) = \frac{1}{12} (21\sqrt{21} - 17\sqrt{17}) = \frac{7}{4}\sqrt{21} - \frac{17}{12}\sqrt{17}
 \end{aligned}$$

16. The sphere intersects the cone in the circle $x^2 + y^2 = \frac{1}{2}$, $z = \frac{1}{\sqrt{2}}$, so S is the portion of the sphere where $z \geq \frac{1}{\sqrt{2}}$.

Using spherical coordinates to parametrize the sphere we have $\mathbf{r}(\phi, \theta) = \sin \phi \cos \theta \mathbf{i} + \sin \phi \sin \theta \mathbf{j} + \cos \phi \mathbf{k}$, and

$|\mathbf{r}_\phi \times \mathbf{r}_\theta| = \sin \phi$ (as in Example 1). The portion where $z \geq \frac{1}{\sqrt{2}}$ corresponds to $0 \leq \phi \leq \frac{\pi}{4}$, $0 \leq \theta \leq 2\pi$, so by Formula 2,

$$\begin{aligned}
 \iint_S y^2 dS &= \int_0^{2\pi} \int_0^{\pi/4} (\sin \phi \sin \theta)^2 (\sin \phi) d\phi d\theta = \int_0^{2\pi} \sin^2 \theta d\theta \int_0^{\pi/4} \sin^3 \phi d\phi = \int_0^{2\pi} \sin^2 \theta d\theta \int_0^{\pi/4} (1 - \cos^2 \phi) \sin \phi d\phi \\
 &= \left[\frac{1}{2}\theta - \frac{1}{4} \sin 2\theta \right]_0^{2\pi} \left[\frac{1}{3} \cos^3 \phi - \cos \phi \right]_0^{\pi/4} = \pi \left(\frac{\sqrt{2}}{12} - \frac{\sqrt{2}}{2} - \frac{1}{3} + 1 \right) = \left(\frac{2}{3} - \frac{5\sqrt{2}}{12} \right) \pi
 \end{aligned}$$

17. Using spherical coordinates to parametrize the sphere (see Example 16.6.4), we have

$\mathbf{r}(\phi, \theta) = 2 \sin \phi \cos \theta \mathbf{i} + 2 \sin \phi \sin \theta \mathbf{j} + 2 \cos \phi \mathbf{k}$ and $|\mathbf{r}_\phi \times \mathbf{r}_\theta| = 4 \sin \phi$ (see Example 16.6.10). Here S is the portion of the sphere corresponding to $0 \leq \phi \leq \pi/2$, so by Formula 2,

$$\begin{aligned}
 \iint_S (x^2 z + y^2 z) dS &= \iint_S (x^2 + y^2) z dS = \int_0^{2\pi} \int_0^{\pi/2} (4 \sin^2 \phi) (2 \cos \phi) (4 \sin \phi) d\phi d\theta \\
 &= 32 \int_0^{2\pi} d\theta \int_0^{\pi/2} \sin^3 \phi \cos \phi d\phi = 32 (2\pi) \left[\frac{1}{4} \sin^4 \phi \right]_0^{\pi/2} = 16\pi (1 - 0) = 16\pi
 \end{aligned}$$

18. S is given by $\mathbf{r}(u, v) = \cos v \mathbf{i} + u \mathbf{j} + \sin v \mathbf{k}$, $0 \leq u \leq 2$, $0 \leq v \leq \pi$ (see Example 16.6.5). Then

$$\mathbf{r}_u \times \mathbf{r}_v = \mathbf{j} \times (-\sin v \mathbf{i} + \cos v \mathbf{k}) = \cos v \mathbf{i} + \sin v \mathbf{k} \text{ and } |\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{\cos^2 v + \sin^2 v} = 1, \text{ so}$$

$$\begin{aligned} \iint_S (x + y + z) dS &= \int_0^\pi \int_0^2 (\cos v + u + \sin v)(1) du dv = \int_0^\pi \left[u(\cos v + \sin v) + \frac{1}{2}u^2 \right]_{u=0}^{u=2} dv \\ &= \int_0^\pi (2 \cos v + 2 \sin v + 2) dv = [2 \sin v - 2 \cos v + 2v]_0^\pi = 2 + 2\pi + 2 = 4 + 2\pi \end{aligned}$$

19. Here S consists of three surfaces: S_1 , the lateral surface of the cylinder; S_2 , the front formed by the plane $x + y = 5$; and the back, S_3 , in the plane $x = 0$.

On S_1 : the surface is given by $\mathbf{r}(u, v) = u \mathbf{i} + 3 \cos v \mathbf{j} + 3 \sin v \mathbf{k}$, $0 \leq v \leq 2\pi$ (see Example 16.6.5), and

$$0 \leq x \leq 5 - y \Rightarrow 0 \leq u \leq 5 - 3 \cos v. \text{ Then } \mathbf{r}_u \times \mathbf{r}_v = -3 \cos v \mathbf{j} - 3 \sin v \mathbf{k} \text{ and}$$

$$|\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{9 \cos^2 v + 9 \sin^2 v} = 3, \text{ so}$$

$$\begin{aligned} \iint_{S_1} xz dS &= \int_0^{2\pi} \int_0^{5-3\cos v} u(3 \sin v)(3) du dv = 9 \int_0^{2\pi} \left[\frac{1}{2}u^2 \right]_{u=0}^{u=5-3\cos v} \sin v dv \\ &= \frac{9}{2} \int_0^{2\pi} (5 - 3 \cos v)^2 \sin v dv = \frac{9}{2} \left[\frac{1}{9}(5 - 3 \cos v)^3 \right]_0^{2\pi} = 0. \end{aligned}$$

On S_2 : $\mathbf{r}(y, z) = (5 - y) \mathbf{i} + y \mathbf{j} + z \mathbf{k}$ and $|\mathbf{r}_y \times \mathbf{r}_z| = |\mathbf{i} + \mathbf{j}| = \sqrt{2}$, where $y^2 + z^2 \leq 9$ and

$$\begin{aligned} \iint_{S_2} xz dS &= \iint_{y^2+z^2 \leq 9} (5 - y)z \sqrt{2} dA = \sqrt{2} \int_0^{2\pi} \int_0^3 (5 - r \cos \theta)(r \sin \theta) r dr d\theta \\ &= \sqrt{2} \int_0^{2\pi} \int_0^3 (5r^2 - r^3 \cos \theta)(\sin \theta) dr d\theta = \sqrt{2} \int_0^{2\pi} \left[\frac{5}{3}r^3 - \frac{1}{4}r^4 \cos \theta \right]_{r=0}^{r=3} \sin \theta d\theta \\ &= \sqrt{2} \int_0^{2\pi} \left(45 - \frac{81}{4} \cos \theta \right) \sin \theta d\theta = \sqrt{2} \left(\frac{4}{81} \right) \cdot \frac{1}{2} \left(45 - \frac{81}{4} \cos \theta \right)^2 \Big|_0^{2\pi} = 0 \end{aligned}$$

On S_3 : $x = 0$ so $\iint_{S_3} xz dS = 0$. Hence $\iint_S xz dS = 0 + 0 + 0 = 0$.

20. Let S_1 be the lateral surface, S_2 the top disk, and S_3 the bottom disk.

On S_1 : $\mathbf{r}(\theta, z) = 3 \cos \theta \mathbf{i} + 3 \sin \theta \mathbf{j} + z \mathbf{k}$, $0 \leq \theta \leq 2\pi$, $0 \leq z \leq 2$, $|\mathbf{r}_\theta \times \mathbf{r}_z| = 3$,

$$\iint_{S_1} (x^2 + y^2 + z^2) dS = \int_0^{2\pi} \int_0^2 (9 + z^2) 3 dz d\theta = 2\pi(54 + 8) = 124\pi.$$

On S_2 : $\mathbf{r}(\theta, r) = r \cos \theta \mathbf{i} + r \sin \theta \mathbf{j} + 2 \mathbf{k}$, $0 \leq r \leq 3$, $0 \leq \theta \leq 2\pi$, $|\mathbf{r}_\theta \times \mathbf{r}_r| = r$,

$$\iint_{S_2} (x^2 + y^2 + z^2) dS = \int_0^{2\pi} \int_0^3 (r^2 + 4) r dr d\theta = 2\pi \left(\frac{81}{4} + 18 \right) = \frac{153}{2}\pi.$$

On S_3 : $\mathbf{r}(\theta, r) = r \cos \theta \mathbf{i} + r \sin \theta \mathbf{j}$, $0 \leq r \leq 3$, $0 \leq \theta \leq 2\pi$, $|\mathbf{r}_\theta \times \mathbf{r}_r| = r$,

$$\iint_{S_3} (x^2 + y^2 + z^2) dS = \int_0^{2\pi} \int_0^3 (r^2 + 0) r dr d\theta = 2\pi \left(\frac{81}{4} \right) = \frac{81}{2}\pi.$$

$$\text{Hence } \iint_S (x^2 + y^2 + z^2) dS = 124\pi + \frac{153}{2}\pi + \frac{81}{2}\pi = 241\pi.$$

21. From Exercise 5, $\mathbf{r}(u, v) = (u + v) \mathbf{i} + (u - v) \mathbf{j} + (1 + 2u + v) \mathbf{k}$, $0 \leq u \leq 2$, $0 \leq v \leq 1$, and $\mathbf{r}_u \times \mathbf{r}_v = 3 \mathbf{i} + \mathbf{j} - 2 \mathbf{k}$.

Then

$$\begin{aligned} \mathbf{F}(\mathbf{r}(u, v)) &= (1 + 2u + v)e^{(u+v)(u-v)} \mathbf{i} - 3(1 + 2u + v)e^{(u+v)(u-v)} \mathbf{j} + (u + v)(u - v) \mathbf{k} \\ &= (1 + 2u + v)e^{u^2-v^2} \mathbf{i} - 3(1 + 2u + v)e^{u^2-v^2} \mathbf{j} + (u^2 - v^2) \mathbf{k} \end{aligned}$$

[continued]

Because the z -component of $\mathbf{r}_u \times \mathbf{r}_v$ is negative we use $-(\mathbf{r}_u \times \mathbf{r}_v)$ in Formula 9 for the upward orientation:

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_D \mathbf{F} \cdot (-\mathbf{r}_u \times \mathbf{r}_v) dA = \int_0^1 \int_0^2 \left[-3(1+2u+v)e^{u^2-v^2} + 3(1+2u+v)e^{u^2-v^2} + 2(u^2-v^2) \right] du dv \\ &= \int_0^1 \int_0^2 2(u^2-v^2) du dv = 2 \int_0^1 \left[\frac{1}{3}u^3 - uv^2 \right]_{u=0}^{u=2} dv = 2 \int_0^1 \left(\frac{8}{3} - 2v^2 \right) dv \\ &= 2 \left[\frac{8}{3}v - \frac{2}{3}v^3 \right]_0^1 = 2 \left(\frac{8}{3} - \frac{2}{3} \right) = 4\end{aligned}$$

22. $\mathbf{r}(u, v) = \langle u \cos v, u \sin v, v \rangle$, $0 \leq u \leq 1$, $0 \leq v \leq \pi$ and

$$\mathbf{r}_u \times \mathbf{r}_v = \langle \cos v, \sin v, 0 \rangle \times \langle -u \sin v, u \cos v, 1 \rangle = \langle \sin v, -\cos v, u \rangle. \text{ Since } \mathbf{F}(x, y, z) = z \mathbf{i} + y \mathbf{j} + x \mathbf{k},$$

$\mathbf{F}(\mathbf{r}(u, v)) = v \mathbf{i} + u \sin v \mathbf{j} + u \cos v \mathbf{k}$, and by Formula 9,

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) dA = \int_0^1 \int_0^\pi (v \sin v - u \sin v \cos v + u^2 \cos v) dv du \\ &= \int_0^1 \left[\sin v - v \cos v - \frac{1}{2}u \sin^2 v + u^2 \sin v \right]_{v=0}^{v=\pi} du = \int_0^1 \pi du = \pi u \Big|_0^1 = \pi\end{aligned}$$

23. $\mathbf{F}(x, y, z) = xy \mathbf{i} + yz \mathbf{j} + zx \mathbf{k}$, $z = g(x, y) = 4 - x^2 - y^2$, and D is the square $[0, 1] \times [0, 1]$, so by Equation 10

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_D [-xy(-2x) - yz(-2y) + zx] dA = \int_0^1 \int_0^1 [2x^2y + 2y^2(4 - x^2 - y^2) + x(4 - x^2 - y^2)] dy dx \\ &= \int_0^1 \left[x^2y^2 + \frac{8}{3}y^3 - \frac{2}{3}x^2y^3 - \frac{2}{5}y^5 + 4xy - x^3y - \frac{1}{3}xy^3 \right]_{y=0}^{y=1} dx \\ &= \int_0^1 \left(\frac{1}{3}x^2 + \frac{11}{3}x - x^3 + \frac{34}{15} \right) dx = \left[\frac{1}{9}x^3 + \frac{11}{6}x^2 - \frac{1}{4}x^4 + \frac{34}{15}x \right]_0^1 = \frac{713}{180}\end{aligned}$$

24. $\mathbf{F}(x, y, z) = -x \mathbf{i} - y \mathbf{j} + z^3 \mathbf{k}$, $z = g(x, y) = \sqrt{x^2 + y^2}$, and D is the annular region $\{(x, y) \mid 1 \leq x^2 + y^2 \leq 9\}$. Since S has downward orientation, we have, by Equation 10,

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= - \iint_D \left[-(-x) \left(\frac{x}{\sqrt{x^2 + y^2}} \right) - (-y) \left(\frac{y}{\sqrt{x^2 + y^2}} \right) + z^3 \right] dA \\ &= - \iint_D \left[\frac{x^2 + y^2}{\sqrt{x^2 + y^2}} + \left(\sqrt{x^2 + y^2} \right)^3 \right] dA = - \int_0^{2\pi} \int_1^3 \left(\frac{r^2}{r} + r^3 \right) r dr d\theta \\ &= - \int_0^{2\pi} d\theta \int_1^3 (r^2 + r^4) dr = - \left[\theta \right]_0^{2\pi} \left[\frac{1}{3}r^3 + \frac{1}{5}r^5 \right]_1^3 \\ &= -2\pi \left(9 + \frac{243}{5} - \frac{1}{3} - \frac{1}{5} \right) = -\frac{1712}{15}\pi\end{aligned}$$

25. $\mathbf{F}(x, y, z) = x \mathbf{i} + y \mathbf{j} + z^2 \mathbf{k}$, and using spherical coordinates, S is given by $x = \sin \phi \cos \theta$, $y = \sin \phi \sin \theta$, $z = \cos \phi$,

$$0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \pi. \quad \mathbf{F}(\mathbf{r}(\phi, \theta)) = (\sin \phi \cos \theta) \mathbf{i} + (\sin \phi \sin \theta) \mathbf{j} + (\cos^2 \phi) \mathbf{k} \text{ and, from Example 4,}$$

$$\mathbf{r}_\phi \times \mathbf{r}_\theta = \sin^2 \phi \cos \theta \mathbf{i} + \sin^2 \phi \sin \theta \mathbf{j} + \sin \phi \cos \phi \mathbf{k}. \text{ Thus,}$$

$$\mathbf{F}(\mathbf{r}(\phi, \theta)) \cdot (\mathbf{r}_\phi \times \mathbf{r}_\theta) = \sin^3 \phi \cos^2 \theta + \sin^3 \phi \sin^2 \theta + \sin \phi \cos^3 \phi = \sin^3 \phi + \sin \phi \cos^3 \phi$$

and

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_D [\mathbf{F}(\mathbf{r}(\phi, \theta)) \cdot (\mathbf{r}_\phi \times \mathbf{r}_\theta)] dA = \int_0^{2\pi} \int_0^\pi (\sin^3 \phi + \sin \phi \cos^3 \phi) d\phi d\theta \\ &= \int_0^{2\pi} d\theta \int_0^\pi (1 - \cos^2 \phi + \cos^3 \phi) \sin \phi d\phi = (2\pi) \left[-\cos \phi + \frac{1}{3} \cos^3 \phi - \frac{1}{4} \cos^4 \phi \right]_0^\pi \\ &= 2\pi \left(1 - \frac{1}{3} - \frac{1}{4} + 1 - \frac{1}{3} + \frac{1}{4} \right) = \frac{8}{3}\pi\end{aligned}$$

26. $\mathbf{F}(x, y, z) = y\mathbf{i} - x\mathbf{j} + 2z\mathbf{k}$, $z = g(x, y) = \sqrt{4 - x^2 - y^2}$ and D is the disk $\{(x, y) \mid x^2 + y^2 \leq 4\}$. S has downward orientation, so by Equation 10,

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= -\iint_D \left[-y \cdot \frac{1}{2}(4 - x^2 - y^2)^{-1/2}(-2x) - (-x) \cdot \frac{1}{2}(4 - x^2 - y^2)^{-1/2}(-2y) + 2z \right] dA \\ &= -\iint_D \left(\frac{xy}{\sqrt{4 - x^2 - y^2}} - \frac{xy}{\sqrt{4 - x^2 - y^2}} + 2\sqrt{4 - x^2 - y^2} \right) dA \\ &= -\iint_D 2\sqrt{4 - x^2 - y^2} dA = -2 \int_0^{2\pi} \int_0^2 \sqrt{4 - r^2} r dr d\theta = -2 \int_0^{2\pi} d\theta \int_0^2 r\sqrt{4 - r^2} dr \\ &= -2(2\pi) \left[-\frac{1}{2} \cdot \frac{2}{3}(4 - r^2)^{3/2} \right]_0^2 = -4\pi \left[0 + \frac{1}{3}(4)^{3/2} \right] = -4\pi \cdot \frac{8}{3} = -\frac{32}{3}\pi\end{aligned}$$

27. $\mathbf{F}(x, y, z) = y\mathbf{j} - z\mathbf{k}$. Let S_1 be the paraboloid $y = x^2 + z^2$, $0 \leq y \leq 1$ and S_2 the disk $x^2 + z^2 \leq 1$, $y = 1$. On S_1 we have $\mathbf{r}(x, z) = x\mathbf{i} + (x^2 + z^2)\mathbf{j} + z\mathbf{k}$. Since S is a closed surface, we use the outward orientation.

On S_1 : $\mathbf{F}(\mathbf{r}(x, z)) = (x^2 + z^2)\mathbf{j} - z\mathbf{k}$ and $\mathbf{r}_x \times \mathbf{r}_z = 2x\mathbf{i} - \mathbf{j} + 2z\mathbf{k}$ (since the $\mathbf{j}$ -component must be negative on S_1).

Then by Formula 9,

$$\begin{aligned}\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} &= \iint_{x^2 + z^2 \leq 1} [0 - (x^2 + z^2) - 2z^2] dA = -\int_0^{2\pi} \int_0^1 (r^2 + 2r^2 \sin^2 \theta) r dr d\theta \\ &= -\int_0^{2\pi} \int_0^1 r^3 (1 + 2 \sin^2 \theta) dr d\theta = -\int_0^{2\pi} (1 + 1 - \cos 2\theta) d\theta \int_0^1 r^3 dr \\ &= -\left[2\theta - \frac{1}{2} \sin 2\theta \right]_0^{2\pi} \left[\frac{1}{4} r^4 \right]_0^1 = -4\pi \cdot \frac{1}{4} = -\pi\end{aligned}$$

On S_2 : $\mathbf{F}(\mathbf{r}(x, z)) = \mathbf{j} - z\mathbf{k}$ and $\mathbf{r}_z \times \mathbf{r}_x = \mathbf{j}$. Then $\iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \iint_{x^2 + z^2 \leq 1} (1) dA = \pi$.

Hence $\iint_S \mathbf{F} \cdot d\mathbf{S} = -\pi + \pi = 0$.

28. $\mathbf{F}(x, y, z) = yz\mathbf{i} + zx\mathbf{j} + xy\mathbf{k}$, $z = g(x, y) = x \sin y$, and D is the rectangle $[0, 2] \times [0, \pi]$, so by Equation 10

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_D [-yz(\sin y) - zx(x \cos y) + xy] dA = \int_0^\pi \int_0^2 (-xy \sin^2 y - x^3 \sin y \cos y + xy) dx dy \\ &= \int_0^\pi \left[-\frac{1}{2}x^2 y \sin^2 y - \frac{1}{4}x^4 \sin y \cos y + \frac{1}{2}x^2 y \right]_{x=0}^{x=2} dy \\ &= \int_0^\pi (-2y \sin^2 y - 4 \sin y \cos y + 2y) dy \quad [\text{integrate by parts in the first term}] \\ &= \left[\left(-\frac{1}{2}y^2 + \frac{1}{2}y \sin 2y + \frac{1}{4} \cos 2y \right) - 2 \sin^2 y + y^2 \right]_0^\pi = -\frac{1}{2}\pi^2 + \frac{1}{4} + \pi^2 - \frac{1}{4} = \frac{1}{2}\pi^2\end{aligned}$$

29. Here S consists of the six faces of the cube as labeled in the figure. On S_1 :

$$\mathbf{F} = \mathbf{i} + 2y\mathbf{j} + 3z\mathbf{k}, \mathbf{r}_y \times \mathbf{r}_z = \mathbf{i} \text{ and } \iint_{S_1} \mathbf{F} \cdot d\mathbf{S} = \int_{-1}^1 \int_{-1}^1 dy dz = 4;$$

$$S_2: \mathbf{F} = x\mathbf{i} + 2\mathbf{j} + 3z\mathbf{k}, \mathbf{r}_z \times \mathbf{r}_x = \mathbf{j} \text{ and } \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \int_{-1}^1 \int_{-1}^1 2 dx dz = 8;$$

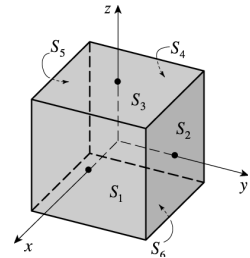
$$S_3: \mathbf{F} = x\mathbf{i} + 2y\mathbf{j} + 3\mathbf{k}, \mathbf{r}_x \times \mathbf{r}_y = \mathbf{k} \text{ and } \iint_{S_3} \mathbf{F} \cdot d\mathbf{S} = \int_{-1}^1 \int_{-1}^1 3 dx dy = 12;$$

$$S_4: \mathbf{F} = -\mathbf{i} + 2y\mathbf{j} + 3z\mathbf{k}, \mathbf{r}_z \times \mathbf{r}_y = -\mathbf{i} \text{ and } \iint_{S_4} \mathbf{F} \cdot d\mathbf{S} = 4;$$

$$S_5: \mathbf{F} = x\mathbf{i} - 2\mathbf{j} + 3z\mathbf{k}, \mathbf{r}_x \times \mathbf{r}_z = -\mathbf{j} \text{ and } \iint_{S_5} \mathbf{F} \cdot d\mathbf{S} = 8;$$

$$S_6: \mathbf{F} = x\mathbf{i} + 2y\mathbf{j} - 3\mathbf{k}, \mathbf{r}_y \times \mathbf{r}_x = -\mathbf{k} \text{ and } \iint_{S_6} \mathbf{F} \cdot d\mathbf{S} = \int_{-1}^1 \int_{-1}^1 3 dx dy = 12.$$

$$\text{Hence } \iint_S \mathbf{F} \cdot d\mathbf{S} = \sum_{i=1}^6 \iint_{S_i} \mathbf{F} \cdot d\mathbf{S} = 48.$$



30. $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + 5\mathbf{k}$. Here S consists of three surfaces: S_1 , the lateral surface of the cylinder $x^2 + z^2 = 1$; S_2 , the front formed by the plane $x + y = 2$; and the back, S_3 , in the plane $y = 0$.

On S_1 : $\mathbf{r}(\theta, y) = \sin\theta\mathbf{i} + y\mathbf{j} + \cos\theta\mathbf{k}$. $\mathbf{F}(\mathbf{r}(\theta, y)) = \sin\theta\mathbf{i} + y\mathbf{j} + 5\mathbf{k}$ and $\mathbf{r}_\theta \times \mathbf{r}_y = \sin\theta\mathbf{i} + \cos\theta\mathbf{k} \Rightarrow$

$$\begin{aligned}\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} &= \int_0^{2\pi} \int_0^{2-\sin\theta} (\sin^2\theta + 5\cos\theta) dy d\theta \\ &= \int_0^{2\pi} (2\sin^2\theta + 10\cos\theta - \sin^3\theta - 5\sin\theta\cos\theta) d\theta = 2\pi\end{aligned}$$

On S_2 : $\mathbf{r}(x, z) = x\mathbf{i} + (2-x)\mathbf{j} + z\mathbf{k}$. $\mathbf{F}(\mathbf{r}(x, z)) = x\mathbf{i} + (2-x)\mathbf{j} + 5\mathbf{k}$ and $\mathbf{r}_x \times \mathbf{r}_z = \mathbf{i} + \mathbf{j}$.

$$\iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \iint_{x^2+z^2 \leq 1} [x + (2-x)] dA = 2\pi$$

On S_3 : $\mathbf{F}(\mathbf{r}(x, z)) = x\mathbf{i} + 5\mathbf{k}$. The surface is oriented in the negative y -direction so that $\mathbf{n} = -\mathbf{j}$ and by (8),

$$\iint_{S_3} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_3} \mathbf{F} \cdot \mathbf{n} dS = 0. \text{ Hence, } \iint_S \mathbf{F} \cdot d\mathbf{S} = 4\pi.$$

31. $\mathbf{F}(x, y, z) = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$. Here S consists of four surfaces: S_1 , the top surface (a portion of the circular cylinder $y^2 + z^2 = 1$); S_2 , the bottom surface (a portion of the xy -plane); S_3 , the front half-disk in the plane $x = 2$, and S_4 , the back half-disk in the plane $x = 0$.

On S_1 : The surface is $z = \sqrt{1-y^2}$ for $0 \leq x \leq 2$, $-1 \leq y \leq 1$ with upward orientation, so by Equation 10,

$$\begin{aligned}\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} &= \int_0^2 \int_{-1}^1 \left[-x^2(0) - y^2 \left(-\frac{y}{\sqrt{1-y^2}} \right) + z^2 \right] dy dx = \int_0^2 \int_{-1}^1 \left(\frac{y^3}{\sqrt{1-y^2}} + 1 - y^2 \right) dy dx \\ &= \int_0^2 \left[-\sqrt{1-y^2} + \frac{1}{3}(1-y^2)^{3/2} + y - \frac{1}{3}y^3 \right]_{y=-1}^{y=1} dx = \int_0^2 \frac{4}{3} dx = \frac{8}{3}\end{aligned}$$

On S_2 : The surface is $z = 0$ for $0 \leq x \leq 2$, $-1 \leq y \leq 1$ with downward orientation, so that $\mathbf{n} = -\mathbf{k}$ and by (8),

$$\iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_2} \mathbf{F} \cdot \mathbf{n} dS = \int_0^2 \int_{-1}^1 (-z^2) dy dx = \int_0^2 \int_{-1}^1 (0) dy dx = 0$$

On S_3 : The surface is $x = 2$ for $-1 \leq y \leq 1$, $0 \leq z \leq \sqrt{1-y^2}$, oriented in the positive x -direction, so that $\mathbf{n} = \mathbf{i}$ and by (8),

$$\iint_{S_3} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_3} \mathbf{F} \cdot \mathbf{n} dS = \int_{-1}^1 \int_0^{\sqrt{1-y^2}} x^2 dz dy = \int_{-1}^1 \int_0^{\sqrt{1-y^2}} 4 dz dy = 4A(S_3) = 2\pi$$

On S_4 : The surface is $x = 0$ for $-1 \leq y \leq 1$, $0 \leq z \leq \sqrt{1-y^2}$, oriented in the negative x -direction, so that $\mathbf{n} = -\mathbf{i}$ and by (8),

$$\iint_{S_4} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_4} \mathbf{F} \cdot \mathbf{n} dS = \int_{-1}^1 \int_0^{\sqrt{1-y^2}} (-x^2) dz dy = \int_{-1}^1 \int_0^{\sqrt{1-y^2}} (0) dz dy = 0$$

Thus, $\iint_S \mathbf{F} \cdot d\mathbf{S} = \frac{8}{3} + 0 + 2\pi + 0 = 2\pi + \frac{8}{3}$.

32. $\mathbf{F}(x, y, z) = y\mathbf{i} + (z-y)\mathbf{j} + x\mathbf{k}$. Here S consists of four surfaces: S_1 , the triangular face with vertices $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$; S_2 , the face of the tetrahedron in the xy -plane; S_3 , the face in the xz -plane; and S_4 , the face in the yz -plane.

On S_1 : The face is the portion of the plane $z = 1 - x - y$ for $0 \leq x \leq 1$, $0 \leq y \leq 1 - x$ with upward orientation,

so by Equation 10,

$$\begin{aligned}\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} &= \int_0^1 \int_0^{1-x} [-y(-1) - (z-y)(-1) + x] dy dx = \int_0^1 \int_0^{1-x} (z+x) dy dx = \int_0^1 \int_0^{1-x} (1-y) dy dx \\ &= \int_0^1 \left[y - \frac{1}{2}y^2 \right]_{y=0}^{y=1-x} dx = \frac{1}{2} \int_0^1 (1-x^2) dx = \frac{1}{2} \left[x - \frac{1}{3}x^3 \right]_0^1 = \frac{1}{3}\end{aligned}$$

On S_2 : The surface is $z = 0$ for $0 \leq x \leq 1$, $0 \leq y \leq 1-x$ with downward orientation, so that $\mathbf{n} = -\mathbf{k}$ and by (8),

$$\iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_2} \mathbf{F} \cdot \mathbf{n} dS = \int_0^1 \int_0^{1-x} (-x) dy dx = -\int_0^1 x(1-x) dx = -\left[\frac{1}{2}x^2 - \frac{1}{3}x^3 \right]_0^1 = -\frac{1}{6}$$

On S_3 : The surface is $y = 0$ for $0 \leq x \leq 1$, $0 \leq z \leq 1-x$, oriented in the negative y -direction, so that $\mathbf{n} = -\mathbf{j}$ and by (8),

$$\begin{aligned}\iint_{S_3} \mathbf{F} \cdot d\mathbf{S} &= \iint_{S_3} \mathbf{F} \cdot \mathbf{n} dS = \int_0^1 \int_0^{1-x} -(z-y) dz dx = -\int_0^1 \int_0^{1-x} z dz dx = -\int_0^1 \left[\frac{1}{2}z^2 \right]_{z=0}^{z=1-x} dx \\ &= -\frac{1}{2} \int_0^1 (1-x)^2 dx = \frac{1}{6} [(1-x)^3]_0^1 = -\frac{1}{6}\end{aligned}$$

On S_4 : The surface is $x = 0$ for $0 \leq y \leq 1$, $0 \leq z \leq 1-y$, oriented in the negative x -direction, so that $\mathbf{n} = -\mathbf{i}$ and by (8),

$$\iint_{S_4} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_4} \mathbf{F} \cdot \mathbf{n} dS = \int_0^1 \int_0^{1-y} (-y) dz dy = -\int_0^1 y(1-y) dy = -\left[\frac{1}{2}y^2 - \frac{1}{3}y^3 \right]_0^1 = -\frac{1}{6}$$

Thus, $\iint_S \mathbf{F} \cdot d\mathbf{S} = \frac{1}{3} - \frac{1}{6} - \frac{1}{6} - \frac{1}{6} = -\frac{1}{6}$.

33. $z = xe^y \Rightarrow \partial z / \partial x = e^y$, $\partial z / \partial y = xe^y$, so by Formula 4, a CAS gives

$$\iint_S (x^2 + y^2 + z^2) dS = \int_0^1 \int_0^1 (x^2 + y^2 + x^2 e^{2y}) \sqrt{e^{2y} + x^2 e^{2y} + 1} dx dy \approx 4.5822.$$

34. $z = x^2 y^2 \Rightarrow \partial z / \partial x = 2xy^2$, $\partial z / \partial y = 2x^2 y$, so by Formula 4, a CAS gives

$$\begin{aligned}\iint_S xyz dS &= \int_0^2 \int_0^1 xy(x^2 y^2) \sqrt{(2xy^2)^2 + (2x^2 y)^2 + 1} dx dy \\ &= \int_0^2 \int_0^1 x^3 y^3 \sqrt{4x^2 y^4 + 4x^4 y^2 + 1} dx dy = -\frac{151}{33} - \frac{1}{220} \sqrt{3} \pi + \frac{1977}{176} \ln 7 - \frac{9891}{880} \ln 3 + \frac{3}{440} \sqrt{3} \tan^{-1} \frac{5}{\sqrt{3}}\end{aligned}$$

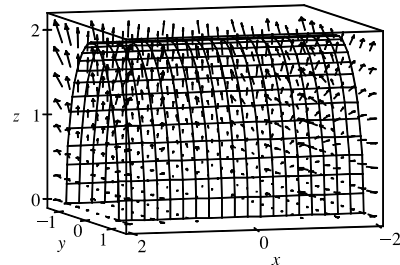
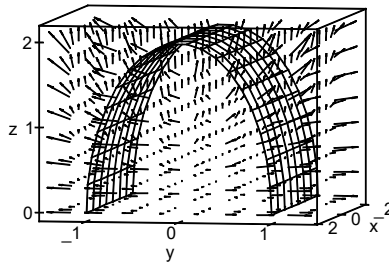
35. We use Formula 4 with $z = 3 - 2x^2 - y^2 \Rightarrow \partial z / \partial x = -4x$, $\partial z / \partial y = -2y$. The boundaries of the region

$3 - 2x^2 - y^2 \geq 0$ are $-\sqrt{\frac{3}{2}} \leq x \leq \sqrt{\frac{3}{2}}$ and $-\sqrt{3 - 2x^2} \leq y \leq \sqrt{3 - 2x^2}$, so we use a CAS (with precision reduced to seven or fewer digits; otherwise the calculation may take a long time) to calculate

$$\iint_S x^2 y^2 z^2 dS = \int_{-\sqrt{3/2}}^{\sqrt{3/2}} \int_{-\sqrt{3-2x^2}}^{\sqrt{3-2x^2}} x^2 y^2 (3 - 2x^2 - y^2)^2 \sqrt{16x^2 + 4y^2 + 1} dy dx \approx 3.4895$$

36. The flux of $\mathbf{F}$ across S is given by $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} dS$. Now on S , $z = g(x, y) = 2\sqrt{1-y^2}$, so $\partial g / \partial x = 0$ and $\partial g / \partial y = -2y(1-y^2)^{-1/2}$. Therefore, by Equation 10,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \int_{-2}^2 \int_{-1}^1 \left(-x^2 y \left[-2y(1-y^2)^{-1/2} \right] + \left[2\sqrt{1-y^2} \right]^2 e^{x/5} \right) dy dx = \frac{1}{3} (16\pi + 80e^{2/5} - 80e^{-2/5})$$



37. If S is given by $y = h(x, z)$, then S is also the level surface $f(x, y, z) = y - h(x, z) = 0$.

$\mathbf{n} = \frac{\nabla f(x, y, z)}{|\nabla f(x, y, z)|} = \frac{-h_x \mathbf{i} + \mathbf{j} - h_z \mathbf{k}}{\sqrt{h_x^2 + 1 + h_z^2}}$, and $-\mathbf{n}$ is the unit normal that points to the left. Now we proceed as in the derivation of (10), using Formula 4 to evaluate

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot (-\mathbf{n}) dS = \iint_D (P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) \cdot \frac{\frac{\partial h}{\partial x} \mathbf{i} - \mathbf{j} + \frac{\partial h}{\partial z} \mathbf{k}}{\sqrt{\left(\frac{\partial h}{\partial x}\right)^2 + 1 + \left(\frac{\partial h}{\partial z}\right)^2}} \sqrt{\left(\frac{\partial h}{\partial x}\right)^2 + 1 + \left(\frac{\partial h}{\partial z}\right)^2} dA$$

where D is the projection of S onto the xz -plane. Therefore $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \left(P \frac{\partial h}{\partial x} - Q + R \frac{\partial h}{\partial z} \right) dA$.

38. If S is given by $x = k(y, z)$, then S is also the level surface $f(x, y, z) = x - k(y, z) = 0$.

$\mathbf{n} = \frac{\nabla f(x, y, z)}{|\nabla f(x, y, z)|} = \frac{\mathbf{i} - k_y \mathbf{j} - k_z \mathbf{k}}{\sqrt{1 + k_y^2 + k_z^2}}$, and since the x -component is positive this is the unit normal that points forward.

Now we proceed as in the derivation of (10), using Formula 4 for

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_D (P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) \cdot \frac{\mathbf{i} - \frac{\partial k}{\partial y} \mathbf{j} - \frac{\partial k}{\partial z} \mathbf{k}}{\sqrt{1 + \left(\frac{\partial k}{\partial y}\right)^2 + \left(\frac{\partial k}{\partial z}\right)^2}} \sqrt{1 + \left(\frac{\partial k}{\partial y}\right)^2 + \left(\frac{\partial k}{\partial z}\right)^2} dA$$

where D is the projection of S onto the yz -plane. Therefore $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \left(P - Q \frac{\partial k}{\partial y} - R \frac{\partial k}{\partial z} \right) dA$.

39. $m = \iint_S K dS = K \cdot 4\pi\left(\frac{1}{2}a^2\right) = 2\pi a^2 K$; by symmetry $M_{xz} = M_{yz} = 0$, and

$$M_{xy} = \iint_S zK dS = K \int_0^{2\pi} \int_0^{\pi/2} (a \cos \phi)(a^2 \sin \phi) d\phi d\theta = 2\pi K a^3 \left[-\frac{1}{4} \cos 2\phi\right]_0^{\pi/2} = \pi K a^3.$$

Hence $(\bar{x}, \bar{y}, \bar{z}) = (0, 0, \frac{1}{2}a)$.

40. S is given by $\mathbf{r}(x, y) = x\mathbf{i} + y\mathbf{j} + \sqrt{x^2 + y^2}\mathbf{k}$, $|\mathbf{r}_x \times \mathbf{r}_y| = \sqrt{1 + \frac{x^2 + y^2}{x^2 + y^2}} = \sqrt{2}$ so

$$\begin{aligned} m &= \iint_S (10 - z) dS = \iint_S \left(10 - \sqrt{x^2 + y^2}\right) dS = \iint_{1 \leq x^2 + y^2 \leq 16} \left(10 - \sqrt{x^2 + y^2}\right) \sqrt{2} dA \\ &= \int_0^{2\pi} \int_1^4 \sqrt{2} (10 - r) r dr d\theta = 2\pi \sqrt{2} \left[5r^2 - \frac{1}{3}r^3\right]_1^4 = 108\sqrt{2}\pi \end{aligned}$$

41. (a) $I_z = \iint_S (x^2 + y^2)\rho(x, y, z) dS$ (see 15.6.16).

$$\begin{aligned} \text{(b) } I_z &= \iint_S (x^2 + y^2) \left(10 - \sqrt{x^2 + y^2}\right) dS = \iint_{1 \leq x^2 + y^2 \leq 16} (x^2 + y^2) \left(10 - \sqrt{x^2 + y^2}\right) \sqrt{2} dA \\ &= \int_0^{2\pi} \int_1^4 \sqrt{2} (10r^3 - r^4) dr d\theta = 2\sqrt{2}\pi \left(\frac{4329}{10}\right) = \frac{4329}{5}\sqrt{2}\pi \end{aligned}$$

42. Using spherical coordinates to parametrize the sphere we have $\mathbf{r}(\phi, \theta) = 5 \sin \phi \cos \theta \mathbf{i} + 5 \sin \phi \sin \theta \mathbf{j} + 5 \cos \phi \mathbf{k}$, and

$|\mathbf{r}_\phi \times \mathbf{r}_\theta| = 25 \sin \phi$ (see Example 16.6.10). S is the portion of the sphere where $z \geq 4$, so $0 \leq \phi \leq \tan^{-1}(3/4)$ and

$0 \leq \theta \leq 2\pi$.

[continued]

$$\begin{aligned}
 \text{(a) } m &= \iint_S \rho(x, y, z) dS = \int_0^{2\pi} \int_0^{\tan^{-1}(3/4)} k(25 \sin \phi) d\phi d\theta = 25k \int_0^{2\pi} d\theta \int_0^{\tan^{-1}(3/4)} \sin \phi d\phi \\
 &= 25k(2\pi) \left[-\cos \left(\tan^{-1} \frac{3}{4} \right) + 1 \right] = 50\pi k \left(-\frac{4}{5} + 1 \right) = 10\pi k.
 \end{aligned}$$

Because S has constant density, $\bar{x} = \bar{y} = 0$ by symmetry, and

$$\begin{aligned}
 \bar{z} &= \frac{1}{m} \iint_S z \rho(x, y, z) dS = \frac{1}{10\pi k} \int_0^{2\pi} \int_0^{\tan^{-1}(3/4)} k(5 \cos \phi)(25 \sin \phi) d\phi d\theta \\
 &= \frac{1}{10\pi k} (125k) \int_0^{2\pi} d\theta \int_0^{\tan^{-1}(3/4)} \sin \phi \cos \phi d\phi = \frac{1}{10\pi k} (125k) (2\pi) \left[\frac{1}{2} \sin^2 \phi \right]_0^{\tan^{-1}(3/4)} = 25 \cdot \frac{1}{2} \left(\frac{3}{5} \right)^2 = \frac{9}{2},
 \end{aligned}$$

so the center of mass is $(\bar{x}, \bar{y}, \bar{z}) = (0, 0, \frac{9}{2})$.

$$\begin{aligned}
 \text{(b) } I_z &= \iint_S (x^2 + y^2) \rho(x, y, z) dS = \int_0^{2\pi} \int_0^{\tan^{-1}(3/4)} k(25 \sin^2 \phi)(25 \sin \phi) d\phi d\theta \\
 &= 625k \int_0^{2\pi} d\theta \int_0^{\tan^{-1}(3/4)} \sin^3 \phi d\phi = 625k(2\pi) \left[\frac{1}{3} \cos^3 \phi - \cos \phi \right]_0^{\tan^{-1}(3/4)} \\
 &= 1250\pi k \left[\frac{1}{3} \left(\frac{4}{5} \right)^3 - \frac{4}{5} - \frac{1}{3} + 1 \right] = 1250\pi k \left(\frac{14}{375} \right) = \frac{140}{3} \pi k
 \end{aligned}$$

43. The rate of flow through the cylinder is the flux $\iint_S \rho \mathbf{v} \cdot \mathbf{n} dS = \iint_S \rho \mathbf{v} \cdot d\mathbf{S}$ (see Formula 7). We use the parametric representation of the cylinder $\mathbf{r}(u, v) = 2 \cos u \mathbf{i} + 2 \sin u \mathbf{j} + v \mathbf{k}$ for S , where $0 \leq u \leq 2\pi$, $0 \leq v \leq 1$, so $\mathbf{r}_u = -2 \sin u \mathbf{i} + 2 \cos u \mathbf{j}$, $\mathbf{r}_v = \mathbf{k}$, and the outward orientation is given by $\mathbf{r}_u \times \mathbf{r}_v = 2 \cos u \mathbf{i} + 2 \sin u \mathbf{j}$. Then

$$\begin{aligned}
 \iint_S \rho \mathbf{v} \cdot d\mathbf{S} &= \rho \int_0^{2\pi} \int_0^1 (v \mathbf{i} + 4 \sin^2 u \mathbf{j} + 4 \cos^2 u \mathbf{k}) \cdot (2 \cos u \mathbf{i} + 2 \sin u \mathbf{j}) dv du \\
 &= \rho \int_0^{2\pi} \int_0^1 (2v \cos u + 8 \sin^3 u) dv du = \rho \int_0^{2\pi} (\cos u + 8 \sin^3 u) du \\
 &= \rho \left[\sin u + 8 \left(-\frac{1}{3} \right) (2 + \sin^2 u) \cos u \right]_0^{2\pi} = 0 \text{ kg/s}
 \end{aligned}$$

44. A parametric representation for the hemisphere S is $\mathbf{r}(\phi, \theta) = 3 \sin \phi \cos \theta \mathbf{i} + 3 \sin \phi \sin \theta \mathbf{j} + 3 \cos \phi \mathbf{k}$, $0 \leq \phi \leq \pi/2$, $0 \leq \theta \leq 2\pi$. Then $\mathbf{r}_\phi = 3 \cos \phi \cos \theta \mathbf{i} + 3 \cos \phi \sin \theta \mathbf{j} - 3 \sin \phi \mathbf{k}$, $\mathbf{r}_\theta = -3 \sin \phi \sin \theta \mathbf{i} + 3 \sin \phi \cos \theta \mathbf{j}$, and the outward orientation is given by $\mathbf{r}_\phi \times \mathbf{r}_\theta = 9 \sin^2 \phi \cos \theta \mathbf{i} + 9 \sin^2 \phi \sin \theta \mathbf{j} + 9 \sin \phi \cos \phi \mathbf{k}$. The rate of flow through S is [by (7)]

$$\begin{aligned}
 \iint_S \rho \mathbf{v} \cdot d\mathbf{S} &= \rho \int_0^{\pi/2} \int_0^{2\pi} (3 \sin \phi \sin \theta \mathbf{i} + 3 \sin \phi \cos \theta \mathbf{j}) \cdot (9 \sin^2 \phi \cos \theta \mathbf{i} + 9 \sin^2 \phi \sin \theta \mathbf{j} + 9 \sin \phi \cos \phi \mathbf{k}) d\theta d\phi \\
 &= 27\rho \int_0^{\pi/2} \int_0^{2\pi} (\sin^3 \phi \sin \theta \cos \theta + \sin^3 \phi \sin \theta \cos \theta) d\theta d\phi = 54\rho \int_0^{\pi/2} \sin^3 \phi d\phi \int_0^{2\pi} \sin \theta \cos \theta d\theta \\
 &= 54\rho \left[-\frac{1}{3} (2 + \sin^2 \phi) \cos \phi \right]_0^{\pi/2} \left[\frac{1}{2} \sin^2 \theta \right]_0^{2\pi} = 0 \text{ kg/s}
 \end{aligned}$$

45. S consists of the hemisphere S_1 given by $z = \sqrt{a^2 - x^2 - y^2}$ and the disk S_2 given by $0 \leq x^2 + y^2 \leq a^2$, $z = 0$.

On S_1 : As in Example 4, we use a parametric representation to get $\mathbf{E} = a \sin \phi \cos \theta \mathbf{i} + a \sin \phi \sin \theta \mathbf{j} + 2a \cos \phi \mathbf{k}$,

$\mathbf{T}_\phi \times \mathbf{T}_\theta = a^2 \sin^2 \phi \cos \theta \mathbf{i} + a^2 \sin^2 \phi \sin \theta \mathbf{j} + a^2 \sin \phi \cos \phi \mathbf{k}$. Thus

$$\begin{aligned}
 \iint_{S_1} \mathbf{E} \cdot d\mathbf{S} &= \int_0^{2\pi} \int_0^{\pi/2} (a^3 \sin^3 \phi + 2a^3 \sin \phi \cos^2 \phi) d\phi d\theta \\
 &= \int_0^{2\pi} \int_0^{\pi/2} (a^3 \sin \phi + a^3 \sin \phi \cos^2 \phi) d\phi d\theta = (2\pi)a^3 \left(1 + \frac{1}{3} \right) = \frac{8}{3} \pi a^3
 \end{aligned}$$

On S_2 : $\mathbf{E} = x \mathbf{i} + y \mathbf{j}$, and $\mathbf{r}_y \times \mathbf{r}_x = -\mathbf{k}$ so $\iint_{S_2} \mathbf{E} \cdot d\mathbf{S} = 0$.

Hence, by (11), the total charge is $q = \epsilon_0 \iint_S \mathbf{E} \cdot d\mathbf{S} = \frac{8}{3} \pi a^3 \epsilon_0$.

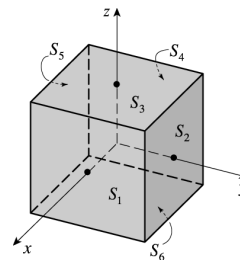
46. Referring to the figure, on

$$S_1: \mathbf{E} = \mathbf{i} + y\mathbf{j} + z\mathbf{k}, \mathbf{r}_y \times \mathbf{r}_z = \mathbf{i} \text{ and } \iint_{S_1} \mathbf{E} \cdot d\mathbf{S} = \int_{-1}^1 \int_{-1}^1 dy dz = 4;$$

$$S_2: \mathbf{E} = x\mathbf{i} + \mathbf{j} + z\mathbf{k}, \mathbf{r}_z \times \mathbf{r}_x = \mathbf{j} \text{ and } \iint_{S_2} \mathbf{E} \cdot d\mathbf{S} = \int_{-1}^1 \int_{-1}^1 dx dz = 4;$$

$$S_3: \mathbf{E} = x\mathbf{i} + y\mathbf{j} + \mathbf{k}, \mathbf{r}_x \times \mathbf{r}_y = \mathbf{k} \text{ and } \iint_{S_3} \mathbf{E} \cdot d\mathbf{S} = \int_{-1}^1 \int_{-1}^1 dx dy = 4;$$

$$S_4: \mathbf{E} = -\mathbf{i} + y\mathbf{j} + z\mathbf{k}, \mathbf{r}_z \times \mathbf{r}_y = -\mathbf{i} \text{ and } \iint_{S_4} \mathbf{E} \cdot d\mathbf{S} = 4.$$



Similarly, $\iint_{S_5} \mathbf{E} \cdot d\mathbf{S} = \iint_{S_6} \mathbf{E} \cdot d\mathbf{S} = 4$. Hence, by (11), $q = \varepsilon_0 \iint_S \mathbf{E} \cdot d\mathbf{S} = \varepsilon_0 \sum_{i=1}^6 \iint_{S_i} \mathbf{E} \cdot d\mathbf{S} = 24\varepsilon_0$.

47. The heat flow for $K = 6.5$ and $u(x, y, z) = 2y^2 + 2z^2$ is $-K \nabla u = -6.5(4y\mathbf{j} + 4z\mathbf{k})$. As in Example 16.6.5, we can parametrize the cylindrical surface $S: y^2 + z^2 = 6, 0 \leq x \leq 4$ as $\mathbf{r}(x, \theta) = x\mathbf{i} + \sqrt{6} \cos \theta \mathbf{j} + \sqrt{6} \sin \theta \mathbf{k}, 0 \leq x \leq 4, 0 \leq \theta \leq 2\pi$. Since we want the inward heat flow, we use $\mathbf{r}_x \times \mathbf{r}_\theta = -\sqrt{6} \cos \theta \mathbf{j} - \sqrt{6} \sin \theta \mathbf{k}$. Then the rate of heat flow inward is given by

$$\begin{aligned} \iint_S (-K \nabla u) \cdot d\mathbf{S} &= -K \iint \nabla u \cdot (\mathbf{r}_x \times \mathbf{r}_\theta) dA \\ &= -6.5 \int_0^{2\pi} \int_0^4 (4\sqrt{6} \cos \theta \mathbf{j} + 4\sqrt{6} \sin \theta \mathbf{k}) \cdot (-\sqrt{6} \cos \theta \mathbf{j} - \sqrt{6} \sin \theta \mathbf{k}) dx d\theta \\ &= -6.5 \int_0^{2\pi} \int_0^4 (-24 \cos^2 \theta - 24 \sin^2 \theta) dx d\theta = -6.5(-24)(4)(2\pi) = 1248\pi \end{aligned}$$

48. $u(x, y, z) = c/\sqrt{x^2 + y^2 + z^2}$,

$$\begin{aligned} \mathbf{F} &= -K \nabla u = -K \left[-\frac{cx}{(x^2 + y^2 + z^2)^{3/2}} \mathbf{i} - \frac{cy}{(x^2 + y^2 + z^2)^{3/2}} \mathbf{j} - \frac{cz}{(x^2 + y^2 + z^2)^{3/2}} \mathbf{k} \right] \\ &= \frac{cK}{(x^2 + y^2 + z^2)^{3/2}} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \end{aligned}$$

and the outward unit normal is $\mathbf{n} = \frac{1}{a} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$.

Thus $\mathbf{F} \cdot \mathbf{n} = \frac{cK}{a(x^2 + y^2 + z^2)^{3/2}} (x^2 + y^2 + z^2)$, but on $S, x^2 + y^2 + z^2 = a^2$ so $\mathbf{F} \cdot \mathbf{n} = \frac{cK}{a^2}$. Hence the rate of heat flow

across S is $\iint_S \mathbf{F} \cdot d\mathbf{S} = \frac{cK}{a^2} \iint_S dS = \frac{cK}{a^2} (4\pi a^2) = 4\pi Kc$.

49. Let S be a sphere of radius a centered at the origin. Then $|\mathbf{r}| = a$ and $\mathbf{F}(\mathbf{r}) = c\mathbf{r}/|\mathbf{r}|^3 = (c/a^3)(x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$. A parametric representation for S is $\mathbf{r}(\phi, \theta) = a \sin \phi \cos \theta \mathbf{i} + a \sin \phi \sin \theta \mathbf{j} + a \cos \phi \mathbf{k}, 0 \leq \phi \leq \pi, 0 \leq \theta \leq 2\pi$. Then $\mathbf{r}_\phi = a \cos \phi \cos \theta \mathbf{i} + a \cos \phi \sin \theta \mathbf{j} - a \sin \phi \mathbf{k}, \mathbf{r}_\theta = -a \sin \phi \sin \theta \mathbf{i} + a \sin \phi \cos \theta \mathbf{j}$, and the outward orientation is given by $\mathbf{r}_\phi \times \mathbf{r}_\theta = a^2 \sin^2 \phi \cos \theta \mathbf{i} + a^2 \sin^2 \phi \sin \theta \mathbf{j} + a^2 \sin \phi \cos \phi \mathbf{k}$. The flux of $\mathbf{F}$ across S is

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \int_0^\pi \int_0^{2\pi} \frac{c}{a^3} (a \sin \phi \cos \theta \mathbf{i} + a \sin \phi \sin \theta \mathbf{j} + a \cos \phi \mathbf{k}) \\ &\quad \cdot (a^2 \sin^2 \phi \cos \theta \mathbf{i} + a^2 \sin^2 \phi \sin \theta \mathbf{j} + a^2 \sin \phi \cos \phi \mathbf{k}) d\theta d\phi \\ &= \frac{c}{a^3} \int_0^\pi \int_0^{2\pi} a^3 (\sin^3 \phi + \sin \phi \cos^2 \phi) d\theta d\phi = c \int_0^\pi \int_0^{2\pi} \sin \phi d\theta d\phi = 4\pi c \end{aligned}$$

Thus the flux does not depend on the radius a .

16.8 Stokes' Theorem

1. Both H and P are oriented piecewise-smooth surfaces that are bounded by the simple, closed, smooth curve $x^2 + y^2 = 4$, $z = 0$ (which we can take to be oriented positively for both surfaces). Then H and P satisfy the hypotheses of Stokes' Theorem, so by (3) we know $\iint_H \text{curl } \mathbf{F} \cdot d\mathbf{S} = \int_C \mathbf{F} \cdot d\mathbf{r} = \iint_P \text{curl } \mathbf{F} \cdot d\mathbf{S}$ (where C is the boundary curve).

2. $\mathbf{F}(x, y, z) = x^2 \sin z \mathbf{i} + y^2 \mathbf{j} + xy \mathbf{k}$. The paraboloid $z = 1 - x^2 - y^2$ intersects the xy -plane in the circle $x^2 + y^2 = 1$, $z = 0$. This boundary curve C should be oriented in the counterclockwise direction when viewed from above, so a vector equation of C is $\mathbf{r}(t) = \cos t \mathbf{i} + \sin t \mathbf{j}$, $0 \leq t \leq 2\pi$. Then $\mathbf{r}'(t) = -\sin t \mathbf{i} + \cos t \mathbf{j}$,

$$\mathbf{F}(\mathbf{r}(t)) = (\cos t)^2(\sin 0) \mathbf{i} + (\sin t)^2 \mathbf{j} + (\cos t)(\sin t) \mathbf{k} = \sin^2 t \mathbf{j} + \sin t \cos t \mathbf{k},$$

and by Stokes' Theorem,

$$\begin{aligned} \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^{2\pi} (\sin^2 t \mathbf{j} + \sin t \cos t \mathbf{k}) \cdot (-\sin t \mathbf{i} + \cos t \mathbf{j}) dt \\ &= \int_0^{2\pi} (0 + \sin^2 t \cos t + 0) dt = \left[\frac{1}{3} \sin^3 t \right]_0^{2\pi} = 0 \end{aligned}$$

3. $\mathbf{F}(x, y, z) = ze^y \mathbf{i} + x \cos y \mathbf{j} + xz \sin y \mathbf{k}$. The boundary curve C is the circle $x^2 + z^2 = 16$, $y = 0$ where the hemisphere intersects the xz -plane. The curve should be oriented in the counterclockwise direction when viewed from the right (from the positive y -axis), so a vector equation of C is $\mathbf{r}(t) = 4 \cos(-t) \mathbf{i} + 4 \sin(-t) \mathbf{k} = 4 \cos t \mathbf{i} - 4 \sin t \mathbf{k}$, $0 \leq t \leq 2\pi$. Then $\mathbf{r}'(t) = -4 \sin t \mathbf{i} - 4 \cos t \mathbf{k}$ and

$$\mathbf{F}(\mathbf{r}(t)) = (-4 \sin t)e^0 \mathbf{i} + (4 \cos t)(\cos 0) \mathbf{j} + (4 \cos t)(-4 \sin t)(\sin 0) \mathbf{k} = -4 \sin t \mathbf{i} + 4 \cos t \mathbf{j},$$

and by Stokes' Theorem,

$$\begin{aligned} \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^{2\pi} (-4 \sin t \mathbf{i} + 4 \cos t \mathbf{j}) \cdot (-4 \sin t \mathbf{i} - 4 \cos t \mathbf{k}) dt \\ &= \int_0^{2\pi} (16 \sin^2 t + 0 + 0) dt = 16 \left[\frac{1}{2} t - \frac{1}{4} \sin 2t \right]_0^{2\pi} = 16\pi \end{aligned}$$

4. $\mathbf{F}(x, y, z) = \tan^{-1}(x^2 y z^2) \mathbf{i} + x^2 y \mathbf{j} + x^2 z^2 \mathbf{k}$. The boundary curve C is the circle $y^2 + z^2 = 4$, $x = 2$ which should be oriented in the counterclockwise direction when viewed from the front, so a vector equation of C is

$$\mathbf{r}(t) = 2 \mathbf{i} + 2 \cos t \mathbf{j} + 2 \sin t \mathbf{k}, \quad 0 \leq t \leq 2\pi. \text{ Then } \mathbf{F}(\mathbf{r}(t)) = \tan^{-1}(32 \cos t \sin^2 t) \mathbf{i} + 8 \cos t \mathbf{j} + 16 \sin^2 t \mathbf{k},$$

$$\mathbf{r}'(t) = -2 \sin t \mathbf{j} + 2 \cos t \mathbf{k}, \text{ and } \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = -16 \sin t \cos t + 32 \sin^2 t \cos t. \text{ Thus,}$$

$$\begin{aligned} \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^{2\pi} (-16 \sin t \cos t + 32 \sin^2 t \cos t) dt \\ &= \left[-8 \sin^2 t + \frac{32}{3} \sin^3 t \right]_0^{2\pi} = 0 \end{aligned}$$

5. $\mathbf{F}(x, y, z) = xyz \mathbf{i} + xy \mathbf{j} + x^2 y z \mathbf{k}$. C is the square in the plane $z = -1$. Rather than evaluating a line integral around C we can use Equation 3: $\iint_{S_1} \text{curl } \mathbf{F} \cdot d\mathbf{S} = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_{S_2} \text{curl } \mathbf{F} \cdot d\mathbf{S}$ where S_1 is the original cube without the bottom and S_2 is the bottom face of the cube. $\text{curl } \mathbf{F} = x^2 z \mathbf{i} + (xy - 2xyz) \mathbf{j} + (y - xz) \mathbf{k}$. For S_2 , we choose $\mathbf{n} = \mathbf{k}$ so that C has the same orientation for both surfaces. Then $\text{curl } \mathbf{F} \cdot \mathbf{n} = y - xz = x + y$ on S_2 , where $z = -1$. Thus, by (16.7.8),

$$\iint_{S_2} \text{curl } \mathbf{F} \cdot d\mathbf{S} = \int_{-1}^1 \int_{-1}^1 (x + y) dx dy = 0 \text{ so } \iint_{S_1} \text{curl } \mathbf{F} \cdot d\mathbf{S} = 0.$$

6. $\mathbf{F}(x, y, z) = e^{xy} \mathbf{i} + e^{xz} \mathbf{j} + x^2 z \mathbf{k}$. The boundary curve C is the circle $x^2 + z^2 = 1, y = 0$ which should be oriented in the counterclockwise direction when viewed from the right, so a vector equation of C is

$$\mathbf{r}(t) = \cos(-t) \mathbf{i} + \sin(-t) \mathbf{k} = \cos t \mathbf{i} - \sin t \mathbf{k}, 0 \leq t \leq 2\pi. \text{ Then } \mathbf{F}(\mathbf{r}(t)) = \mathbf{i} + e^{-\cos t \sin t} \mathbf{j} - \cos^2 t \sin t \mathbf{k},$$

$$\mathbf{r}'(t) = -\sin t \mathbf{i} - \cos t \mathbf{k}, \text{ and } \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = -\sin t + \cos^3 t \sin t. \text{ Thus,}$$

$$\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^{2\pi} (-\sin t + \cos^3 t \sin t) dt = [\cos t - \frac{1}{4} \cos^4 t]_0^{2\pi} = 0.$$

7. $\mathbf{F}(x, y, z) = (x + y^2) \mathbf{i} + (y + z^2) \mathbf{j} + (z + x^2) \mathbf{k}$ and $\text{curl } \mathbf{F} = -2z \mathbf{i} - 2x \mathbf{j} - 2y \mathbf{k}$. We take the surface to be the planar region enclosed by C and D to be the projection of S onto the xy -plane, so S is the portion of the plane $x + y + z = 1$ over $D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}$. Since C is oriented counterclockwise, we orient S upward. Using Equation 16.7.10, we have $z = g(x, y) = 1 - x - y, P = -2z, Q = -2x, R = -2y$, and

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_D [-(-2z)(-1) - (-2x)(-1) + (-2y)] dA \\ &= \int_0^1 \int_0^{1-x} (-2) dy dx = -2 \int_0^1 (1-x) dx = -1 \end{aligned}$$

8. $\mathbf{F}(x, y, z) = \mathbf{i} + (x + yz) \mathbf{j} + (xy - \sqrt{z}) \mathbf{k}$ and $\text{curl } \mathbf{F} = (x - y) \mathbf{i} - y \mathbf{j} + \mathbf{k}$. We take the surface S to be the planar region enclosed by C and D to be the projection of S onto the xy -plane, so S is the portion of the plane $3x + 2y + z = 1$ over $D = \{(x, y) \mid 0 \leq x \leq \frac{1}{3}, 0 \leq y \leq \frac{1}{2}(1 - 3x)\}$. We orient S upward and use Equation 16.7.10 with $z = g(x, y) = 1 - 3x - 2y$:

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_D [-(x - y)(-3) - (-y)(-2) + 1] dA = \int_0^{1/3} \int_0^{(1-3x)/2} (1 + 3x - 5y) dy dx \\ &= \int_0^{1/3} \left[(1 + 3x)y - \frac{5}{2}y^2 \right]_{y=0}^{y=(1-3x)/2} dx = \int_0^{1/3} \left[\frac{1}{2}(1 + 3x)(1 - 3x) - \frac{5}{2} \cdot \frac{1}{4}(1 - 3x)^2 \right] dx \\ &= \int_0^{1/3} \left(-\frac{81}{8}x^2 + \frac{15}{4}x - \frac{1}{8} \right) dx = \left[-\frac{27}{8}x^3 + \frac{15}{8}x^2 - \frac{1}{8}x \right]_0^{1/3} = -\frac{1}{8} + \frac{5}{24} - \frac{1}{24} = \frac{1}{24} \end{aligned}$$

9. $\mathbf{F}(x, y, z) = xy \mathbf{i} + yz \mathbf{j} + zx \mathbf{k}$. $\text{curl } \mathbf{F} = -y \mathbf{i} - z \mathbf{j} - x \mathbf{k}$ and we take S to be the part of the paraboloid $z = 1 - x^2 - y^2$ in the first octant. Since C is oriented counterclockwise (from above), we orient S upward. Then using Equation 16.7.10 with $z = g(x, y) = 1 - x^2 - y^2$ we have

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_D [-(-y)(-2x) - (-z)(-2y) + (-x)] dA \\ &= \iint_D [-2xy - 2y(1 - x^2 - y^2) - x] dA \\ &= \int_0^{\pi/2} \int_0^1 [-2(r \cos \theta)(r \sin \theta) - 2(r \sin \theta)(1 - r^2) - r \cos \theta] r dr d\theta \\ &= \int_0^{\pi/2} \int_0^1 [-2r^3 \sin \theta \cos \theta - 2(r^2 - r^4) \sin \theta - r^2 \cos \theta] dr d\theta \\ &= \int_0^{\pi/2} \left[-\frac{1}{2}r^4 \sin \theta \cos \theta - 2 \left(\frac{1}{3}r^3 - \frac{1}{5}r^5 \right) \sin \theta - \frac{1}{3}r^3 \cos \theta \right]_{r=0}^{r=1} d\theta \\ &= \int_0^{\pi/2} \left(-\frac{1}{2} \sin \theta \cos \theta - \frac{4}{15} \sin \theta - \frac{1}{3} \cos \theta \right) d\theta = \left[-\frac{1}{4} \sin^2 \theta + \frac{4}{15} \cos \theta - \frac{1}{3} \sin \theta \right]_0^{\pi/2} \\ &= -\frac{1}{4} - \frac{4}{15} - \frac{1}{3} = -\frac{17}{20} \end{aligned}$$

10. $\mathbf{F}(x, y, z) = 2y\mathbf{i} + xz\mathbf{j} + (x + y)\mathbf{k}$. The curve of intersection is an ellipse in the plane $z = y + 2$.

$\text{curl } \mathbf{F} = (1 - x)\mathbf{i} - \mathbf{j} + (z - 2)\mathbf{k}$ and we take the surface S to be the planar region enclosed by C with upward orientation.

From Equation 16.7.10 with $z = g(x, y) = y + 2$ we have

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_{x^2+y^2 \leq 1} [-(1-x)(0) - (-1)(1) + (y+2-2)] dA \\ &= \iint_{x^2+y^2 \leq 1} (y+1) dA = \int_0^{2\pi} \int_0^1 (r \sin \theta + 1) r dr d\theta = \int_0^{2\pi} \left[\frac{1}{3} r^3 \sin \theta + \frac{1}{2} r^2 \right]_{r=0}^{r=1} d\theta \\ &= \int_0^{2\pi} \left(\frac{1}{3} \sin \theta + \frac{1}{2} \right) d\theta = \left[-\frac{1}{3} \cos \theta + \frac{1}{2} \theta \right]_0^{2\pi} = \pi\end{aligned}$$

11. $\mathbf{F}(x, y, z) = \langle -yx^2, xy^2, e^{xy} \rangle$ and $\text{curl } \mathbf{F} = xe^{xy}\mathbf{i} - ye^{xy}\mathbf{j} + (x^2 + y^2)\mathbf{k}$. C is the circle in the xy -plane centered at the origin with radius 2. Choose S to be the portion of the xy -plane enclosed by C . So $S = D = \{(x, y) \mid x^2 + y^2 \leq 4\}$. C is oriented counterclockwise, so we orient S upward and the normal vector to S is $\mathbf{n} = \mathbf{k}$. By Stokes' Theorem (See Solution 2 of Example 2), we get

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = \iint_S [xe^{xy}\mathbf{i} - ye^{xy}\mathbf{j} + (x^2 + y^2)\mathbf{k}] \cdot \mathbf{k} dS \\ &= \iint_D (x^2 + y^2) dA = \int_0^2 \int_0^{2\pi} r^2 r d\theta dr = 2\pi \int_0^2 r^3 dr = 2\pi \left[\frac{r^4}{4} \right]_0^2 = 8\pi\end{aligned}$$

12. $\mathbf{F}(x, y, z) = ze^x\mathbf{i} + (z - y^3)\mathbf{j} + (x - z^3)\mathbf{k}$ and $\text{curl } \mathbf{F} = -\mathbf{i} - (1 - e^x)\mathbf{j}$. C is the circle $y^2 + z^2 = 4$, $x = 3$, and we choose the surface S to be the portion of the plane $x = 3$ enclosed by C . The projection of S onto the yz -plane is the disk $D = \{(y, z) \mid y^2 + z^2 \leq 4\}$. C is oriented clockwise, so we orient S to have normal vector $\mathbf{n} = \mathbf{i}$. By Stokes' Theorem (see Solution 2 of Example 2), we get

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = \iint_S [-\mathbf{i} - (1 - e^x)\mathbf{j}] \cdot \mathbf{i} dS \\ &= -\iint_D dA = -A(D) = -\pi(2^2) = -4\pi\end{aligned}$$

13. $\mathbf{F}(x, y, z) = x^2y\mathbf{i} + x^3\mathbf{j} + e^z \tan^{-1} z\mathbf{k}$ and $\text{curl } \mathbf{F} = 2x^2\mathbf{k}$. Note that the curve $C = \langle \cos t, \sin t, \sin t \rangle$ is contained in the plane $y = z$ because the $\mathbf{j}$ and $\mathbf{k}$ components of the curve are equal. We choose the surface S to be the portion of the plane $y = z$ enclosed by C . The projection of C onto the xy -plane is the circle $\langle \cos t, \sin t, 0 \rangle$, and D is the disk in the xy -plane enclosed by C . We orient S upward and use Equation 16.7.10 with $z = g(x, y) = y$:

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_D [-(0)(0) - (0)(1) + 2x^2] dA = \int_0^{2\pi} \int_0^1 2(r \cos \theta)^2 r dr d\theta \\ &= \int_0^{2\pi} (1 + \cos 2\theta) d\theta \int_0^1 r^3 dr = \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{2\pi} \left[\frac{r^4}{4} \right]_0^1 = (2\pi + 0 - 0) \frac{1}{4} = \frac{\pi}{2}\end{aligned}$$

14. $\mathbf{F}(x, y, z) = \langle x^3 - z, xy, y + z^2 \rangle$ and $\text{curl } \mathbf{F} = \mathbf{i} - \mathbf{j} + y\mathbf{k}$. C is the curve of intersection of the paraboloid $z = x^2 + y^2$ and the plane $z = x$. Let S be the portion of the plane $z = x$ enclosed by C . To find the projection of S onto the xy -plane, note that $x = x^2 + y^2$. Converting to polar coordinates, we get $r \cos \theta = r^2 \Rightarrow r = \cos \theta$. So D is the region in the

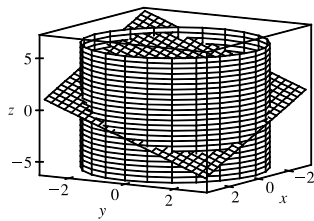
xy -plane enclosed by the circle $r = \cos \theta$; that is, $D = \{(r, \theta) \mid 0 \leq r \leq \cos \theta, 0 \leq \theta \leq \pi\}$. We orient S upward and use Equation 16.7.10 with $z = g(x, y) = x$:

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \iint_D [-(1)(1) - (-1)(0) + y] dA = \iint_D (-1 + y) dA \\ &= \int_0^\pi \int_0^{\cos \theta} (-1 + r \sin \theta) r dr d\theta = \int_0^\pi \left[-\frac{r^2}{2} + \frac{r^3}{3} \sin \theta \right]_0^{\cos \theta} d\theta \\ &= \int_0^\pi \left(-\frac{\cos^2 \theta}{2} + \frac{\cos^3 \theta}{3} \sin \theta \right) d\theta = \int_0^\pi \left(-\frac{1 + \cos 2\theta}{4} + \frac{\cos^3 \theta}{3} \sin \theta \right) d\theta \\ &= \left[-\frac{1}{4}\theta - \frac{\sin 2\theta}{8} - \frac{\cos^4 \theta}{12} \right]_0^\pi = \left(-\frac{\pi}{4} - 0 - \frac{1}{12} \right) - \left(0 - 0 - \frac{1}{12} \right) = -\frac{\pi}{4}\end{aligned}$$

15. (a) $\mathbf{F}(x, y, z) = x^2 z \mathbf{i} + xy^2 \mathbf{j} + z^2 \mathbf{k}$. The curve of intersection is an ellipse in the plane $x + y + z = 1$. The unit normal is $\mathbf{n} = \frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} + \mathbf{k})$, $\operatorname{curl} \mathbf{F} = x^2 \mathbf{j} + y^2 \mathbf{k}$, and $\operatorname{curl} \mathbf{F} \cdot \mathbf{n} = \frac{1}{\sqrt{3}}(x^2 + y^2)$. Then, by Stokes' Theorem,

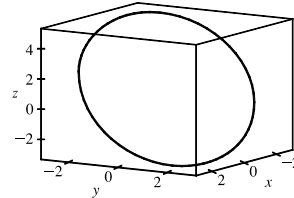
$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n} dS = \iint_S \frac{1}{\sqrt{3}}(x^2 + y^2) dS \\ &= \iint_{x^2 + y^2 \leq 9} (x^2 + y^2) dx dy = \int_0^{2\pi} \int_0^3 r^3 dr d\theta = 2\pi \left(\frac{81}{4} \right) = \frac{81\pi}{2}\end{aligned}$$

(b)



(c) One possible parametrization is $x = 3 \cos t$, $y = 3 \sin t$,

$$z = 1 - 3 \cos t - 3 \sin t, 0 \leq t \leq 2\pi.$$

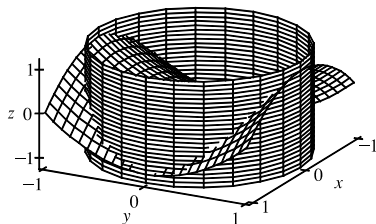


16. (a) $\mathbf{F}(x, y, z) = x^2 y \mathbf{i} + \frac{1}{3} x^3 \mathbf{j} + xy \mathbf{k}$. S is the part of the surface $z = y^2 - x^2$ that lies above the unit disk D .

$\operatorname{curl} \mathbf{F} = x \mathbf{i} - y \mathbf{j} + (x^2 - x^2) \mathbf{k} = x \mathbf{i} - y \mathbf{j}$. Using Equation 16.7.10 with $z = g(x, y) = y^2 - x^2$, by Stokes' Theorem we have

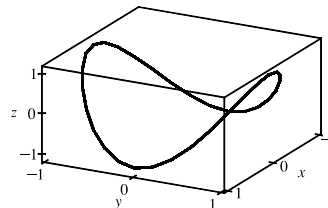
$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \iint_D [-x(-2x) - (-y)(2y)] dA = 2 \iint_D (x^2 + y^2) dA \\ &= 2 \int_0^{2\pi} \int_0^1 r^2 r dr d\theta = 2(2\pi) \left[\frac{1}{4} r^4 \right]_0^1 = \pi\end{aligned}$$

(b)



(c) One possible set of parametric equations is $x = \cos t$,

$$y = \sin t, z = \sin^2 t - \cos^2 t, 0 \leq t \leq 2\pi.$$



17. $\mathbf{F}(x, y, z) = -y\mathbf{i} + x\mathbf{j} - 2\mathbf{k}$. The boundary curve C is the circle $x^2 + y^2 = 16$, $z = 4$ oriented in the clockwise direction as viewed from above (since S is oriented downward). We can parametrize C by $\mathbf{r}(t) = 4\cos t\mathbf{i} - 4\sin t\mathbf{j} + 4\mathbf{k}$,

$0 \leq t \leq 2\pi$, and then $\mathbf{r}'(t) = -4\sin t\mathbf{i} - 4\cos t\mathbf{j}$. Thus $\mathbf{F}(\mathbf{r}(t)) = 4\sin t\mathbf{i} + 4\cos t\mathbf{j} - 2\mathbf{k}$,

$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = -16\sin^2 t - 16\cos^2 t = -16$, and

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^{2\pi} (-16) dt = -16(2\pi) = -32\pi$$

Now $\text{curl } \mathbf{F} = 2\mathbf{k}$, and the projection D of S on the xy -plane is the disk $x^2 + y^2 \leq 16$, so by Equation 16.7.10 with

$z = g(x, y) = \sqrt{x^2 + y^2}$ [and multiplying by -1 for the downward orientation] we have

$$\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = -\iint_D (-0 - 0 + 2) dA = -2 \cdot A(D) = -2 \cdot \pi(4^2) = -32\pi$$

18. $\mathbf{F}(x, y, z) = -2yz\mathbf{i} + y\mathbf{j} + 3x\mathbf{k}$. The paraboloid intersects the plane $z = 1$ when $1 = 5 - x^2 - y^2 \Leftrightarrow x^2 + y^2 = 4$, so the boundary curve C is the circle $x^2 + y^2 = 4$, $z = 1$ oriented in the counterclockwise direction as viewed from above. We can parametrize C by $\mathbf{r}(t) = 2\cos t\mathbf{i} + 2\sin t\mathbf{j} + \mathbf{k}$, $0 \leq t \leq 2\pi$, and then $\mathbf{r}'(t) = -2\sin t\mathbf{i} + 2\cos t\mathbf{j}$. Thus

$\mathbf{F}(\mathbf{r}(t)) = -4\sin t\mathbf{i} + 2\sin t\mathbf{j} + 6\cos t\mathbf{k}$, $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 8\sin^2 t + 4\sin t\cos t$, and

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} (8\sin^2 t + 4\sin t\cos t) dt = 8\left(\frac{1}{2}t - \frac{1}{4}\sin 2t\right) + 2\sin^2 t \Big|_0^{2\pi} = 8\pi$$

Now $\text{curl } \mathbf{F} = (-3 - 2y)\mathbf{j} + 2z\mathbf{k}$, and the projection D of S on the xy -plane is the disk $x^2 + y^2 \leq 4$, so by Equation 16.7.10 with $z = g(x, y) = 5 - x^2 - y^2$ we have

$$\begin{aligned} \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} &= \iint_D [-0 - (-3 - 2y)(-2y) + 2z] dA = \iint_D [-6y - 4y^2 + 2(5 - x^2 - y^2)] dA \\ &= \int_0^{2\pi} \int_0^2 [-6r\sin\theta - 4r^2\sin^2\theta + 2(5 - r^2)] r dr d\theta \\ &= \int_0^{2\pi} \left[-2r^3\sin\theta - r^4\sin^2\theta + 5r^2 - \frac{1}{2}r^4\right]_{r=0}^{r=2} d\theta = \int_0^{2\pi} (-16\sin\theta - 16\sin^2\theta + 20 - 8) d\theta \\ &= 16\cos\theta - 16\left(\frac{1}{2}\theta - \frac{1}{4}\sin 2\theta\right) + 12\theta \Big|_0^{2\pi} = 8\pi \end{aligned}$$

19. $\mathbf{F}(x, y, z) = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$. The boundary curve C is the circle $x^2 + z^2 = 1$, $y = 0$ oriented in the counterclockwise direction as viewed from the positive y -axis. Then C can be described by $\mathbf{r}(t) = \cos t\mathbf{i} - \sin t\mathbf{k}$, $0 \leq t \leq 2\pi$, and $\mathbf{r}'(t) = -\sin t\mathbf{i} - \cos t\mathbf{k}$. Thus $\mathbf{F}(\mathbf{r}(t)) = -\sin t\mathbf{j} + \cos t\mathbf{k}$, $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = -\cos^2 t$, and

$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} (-\cos^2 t) dt = -\frac{1}{2}t - \frac{1}{4}\sin 2t \Big|_0^{2\pi} = -\pi$. Now $\text{curl } \mathbf{F} = -\mathbf{i} - \mathbf{j} - \mathbf{k}$, and S can be parametrized

(see Example 16.6.10) by $\mathbf{r}(\phi, \theta) = \sin\phi\cos\theta\mathbf{i} + \sin\phi\sin\theta\mathbf{j} + \cos\phi\mathbf{k}$, $0 \leq \theta \leq \pi$, $0 \leq \phi \leq \pi$. Then

$\mathbf{r}_\phi \times \mathbf{r}_\theta = \sin^2\phi\cos\theta\mathbf{i} + \sin^2\phi\sin\theta\mathbf{j} + \sin\phi\cos\phi\mathbf{k}$ and

$$\begin{aligned} \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} &= \iint_{x^2+z^2 \leq 1} \text{curl } \mathbf{F} \cdot (\mathbf{r}_\phi \times \mathbf{r}_\theta) dA = \int_0^\pi \int_0^\pi (-\sin^2\phi\cos\theta - \sin^2\phi\sin\theta - \sin\phi\cos\phi) d\theta d\phi \\ &= \int_0^\pi (-2\sin^2\phi - \pi\sin\phi\cos\phi) d\phi = \left[\frac{1}{2}\sin 2\phi - \phi - \frac{\pi}{2}\sin^2\phi\right]_0^\pi = -\pi \end{aligned}$$

20. Let S be the surface in the plane $x + y + z = 1$ with upward orientation enclosed by C . Then an upward unit normal vector for S is $\mathbf{n} = \frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} + \mathbf{k})$. Orient C in the counterclockwise direction, as viewed from above. $\int_C z dx - 2x dy + 3y dz$ is equivalent to $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F}(x, y, z) = z\mathbf{i} - 2x\mathbf{j} + 3y\mathbf{k}$, and the components of $\mathbf{F}$ are polynomials, which have continuous

partial derivatives throughout $\mathbb{R}^3$. We have $\text{curl } \mathbf{F} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$, so by Stokes' Theorem,

$$\begin{aligned}\int_C z \, dx - 2x \, dy + 3y \, dz &= \int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = \iint_S (3\mathbf{i} + \mathbf{j} - 2\mathbf{k}) \cdot \frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} + \mathbf{k}) \, dS \\ &= \frac{2}{\sqrt{3}} \iint_S dS = \frac{2}{\sqrt{3}} (\text{surface area of } S)\end{aligned}$$

Thus the value of $\int_C z \, dx - 2x \, dy + 3y \, dz$ is always $\frac{2}{\sqrt{3}}$ times the area of the region enclosed by C , regardless of its shape or location. [Notice that because $\mathbf{n}$ is normal to a plane, it is constant. But $\text{curl } \mathbf{F}$ is also constant, so the dot product $\text{curl } \mathbf{F} \cdot \mathbf{n}$ is constant and we could have simply argued that $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS$ is a constant multiple of $\iint_S dS$, the surface area of S .]

21. $\mathbf{F}(x, y, z) = z^2\mathbf{i} + 2xy\mathbf{j} + 4y^2\mathbf{k}$. It is easier to use Stokes' Theorem than to compute the work directly. Let S be the planar region enclosed by the path of the particle, so S is the portion of the plane $z = \frac{1}{2}y$ for $0 \leq x \leq 1$, $0 \leq y \leq 2$, with upward orientation. $\text{curl } \mathbf{F} = 8y\mathbf{i} + 2z\mathbf{j} + 2y\mathbf{k}$ and by Stokes' Theorem and Equation 16.7.10, the work done is

$$\begin{aligned}W &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_D [-8y(0) - 2z(\frac{1}{2}) + 2y] \, dA \\ &= \int_0^1 \int_0^2 (2y - \frac{1}{2}y) \, dy \, dx = \int_0^1 \int_0^2 \frac{3}{2}y \, dy \, dx = \int_0^1 [\frac{3}{4}y^2]_{y=0}^{y=2} \, dx = \int_0^1 3 \, dx = 3\end{aligned}$$

22. $\int_C (y + \sin x) \, dx + (z^2 + \cos y) \, dy + x^3 \, dz = \int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y, z) = (y + \sin x)\mathbf{i} + (z^2 + \cos y)\mathbf{j} + x^3\mathbf{k} \Rightarrow \text{curl } \mathbf{F} = -2z\mathbf{i} - 3x^2\mathbf{j} - \mathbf{k}$. Since $\sin 2t = 2 \sin t \cos t$, C lies on the surface $z = 2xy$. Let S be the part of this surface that is bounded by C . Then the projection of S onto the xy -plane is the unit disk D [$x^2 + y^2 \leq 1$]. C is traversed clockwise (when viewed from above) so S is oriented downward. Using Equation 16.7.10 with $g(x, y) = 2xy$,

$P = -2z = -2(2xy) = -4xy$, $Q = -3x^2$, $R = -1$ and multiplying by -1 for the downward orientation, we have

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= -\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = -\iint_D [-(-4xy)(2y) - (-3x^2)(2x) - 1] \, dA \\ &= -\iint_D (8xy^2 + 6x^3 - 1) \, dA = -\int_0^{2\pi} \int_0^1 (8r^3 \cos \theta \sin^2 \theta + 6r^3 \cos^3 \theta - 1) r \, dr \, d\theta \\ &= -\int_0^{2\pi} (\frac{8}{5} \cos \theta \sin^2 \theta + \frac{6}{5} \cos^3 \theta - \frac{1}{2}) \, d\theta = -[\frac{8}{15} \sin^3 \theta + \frac{6}{5} (\sin \theta - \frac{1}{3} \sin^3 \theta) - \frac{1}{2} \theta]_0^{2\pi} = \pi\end{aligned}$$

23. Assume S is a sphere centered at the origin with radius a and let H_1 and H_2 be the upper and lower hemispheres, respectively, of S . Then $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_{H_1} \text{curl } \mathbf{F} \cdot d\mathbf{S} + \iint_{H_2} \text{curl } \mathbf{F} \cdot d\mathbf{S} = \oint_{C_1} \mathbf{F} \cdot d\mathbf{r} + \oint_{C_2} \mathbf{F} \cdot d\mathbf{r}$ by Stokes' Theorem. But C_1 is the circle $x^2 + y^2 = a^2$ oriented in the counterclockwise direction while C_2 is the same circle oriented in the clockwise direction. Hence $\oint_{C_2} \mathbf{F} \cdot d\mathbf{r} = -\oint_{C_1} \mathbf{F} \cdot d\mathbf{r}$ so $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = 0$ as desired.

24. (a) By Exercise 16.5.28, $\text{curl}(f\nabla g) = f \text{curl}(\nabla g) + \nabla f \times \nabla g = \nabla f \times \nabla g$ since $\text{curl}(\nabla g) = \mathbf{0}$. Hence by Stokes' Theorem $\int_C (f\nabla g) \cdot d\mathbf{r} = \iint_S (\nabla f \times \nabla g) \cdot d\mathbf{S}$.

(b) As in (a), $\text{curl}(f\nabla f) = \nabla f \times \nabla f = \mathbf{0}$, so by Stokes' Theorem, $\int_C (f\nabla f) \cdot d\mathbf{r} = \iint_S [\text{curl}(f\nabla f)] \cdot d\mathbf{S} = 0$.

(c) As in part (a),

$$\begin{aligned}\text{curl}(f\nabla g + g\nabla f) &= \text{curl}(f\nabla g) + \text{curl}(g\nabla f) && \text{[by Exercise 16.5.26]} \\ &= (\nabla f \times \nabla g) + (\nabla g \times \nabla f) = \mathbf{0} && \text{[since } \mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})\text{]}\end{aligned}$$

Hence by Stokes' Theorem, $\int_C (f\nabla g + g\nabla f) \cdot d\mathbf{r} = \iint_S \text{curl}(f\nabla g + g\nabla f) \cdot d\mathbf{S} = 0$.

16.9 The Divergence Theorem

- 1.
- $\mathbf{F}(x, y, z) = 3x\mathbf{i} + xy\mathbf{j} + 2xz\mathbf{k} \Rightarrow \operatorname{div} \mathbf{F} = 3 + x + 2x = 3 + 3x$
- , so

$\iiint_E \operatorname{div} \mathbf{F} dV = \int_0^1 \int_0^1 \int_0^1 (3x + 3) dx dy dz = \frac{9}{2}$ (notice the triple integral is three times the volume of the cube plus three times $\bar{x}$).

To compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$, on

$$S_1: x = 1, \mathbf{F} = 3\mathbf{i} + y\mathbf{j} + 2z\mathbf{k}, \mathbf{n} = \mathbf{i}, \text{ and } \iint_{S_1} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_1} \mathbf{F} \cdot \mathbf{i} dS = \iint_{S_1} 3 dS = 3;$$

$$S_2: y = 1, \mathbf{F} = 3x\mathbf{i} + x\mathbf{j} + 2xz\mathbf{k}, \mathbf{n} = \mathbf{j}, \text{ and } \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_2} \mathbf{F} \cdot \mathbf{j} dS = \iint_{S_2} x dS = \frac{1}{2};$$

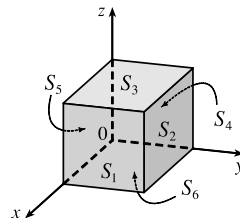
$$S_3: z = 1, \mathbf{F} = 3x\mathbf{i} + xy\mathbf{j} + 2x\mathbf{k}, \mathbf{n} = \mathbf{k}, \text{ and } \iint_{S_3} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_3} \mathbf{F} \cdot \mathbf{k} dS = \iint_{S_3} 2x dS = 1;$$

$$S_4: x = 0, \mathbf{F} = \mathbf{0}, \mathbf{n} = -\mathbf{i}, \iint_{S_4} \mathbf{F} \cdot d\mathbf{S} = 0;$$

$$S_5: y = 0, \mathbf{F} = 3x\mathbf{i} + 2xz\mathbf{k}, \mathbf{n} = -\mathbf{j}, \text{ and } \iint_{S_5} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_5} \mathbf{F} \cdot (-\mathbf{j}) dS = \iint_{S_5} 0 dS = 0;$$

$$S_6: z = 0, \mathbf{F} = 3x\mathbf{i} + xy\mathbf{j}, \mathbf{n} = -\mathbf{k}, \text{ and } \iint_{S_6} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_6} \mathbf{F} \cdot (-\mathbf{k}) dS = \iint_{S_6} 0 dS = 0.$$

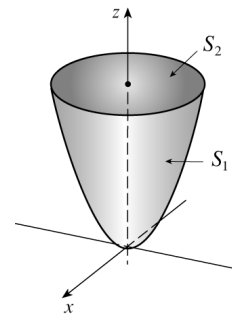
$$\text{Thus, } \iint_S \mathbf{F} \cdot d\mathbf{S} = 3 + \frac{1}{2} + 1 + 0 + 0 + 0 = \frac{9}{2}.$$



- 2.
- $\mathbf{F}(x, y, z) = y^2z^3\mathbf{i} + 2yz^3\mathbf{j} + 4z^2\mathbf{k} \Rightarrow \operatorname{div} \mathbf{F} = 0 + 2z + 8z = 10z$
- ,

so, using cylindrical coordinates,

$$\begin{aligned} \iiint_E \operatorname{div} \mathbf{F} dV &= \iiint_E 10z dV = \int_0^{2\pi} \int_0^3 \int_{r^2}^9 (10z) r dz dr d\theta \\ &= \int_0^{2\pi} \int_0^3 [5rz^2]_{z=r^2}^{z=9} dr d\theta = \int_0^{2\pi} \int_0^3 (405r - 5r^5) dr d\theta \\ &= \int_0^{2\pi} d\theta \int_0^3 (405r - 5r^5) dr = [\theta]_0^{2\pi} \left[\frac{405}{2}r^2 - \frac{5}{6}r^6 \right]_0^3 \\ &= 2\pi \left(\frac{3645}{2} - \frac{1215}{2} \right) = 2430\pi \end{aligned}$$



On S_1 : The surface is $z = x^2 + y^2$, $x^2 + y^2 \leq 9$, with downward orientation. Then, by Equation 16.7.10,

$$\begin{aligned} \iint_{S_1} \mathbf{F} \cdot d\mathbf{S} &= - \iint_D [-(y^2z^3)(2x) - (2yz^3)(2y) + (4z^2)] dA \\ &= \iint_D [2xy^2(x^2 + y^2)^3 + 4y^2(x^2 + y^2) - 4(x^2 + y^2)^2] dA \\ &= \int_0^{2\pi} \int_0^3 (2r^3 \cos \theta \sin^2 \theta \cdot r^6 + 4r^2 \sin^2 \theta \cdot r^2 - 4r^4) r dr d\theta \\ &= \int_0^{2\pi} \int_0^3 (2r^{10} \sin^2 \theta \cos \theta + 4r^5 \sin^2 \theta - 4r^5) dr d\theta \\ &= \int_0^{2\pi} \left[\frac{2}{11} r^{11} \sin^2 \theta \cos \theta + \frac{2}{3} r^6 \sin^2 \theta - \frac{2}{3} r^6 \right]_{r=0}^{r=3} d\theta \\ &= \int_0^{2\pi} \left(\frac{354,294}{11} \sin^2 \theta \cos \theta + 486 \sin^2 \theta - 486 \right) d\theta \\ &= \left[\frac{354,294}{11} \cdot \frac{1}{3} \sin^3 \theta + 486 \left(\frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right) - 486\theta \right]_0^{2\pi} \\ &= 0 + 486(\pi - 0) - 486(2\pi) = -486\pi \end{aligned}$$

On S_2 : The surface is $z = 9$, $x^2 + y^2 \leq 9$, with upward orientation and $\mathbf{n} = \mathbf{k}$, so

$$\begin{aligned} \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} &= \iint_{S_2} \mathbf{F} \cdot \mathbf{n} dS = \iint_{S_2} 4z^2 dS = \iint_{x^2 + y^2 \leq 9} 4(9)^2 dA \\ &= 324(\text{area of circle}) = 324 \cdot \pi(3)^2 = 2916\pi \end{aligned}$$

$$\text{Thus } \iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_{S_1} \mathbf{F} \cdot d\mathbf{S} + \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = -486\pi + 2916\pi = 2430\pi.$$

3. $\mathbf{F}(x, y, z) = \langle z, y, x \rangle \Rightarrow \operatorname{div} \mathbf{F} = 0 + 1 + 0 = 1$, so $\iiint_E \operatorname{div} \mathbf{F} dV = \iiint_E 1 dV = V(E) = \frac{4}{3}\pi \cdot 4^3 = \frac{256}{3}\pi$. S is a sphere of radius 4 centered at the origin which can be parametrized by $\mathbf{r}(\phi, \theta) = \langle 4 \sin \phi \cos \theta, 4 \sin \phi \sin \theta, 4 \cos \phi \rangle$, $0 \leq \phi \leq \pi$, $0 \leq \theta \leq 2\pi$ (similar to Example 16.6.10). Then

$$\begin{aligned}\mathbf{r}_\phi \times \mathbf{r}_\theta &= \langle 4 \cos \phi \cos \theta, 4 \cos \phi \sin \theta, -4 \sin \phi \rangle \times \langle -4 \sin \phi \sin \theta, 4 \sin \phi \cos \theta, 0 \rangle \\ &= \langle 16 \sin^2 \phi \cos \theta, 16 \sin^2 \phi \sin \theta, 16 \cos \phi \sin \phi \rangle\end{aligned}$$

and $\mathbf{F}(\mathbf{r}(\phi, \theta)) = \langle 4 \cos \phi, 4 \sin \phi \sin \theta, 4 \sin \phi \cos \theta \rangle$. Thus,

$$\begin{aligned}\mathbf{F} \cdot (\mathbf{r}_\phi \times \mathbf{r}_\theta) &= 64 \cos \phi \sin^2 \phi \cos \theta + 64 \sin^3 \phi \sin^2 \theta + 64 \cos \phi \sin^2 \phi \cos \theta \\ &= 128 \cos \phi \sin^2 \phi \cos \theta + 64 \sin^3 \phi \sin^2 \theta\end{aligned}$$

and

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_D \mathbf{F} \cdot (\mathbf{r}_\phi \times \mathbf{r}_\theta) dA = \int_0^{2\pi} \int_0^\pi (128 \cos \phi \sin^2 \phi \cos \theta + 64 \sin^3 \phi \sin^2 \theta) d\phi d\theta \\ &= \int_0^{2\pi} \left[\frac{128}{3} \sin^3 \phi \cos \theta + 64 \left(\frac{1}{3} \cos^3 \phi - \cos \phi \right) \sin^2 \theta \right]_{\phi=0}^{\phi=\pi} d\theta \\ &= \int_0^{2\pi} \frac{256}{3} \sin^2 \theta d\theta = \frac{256}{3} \left[\frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right]_0^{2\pi} = \frac{256}{3} \pi\end{aligned}$$

4. $F(x, y, z) = \langle x^2, -y, z \rangle \Rightarrow \operatorname{div} \mathbf{F} = 2x - 1 + 1 = 2x$, so

$$\iiint_E \operatorname{div} \mathbf{F} dV = \iint_{y^2+z^2 \leq 9} \left[\int_0^2 2x dx \right] dA = \iint_{y^2+z^2 \leq 9} 4 dA = 4(\text{area of circle}) = 4(\pi \cdot 3^2) = 36\pi$$

Let S_1 be the front of the cylinder (in the plane $x = 2$), S_2 the back (in the yz -plane), and S_3 the lateral surface of the cylinder.

S_1 is the disk $x = 2$, $y^2 + z^2 \leq 9$. A unit normal vector is $\mathbf{n} = \langle 1, 0, 0 \rangle$ and $\mathbf{F} = \langle 4, -y, z \rangle$ on S_1 , so

$$\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_1} \mathbf{F} \cdot \mathbf{n} dS = \iint_{S_1} 4 dS = 4(\text{surface area of } S_1) = 4(\pi \cdot 3^2) = 36\pi. S_2 \text{ is the disk } x = 0, y^2 + z^2 \leq 9.$$

Here $\mathbf{n} = \langle -1, 0, 0 \rangle$ and $\mathbf{F} = \langle 0, -y, z \rangle$, so $\iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_2} \mathbf{F} \cdot \mathbf{n} dS = \iint_{S_2} 0 dS = 0$.

S_3 can be parametrized by $\mathbf{r}(x, \theta) = \langle x, 3 \cos \theta, 3 \sin \theta \rangle$, $0 \leq x \leq 2$, $0 \leq \theta \leq 2\pi$. Then

$\mathbf{r}_x \times \mathbf{r}_\theta = \langle 1, 0, 0 \rangle \times \langle 0, -3 \sin \theta, 3 \cos \theta \rangle = \langle 0, -3 \cos \theta, -3 \sin \theta \rangle$. For the outward (positive) orientation we use $-(\mathbf{r}_x \times \mathbf{r}_\theta)$ and $\mathbf{F}(\mathbf{r}(x, \theta)) = \langle x^2, -3 \cos \theta, 3 \sin \theta \rangle$, so

$$\begin{aligned}\iint_{S_3} \mathbf{F} \cdot d\mathbf{S} &= \iint_D \mathbf{F} \cdot (-\mathbf{r}_x \times \mathbf{r}_\theta) dA = \int_0^2 \int_0^{2\pi} (0 - 9 \cos^2 \theta + 9 \sin^2 \theta) d\theta dx \\ &= -9 \int_0^2 dx \int_0^{2\pi} \cos 2\theta d\theta = -9(2) \left[\frac{1}{2} \sin 2\theta \right]_0^{2\pi} = 0\end{aligned}$$

Thus $\iint_S \mathbf{F} \cdot d\mathbf{S} = 36\pi + 0 + 0 = 36\pi$.

5. $\mathbf{F}(x, y, z) = xye^z \mathbf{i} + xy^2z^3 \mathbf{j} - ye^z \mathbf{k} \Rightarrow$

$\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(xye^z) + \frac{\partial}{\partial y}(xy^2z^3) + \frac{\partial}{\partial z}(-ye^z) = ye^z + 2xyz^3 - ye^z = 2xyz^3$, so by the Divergence Theorem,

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E \operatorname{div} \mathbf{F} dV = \int_0^3 \int_0^2 \int_0^1 2xyz^3 dz dy dx = 2 \int_0^3 x dx \int_0^2 y dy \int_0^1 z^3 dz \\ &= 2 \left[\frac{1}{2} x^2 \right]_0^3 \left[\frac{1}{2} y^2 \right]_0^2 \left[\frac{1}{4} z^4 \right]_0^1 = 2 \left(\frac{9}{2} \right) (2) \left(\frac{1}{4} \right) = \frac{9}{2}\end{aligned}$$

$$6. \mathbf{F}(x, y, z) = x^2yz \mathbf{i} + xy^2z \mathbf{j} + xyz^2 \mathbf{k} \Rightarrow$$

$\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(x^2yz) + \frac{\partial}{\partial y}(xy^2z) + \frac{\partial}{\partial z}(xyz^2) = 2xyz + 2xyz + 2xyz = 6xyz$, so by the Divergence Theorem,

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E \operatorname{div} \mathbf{F} dV = \int_0^a \int_0^b \int_0^c 6xyz dz dy dx = 6 \int_0^a x dx \int_0^b y dy \int_0^c z dz \\ &= 6 \left[\frac{1}{2}x^2 \right]_0^a \left[\frac{1}{2}y^2 \right]_0^b \left[\frac{1}{2}z^2 \right]_0^c = 6 \left(\frac{1}{2}a^2 \right) \left(\frac{1}{2}b^2 \right) \left(\frac{1}{2}c^2 \right) = \frac{3}{4}a^2b^2c^2 \end{aligned}$$

$$7. \mathbf{F}(x, y, z) = 3xy^2 \mathbf{i} + xe^z \mathbf{j} + z^3 \mathbf{k} \Rightarrow \operatorname{div} \mathbf{F} = 3y^2 + 0 + 3z^2, \text{ so using cylindrical coordinates with } y = r \cos \theta, \\ z = r \sin \theta, x = x \text{ we have, by the Divergence Theorem,}$$

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E (3y^2 + 3z^2) dV = \int_0^{2\pi} \int_0^1 \int_{-1}^2 (3r^2 \cos^2 \theta + 3r^2 \sin^2 \theta) r dx dr d\theta \\ &= 3 \int_0^{2\pi} d\theta \int_0^1 r^3 dr \int_{-1}^2 dx = 3 \left[\theta \right]_0^{2\pi} \left[\frac{1}{4}r^4 \right]_0^1 \left[x \right]_{-1}^2 = 3(2\pi) \left(\frac{1}{4} \right) (3) = \frac{9\pi}{2} \end{aligned}$$

$$8. \mathbf{F}(x, y, z) = (x^3 + y^3) \mathbf{i} + (y^3 + z^3) \mathbf{j} + (z^3 + x^3) \mathbf{k} \Rightarrow \operatorname{div} \mathbf{F} = 3x^2 + 3y^2 + 3z^2, \text{ so by the Divergence Theorem,}$$

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E 3(x^2 + y^2 + z^2) dV = \int_0^\pi \int_0^{2\pi} \int_0^2 3\rho^2 \cdot \rho^2 \sin \phi d\rho d\theta d\phi = 3 \int_0^\pi \sin \phi d\phi \int_0^{2\pi} d\theta \int_0^2 \rho^4 d\rho \\ &= 3 [-\cos \phi]_0^\pi \left[\theta \right]_0^{2\pi} \left[\frac{1}{5}\rho^5 \right]_0^2 = 3(2)(2\pi) \left(\frac{32}{5} \right) = \frac{384}{5}\pi \end{aligned}$$

$$9. \mathbf{F}(x, y, z) = xe^y \mathbf{i} + (z - e^y) \mathbf{j} - xy \mathbf{k} \Rightarrow \operatorname{div} \mathbf{F} = e^y + (-e^y) + 0 = 0, \text{ so by the Divergence Theorem,}$$

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E 0 dV = 0.$$

$$10. \mathbf{F}(x, y, z) = e^y \tan z \mathbf{i} + x^2y \mathbf{j} + e^x \cos y \mathbf{k} \Rightarrow \operatorname{div} \mathbf{F} = x^2, \text{ so by the Divergence Theorem,}$$

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E \operatorname{div} \mathbf{F} dV = \int_{-1}^1 \int_{-1}^1 \int_0^{2-x-y^3} x^2 dz dx dy = \int_{-1}^1 \int_{-1}^1 x^2(2-x-y^3) dx dy \\ &= \int_{-1}^1 \int_{-1}^1 (2x^2 - x^3 - x^2y^3) dx dy = \int_{-1}^1 \int_{-1}^1 (2x^2 - x^2y^3) dx dy \quad [x^3 \text{ is odd}] \\ &= \int_{-1}^1 \left[\frac{2}{3}x^3 - \frac{x^3}{3}y^3 \right]_{x=-1}^{x=1} dy = \int_{-1}^1 \left(\frac{4}{3} - \frac{2}{3}y^3 \right) dy = 2 \int_0^1 \frac{4}{3} dy \quad \left[\begin{array}{l} \frac{4}{3} \text{ is even,} \\ y^3 \text{ is odd} \end{array} \right] \\ &= \frac{8}{3} \end{aligned}$$

$$11. \mathbf{F}(x, y, z) = (2x^3 + y^3) \mathbf{i} + (y^3 + z^3) \mathbf{j} + 3y^2z \mathbf{k} \Rightarrow \operatorname{div} \mathbf{F} = 6x^2 + 3y^2 + 3y^2 = 6x^2 + 6y^2, \text{ so}$$

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E 6(x^2 + y^2) dV = \int_0^{2\pi} \int_0^1 \int_0^{1-r^2} 6r^2 \cdot r dz dr d\theta = \int_0^{2\pi} \int_0^1 6r^3(1-r^2) dr d\theta \\ &= \int_0^{2\pi} d\theta \int_0^1 (6r^3 - 6r^5) dr = \left[\theta \right]_0^{2\pi} \left[\frac{3}{2}r^4 - r^6 \right]_0^1 = 2\pi \left(\frac{3}{2} - 1 \right) = \pi \end{aligned}$$

$$12. \mathbf{F}(x, y, z) = (xy + 2xz) \mathbf{i} + (x^2 + y^2) \mathbf{j} + (xy - z^2) \mathbf{k}. \text{ For } x^2 + y^2 \leq 4 \text{ the plane } z = y - 2 \text{ is below the } xy\text{-plane, so the} \\ \text{solid } E \text{ bounded by } S \text{ is } E = \{(x, y, z) \mid x^2 + y^2 \leq 4, y - 2 \leq z \leq 0\}. \text{ Here } \operatorname{div} \mathbf{F} = y + 2z + 2y - 2z = 3y, \text{ so}$$

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E 3y dV = \int_0^{2\pi} \int_0^2 \int_{r \sin \theta - 2}^0 (3r \sin \theta) r dz dr d\theta \\ &= \int_0^{2\pi} \int_0^2 (3r^2 \sin \theta)(0 - r \sin \theta + 2) dr d\theta = \int_0^{2\pi} \int_0^2 (-3r^3 \sin^2 \theta + 6r^2 \sin \theta) dr d\theta \\ &= \int_0^{2\pi} \left[-\frac{3}{4}r^4 \sin^2 \theta + 2r^3 \sin \theta \right]_{r=0}^{r=2} d\theta = \int_0^{2\pi} (-12 \sin^2 \theta + 16 \sin \theta) d\theta \\ &= \left[-12 \left(\frac{1}{2}\theta - \frac{1}{4} \sin 2\theta \right) - 16 \cos \theta \right]_0^{2\pi} = -12\pi - 16 + 16 = -12\pi \end{aligned}$$

13. $\mathbf{F}(x, y, z) = x^2z\mathbf{i} + xz^3\mathbf{j} + y\ln(x+1)\mathbf{k} \Rightarrow \operatorname{div} \mathbf{F} = 2xz$, so by the Divergence Theorem,

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E \operatorname{div} \mathbf{F} \, dV = \int_0^4 \int_0^3 \int_0^{2-(1/2)x} 2xz \, dz \, dy \, dx = \int_0^4 \int_0^3 x(2 - \tfrac{1}{2}x)^2 \, dy \, dx \\ &= \int_0^4 \int_0^3 (4x - 2x^2 + \tfrac{1}{4}x^3) \, dy \, dx = 3 \int_0^4 (4x - 2x^2 + \tfrac{1}{4}x^3) \, dx \\ &= 3[2x^2 - \tfrac{2}{3}x^3 + \tfrac{1}{16}x^4]_0^4 = 3(32 - \tfrac{128}{3} + 16) = 16\end{aligned}$$

14. $\mathbf{F}(x, y, z) = (xy - z^2)\mathbf{i} + x^3\sqrt{z}\mathbf{j} + (xy + z^2)\mathbf{k} \Rightarrow \operatorname{div} \mathbf{F} = y + 2z$, so by the Divergence Theorem,

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E \operatorname{div} \mathbf{F} \, dV = \int_{-1}^1 \int_{y^2}^1 \int_0^{1-x} (y + 2z) \, dz \, dx \, dy \\ &= \int_{-1}^1 \int_{y^2}^1 [y(1-x) + (1-x)^2] \, dx \, dz = \int_{-1}^1 \int_{y^2}^1 (y - yx + 1 - 2x + x^2) \, dx \, dy \\ &= \int_{-1}^1 \left[yx - y\frac{x^2}{2} + x - x^2 + \frac{x^3}{3} \right]_{x=y^2}^{x=1} dy = \int_{-1}^1 \left(-\frac{y^6}{3} + \frac{y^5}{2} + y^4 - y^3 - y^2 + \frac{y}{2} + \frac{1}{3} \right) dy \\ &= 2 \int_0^1 \left(-\frac{y^6}{3} + y^4 - y^2 + \frac{1}{3} \right) dy = 2 \left[-\frac{y^7}{21} + \frac{y^5}{5} - \frac{y^3}{3} + \frac{y}{3} \right]_0^1 = \frac{32}{105}\end{aligned}$$

15. The tetrahedron has vertices $(0, 0, 0)$, $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$ and is described by

$$E = \left\{ (x, y, z) \mid 0 \leq x \leq a, 0 \leq y \leq b\left(1 - \frac{x}{a}\right), 0 \leq z \leq c\left(1 - \frac{x}{a} - \frac{y}{b}\right) \right\}$$

Here we have $\mathbf{F}(x, y, z) = z\mathbf{i} + y\mathbf{j} + zx\mathbf{k} \Rightarrow \operatorname{div} \mathbf{F} = 0 + 1 + x = x + 1$, so

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E (x+1) \, dV = \int_0^a \int_0^{b(1-\frac{x}{a})} \int_0^{c(1-\frac{x}{a}-\frac{y}{b})} (x+1) \, dz \, dy \, dx \\ &= \int_0^a \int_0^{b(1-\frac{x}{a})} (x+1) \left[c\left(1 - \frac{x}{a} - \frac{y}{b}\right) \right] dy \, dx \\ &= c \int_0^a (x+1) \left[\left(1 - \frac{x}{a}\right)y - \frac{1}{2b}y^2 \right]_{y=0}^{y=b(1-\frac{x}{a})} dx \\ &= c \int_0^a (x+1) \left[\left(1 - \frac{x}{a}\right) \cdot b\left(1 - \frac{x}{a}\right) - \frac{1}{2b} \cdot b^2\left(1 - \frac{x}{a}\right)^2 \right] dx = \frac{1}{2}bc \int_0^a (x+1) \left(1 - \frac{x}{a}\right)^2 dx \\ &= \frac{1}{2}bc \int_0^a \left(\frac{1}{a^2}x^3 + \frac{1}{a^2}x^2 - \frac{2}{a}x + 1 \right) dx \\ &= \frac{1}{2}bc \left[\frac{1}{4a^2}x^4 + \frac{1}{3a^2}x^3 - \frac{2}{a}x^2 + \frac{1}{2}x^2 - \frac{1}{a}x + x \right]_0^a \\ &= \frac{1}{2}bc \left(\frac{1}{4}a^2 + \frac{1}{3}a - \frac{2}{3}a^2 + \frac{1}{2}a^2 - a + a \right) = \frac{1}{2}bc \left(\frac{1}{12}a^2 + \frac{1}{3}a \right) = \frac{1}{24}abc(a+4)\end{aligned}$$

16. $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \Rightarrow |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2} \Rightarrow$

$$\mathbf{F}(x, y, z) = |\mathbf{r}|^2 \mathbf{r} = x(x^2 + y^2 + z^2)\mathbf{i} + y(x^2 + y^2 + z^2)\mathbf{j} + z(x^2 + y^2 + z^2)\mathbf{k} \Rightarrow$$

$\operatorname{div} \mathbf{F} = x \cdot 2x + (x^2 + y^2 + z^2) + y \cdot 2y + (x^2 + y^2 + z^2) + z \cdot 2z + (x^2 + y^2 + z^2) = 5(x^2 + y^2 + z^2)$. Then

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E 5(x^2 + y^2 + z^2) \, dV = \int_0^\pi \int_0^{2\pi} \int_0^R 5\rho^2 \cdot \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi \\ &= 5 \int_0^\pi \sin \phi \, d\phi \int_0^{2\pi} d\theta \int_0^R \rho^4 \, d\rho = 5 [-\cos \phi]_0^\pi [\theta]_0^{2\pi} \left[\frac{1}{5}\rho^5 \right]_0^R = 5(2)(2\pi) \left(\frac{1}{5}R^5 \right) = 4\pi R^5\end{aligned}$$

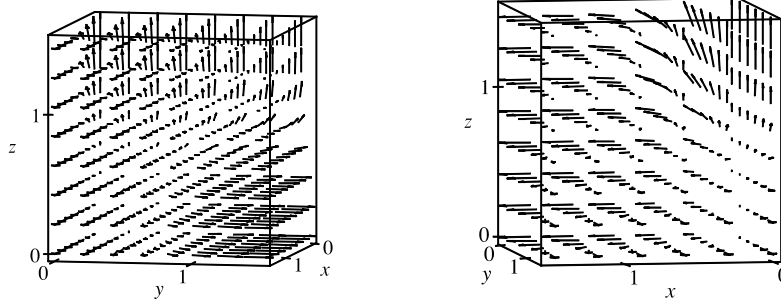
$$17. \mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \Rightarrow |\mathbf{r}| = \sqrt{x^2 + y^2 + z^2} \Rightarrow$$

$$\mathbf{F}(x, y, z) = |\mathbf{r}| \mathbf{r} = x\sqrt{x^2 + y^2 + z^2} \mathbf{i} + y\sqrt{x^2 + y^2 + z^2} \mathbf{j} + z\sqrt{x^2 + y^2 + z^2} \mathbf{k} \Rightarrow$$

$$\begin{aligned} \operatorname{div} \mathbf{F} &= x \cdot \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2}(2x) + (x^2 + y^2 + z^2)^{1/2} \\ &\quad + y \cdot \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2}(2y) + (x^2 + y^2 + z^2)^{1/2} \\ &\quad + z \cdot \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2}(2z) + (x^2 + y^2 + z^2)^{1/2} \\ &= (x^2 + y^2 + z^2)^{-1/2} [x^2 + (x^2 + y^2 + z^2) + y^2 + (x^2 + y^2 + z^2) + z^2 + (x^2 + y^2 + z^2)] \\ &= \frac{4(x^2 + y^2 + z^2)}{\sqrt{x^2 + y^2 + z^2}} = 4\sqrt{x^2 + y^2 + z^2} \end{aligned}$$

$$\begin{aligned} \text{Then } \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E 4\sqrt{x^2 + y^2 + z^2} dV = \int_0^{\pi/2} \int_0^{2\pi} \int_0^1 4\sqrt{\rho^2} \cdot \rho^2 \sin \phi d\rho d\theta d\phi \\ &= \int_0^{\pi/2} \sin \phi d\phi \int_0^{2\pi} d\theta \int_0^1 4\rho^3 d\rho = [-\cos \phi]_0^{\pi/2} [\theta]_0^{2\pi} [\rho^4]_0^1 = (1)(2\pi)(1) = 2\pi \end{aligned}$$

18.



By the Divergence Theorem, the flux of $\mathbf{F}$ across the surface of the cube is

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \int_0^{\pi/2} \int_0^{2\pi} \int_0^1 [\cos x \cos^2 y + 3 \sin^2 y \cos y \cos^4 z + 5 \sin^4 z \cos z \cos^6 x] dz dy dx = \frac{19}{64} \pi^2$$

19. For S_1 we have $\mathbf{n} = -\mathbf{k}$, so $\mathbf{F} \cdot \mathbf{n} = \mathbf{F} \cdot (-\mathbf{k}) = -x^2 z - y^2 = -y^2$ (since $z = 0$ on S_1). So if D is the unit disk, we get

$$\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_1} \mathbf{F} \cdot \mathbf{n} dS = \iint_D (-y^2) dA = -\int_0^{2\pi} \int_0^1 r^2 (\sin^2 \theta) r dr d\theta = -\frac{1}{4} \pi.$$

Now since S_2 is closed, we can use the Divergence Theorem. Since $\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(z^2 x) + \frac{\partial}{\partial y}(\frac{1}{3}y^3 + \tan^{-1} z) + \frac{\partial}{\partial z}(x^2 z + y^2) = z^2 + y^2 + x^2$, we use

spherical coordinates to get $\iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F} dV = \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 \rho^2 \cdot \rho^2 \sin \phi d\rho d\phi d\theta = \frac{2}{5} \pi$. Finally,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} - \iint_{S_1} \mathbf{F} \cdot d\mathbf{S} = \frac{2}{5} \pi - (-\frac{1}{4} \pi) = \frac{13}{20} \pi.$$

20. As in the hint to Exercise 19, we create a closed surface $S_2 = S \cup S_1$, where S is the part of the paraboloid $x^2 + y^2 + z = 2$

that lies above the plane $z = 1$, and S_1 is the disk $x^2 + y^2 = 1$ on the plane $z = 1$ oriented downward, and we then apply the

Divergence Theorem. Since the disk S_1 is oriented downward, its unit normal vector is $\mathbf{n} = -\mathbf{k}$ and $\mathbf{F} \cdot (-\mathbf{k}) = -z = -1$

on S_1 . So $\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_1} \mathbf{F} \cdot \mathbf{n} dS = \iint_{S_1} (-1) dS = -A(S_1) = -\pi$. Let E be the region bounded by S_2 . Then

$$\iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F} dV = \iiint_E 1 dV = \int_0^1 \int_0^{2\pi} \int_1^{2-r^2} r dz d\theta dr = \int_0^1 \int_0^{2\pi} (r - r^3) d\theta dr = (2\pi) \frac{1}{4} = \frac{\pi}{2}.$$

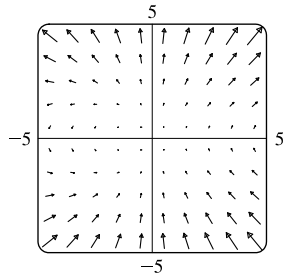
Thus the flux of $\mathbf{F}$ across S is $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} - \iint_{S_1} \mathbf{F} \cdot d\mathbf{S} = \frac{\pi}{2} - (-\pi) = \frac{3\pi}{2}$.

21. The vectors that end near P_1 are longer than the vectors that start near P_1 , so the net flow is inward near P_1 and $\operatorname{div} \mathbf{F}(P_1)$ is negative. The vectors that end near P_2 are shorter than the vectors that start near P_2 , so the net flow is outward near P_2 and $\operatorname{div} \mathbf{F}(P_2)$ is positive.

22. (a) The vectors that end near P_1 are shorter than the vectors that start near P_1 , so the net flow is outward and P_1 is a source. The vectors that end near P_2 are longer than the vectors that start near P_2 , so the net flow is inward and P_2 is a sink.

(b) $\mathbf{F}(x, y) = \langle x, y^2 \rangle \Rightarrow \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = 1 + 2y$. The y -value at P_1 is positive, so $\operatorname{div} \mathbf{F} = 1 + 2y$ is positive, thus P_1 is a source. At P_2 , $y < -1$, so $\operatorname{div} \mathbf{F} = 1 + 2y$ is negative, and P_2 is a sink.

23.



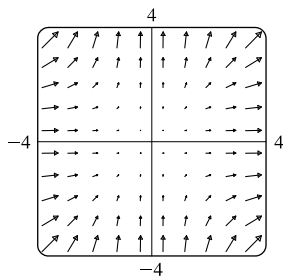
From the graph it appears that for points above the x -axis, vectors starting near a particular point are longer than vectors ending there, so divergence is positive.

The opposite is true at points below the x -axis, where divergence is negative.

$$\mathbf{F}(x, y) = \langle xy, x + y^2 \rangle \Rightarrow \operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(x + y^2) = y + 2y = 3y.$$

Thus $\operatorname{div} \mathbf{F} > 0$ for $y > 0$, and $\operatorname{div} \mathbf{F} < 0$ for $y < 0$.

24.



From the graph it appears that for points above the line $y = -x$, vectors starting near a particular point are longer than vectors ending there, so divergence is positive. The opposite is true at points below the line $y = -x$, where divergence is negative.

$$\mathbf{F}(x, y) = \langle x^2, y^2 \rangle \Rightarrow \operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(y^2) = 2x + 2y.$$

Then $\operatorname{div} \mathbf{F} > 0$ for $2x + 2y > 0 \Rightarrow y > -x$, and $\operatorname{div} \mathbf{F} < 0$ for $y < -x$.

25. Since $\frac{\mathbf{x}}{|\mathbf{x}|^3} = \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{(x^2 + y^2 + z^2)^{3/2}}$ and $\frac{\partial}{\partial x} \left(\frac{x}{(x^2 + y^2 + z^2)^{3/2}} \right) = \frac{(x^2 + y^2 + z^2) - 3x^2}{(x^2 + y^2 + z^2)^{5/2}}$ with similar expressions

for $\frac{\partial}{\partial y} \left(\frac{y}{(x^2 + y^2 + z^2)^{3/2}} \right)$ and $\frac{\partial}{\partial z} \left(\frac{z}{(x^2 + y^2 + z^2)^{3/2}} \right)$, we have

$$\operatorname{div} \left(\frac{\mathbf{x}}{|\mathbf{x}|^3} \right) = \frac{3(x^2 + y^2 + z^2) - 3(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^{5/2}} = 0, \text{ except at } (0, 0, 0) \text{ where it is undefined.}$$

26. We first need to find $\mathbf{F}$ so that $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_S (2x + 2y + z^2) \, dS$, so $\mathbf{F} \cdot \mathbf{n} = 2x + 2y + z^2$. But for S ,

$$\mathbf{n} = \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{\sqrt{x^2 + y^2 + z^2}} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}. \text{ Thus } \mathbf{F} = 2\mathbf{i} + 2\mathbf{j} + z\mathbf{k} \text{ and } \operatorname{div} \mathbf{F} = 1.$$

If $B = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$, then, by the Divergence Theorem,

$$\iint_S (2x + 2y + z^2) \, dS = \iiint_B \operatorname{div} \mathbf{F} \, dV = \iiint_B 1 \, dV = V(B) = \frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi$$

27. $\iint_S \mathbf{a} \cdot \mathbf{n} \, dS = \iiint_E \operatorname{div} \mathbf{a} \, dV = 0$ since $\operatorname{div} \mathbf{a} = 0$.

$$28. \frac{1}{3} \iint_S \mathbf{F} \cdot d\mathbf{S} = \frac{1}{3} \iiint_E \operatorname{div} \mathbf{F} \, dV = \frac{1}{3} \iiint_E 3 \, dV = V(E)$$

$$29. \iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div}(\operatorname{curl} \mathbf{F}) \, dV = 0 \text{ by Theorem 16.5.11.}$$

$$30. \iint_S D_{\mathbf{n}} f \, dS = \iint_S (\nabla f \cdot \mathbf{n}) \, dS = \iiint_E \operatorname{div}(\nabla f) \, dV = \iiint_E \nabla^2 f \, dV$$

$$\begin{aligned} 31. \iint_S (f \nabla g) \cdot \mathbf{n} \, dS &= \iiint_E \operatorname{div}(f \nabla g) \, dV \quad [\text{by (1) with } \mathbf{F} = f \nabla g] \\ &= \iiint_E [f \operatorname{div}(\nabla g) + \nabla g \cdot \nabla f] \, dV \quad [\text{by Exercise 16.5.27}] \\ &= \iiint_E (f \nabla^2 g + \nabla f \cdot \nabla g) \, dV \end{aligned}$$

$$32. \iint_S (f \nabla g - g \nabla f) \cdot \mathbf{n} \, dS = \iiint_E [(f \nabla^2 g + \nabla f \cdot \nabla g) - (g \nabla^2 f + \nabla g \cdot \nabla f)] \, dV \quad [\text{by Exercise 31}].$$

$$\text{But } \nabla g \cdot \nabla f = \nabla f \cdot \nabla g, \text{ so that } \iint_S (f \nabla g - g \nabla f) \cdot \mathbf{n} \, dS = \iiint_E (f \nabla^2 g - g \nabla^2 f) \, dV.$$

33. If $\mathbf{c} = c_1 \mathbf{i} + c_2 \mathbf{j} + c_3 \mathbf{k}$ is an arbitrary constant vector, we define $\mathbf{F} = f\mathbf{c} = f c_1 \mathbf{i} + f c_2 \mathbf{j} + f c_3 \mathbf{k}$. Then

$$\operatorname{div} \mathbf{F} = \operatorname{div} f\mathbf{c} = \frac{\partial f}{\partial x} c_1 + \frac{\partial f}{\partial y} c_2 + \frac{\partial f}{\partial z} c_3 = \nabla f \cdot \mathbf{c} \text{ and the Divergence Theorem says } \iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F} \, dV \Rightarrow$$

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_E \nabla f \cdot \mathbf{c} \, dV. \text{ In particular, if } \mathbf{c} = \mathbf{i} \text{ then } \iint_S f \mathbf{i} \cdot \mathbf{n} \, dS = \iiint_E \nabla f \cdot \mathbf{i} \, dV \Rightarrow$$

$$\iint_S f n_1 \, dS = \iiint_E \frac{\partial f}{\partial x} \, dV \text{ (where } \mathbf{n} = n_1 \mathbf{i} + n_2 \mathbf{j} + n_3 \mathbf{k}). \text{ Similarly, if } \mathbf{c} = \mathbf{j} \text{ we have } \iint_S f n_2 \, dS = \iiint_E \frac{\partial f}{\partial y} \, dV,$$

$$\text{and } \mathbf{c} = \mathbf{k} \text{ gives } \iint_S f n_3 \, dS = \iiint_E \frac{\partial f}{\partial z} \, dV. \text{ Then}$$

$$\begin{aligned} \iint_S f \mathbf{n} \, dS &= (\iint_S f n_1 \, dS) \mathbf{i} + (\iint_S f n_2 \, dS) \mathbf{j} + (\iint_S f n_3 \, dS) \mathbf{k} \\ &= \left(\iiint_E \frac{\partial f}{\partial x} \, dV \right) \mathbf{i} + \left(\iiint_E \frac{\partial f}{\partial y} \, dV \right) \mathbf{j} + \left(\iiint_E \frac{\partial f}{\partial z} \, dV \right) \mathbf{k} = \iiint_E \left(\frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} \right) dV \\ &= \iiint_E \nabla f \, dV \quad \text{as desired.} \end{aligned}$$

34. By Exercise 33, $\iint_S p \mathbf{n} \, dS = \iiint_E \nabla p \, dV$, so

$$\mathbf{F} = - \iint_S p \mathbf{n} \, dS = - \iiint_E \nabla p \, dV = - \iiint_E \nabla(\rho g z) \, dV = - \iiint_E (\rho g \mathbf{k}) \, dV = - \rho g (\iiint_E dV) \mathbf{k} = - \rho g V(E) \mathbf{k}.$$

But the weight of the displaced liquid is volume $\times$ density $\times g = \rho g V(E)$, thus $\mathbf{F} = -W\mathbf{k}$ as desired.

16 Review

TRUE-FALSE QUIZ

1. False. $\operatorname{div} \mathbf{F}$ is a scalar field.
2. True. See Definition 16.5.1.
3. True. Use Theorem 16.5.3 and the fact that $\operatorname{div} \mathbf{0} = 0$.
4. True. See Theorem 16.3.2.

5. False. See Exercise 16.3.41. (But the assertion is true if D is simply-connected; see Theorem 16.3.6.)
6. False. See the discussion accompanying Figure 8 in Section 16.2.
7. False. For example, $\operatorname{div}(y \mathbf{i}) = 0 = \operatorname{div}(x \mathbf{j})$ but $y \mathbf{i} \neq x \mathbf{j}$.
8. True. Line integrals of conservative vector fields are independent of path, and by Theorem 16.3.3, $\text{work} = \int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for any closed path C .
9. True. See Exercise 16.5.26.
10. False. $\mathbf{F} \cdot \mathbf{G}$ is a scalar field, so $\operatorname{curl}(\mathbf{F} \cdot \mathbf{G})$ has no meaning.
11. True. Apply the Divergence Theorem and use the fact that $\operatorname{div} \mathbf{F} = 0$ since $\mathbf{F}$ is a constant vector field.
12. False. See Theorem 16.5.11. If the statement were true, then $\operatorname{div} \operatorname{curl} \mathbf{F}$ would equal $1 + 1 + 1 = 3 \neq 0$.
13. False. By Formulas 16.4.5, the area is given by $-\oint_C y \, dx$ or $\oint_C x \, dy$.

EXERCISES

1. (a) Vectors starting on C point in roughly the direction opposite to C , so the tangential component $\mathbf{F} \cdot \mathbf{T}$ is negative.

Thus $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T} \, ds$ is negative.

- (b) The vectors that end near P are shorter than the vectors that start near P , so the net flow is outward near P and $\operatorname{div} \mathbf{F}(P)$ is positive.

2. We can parametrize C by $x = x$, $y = x^2$, $0 \leq x \leq 1$ so

$$\int_C x \, ds = \int_0^1 x \sqrt{1 + (2x)^2} \, dx = \frac{1}{12} (1 + 4x^2)^{3/2} \Big|_0^1 = \frac{1}{12} (5\sqrt{5} - 1).$$

3. $\int_C yz \cos x \, ds = \int_0^\pi (3 \cos t) (3 \sin t) \cos t \sqrt{(1)^2 + (-3 \sin t)^2 + (3 \cos t)^2} \, dt = \int_0^\pi (9 \cos^2 t \sin t) \sqrt{10} \, dt$
 $= 9\sqrt{10} \left(-\frac{1}{3} \cos^3 t\right) \Big|_0^\pi = -3\sqrt{10}(-2) = 6\sqrt{10}$

4. $x = 3 \cos t \Rightarrow dx = -3 \sin t \, dt$, $y = 2 \sin t \Rightarrow dy = 2 \cos t \, dt$, $0 \leq t \leq 2\pi$, so

$$\begin{aligned} \int_C y \, dx + (x + y^2) \, dy &= \int_0^{2\pi} [(2 \sin t)(-3 \sin t) + (3 \cos t + 4 \sin^2 t)(2 \cos t)] \, dt \\ &= \int_0^{2\pi} (-6 \sin^2 t + 6 \cos^2 t + 8 \sin^2 t \cos t) \, dt = \int_0^{2\pi} [6(\cos^2 t - \sin^2 t) + 8 \sin^2 t \cos t] \, dt \\ &= \int_0^{2\pi} (6 \cos 2t + 8 \sin^2 t \cos t) \, dt = 3 \sin 2t + \frac{8}{3} \sin^3 t \Big|_0^{2\pi} = 0 \end{aligned}$$

Or: Notice that $\frac{\partial}{\partial y}(y) = 1 = \frac{\partial}{\partial x}(x + y^2)$, so $\mathbf{F}(x, y) = \langle y, x + y^2 \rangle$ is a conservative vector field. Since C is a closed curve, $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C y \, dx + (x + y^2) \, dy = 0$.

$$\begin{aligned} 5. \int_C y^3 dx + x^2 dy &= \int_{-1}^1 [y^3(-2y) + (1-y^2)^2] dy = \int_{-1}^1 (-y^4 - 2y^2 + 1) dy \\ &= \left[-\frac{1}{5}y^5 - \frac{2}{3}y^3 + y\right]_{-1}^1 = -\frac{1}{5} - \frac{2}{3} + 1 - \frac{1}{5} - \frac{2}{3} + 1 = \frac{4}{15} \end{aligned}$$

$$\begin{aligned} 6. \int_C \sqrt{xy} dx + e^y dy + xz dz &= \int_0^1 \left(\sqrt{t^4 \cdot t^2} \cdot 4t^3 + e^{t^2} \cdot 2t + t^4 \cdot t^3 \cdot 3t^2 \right) dt = \int_0^1 (4t^6 + 2te^{t^2} + 3t^9) dt \\ &= \left[\frac{4}{7}t^7 + e^{t^2} + \frac{3}{10}t^{10} \right]_0^1 = e - \frac{9}{70} \end{aligned}$$

$$7. C: x = 1 + 2t \Rightarrow dx = 2dt, y = 4t \Rightarrow dy = 4dt, z = -1 + 3t \Rightarrow dz = 3dt, 0 \leq t \leq 1.$$

$$\begin{aligned} \int_C xy dx + y^2 dy + yz dz &= \int_0^1 [(1+2t)(4t)(2) + (4t)^2(4) + (4t)(-1+3t)(3)] dt \\ &= \int_0^1 (116t^2 - 4t) dt = \left[\frac{116}{3}t^3 - 2t^2 \right]_0^1 = \frac{116}{3} - 2 = \frac{110}{3} \end{aligned}$$

$$\begin{aligned} 8. \mathbf{F}(x, y) &= xy \mathbf{i} + x^2 \mathbf{j} \text{ and } \mathbf{r}(t) = \sin t \mathbf{i} + (1+t) \mathbf{j}, 0 \leq t \leq \pi \Rightarrow \mathbf{F}(\mathbf{r}(t)) = (\sin t)(1+t) \mathbf{i} + (\sin^2 t) \mathbf{j}, \\ \mathbf{r}'(t) &= \cos t \mathbf{i} + \mathbf{j} \text{ and} \end{aligned}$$

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^\pi ((1+t) \sin t \cos t + \sin^2 t) dt = \int_0^\pi \left(\frac{1}{2}(1+t) \sin 2t + \sin^2 t \right) dt \\ &= \left[\frac{1}{2} \left((1+t) \left(-\frac{1}{2} \cos 2t \right) + \frac{1}{4} \sin 2t \right) + \frac{1}{2}t - \frac{1}{4} \sin 2t \right]_0^\pi = \frac{\pi}{4} \end{aligned}$$

$$9. \mathbf{F}(x, y, z) = e^z \mathbf{i} + xz \mathbf{j} + (x+y) \mathbf{k} \text{ and } \mathbf{r}(t) = t^2 \mathbf{i} + t^3 \mathbf{j} - t \mathbf{k}, 0 \leq t \leq 1 \Rightarrow$$

$$\mathbf{F}(\mathbf{r}(t)) = e^{-t} \mathbf{i} + t^2(-t) \mathbf{j} + (t^2 + t^3) \mathbf{k}, \mathbf{r}'(t) = 2t \mathbf{i} + 3t^2 \mathbf{j} - \mathbf{k} \text{ and}$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 (2te^{-t} - 3t^5 - (t^2 + t^3)) dt = \left[-2te^{-t} - 2e^{-t} - \frac{1}{2}t^6 - \frac{1}{3}t^3 - \frac{1}{4}t^4 \right]_0^1 = \frac{11}{12} - \frac{4}{e}.$$

$$10. (a) \mathbf{F}(x, y, z) = z \mathbf{i} + x \mathbf{j} + y \mathbf{k} \text{ and } C: x = 3 - 3t, y = \frac{\pi}{2}t, z = 3t, 0 \leq t \leq 1. \text{ Then}$$

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 [3t \mathbf{i} + (3-3t) \mathbf{j} + \frac{\pi}{2}t \mathbf{k}] \cdot [-3 \mathbf{i} + \frac{\pi}{2} \mathbf{j} + 3 \mathbf{k}] dt = \int_0^1 \left[-9t + \frac{3\pi}{2} \right] dt = \frac{1}{2}(3\pi - 9).$$

$$\begin{aligned} (b) W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{\pi/2} (3 \sin t \mathbf{i} + 3 \cos t \mathbf{j} + t \mathbf{k}) \cdot (-3 \sin t \mathbf{i} + \mathbf{j} + 3 \cos t \mathbf{k}) dt \\ &= \int_0^{\pi/2} (-9 \sin^2 t + 3 \cos t + 3t \cos t) dt = \left[-9 \left(\frac{1}{2}t - \frac{1}{4} \sin 2t \right) + 3 \sin t + 3(t \sin t + \cos t) \right]_0^{\pi/2} \\ &= -\frac{9\pi}{4} + 3 + \frac{3\pi}{2} - 3 = -\frac{3\pi}{4} \end{aligned}$$

$$11. \mathbf{F}(x, y) = (1+xy)e^{xy} \mathbf{i} + (e^y + x^2e^{xy}) \mathbf{j} \Rightarrow \frac{\partial}{\partial y} [(1+xy)e^{xy}] = 2xe^{xy} + x^2ye^{xy} = \frac{\partial}{\partial x} [e^y + x^2e^{xy}] \text{ and the domain}$$

of $\mathbf{F}$ is $\mathbb{R}^2$, so $\mathbf{F}$ is conservative. Thus there exists a function f such that $\mathbf{F} = \nabla f$. Then $f_y(x, y) = e^y + x^2e^{xy}$ implies

$f(x, y) = e^y + xe^{xy} + g(x)$ and then $f_x(x, y) = xye^{xy} + e^{xy} + g'(x) = (1+xy)e^{xy} + g'(x)$. But

$f_x(x, y) = (1+xy)e^{xy}$, so $g'(x) = 0 \Rightarrow g(x) = K$. Thus $f(x, y) = e^y + xe^{xy} + K$ is a potential function for $\mathbf{F}$.

$$12. \mathbf{F}(x, y, z) = \sin y \mathbf{i} + x \cos y \mathbf{j} - \sin z \mathbf{k} \text{ is defined on all of } \mathbb{R}^3, \text{ its components have continuous partial derivatives, and}$$

$\text{curl } \mathbf{F} = (0-0) \mathbf{i} - (0-0) \mathbf{j} + (\cos y - \cos y) \mathbf{k} = \mathbf{0}$, so $\mathbf{F}$ is conservative by Theorem 16.5.4. Thus there exists a

function f such that $\nabla f = \mathbf{F}$. Then $f_x(x, y, z) = \sin y$ implies $f(x, y, z) = x \sin y + g(y, z)$ and then

$f_y(x, y, z) = x \cos y + g_y(y, z)$. But $f_y(x, y, z) = x \cos y$, so $g_y(y, z) = 0 \Rightarrow g(y, z) = h(z)$. Then

$f(x, y, z) = x \sin y + h(z)$ implies $f_z(x, y, z) = h'(z)$. But $f_z(x, y, z) = -\sin z$, so $h(z) = \cos z + K$. Thus a potential

function for $\mathbf{F}$ is $f(x, y, z) = x \sin y + \cos z + K$.

13. $\mathbf{F}(x, y) = (4x^3y^2 - 2xy^3)\mathbf{i} + (2x^4y - 3x^2y^2 + 4y^3)\mathbf{j} \Rightarrow$

$\frac{\partial}{\partial y}(4x^3y^2 - 2xy^3) = 8x^3y - 6xy^2 = \frac{\partial}{\partial x}(2x^4y - 3x^2y^2 + 4y^3)$ and the domain of $\mathbf{F}$ is $\mathbb{R}^2$, so $\mathbf{F}$ is conservative.

Furthermore, $f(x, y) = x^4y^2 - x^2y^3 + y^4$ is a potential function for $\mathbf{F}$. $t = 0$ corresponds to the point $(0, 1)$ and $t = 1$

corresponds to $(1, 1)$, so $\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, 1) - f(0, 1) = 1 - 1 = 0$.

14. $\mathbf{F}(x, y, z) = e^y\mathbf{i} + (xe^y + e^z)\mathbf{j} + ye^z\mathbf{k} \Rightarrow \text{curl } \mathbf{F} = (e^z - e^z)\mathbf{i} - (0 - 0)\mathbf{j} + (e^y - e^y)\mathbf{k} = \mathbf{0}$. The domain of $\mathbf{F}$ is $\mathbb{R}^3$

and the components of $\mathbf{F}$ have continuous partial derivatives, so $\mathbf{F}$ is conservative. Furthermore, we can show that

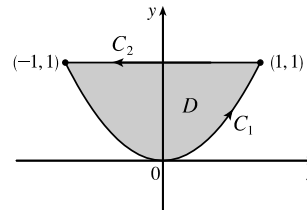
$f(x, y, z) = xe^y + ye^z$ is a potential function for $\mathbf{F}$. Then $\int_C \mathbf{F} \cdot d\mathbf{r} = f(4, 0, 3) - f(0, 2, 0) = 4 - 2 = 2$.

15. $C_1: \mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j}, -1 \leq t \leq 1;$

$C_2: \mathbf{r}(t) = -t\mathbf{i} + \mathbf{j}, -1 \leq t \leq 1$.

Then

$$\begin{aligned} \int_C xy^2 dx - x^2y dy &= \int_{-1}^1 [t(t^2)^2(1) - t^2(t^2)(2t)] dt \\ &\quad + \int_{-1}^1 [(-t)(1)^2(-1) - (-t)^2(1)(0)] dt \\ &= \int_{-1}^1 (-t^5) dt + \int_{-1}^1 t dt \\ &= 0 \quad [-t^5 \text{ and } t \text{ are odd}] \end{aligned}$$



Using Green's Theorem, we have

$$\begin{aligned} \int_C xy^2 dx - x^2y dy &= \iint_D \left[\frac{\partial}{\partial x}(-x^2y) - \frac{\partial}{\partial y}(xy^2) \right] dA = \iint_D (-2xy - 2xy) dA = \int_{-1}^1 \int_{x^2}^1 -4xy dy dx \\ &= \int_{-1}^1 [-2xy^2]_{y=x^2}^{y=1} dx = \int_{-1}^1 (2x^5 - 2x) dx = 0 \quad [2x^5 - 2x \text{ is odd}] \end{aligned}$$

16. $\int_C \sqrt{1+x^3} dx + 2xy dy = \iint_D \left[\frac{\partial}{\partial x}(2xy) - \frac{\partial}{\partial y}(\sqrt{1+x^3}) \right] dA = \int_0^1 \int_0^{3x} (2y - 0) dy dx = \int_0^1 9x^2 dx = [3x^3]_0^1 = 3$

17. $\int_C x^2y dx - xy^2 dy = \iint_{x^2+y^2 \leq 4} \left[\frac{\partial}{\partial x}(-xy^2) - \frac{\partial}{\partial y}(x^2y) \right] dA = \iint_{x^2+y^2 \leq 4} (-y^2 - x^2) dA = -\int_0^{2\pi} \int_0^2 r^3 dr d\theta = -8\pi$

18. $\mathbf{F}(x, y, z) = e^{-x} \sin y \mathbf{i} + e^{-y} \sin z \mathbf{j} + e^{-z} \sin x \mathbf{k} \Rightarrow$

$\text{curl } \mathbf{F} = (0 - e^{-y} \cos z)\mathbf{i} - (e^{-z} \cos x - 0)\mathbf{j} + (0 - e^{-x} \cos y)\mathbf{k} = -e^{-y} \cos z \mathbf{i} - e^{-z} \cos x \mathbf{j} - e^{-x} \cos y \mathbf{k}$, and

$\text{div } \mathbf{F} = -e^{-x} \sin y - e^{-y} \sin z - e^{-z} \sin x$.

19. If we assume there is such a vector field $\mathbf{G}$, then $\text{div}(\text{curl } \mathbf{G}) = \text{div}(2x\mathbf{i} + 3yz\mathbf{j} - xz^2\mathbf{k}) = 2 + 3z - 2xz$. Since

$\text{div}(\text{curl } \mathbf{F}) = 0$ for all vector fields $\mathbf{F}$ [by (16.5.11)], such a $\mathbf{G}$ cannot exist.

20. Let $\mathbf{F} = P_1 \mathbf{i} + Q_1 \mathbf{j} + R_1 \mathbf{k}$ and $\mathbf{G} = P_2 \mathbf{i} + Q_2 \mathbf{j} + R_2 \mathbf{k}$ be vector fields whose first partials exist and are continuous. Then

$$\begin{aligned} \mathbf{F} \operatorname{div} \mathbf{G} - \mathbf{G} \operatorname{div} \mathbf{F} = & \left[P_1 \left(\frac{\partial P_2}{\partial x} + \frac{\partial Q_2}{\partial y} + \frac{\partial R_2}{\partial z} \right) \mathbf{i} + Q_1 \left(\frac{\partial P_2}{\partial x} + \frac{\partial Q_2}{\partial y} + \frac{\partial R_2}{\partial z} \right) \mathbf{j} + R_1 \left(\frac{\partial P_2}{\partial x} + \frac{\partial Q_2}{\partial y} + \frac{\partial R_2}{\partial z} \right) \mathbf{k} \right] \\ & - \left[P_2 \left(\frac{\partial P_1}{\partial x} + \frac{\partial Q_1}{\partial y} + \frac{\partial R_1}{\partial z} \right) \mathbf{i} + Q_2 \left(\frac{\partial P_1}{\partial x} + \frac{\partial Q_1}{\partial y} + \frac{\partial R_1}{\partial z} \right) \mathbf{j} \right. \\ & \left. + R_2 \left(\frac{\partial P_1}{\partial x} + \frac{\partial Q_1}{\partial y} + \frac{\partial R_1}{\partial z} \right) \mathbf{k} \right] \end{aligned}$$

and

$$\begin{aligned} (\mathbf{G} \cdot \nabla) \mathbf{F} - (\mathbf{F} \cdot \nabla) \mathbf{G} = & \left[\left(P_2 \frac{\partial P_1}{\partial x} + Q_2 \frac{\partial P_1}{\partial y} + R_2 \frac{\partial P_1}{\partial z} \right) \mathbf{i} + \left(P_2 \frac{\partial Q_1}{\partial x} + Q_2 \frac{\partial Q_1}{\partial y} + R_2 \frac{\partial Q_1}{\partial z} \right) \mathbf{j} \right. \\ & \left. + \left(P_2 \frac{\partial R_1}{\partial x} + Q_2 \frac{\partial R_1}{\partial y} + R_2 \frac{\partial R_1}{\partial z} \right) \mathbf{k} \right] \\ & - \left[\left(P_1 \frac{\partial P_2}{\partial x} + Q_1 \frac{\partial P_2}{\partial y} + R_1 \frac{\partial P_2}{\partial z} \right) \mathbf{i} + \left(P_1 \frac{\partial Q_2}{\partial x} + Q_1 \frac{\partial Q_2}{\partial y} + R_1 \frac{\partial Q_2}{\partial z} \right) \mathbf{j} \right. \\ & \left. + \left(P_1 \frac{\partial R_2}{\partial x} + Q_1 \frac{\partial R_2}{\partial y} + R_1 \frac{\partial R_2}{\partial z} \right) \mathbf{k} \right] \end{aligned}$$

Hence

$$\begin{aligned} \mathbf{F} \operatorname{div} \mathbf{G} - \mathbf{G} \operatorname{div} \mathbf{F} + (\mathbf{G} \cdot \nabla) \mathbf{F} - (\mathbf{F} \cdot \nabla) \mathbf{G} &= \left[\left(P_1 \frac{\partial Q_2}{\partial y} + Q_2 \frac{\partial P_1}{\partial x} \right) - \left(P_2 \frac{\partial Q_1}{\partial y} + Q_1 \frac{\partial P_2}{\partial y} \right) \right. \\ & \quad \left. - \left(P_2 \frac{\partial R_1}{\partial z} + R_1 \frac{\partial P_2}{\partial z} \right) + \left(P_1 \frac{\partial R_2}{\partial z} + R_2 \frac{\partial P_1}{\partial z} \right) \right] \mathbf{i} \\ & + \left[\left(Q_1 \frac{\partial R_2}{\partial z} + R_2 \frac{\partial Q_1}{\partial z} \right) - \left(Q_2 \frac{\partial R_1}{\partial z} + R_1 \frac{\partial Q_2}{\partial z} \right) \right. \\ & \quad \left. - \left(P_1 \frac{\partial Q_2}{\partial x} + Q_2 \frac{\partial P_1}{\partial x} \right) + \left(P_2 \frac{\partial Q_1}{\partial x} + Q_1 \frac{\partial P_2}{\partial x} \right) \right] \mathbf{j} \\ & + \left[\left(P_2 \frac{\partial R_1}{\partial x} + R_1 \frac{\partial P_2}{\partial x} \right) - \left(P_1 \frac{\partial R_2}{\partial x} + R_2 \frac{\partial P_1}{\partial x} \right) \right. \\ & \quad \left. - \left(Q_1 \frac{\partial R_2}{\partial y} + R_2 \frac{\partial Q_1}{\partial y} \right) + \left(Q_2 \frac{\partial R_1}{\partial y} + R_1 \frac{\partial Q_2}{\partial y} \right) \right] \mathbf{k} \\ &= \left[\frac{\partial}{\partial y} (P_1 Q_2 - P_2 Q_1) - \frac{\partial}{\partial z} (P_2 R_1 - P_1 R_2) \right] \mathbf{i} \\ & + \left[\frac{\partial}{\partial z} (Q_1 R_2 - Q_2 R_1) - \frac{\partial}{\partial x} (P_1 Q_2 - P_2 Q_1) \right] \mathbf{j} \\ & + \left[\frac{\partial}{\partial x} (P_2 R_1 - P_1 R_2) - \frac{\partial}{\partial y} (Q_1 R_2 - Q_2 R_1) \right] \mathbf{k} \\ &= \operatorname{curl}(\mathbf{F} \times \mathbf{G}) \end{aligned}$$

21. For any piecewise-smooth simple closed plane curve C bounding a region D , we can apply Green's Theorem to

$$\mathbf{F}(x, y) = f(x) \mathbf{i} + g(y) \mathbf{j} \text{ to get } \int_C f(x) dx + g(y) dy = \iint_D \left[\frac{\partial}{\partial x} g(y) - \frac{\partial}{\partial y} f(x) \right] dA = \iint_D 0 dA = 0.$$

$$\begin{aligned}
22. \nabla^2(fg) &= \frac{\partial^2(fg)}{\partial x^2} + \frac{\partial^2(fg)}{\partial y^2} + \frac{\partial^2(fg)}{\partial z^2} \\
&= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} g + f \frac{\partial g}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} g + f \frac{\partial g}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial z} g + f \frac{\partial g}{\partial z} \right) \quad [\text{Product Rule}] \\
&= \frac{\partial^2 f}{\partial x^2} g + 2 \frac{\partial f}{\partial x} \frac{\partial g}{\partial x} + f \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} g + 2 \frac{\partial f}{\partial y} \frac{\partial g}{\partial y} \\
&\quad + f \frac{\partial^2 g}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} g + 2 \frac{\partial f}{\partial z} \frac{\partial g}{\partial z} + f \frac{\partial^2 g}{\partial z^2} \quad [\text{Product Rule}] \\
&= f \left(\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} + \frac{\partial^2 g}{\partial z^2} \right) + g \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \right) + 2 \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle \cdot \left\langle \frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right\rangle \\
&= f \nabla^2 g + g \nabla^2 f + 2 \nabla f \cdot \nabla g
\end{aligned}$$

Another method: Using the rules in Exercises 14.6.43(b), 16.5.25, and 16.5.27 [with $\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F}$], we have

$$\begin{aligned}
\nabla^2(fg) &= \nabla \cdot \nabla(fg) = \nabla \cdot (f \nabla g + g \nabla f) = f \nabla^2 g + \nabla g \cdot \nabla f + g \nabla^2 f + \nabla f \cdot \nabla g \\
&= f \nabla^2 g + g \nabla^2 f + 2 \nabla f \cdot \nabla g
\end{aligned}$$

23. $\nabla^2 f = 0$ means that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$. Now if $\mathbf{F} = f_y \mathbf{i} - f_x \mathbf{j}$ and C is any closed path in D , then applying Green's

Theorem, we get

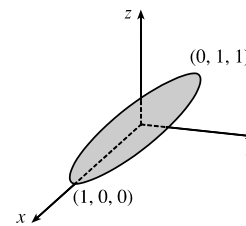
$$\begin{aligned}
\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C f_y dx - f_x dy = \iint_D \left[\frac{\partial}{\partial x}(-f_x) - \frac{\partial}{\partial y}(f_y) \right] dA \\
&= -\iint_D (f_{xx} + f_{yy}) dA = -\iint_D 0 dA = 0
\end{aligned}$$

Therefore the line integral is independent of path, by Theorem 16.3.3.

24. (a) $x^2 + y^2 = \cos^2 t + \sin^2 t = 1$, so C lies on the circular cylinder $x^2 + y^2 = 1$.

But also $y = z$, so C lies on the plane $y = z$. Thus C is contained in the

intersection of the plane $y = z$ and the cylinder $x^2 + y^2 = 1$; with $0 \leq t \leq 2\pi$ we get the entire intersection (an ellipse).



- (b) Apply Stokes' Theorem, $\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S}$:

$$\operatorname{curl} \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2xe^{2y} & 2x^2e^{2y} + 2y \cot z & -y^2 \csc^2 z \end{vmatrix} = \langle -2y \csc^2 z - (-2y \csc^2 z), 0, 4xe^{2y} - 4xe^{2y} \rangle = \mathbf{0}$$

Therefore $\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \mathbf{0} \cdot d\mathbf{S} = 0$.

25. $z = f(x, y) = x^2 + 2y$ with $0 \leq x \leq 1$, $0 \leq y \leq 2x$. Thus

$$A(S) = \iint_D \sqrt{1 + 4x^2 + 4} dA = \int_0^1 \int_0^{2x} \sqrt{5 + 4x^2} dy dx = \int_0^1 2x \sqrt{5 + 4x^2} dx = \frac{1}{6} (5 + 4x^2)^{3/2} \Big|_0^1 = \frac{1}{6} (27 - 5\sqrt{5}).$$

26. (a) $\mathbf{r}(u, v) = v^2 \mathbf{i} - uv \mathbf{j} + u^2 \mathbf{k}$, $0 \leq u \leq 3$, $-3 \leq v \leq 3 \Rightarrow$ (b)

$$\mathbf{r}_u = -v \mathbf{j} + 2u \mathbf{k}, \mathbf{r}_v = 2v \mathbf{i} - u \mathbf{j} \text{ and}$$

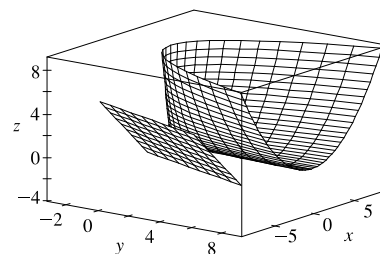
$$\mathbf{r}_u \times \mathbf{r}_v = 2u^2 \mathbf{i} + 4uv \mathbf{j} + 2v^2 \mathbf{k}. \text{ Since the point } (4, -2, 1)$$

corresponds to $u = 1$, $v = 2$ (or $u = -1$, $v = -2$ but $\mathbf{r}_u \times \mathbf{r}_v$

is the same for both), a normal vector to the surface at $(4, -2, 1)$

is $2\mathbf{i} + 8\mathbf{j} + 8\mathbf{k}$ and an equation of the tangent plane is

$$2x + 8y + 8z = 0 \text{ or } x + 4y + 4z = 0.$$



- (c) By Definition 16.6.6, the area of S is given by

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| dA = \int_0^3 \int_{-3}^3 \sqrt{(2u^2)^2 + (4uv)^2 + (2v^2)^2} dv du = 2 \int_0^3 \int_{-3}^3 \sqrt{u^4 + 4u^2v^2 + v^4} dv du.$$

- (d) $\mathbf{F}(x, y, z) = \frac{z^2}{1+x^2} \mathbf{i} + \frac{x^2}{1+y^2} \mathbf{j} + \frac{y^2}{1+z^2} \mathbf{k}$. By Equation 16.7.9, the surface integral is

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) dA = \int_0^3 \int_{-3}^3 \left\langle \frac{(u^2)^2}{1+(v^2)^2}, \frac{(v^2)^2}{1+(-uv)^2}, \frac{(-uv)^2}{1+(u^2)^2} \right\rangle \cdot \langle 2u^2, 4uv, 2v^2 \rangle dv du \\ &= \int_0^3 \int_{-3}^3 \left(\frac{2u^6}{1+v^4} + \frac{4uv^5}{1+u^2v^2} + \frac{2u^2v^4}{1+u^4} \right) dv du \approx 1524.0190 \end{aligned}$$

27. $z = g(x, y) = x^2 + y^2$ with $0 \leq x^2 + y^2 \leq 4$. By Formula 16.7.4,

$$\begin{aligned} \iint_S z dS &= \iint_{x^2+y^2 \leq 4} (x^2 + y^2) \sqrt{(2x)^2 + (2y)^2 + 1} dA = \int_0^{2\pi} \int_0^2 r^2 \sqrt{1+4r^2} r dr d\theta \\ &= \int_0^{2\pi} \int_1^{17} \left(\frac{u-1}{4} \right) \sqrt{u} \left(\frac{1}{8} du \right) d\theta \quad \left[\begin{matrix} u = 1 + 4r^2, \\ du = 8r dr \end{matrix} \right] = \frac{1}{32} (2\pi) \left[\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right]_1^{17} \\ &= \frac{\pi}{16} \cdot \frac{2}{15} \left[(3u^2 - 5u) \sqrt{u} \right]_1^{17} = \frac{\pi}{120} [782\sqrt{17} + 2] = \frac{1}{60} \pi (391\sqrt{17} + 1) \end{aligned}$$

28. $z = g(x, y) = 4 + x + y$ with $0 \leq x^2 + y^2 \leq 4$. By Formula 16.7.4,

$$\begin{aligned} \iint_S (x^2 z + y^2 z) dS &= \iint_{x^2+y^2 \leq 4} (x^2 + y^2)(4 + x + y) \sqrt{1^2 + 1^2 + 1} dA \\ &= \int_0^{2\pi} \int_0^2 \sqrt{3} r^3 (4 + r \cos \theta + r \sin \theta) d\theta dr = \int_0^2 8\pi \sqrt{3} r^3 dr = 32\pi \sqrt{3} \end{aligned}$$

29. $\mathbf{F}(x, y, z) = xz \mathbf{i} - 2y \mathbf{j} + 3x \mathbf{k}$. Since the sphere bounds a simple solid region, the Divergence Theorem applies and

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_E \operatorname{div} \mathbf{F} dV = \iiint_E (z - 2) dV = \iiint_E z dV - 2 \iiint_E dV \\ &= 0 \quad \left[\begin{matrix} \text{odd function in } z \\ \text{and } E \text{ is symmetric} \end{matrix} \right] - 2 \cdot V(E) = -2 \cdot \frac{4}{3} \pi (2)^3 = -\frac{64}{3} \pi \end{aligned}$$

Alternate solution: $\mathbf{F}(\mathbf{r}(\phi, \theta)) = 4 \sin \phi \cos \theta \cos \phi \mathbf{i} - 4 \sin \phi \sin \theta \mathbf{j} + 6 \sin \phi \cos \theta \mathbf{k}$,

$\mathbf{r}_\phi \times \mathbf{r}_\theta = 4 \sin^2 \phi \cos \theta \mathbf{i} + 4 \sin^2 \phi \sin \theta \mathbf{j} + 4 \sin \phi \cos \phi \mathbf{k}$, and

$\mathbf{F} \cdot (\mathbf{r}_\phi \times \mathbf{r}_\theta) = 16 \sin^3 \phi \cos^2 \theta \cos \phi - 16 \sin^3 \phi \sin^2 \theta + 24 \sin^2 \phi \cos \phi \cos \theta$. Then

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \int_0^{2\pi} \int_0^\pi (16 \sin^3 \phi \cos \phi \cos^2 \theta - 16 \sin^3 \phi \sin^2 \theta + 24 \sin^2 \phi \cos \phi \cos \theta) d\phi d\theta \\ &= \int_0^{2\pi} \frac{4}{3} (-16 \sin^2 \theta) d\theta = -\frac{64}{3} \pi \end{aligned}$$

30. $\mathbf{F}(x, y, z) = x^2 \mathbf{i} + xy \mathbf{j} + z \mathbf{k}$, $z = f(x, y) = x^2 + y^2$, $\mathbf{r}(x, y) = x \mathbf{i} + y \mathbf{j} + (x^2 + y^2) \mathbf{k}$, $\mathbf{r}_x \times \mathbf{r}_y = -2x \mathbf{i} - 2y \mathbf{j} + \mathbf{k}$ (because of upward orientation) and $\mathbf{F}(\mathbf{r}(x, y)) \cdot (\mathbf{r}_x \times \mathbf{r}_y) = -2x^3 - 2xy^2 + x^2 + y^2$. Then, by Formula 16.7.9,

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_{x^2+y^2 \leq 1} (-2x^3 - 2xy^2 + x^2 + y^2) dA \\ &= \int_0^1 \int_0^{2\pi} (-2r^3 \cos^3 \theta - 2r^3 \cos \theta \sin^2 \theta + r^2) r dr d\theta = \int_0^1 r^3 (2\pi) dr = \frac{\pi}{2} \end{aligned}$$

31. $F(x, y, z) = x^2 \mathbf{i} + y^2 \mathbf{j} + z^2 \mathbf{k}$. Since $\text{curl } \mathbf{F} = \mathbf{0}$, $\iint_S (\text{curl } \mathbf{F}) \cdot d\mathbf{S} = 0$. We parametrize

C : $\mathbf{r}(t) = \cos t \mathbf{i} + \sin t \mathbf{j}$, $0 \leq t \leq 2\pi$. Then $\mathbf{r}'(t) = -\sin t \mathbf{i} + \cos t \mathbf{j}$ and

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^{2\pi} (-\cos^2 t \sin t + \sin^2 t \cos t) dt = \left[\frac{1}{3} \cos^3 t + \frac{1}{3} \sin^3 t \right]_0^{2\pi} = 0.$$

32. By Stokes' Theorem, $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \oint_C \mathbf{F} \cdot d\mathbf{r}$, where C is $\mathbf{r}(t) = 2 \cos t \mathbf{i} + 2 \sin t \mathbf{j} + \mathbf{k}$, $0 \leq t \leq 2\pi$.

So $\mathbf{r}'(t) = -2 \sin t \mathbf{i} + 2 \cos t \mathbf{j}$, $\mathbf{F}(\mathbf{r}(t)) = 8 \cos^2 t \sin t \mathbf{i} + 2 \sin t \mathbf{j} + e^{4 \cos t \sin t} \mathbf{k}$, and

$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = -16 \cos^2 t \sin^2 t + 4 \sin t \cos t$. Thus,

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{2\pi} (-16 \cos^2 t \sin^2 t + 4 \sin t \cos t) dt \\ &= \left[-16 \left(-\frac{1}{4} \sin t \cos^3 t + \frac{1}{16} \sin 2t + \frac{1}{8} t \right) + 2 \sin^2 t \right]_0^{2\pi} = -4\pi \end{aligned}$$

33. $\mathbf{F}(x, y, z) = xy \mathbf{i} + yz \mathbf{j} + zx \mathbf{k}$. The surface is given by $x + y + z = 1$ or $z = 1 - x - y$, $0 \leq x \leq 1$,

$0 \leq y \leq 1 - x$. $\mathbf{r}(x, y) = x \mathbf{i} + y \mathbf{j} + (1 - x - y) \mathbf{k}$ and $\mathbf{r}_x \times \mathbf{r}_y = \mathbf{i} + \mathbf{j} + \mathbf{k}$. Then, by Formula 16.7.9,

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_D (-y \mathbf{i} - z \mathbf{j} - x \mathbf{k}) \cdot (\mathbf{i} + \mathbf{j} + \mathbf{k}) dA \\ &= \iint_D (-1) dA = -(\text{area of } D) = -\frac{1}{2}. \end{aligned}$$

34. By the Divergence Theorem,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E 3(x^2 + y^2 + z^2) dV = \int_0^{2\pi} \int_0^1 \int_0^2 (3r^2 + 3z^2) r dz dr d\theta = 2\pi \int_0^1 (6r^3 + 8r) dr = 11\pi.$$

35. $\iiint_E \text{div } \mathbf{F} dV = \iiint_{x^2+y^2+z^2 \leq 1} 3 dV = 3(\text{volume of sphere}) = 4\pi$. Then

$\mathbf{F}(\mathbf{r}(\phi, \theta)) \cdot (\mathbf{r}_\phi \times \mathbf{r}_\theta) = \sin^3 \phi \cos^2 \theta + \sin^3 \phi \sin^2 \theta + \sin \phi \cos^2 \phi = \sin \phi$ and

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \int_0^{2\pi} \int_0^\pi \sin \phi d\phi d\theta = (2\pi)(2) = 4\pi.$$

36. Here we must use Equation 16.9.7 since $\mathbf{F}$ is not defined at the origin. Let S_1 be the sphere of radius 1 with center at the origin and outer unit normal $\mathbf{n}_1 = \mathbf{r}/|\mathbf{r}|$. Let S_2 be the surface of the ellipsoid with outer unit normal $\mathbf{n}_2$ and let E be the solid region between S_1 and S_2 . Then the outward flux of $\mathbf{F}$ through the ellipsoid is given by

$$\iint_{S_2} \mathbf{F} \cdot \mathbf{n}_2 dS = -\iint_{S_1} \mathbf{F} \cdot (-\mathbf{n}_1) dS + \iiint_E \text{div } \mathbf{F} dV. \text{ But } \mathbf{F} = \mathbf{r}/|\mathbf{r}|^3 \text{ with } \mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k} \text{ and } \mathbf{r} = |\mathbf{r}|, \text{ so}$$

$$\text{div } \mathbf{F} = \nabla \cdot (|\mathbf{r}|^{-3} \mathbf{r}) = |\mathbf{r}|^{-3} (\nabla \cdot \mathbf{r}) + \mathbf{r} \cdot (\nabla |\mathbf{r}|^{-3}) = |\mathbf{r}|^{-3} (3) + \mathbf{r} \cdot (-3|\mathbf{r}|^{-4}) (\mathbf{r} |\mathbf{r}|^{-1}) = 0$$

[Exercises 16.5.32 and 16.5.33]. And $\mathbf{F} \cdot \mathbf{n}_1 = \frac{\mathbf{r}}{|\mathbf{r}|^3} \cdot \frac{\mathbf{r}}{|\mathbf{r}|} = |\mathbf{r}|^{-2} = 1$ on S_1 .

$$\text{Thus } \iint_{S_2} \mathbf{F} \cdot \mathbf{n}_2 dS = \iint_{S_1} dS + \iiint_E 0 dV = (\text{surface area of the unit sphere}) = 4\pi(1)^2 = 4\pi.$$

37. Because $\text{curl } \mathbf{F} = \mathbf{0}$, $\mathbf{F}$ is conservative, so there exists a function f such that $\nabla f = \mathbf{F}$. Then $f_x(x, y, z) = 3x^2yz - 3y$ implies $f(x, y, z) = x^3yz - 3xy + g(y, z) \Rightarrow f_y(x, y, z) = x^3z - 3x + g_y(y, z)$. But $f_y(x, y, z) = x^3z - 3x$, so $g(y, z) = h(z)$ and $f(x, y, z) = x^3yz - 3xy + h(z)$. Then $f_z(x, y, z) = x^3y + h'(z)$ but $f_z(x, y, z) = x^3y + 2z$, so $h(z) = z^2 + K$ and a potential function for $\mathbf{F}$ is $f(x, y, z) = x^3yz - 3xy + z^2$. Hence
- $$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(0, 3, 0) - f(0, 0, 2) = 0 - 4 = -4.$$

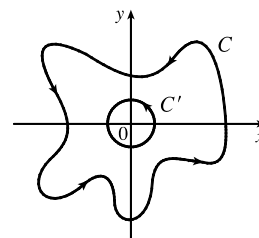
38. Let C' be the circle with center at the origin and radius a as in the figure.

Let D be the region bounded by C and C' . Then D 's positively oriented boundary is $C \cup (-C')$. Hence by Green's Theorem

$$\int_C \mathbf{F} \cdot d\mathbf{r} + \int_{-C'} \mathbf{F} \cdot d\mathbf{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = 0,$$

so

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= -\int_{-C'} \mathbf{F} \cdot d\mathbf{r} = \int_{C'} \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \quad [\mathbf{r}(t) = a \cos t \mathbf{i} + a \sin t \mathbf{j}] \\ &= \int_0^{2\pi} \left[\frac{2a^3 \cos^3 t + 2a^3 \cos t \sin^2 t - 2a \sin t}{a^2} (-a \sin t) + \frac{2a^3 \sin^3 t + 2a^3 \cos^2 t \sin t + 2a \cos t}{a^2} (a \cos t) \right] dt \\ &= \int_0^{2\pi} \frac{2a^2}{a^2} dt = 4\pi \end{aligned}$$



39. By the Divergence Theorem, $\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_E \text{div } \mathbf{F} dV = 3(\text{volume of } E) = 3(8 - 1) = 21$.
40. The stated conditions allow us to use the Divergence Theorem. Hence $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iiint_E \text{div}(\text{curl } \mathbf{F}) dV = 0$ since $\text{div}(\text{curl } \mathbf{F}) = 0$.
41. Let $\mathbf{F} = \mathbf{a} \times \mathbf{r} = \langle a_1, a_2, a_3 \rangle \times \langle x, y, z \rangle = \langle a_2z - a_3y, a_3x - a_1z, a_1y - a_2x \rangle$. Then $\text{curl } \mathbf{F} = \langle 2a_1, 2a_2, 2a_3 \rangle = 2\mathbf{a}$, and $\iint_S 2\mathbf{a} \cdot d\mathbf{S} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (\mathbf{a} \times \mathbf{r}) \cdot d\mathbf{r}$ by Stokes' Theorem.

□ PROBLEMS PLUS

1. Let S_1 be the portion of $\Omega(S)$ between $S(a)$ and S , and let ∂S_1 be its boundary. Also let S_L be the lateral surface of S_1 [that is, the surface of S_1 except S and $S(a)$]. Applying the Divergence Theorem we have $\iint_{\partial S_1} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3} dS = \iiint_{S_1} \nabla \cdot \frac{\mathbf{r}}{r^3} dV$.

But

$$\begin{aligned} \nabla \cdot \frac{\mathbf{r}}{r^3} &= \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \cdot \left\langle \frac{x}{(x^2 + y^2 + z^2)^{3/2}}, \frac{y}{(x^2 + y^2 + z^2)^{3/2}}, \frac{z}{(x^2 + y^2 + z^2)^{3/2}} \right\rangle \\ &= \frac{(x^2 + y^2 + z^2 - 3x^2) + (x^2 + y^2 + z^2 - 3y^2) + (x^2 + y^2 + z^2 - 3z^2)}{(x^2 + y^2 + z^2)^{5/2}} = 0 \end{aligned}$$

$$\Rightarrow \iint_{\partial S_1} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3} dS = \iiint_{S_1} 0 dV = 0. \text{ On the other hand, notice that for the surfaces of } \partial S_1 \text{ other than } S(a) \text{ and } S,$$

$$\mathbf{r} \cdot \mathbf{n} = 0 \Rightarrow$$

$$0 = \iint_{\partial S_1} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3} dS = \iint_S \frac{\mathbf{r} \cdot \mathbf{n}}{r^3} dS + \iint_{S(a)} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3} dS + \iint_{S_L} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3} dS = \iint_S \frac{\mathbf{r} \cdot \mathbf{n}}{r^3} dS + \iint_{S(a)} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3} dS \Rightarrow$$

$$\iint_S \frac{\mathbf{r} \cdot \mathbf{n}}{r^3} dS = - \iint_{S(a)} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3} dS. \text{ Notice that on } S(a), r = a \Rightarrow \mathbf{n} = -\frac{\mathbf{r}}{r} = -\frac{\mathbf{r}}{a} \text{ and } \mathbf{r} \cdot \mathbf{r} = r^2 = a^2, \text{ so}$$

$$\text{that } - \iint_{S(a)} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3} dS = \iint_{S(a)} \frac{\mathbf{r} \cdot \mathbf{r}}{a^4} dS = \iint_{S(a)} \frac{a^2}{a^4} dS = \frac{1}{a^2} \iint_{S(a)} dS = \frac{\text{area of } S(a)}{a^2} = |\Omega(S)|.$$

$$\text{Therefore } |\Omega(S)| = \iint_S \frac{\mathbf{r} \cdot \mathbf{n}}{r^3} dS.$$

2. By Green's Theorem

$$\int_C (y^3 - y) dx - 2x^3 dy = \iint_D \left[\frac{\partial(-2x^3)}{\partial x} - \frac{\partial(y^3 - y)}{\partial y} \right] dA = \iint_D (1 - 6x^2 - 3y^2) dA$$

Notice that for $6x^2 + 3y^2 > 1$, the integrand is negative. The integral has maximum value if it is evaluated only in the region where the integrand is positive, which is within the ellipse $6x^2 + 3y^2 = 1$. So the simple closed curve that gives a maximum value for the line integral is the ellipse $6x^2 + 3y^2 = 1$.

3. The given line integral $\frac{1}{2} \int_C (bz - cy) dx + (cx - az) dy + (ay - bx) dz$ can be expressed as $\int_C \mathbf{F} \cdot d\mathbf{r}$ if we define the vector field $\mathbf{F}$ by $\mathbf{F}(x, y, z) = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k} = \frac{1}{2}(bz - cy)\mathbf{i} + \frac{1}{2}(cx - az)\mathbf{j} + \frac{1}{2}(ay - bx)\mathbf{k}$. Then define S to be the planar interior of C , so S is an oriented, smooth surface. Stokes' Theorem says $\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS$.

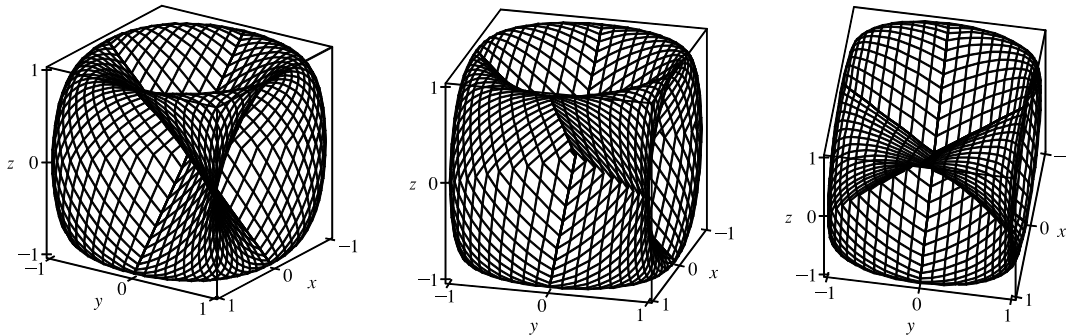
Now

$$\begin{aligned} \text{curl } \mathbf{F} &= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} \\ &= \left(\frac{1}{2}a + \frac{1}{2}a \right) \mathbf{i} + \left(\frac{1}{2}b + \frac{1}{2}b \right) \mathbf{j} + \left(\frac{1}{2}c + \frac{1}{2}c \right) \mathbf{k} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} = \mathbf{n} \end{aligned}$$

so $\text{curl } \mathbf{F} \cdot \mathbf{n} = \mathbf{n} \cdot \mathbf{n} = |\mathbf{n}|^2 = 1$, hence $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = \iint_S dS$ which is simply the surface area of S . Thus,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \frac{1}{2} \int_C (bz - cy) dx + (cx - az) dy + (ay - bx) dz \text{ is the plane area enclosed by } C.$$

4. The surface given by $x = \sin u$, $y = \sin v$, $z = \sin(u + v)$ is difficult to visualize, so we first graph the surface from three different points of view.



The trace in the horizontal plane $z = 0$ is given by $z = \sin(u + v) = 0 \Rightarrow u + v = k\pi$ [k an integer]. Then

we can write $v = k\pi - u$, and the trace is given by the parametric equations $x = \sin u$,

$y = \sin v = \sin(k\pi - u) = \sin k\pi \cos u - \cos k\pi \sin u = \pm \sin u$, and since $\sin u = x$, the trace consists of the two lines $y = \pm x$.

If $z = 1$, $z = \sin(u + v) = 1 \Rightarrow u + v = \frac{\pi}{2} + 2k\pi$. So $v = (\frac{\pi}{2} + 2k\pi) - u$ and the trace in $z = 1$ is given by the parametric equations $x = \sin u$, $y = \sin v = \sin((\frac{\pi}{2} + 2k\pi) - u) = \sin(\frac{\pi}{2} + 2k\pi) \cos u - \cos(\frac{\pi}{2} + 2k\pi) \sin u = \cos u$.

This curve is equivalent to $x^2 + y^2 = 1$, $z = 1$, a circle of radius 1. Similarly, in $z = -1$ we have $z = \sin(u + v) = -1 \Rightarrow$

$u + v = \frac{3\pi}{2} + 2k\pi \Rightarrow v = (\frac{3\pi}{2} + 2k\pi) - u$, so the trace is given by the parametric equations $x = \sin u$,

$y = \sin v = \sin((\frac{3\pi}{2} + 2k\pi) - u) = \sin(\frac{3\pi}{2} + 2k\pi) \cos u - \cos(\frac{3\pi}{2} + 2k\pi) \sin u = -\cos u$, which again is a circle,

$x^2 + y^2 = 1$, $z = -1$.

If $z = \frac{1}{2}$, $z = \sin(u + v) = \frac{1}{2} \Rightarrow u + v = \alpha + 2k\pi$ where $\alpha = \frac{\pi}{6}$ or $\frac{5\pi}{6}$. Then $v = (\alpha + 2k\pi) - u$ and the trace in $z = \frac{1}{2}$ is given by the parametric equations $x = \sin u$,

$y = \sin v = \sin[(\alpha + 2k\pi) - u] = \sin(\alpha + 2k\pi) \cos u - \cos(\alpha + 2k\pi) \sin u = \frac{1}{2} \cos u \pm \frac{\sqrt{3}}{2} \sin u$. In rectangular

coordinates, $x = \sin u$ so $y = \frac{1}{2} \cos u \pm \frac{\sqrt{3}}{2} x \Rightarrow y \pm \frac{\sqrt{3}}{2} x = \frac{1}{2} \cos u \Rightarrow 2y \pm \sqrt{3} x = \cos u$. But then

$x^2 + (2y \pm \sqrt{3} x)^2 = \sin^2 u + \cos^2 u = 1 \Rightarrow x^2 + 4y^2 \pm 4\sqrt{3} xy + 3x^2 = 1 \Rightarrow 4x^2 \pm 4\sqrt{3} xy + 4y^2 = 1$, which

may be recognized as a conic section. In particular, each equation is an ellipse rotated $\pm 45^\circ$ from the standard orientation (see

the following graph). The trace in $z = -\frac{1}{2}$ is similar: $z = \sin(u + v) = -\frac{1}{2} \Rightarrow u + v = \beta + 2k\pi$ where $\beta = \frac{7\pi}{6}$ or $\frac{11\pi}{6}$.

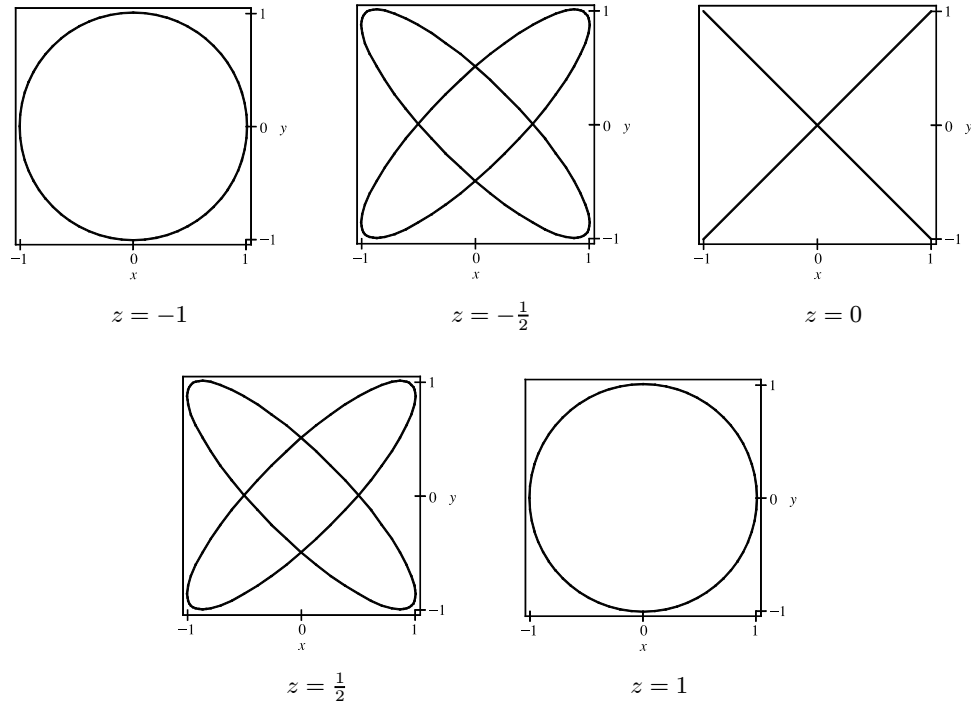
Then $v = (\beta + 2k\pi) - u$ and the trace is given by the parametric equations $x = \sin u$,

$y = \sin v = \sin[(\beta + 2k\pi) - u] = \sin(\beta + 2k\pi) \cos u - \cos(\beta + 2k\pi) \sin u = -\frac{1}{2} \cos u \pm \frac{\sqrt{3}}{2} \sin u$. If we convert to

rectangular coordinates, we arrive at the same pair of equations, $4x^2 \pm 4\sqrt{3} xy + 4y^2 = 1$, so the trace is identical to the trace

in $z = \frac{1}{2}$.

Graphing each of these, we have the following 5 traces.



Visualizing these traces on the surface reveals that horizontal cross sections are pairs of intersecting ellipses whose major axes are perpendicular to each other. At the bottom of the surface, $z = -1$, the ellipses coincide as circles of radius 1. As we move up the surface, the ellipses become narrower until at $z = 0$ they collapse into line segments, after which the process is reversed, and the ellipses widen to again coincide as circles at $z = 1$.

$$\begin{aligned}
 5. \quad (\mathbf{F} \cdot \nabla) \mathbf{G} &= \left(P_1 \frac{\partial}{\partial x} + Q_1 \frac{\partial}{\partial y} + R_1 \frac{\partial}{\partial z} \right) (P_2 \mathbf{i} + Q_2 \mathbf{j} + R_2 \mathbf{k}) \\
 &= \left(P_1 \frac{\partial P_2}{\partial x} + Q_1 \frac{\partial P_2}{\partial y} + R_1 \frac{\partial P_2}{\partial z} \right) \mathbf{i} + \left(P_1 \frac{\partial Q_2}{\partial x} + Q_1 \frac{\partial Q_2}{\partial y} + R_1 \frac{\partial Q_2}{\partial z} \right) \mathbf{j} \\
 &\quad + \left(P_1 \frac{\partial R_2}{\partial x} + Q_1 \frac{\partial R_2}{\partial y} + R_1 \frac{\partial R_2}{\partial z} \right) \mathbf{k} \\
 &= (\mathbf{F} \cdot \nabla P_2) \mathbf{i} + (\mathbf{F} \cdot \nabla Q_2) \mathbf{j} + (\mathbf{F} \cdot \nabla R_2) \mathbf{k}.
 \end{aligned}$$

Similarly, $(\mathbf{G} \cdot \nabla) \mathbf{F} = (\mathbf{G} \cdot \nabla P_1) \mathbf{i} + (\mathbf{G} \cdot \nabla Q_1) \mathbf{j} + (\mathbf{G} \cdot \nabla R_1) \mathbf{k}$. Then

$$\begin{aligned}
 \mathbf{F} \times \text{curl } \mathbf{G} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ P_1 & Q_1 & R_1 \\ \partial R_2 / \partial y - \partial Q_2 / \partial z & \partial P_2 / \partial z - \partial R_2 / \partial x & \partial Q_2 / \partial x - \partial P_2 / \partial y \end{vmatrix} \\
 &= \left(Q_1 \frac{\partial Q_2}{\partial x} - Q_1 \frac{\partial P_2}{\partial y} - R_1 \frac{\partial P_2}{\partial z} + R_1 \frac{\partial R_2}{\partial x} \right) \mathbf{i} + \left(R_1 \frac{\partial R_2}{\partial y} - R_1 \frac{\partial Q_2}{\partial z} - P_1 \frac{\partial Q_2}{\partial x} + P_1 \frac{\partial P_2}{\partial y} \right) \mathbf{j} \\
 &\quad + \left(P_1 \frac{\partial P_2}{\partial z} - P_1 \frac{\partial R_2}{\partial x} - Q_1 \frac{\partial R_2}{\partial y} + Q_1 \frac{\partial Q_2}{\partial z} \right) \mathbf{k}
 \end{aligned}$$

[continued]

and

$$\begin{aligned}\mathbf{G} \times \operatorname{curl} \mathbf{F} = & \left(Q_2 \frac{\partial Q_1}{\partial x} - Q_2 \frac{\partial P_1}{\partial y} - R_2 \frac{\partial P_1}{\partial z} + R_2 \frac{\partial R_1}{\partial x} \right) \mathbf{i} + \left(R_2 \frac{\partial R_1}{\partial y} - R_2 \frac{\partial Q_1}{\partial z} - P_2 \frac{\partial Q_1}{\partial x} + P_2 \frac{\partial P_1}{\partial y} \right) \mathbf{j} \\ & + \left(P_2 \frac{\partial P_1}{\partial z} - P_2 \frac{\partial R_1}{\partial x} - Q_2 \frac{\partial R_1}{\partial y} + Q_2 \frac{\partial Q_1}{\partial z} \right) \mathbf{k}.\end{aligned}$$

Then

$$\begin{aligned}(\mathbf{F} \cdot \nabla) \mathbf{G} + \mathbf{F} \times \operatorname{curl} \mathbf{G} = & \left(P_1 \frac{\partial P_2}{\partial x} + Q_1 \frac{\partial Q_2}{\partial x} + R_1 \frac{\partial R_2}{\partial x} \right) \mathbf{i} + \left(P_1 \frac{\partial P_2}{\partial y} + Q_1 \frac{\partial Q_2}{\partial y} + R_1 \frac{\partial R_2}{\partial y} \right) \mathbf{j} \\ & + \left(P_1 \frac{\partial P_2}{\partial z} + Q_1 \frac{\partial Q_2}{\partial z} + R_1 \frac{\partial R_2}{\partial z} \right) \mathbf{k}\end{aligned}$$

and

$$\begin{aligned}(\mathbf{G} \cdot \nabla) \mathbf{F} + \mathbf{G} \times \operatorname{curl} \mathbf{F} = & \left(P_2 \frac{\partial P_1}{\partial x} + Q_2 \frac{\partial Q_1}{\partial x} + R_2 \frac{\partial R_1}{\partial x} \right) \mathbf{i} + \left(P_2 \frac{\partial P_1}{\partial y} + Q_2 \frac{\partial Q_1}{\partial y} + R_2 \frac{\partial R_1}{\partial y} \right) \mathbf{j} \\ & + \left(P_2 \frac{\partial P_1}{\partial z} + Q_2 \frac{\partial Q_1}{\partial z} + R_2 \frac{\partial R_1}{\partial z} \right) \mathbf{k}.\end{aligned}$$

Hence

$$\begin{aligned}(\mathbf{F} \cdot \nabla) \mathbf{G} + \mathbf{F} \times \operatorname{curl} \mathbf{G} + (\mathbf{G} \cdot \nabla) \mathbf{F} + \mathbf{G} \times \operatorname{curl} \mathbf{F} \\ = & \left[\left(P_1 \frac{\partial P_2}{\partial x} + P_2 \frac{\partial P_1}{\partial x} \right) + \left(Q_1 \frac{\partial Q_2}{\partial x} + Q_2 \frac{\partial Q_1}{\partial y} \right) + \left(R_1 \frac{\partial R_2}{\partial x} + R_2 \frac{\partial R_1}{\partial x} \right) \right] \mathbf{i} \\ & + \left[\left(P_1 \frac{\partial P_2}{\partial y} + P_2 \frac{\partial P_1}{\partial y} \right) + \left(Q_1 \frac{\partial Q_2}{\partial y} + Q_2 \frac{\partial Q_1}{\partial y} \right) + \left(R_1 \frac{\partial R_2}{\partial y} + R_2 \frac{\partial R_1}{\partial y} \right) \right] \mathbf{j} \\ & + \left[\left(P_1 \frac{\partial P_2}{\partial z} + P_2 \frac{\partial P_1}{\partial z} \right) + \left(Q_1 \frac{\partial Q_2}{\partial z} + Q_2 \frac{\partial Q_1}{\partial z} \right) + \left(R_1 \frac{\partial R_2}{\partial z} + R_2 \frac{\partial R_1}{\partial z} \right) \right] \mathbf{k} \\ = & \nabla(P_1 P_2 + Q_1 Q_2 + R_1 R_2) = \nabla(\mathbf{F} \cdot \mathbf{G}).\end{aligned}$$

6. (a) First we place the piston on coordinate axes so the top of the cylinder is at the origin and $x(t) \geq 0$ is the distance from the top of the cylinder to the piston at time t . Let C_1 be the curve traced out by the piston during one four-stroke cycle, so C_1 is given by $\mathbf{r}(t) = x(t) \mathbf{i}$, $a \leq t \leq b$. (Thus, the curve lies on the positive x -axis and reverses direction several times.) The force on the piston is $AP(t) \mathbf{i}$, where A is the area of the top of the piston and $P(t)$ is the pressure in the cylinder at time t . As in Section 16.2, the work done on the piston is $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_a^b AP(t) \mathbf{i} \cdot x'(t) \mathbf{i} dt = \int_a^b AP(t) x'(t) dt$. Here, the volume of the cylinder at time t is $V(t) = Ax(t) \Rightarrow V'(t) = Ax'(t) \Rightarrow \int_a^b AP(t) x'(t) dt = \int_a^b P(t) V'(t) dt$. Since the curve C in the PV -plane corresponds to the values of P and V at time t , $a \leq t \leq b$, we have

$$W = \int_a^b AP(t) x'(t) dt = \int_a^b P(t) V'(t) dt = \int_C P dV$$

Another method: If we divide the time interval $[a, b]$ into n subintervals of equal length Δt , the amount of work done on the piston in the i th time interval is approximately $AP(t_i)[x(t_i) - x(t_{i-1})]$. Thus we estimate the total work done during

one cycle to be $\sum_{i=1}^n AP(t_i)[x(t_i) - x(t_{i-1})]$. If we allow $n \rightarrow \infty$, we have

$$\begin{aligned} W &= \lim_{n \rightarrow \infty} \sum_{i=1}^n AP(t_i)[x(t_i) - x(t_{i-1})] = \lim_{n \rightarrow \infty} \sum_{i=1}^n P(t_i)[Ax(t_i) - Ax(t_{i-1})] \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n P(t_i)[V(t_i) - V(t_{i-1})] = \int_C P dV \end{aligned}$$

(b) Let C_L be the lower loop of the curve C and C_U the upper loop. Then $C = C_L \cup C_U$. C_L is positively oriented, so from Formula 16.4.5 we know the area of the lower loop in the PV -plane is given by $-\oint_{C_L} P dV$. C_U is negatively oriented, so the area of the upper loop is given by $-\left(-\oint_{C_U} P dV\right) = \oint_{C_U} P dV$. From part (a),

$$\begin{aligned} W &= \int_C P dV = \int_{C_L \cup C_U} P dV = \oint_{C_L} P dV + \oint_{C_U} P dV \\ &= \oint_{C_U} P dV - \left(-\oint_{C_L} P dV\right), \end{aligned}$$

the difference of the areas enclosed by the two loops of C .

7. (a) For each value of $u = u_0$, $\mathbf{X}(u_0, v) = \mathbf{r}(u_0) + q \cos v \mathbf{N}(u_0) + q \sin v \mathbf{B}(u_0)$ is a circle of radius q that is perpendicular to the tangent vector, $\mathbf{r}'(u_0)$. Thus, the union of all circles, $a \leq u \leq b$ gives a $\text{Tube}(C, q)$ around the curve C .

(b) $\mathbf{X}(u, v) = \mathbf{r}(u) + q \cos v \mathbf{N}(u) + q \sin v \mathbf{B}(u)$, $a \leq u \leq b$, $0 \leq v \leq 2\pi$.

First, we find $\mathbf{X}_u(u, v)$ and $\mathbf{X}_v(u, v)$. Note that, as $\mathbf{r}(u)$ is parametrized with respect to arc length, $|\mathbf{r}'(u)| = 1$ and thus, $\mathbf{r}'(u) = \mathbf{T}(u)$. With the Frenet-Serret Formulas, this gives

$$\begin{aligned} \mathbf{X}_u(u, v) &= \mathbf{r}'(u) + q \cos v \mathbf{N}'(u) + q \sin v \mathbf{B}'(u) \\ &= \mathbf{T}(u) + q \cos v (-\kappa \mathbf{T}(u) + \tau \mathbf{B}(u)) + q \sin v (-\tau \mathbf{N}(u)) \\ &= \mathbf{T}(u) - q\kappa \cos v \mathbf{T}(u) + q\tau \cos v \mathbf{B}(u) - q\tau \sin v \mathbf{N}(u) \\ &= (1 - q\kappa \cos v) \mathbf{T}(u) + q\tau \cos v \mathbf{B}(u) - q\tau \sin v \mathbf{N}(u) \end{aligned}$$

$$\begin{aligned} \mathbf{X}_v(u, v) &= \mathbf{0} - q \sin v \mathbf{N}(u) + q \cos v \mathbf{B}(u) \\ &= -q \sin v \mathbf{N}(u) + q \cos v \mathbf{B}(u) \end{aligned}$$

Then

$$\begin{aligned} \mathbf{X}_u(u, v) \times \mathbf{X}_v(u, v) &= [(1 - q\kappa \cos v) \mathbf{T}(u) + q\tau \cos v \mathbf{B}(u) - q\tau \sin v \mathbf{N}(u)] \\ &\quad \times [-q \sin v \mathbf{N}(u) + q \cos v \mathbf{B}(u)] \\ &= (1 - q\kappa \cos v)(-q \sin v) \mathbf{T}(u) \times \mathbf{N}(u) + (1 - q\kappa \cos v)(q \cos v) \mathbf{T}(u) \times \mathbf{B}(u) \\ &\quad - q^2 \tau \cos v \sin v \mathbf{B}(u) \times \mathbf{N}(u) + q^2 \tau \cos^2 v \mathbf{B}(u) \times \mathbf{B}(u) \\ &\quad + q^2 \tau \sin^2 v \mathbf{N}(u) \times \mathbf{N}(u) - q^2 \tau \sin v \cos v \mathbf{N}(u) \times \mathbf{B}(u) \end{aligned}$$

The last four terms drop out since $\mathbf{N}(u) \times \mathbf{N}(u) = \mathbf{B}(u) \times \mathbf{B}(u) = \mathbf{0}$ and $\mathbf{N}(u) \times \mathbf{B}(u) = -(\mathbf{B}(u) \times \mathbf{N}(u))$, so

$$\mathbf{X}_u(u, v) \times \mathbf{X}_v(u, v) = q(1 - q\kappa \cos v)[- \sin v \mathbf{B}(u) - \cos v \mathbf{N}(u)] \quad [\mathbf{B} = \mathbf{T} \times \mathbf{N}, \mathbf{N} = -\mathbf{T} \times \mathbf{B}]$$

[continued]

Next,

$$\begin{aligned}
 |\mathbf{X}_u(u, v) \times \mathbf{X}_v(u, v)|^2 &= |q(1 - q\kappa \cos v)[- \sin v \mathbf{B}(u) - \cos v \mathbf{N}(u)]|^2 \\
 &= q^2(1 - q\kappa \cos v)^2 [(- \sin v)^2 |\mathbf{B}(u)|^2 + (- \cos v)^2 |\mathbf{N}(u)|^2] \\
 &\quad \text{[by the Pythagorean Theorem for vectors]} \\
 &= q^2(1 - q\kappa \cos v)^2 [\sin^2 v (1) + \cos^2 v (1)] = q^2(1 - q\kappa \cos v)^2
 \end{aligned}$$

Thus, $|\mathbf{X}_u(u, v) \times \mathbf{X}_v(u, v)| = q(1 - q\kappa \cos v)$. Therefore, the surface area of $\text{Tube}(C, q)$ is

$$\begin{aligned}
 S(q) &= \int_a^b \int_0^{2\pi} |\mathbf{X}_u(u, v) \times \mathbf{X}_v(u, v)| \, dv \, du = \int_a^b \int_0^{2\pi} q(1 - q\kappa \cos v) \, dv \, du \\
 &= q \int_a^b [v - q\kappa \sin v]_{v=0}^{v=2\pi} \, du = q \int_a^b 2\pi \, du = 2\pi q(b - a) = 2\pi qL \quad [\text{as } b - a = L]
 \end{aligned}$$

(c) Volume V of $\text{Tube}(C, r) = \int_0^r S(q) \, dq = \int_0^r 2\pi qL \, dq = [\pi q^2 L]_0^r = \pi r^2 L$

(d) First find the length L of the helix: $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle \Rightarrow \mathbf{r}'(t) = \langle -\sin t, \cos t, 1 \rangle \Rightarrow |\mathbf{r}'(t)| = \sqrt{2} \Rightarrow$

$L = \int_a^b |\mathbf{r}'(t)| \, dt = \int_0^{4\pi} \sqrt{2} \, dt = 4\sqrt{2}\pi$. Then by part (c), with $r = 0.2$ and $L = 4\sqrt{2}\pi$, we have

$V = \pi r^2 L = \pi(0.2)^2 (4\sqrt{2}\pi) = \frac{4}{25}\sqrt{2}\pi^2$.

(e) We create the torus by forming a tube of radius r around the circle of radius R . The length of the curve is $L = 2\pi R$ and by part (c), $V = \pi r^2 L = \pi r^2 (2\pi R) = 2\pi^2 r^2 R$.